A First-Principles Dynamic Model of a Rubber V-Belt CVT for General Transient Simulation

With Mechanical Actuation and Traction-Limited Torque Transfer

Kai Arseneau

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Abstract

This work develops a first-principles dynamic model for a mechanically actuated dry rubber V-belt continuously variable transmission (CVT) and its coupling to the vehicle drivetrain. The model is motivated by a persistent gap in the available literature: while quasi-static Baja-oriented models are common and detailed dynamic models exist for hydraulically actuated metal-belt CVTs, publicly available formulations for mechanically actuated rubber belt CVTs do not generally combine sheave inertia, flyweight actuation, torque-reactive secondary clamping, belt geometric closure, traction-limited torque transmission, and vehicle longitudinal dynamics within a single unified framework.

Starting from pulley geometry and belt-length closure, the derivation develops the CVT kinematics, the rotational dynamics of the primary and secondary pulleys, the axial equation of motion governing shift, and the torque-transfer closure required to distinguish sticking from traction-limited slip. The primary clutch model includes centrifugal flyweight actuation and spring preload, while the secondary model includes helix-generated torque feedback together with spring forces. A centrifugal belt correction is also derived to account for the reduction in effective normal force at speed. These elements are assembled into a closed nonlinear state-space model suitable for direct numerical time integration.

The resulting formulation is intended both as a simulation model and as a derivation document in which each major modeling step remains physically interpretable. Simulation results are used to examine whether the completed model behaves in mechanically sensible ways under transient launch conditions. Comparisons of launch shift trajectories, pulley axial-force balance, and torque admissibility across multiple tuning cases show a consistent mechanism chain from tuning changes, to clamping-force evolution, to stick–slip behavior, to overall launch outcome. The default tuning produces the most balanced launch response, combining strong early admissible coupling torque, controlled low-ratio retention, and a smoother main shift. More aggressive cases reveal renewed slip and visibly underdamped oscillatory transients during shift onset, highlighting both the usefulness of the model and the limitations introduced by idealized loss assumptions.

Overall, the formulation provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behavior during launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behavior remains traceable back to clear mechanical causes.

Contents

Nomenclature and Notation

All quantities are expressed in SI units unless otherwise stated. Angles are expressed in radians. Angular velocities are expressed in rad/s. Torques are expressed in N m, forces in N, lengths in m, and masses in kg.

Symbol Summary

Symbols used throughout the derivation are summarized in Table 1.

Symbol	Meaning
State and kinematics	
$\boldsymbol{x}(t)$	state vector
t	time
$\omega_p,\ \omega_s$	primary and secondary angular speed
$s,\ \dot{s},\ \ddot{s}$	shift position, velocity, and acceleration
$R,\ \dot{R}$	CVT ratio and ratio rate
v_b	belt transport speed
$v_b^*,\ T_b$	slip-branch target belt speed and belt-speed relaxation time constant
v_Δ	slip metric: primary-imposed minus secondary-imposed belt-line speed
Geometry	
$r_{p,\mathrm{outer}}(s),\ r_{s,\mathrm{outer}}(s)$	primary and secondary outer radius
$r_{p,\mathrm{eff}}(s),\ r_{s,\mathrm{eff}}(s)$	primary and secondary effective torque radius
$r_{p,\mathrm{outer,min}}$	minimum primary outer radius
$r_{s,\mathrm{outer,max}}$	maximum secondary outer radius
$s_{\mathrm{dz}},\ s_{\max}$	deadzone threshold and maximum shift
β	sheave face half-angle (radial reference)
C	pulley center distance
L_b	total belt length
ϕ	belt wrap angle used in distributed contact model
$r,\ \dot{r}$	local belt path radius and its time derivative
r_{cm}	belt element center-of-mass radius
Constraint and mapping functions	
$f(r_p, r_s, C)$	belt-length compatibility root function ($f = 0$)
$\mathrm{root}_{r_s}(\cdot)$	numerical root solve for secondary radius
$r_f(s),\ \dfrac{dr_f}{ds}$	primary flyweight radius map and gradient
$\theta_s(s),\ \dfrac{d\theta_s}{ds}$	secondary helix angle map and gradient

Symbol	Meaning
Forces and torques	
$F_{p,\mathrm{ax}},\ F_{s,\mathrm{ax}}$	primary and secondary axial mechanism force
$F_{c,p},\ F_{c,s}$	primary and secondary belt centrifugal axial contribution
F_{ax}	generic axial clamp force at one pulley contact
$\tau_{\mathrm{eng}},\ \tau_{\mathrm{load}}$	engine input torque and reflected load torque
$\tau_p,\ \tau_s$	coupling torque seen at primary and secondary
$\tau,\ \tau_c$	generic interface torque and applied coupling/friction torque
τ_{ns}	no-slip requested coupling torque
$\tau_-^{\mathrm{stick}},\ \tau_+^{\mathrm{stick}}$	aggregate lower and upper coupling-torque bounds for stick admissibility
$\tau_-^{\mathrm{slip}},\ \tau_+^{\mathrm{slip}}$	aggregate lower and upper coupling-torque bounds on slip branches
$\tau_{c,p,\pm}^{\mathrm{stick}},\ \tau_{c,s,\pm}^{\mathrm{stick}}$	primary/secondary pulley-wise stick-branch lower/upper bounds
$\tau_{c,p,\pm}^{\mathrm{slip}},\ \tau_{c,s,\pm}^{\mathrm{slip}}$	primary/secondary pulley-wise slip-branch lower/upper bounds
Mass/inertia and material parameters	
$I_p,\ I_s$	primary and secondary rotational inertia
m_f	effective flyweight mass
$m_{p,m},\ m_{s,m}$	equivalent primary and secondary translating mechanism mass
ρ_b	belt material density
A_b	belt cross-sectional area
μ	belt–pulley traction coefficient
$k_p,\ x_{p,0}$	primary spring stiffness and preload displacement
$k_{s,\theta},\ \theta_{s,0}$	secondary torsional stiffness and preload angle
$k_{s,x},\ x_{s,0}$	secondary axial stiffness and preload displacement

Table 1: Core symbols used throughout the derivation.

Constants and Parameters

All parameters in Table 2 are treated as constant values that do not vary through the model. They are either physical constants, fixed by the physical design of the CVT and vehicle or else are tuning parameters that are set and held constant during a given simulation run.

Symbol	Name	Unit	Description
Geometry and belt			
L_b	Belt length	m	Fixed total belt length
β	Sheave half-angle	rad	Cone half-angle used for radial/axial conversion
h_b	Belt thickness	m	Radial belt thickness
b_{out}	Belt outer width	m	Belt width at outer face
b_{in}	Belt inner width	m	Belt width at inner face
$r_{p,\mathrm{outer,min}}$	Primary minimum outer radius	m	Radius at fully open primary
$r_{s,\mathrm{outer,max}}$	Secondary maximum outer radius	m	Radius at fully closed secondary
s_{dz}	Shift deadzone threshold	m	Onset of effective radial motion

Symbol	Name	Unit	Description
$s_{\max}$	Maximum shift	m	Physical upper travel bound
Primary/secondary actuation			
m_f	Flyweight mass	kg	Effective flyweight mass
$r_{f,0}$	Initial flyweight radius	m	Flyweight radius offset at $s = 0$
k_p	Primary spring stiffness	N/m	Primary compression spring rate
$x_{p,0}$	Primary spring preload	m	Initial compression of primary spring
$k_{s,\theta}$	Secondary torsional stiffness	N m/rad	Torsional spring rate in helix assembly
$\theta_{s,0}$	Secondary torsional preload	rad	Initial torsional deflection preload
r_h	Helix reference radius	m	Effective cylindrical radius in helix kinematics
$k_{s,x}$	Secondary axial stiffness	N/m	Secondary axial compression spring rate
$x_{s,0}$	Secondary axial preload	m	Initial compression of secondary axial spring
Inertia and drivetrain			
I_{eng}	Engine inertia	kg m^2	Engine rotational inertia
I_{prim}	Primary pulley inertia	kg m^2	Primary CVT pulley inertia
I_p	Effective primary inertia	kg m^2	Implemented primary-side inertia in current setup
I_{sec}	Secondary pulley inertia	kg m^2	Secondary CVT pulley inertia
I_{gb}	Gearbox inertia	kg m^2	Gearbox/intermediate shaft inertia
I_{wheel}	Wheel inertia	kg m^2	Wheel/hub inertia about wheel axis
I_s	Effective secondary inertia	kg m^2	Lumped secondary-side inertia about secondary axis
G	Final drive ratio	1	Fixed ratio defined by $\omega_s = G\,\omega_w$
m	Vehicle mass	kg	Total vehicle mass
r_w	Wheel rolling radius	m	Effective tire rolling radius
Contact and environment			
ρ_b	Belt density	kg/m^3	Belt material density
μ_s	Belt–pulley static friction coefficient	1	Static traction coefficient used at impending slip
μ_k	Belt–pulley kinetic friction coefficient	1	Kinetic traction coefficient used on slip branches
C_{rr}	Rolling resistance coefficient	1	Tire-road rolling resistance coefficient
C_d	Drag coefficient	1	Aerodynamic drag coefficient
A_f	Frontal area	m^2	Vehicle aerodynamic reference area
ρ	Air density	kg/m^3	Ambient air density used in drag model
g	Gravitational acceleration	m/s^2	Gravity constant

Table 2: Canonical non-varying parameters used by the CVT model equations.

Chapter 1

Introduction & Background

1.1 Problem and Motivation

Mechanically actuated rubber V-belt continuously variable transmissions (CVTs) are widely used in compact powersport drivetrains, including snowmobiles, all-terrain vehicles (ATVs), and utility terrain vehicles (UTVs) (Messick, 2018), because they can vary the speed ratio automatically while transmitting power through a relatively simple mechanical assembly. Their visible layout is straightforward: two variable-radius pulleys are connected by a belt. Their transient behaviour, however, is not. The operating state of the transmission emerges from the interaction of its internal mechanisms, the belt–pulley contacts, and the mechanical conditions under which it operates.

Several coupled effects determine that response. Engine speed influences the centrifugal actuation of the primary pulley, while transmitted torque influences the torque-reactive response of the secondary pulley. Together, these mechanisms generate the clamping forces that press the belt into the sheave grooves. Clamping force determines the traction available at the belt–pulley contacts, which in turn determines how torque is carried and whether the belt and pulleys can remain coupled without gross slip. At the same time, pulley motion changes the belt radii and therefore the transmission ratio, which changes the speed and torque relationships between the two pulleys. These effects do not occur independently; each changes the conditions under which the others act.

This coupling makes the behaviour of a mechanically actuated CVT difficult to interpret from external measurements alone. A shift curve, pulley-speed trace, or observed loss of traction may reflect the combined effect of primary actuation, secondary reaction, belt traction, and shift motion. The underlying states responsible for that behaviour are largely internal to the transmission and are not usually available directly during testing. This is especially consequential because mechanically actuated CVTs are tuned by changing physical components such as springs, flyweights, ramps, and helices; determining the effect of a change can therefore require repeated hardware changes and testing (Skinner, 2020).

Addressing this problem requires a physically interpretable dynamic framework that preserves the connection between the internal CVT mechanisms and the behaviour they produce. A useful model should do more than reproduce an output trace: it should make it possible to follow how actuation, clamping, belt traction, shift motion, and inertia combine as the transmission responds to changing operating conditions. If an undesirable response can be traced to its mechanical cause, changes to the hardware or operating conditions can be made deliberately rather than through trial and error. The motivation is therefore not only to predict transient CVT behaviour, but to make that behaviour understandable enough to diagnose and improve.

The remainder of this chapter develops the context for that framework. Section 1.2 introduces the physical layout and operating principles of the belt–sheave CVT, including ratio change, traction, and mechanical actuation. Section 1.3 contains a literature review that organizes prior work by modelling emphasis and identifies the foundations available for mechanical actuation, belt contact, and transient CVT dynamics. Given this context, Section 1.4 then states the aim, scope, assumptions, and contributions of the present formulation.

1.2 Conceptual Overview of a Continuously Variable Transmission

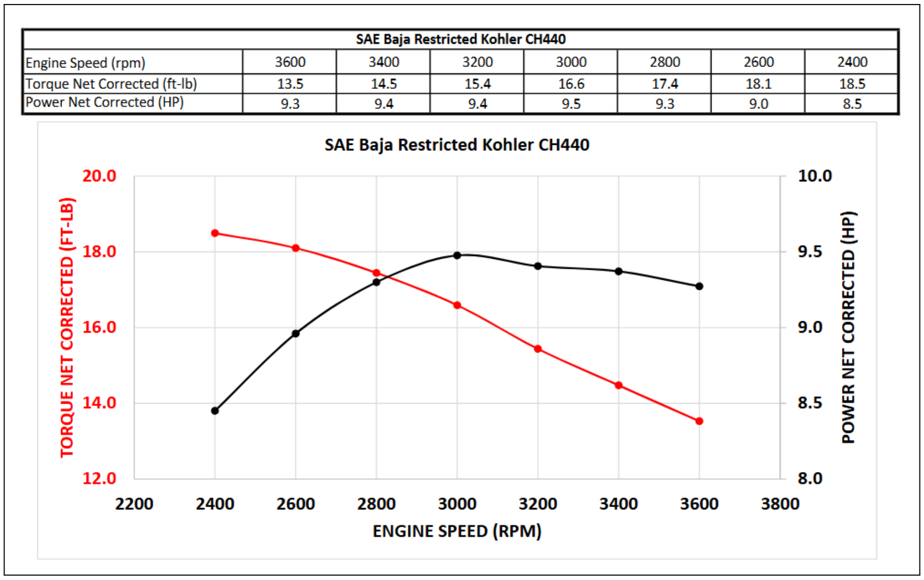
This section introduces the operating principles of the continuously variable transmission (CVT) considered in this work. The goal is not yet to derive the model equations, but to build the mechanical intuition needed to understand them. In particular, this section explains why a CVT is useful, how a rubber V-belt CVT changes ratio, why clamping and traction matter, how the primary and secondary pulleys generate their forces, and how these mechanisms interact to determine the transmission operating state.

1.2.1 Why Use a Continuously Variable Transmission?

A transmission is responsible for matching the operating speed of an engine to the speed required at the vehicle wheels. In a conventional stepped gearbox, this is achieved using a fixed set of discrete gear ratios. Each gear imposes a fixed relationship between engine speed and wheel speed. As the vehicle accelerates, the engine speed rises through one gear, drops when the transmission shifts, and then rises again in the next gear. This repeated sweep through the engine speed range is a direct consequence of having only a small number of available ratios.

This matters because an internal combustion engine (ICE) does not produce the same torque and power at every speed. Instead, the engine has a useful operating region where it can produce power more effectively, as shown conceptually in Figure 1.1. A stepped gearbox can pass through this region, but it cannot generally stay there while vehicle speed continues to increase. Figure 1.2 contrasts this stepped behavior with the smoother operating path made possible by a continuously variable transmission.

A continuously variable transmission (CVT) addresses this limitation by allowing the transmission ratio to vary smoothly within a finite range rather than changing through a fixed set of gear steps. Since the ratio can change continuously, the engine can remain closer to a desired operating region while the vehicle accelerates. This is the main benefit of a CVT: it gives the drivetrain more freedom to match engine behavior to vehicle load.



SAE Baja Restricted Kohler CH440							
Engine Speed (rpm)	3600	3400	3200	3000	2800	2600	2400
Torque Net Corrected (ft-lb)	13.5	14.5	15.4	16.6	17.4	18.1	18.5
Power Net Corrected (HP)	9.3	9.4	9.4	9.5	9.3	9.0	8.5

Figure 1.1: Torque and power of the Kohler CH440 engine as functions of engine speed. Both quantities vary substantially across the operating range, with peak torque occurring at a lower engine speed than peak power. (Baja SAE, 2022)

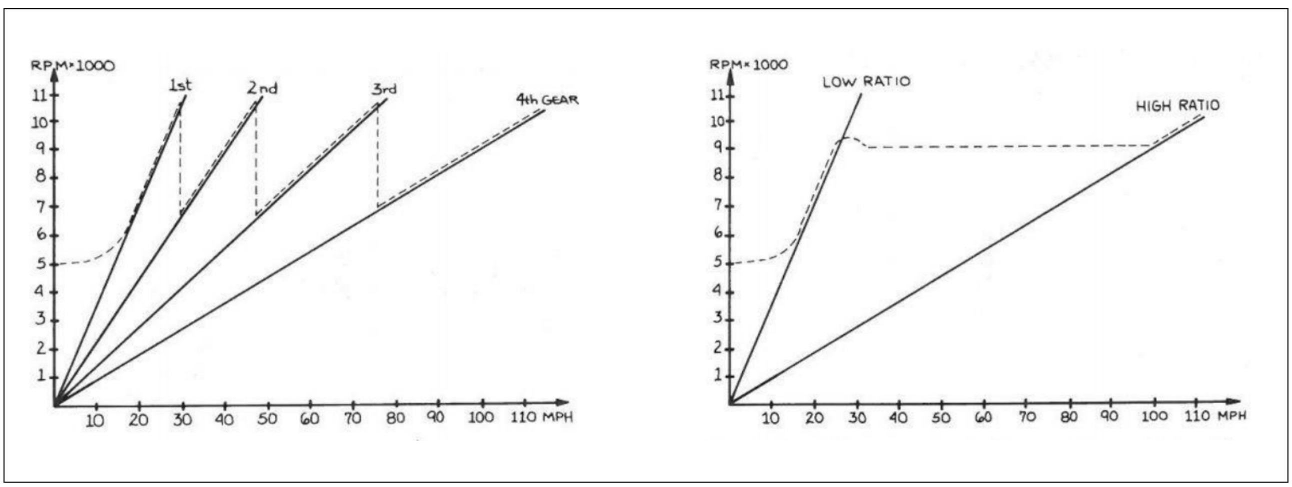


Figure 1.2: Conceptual comparison of engine-speed behavior during acceleration. Solid lines show the engine-speed–vehicle-speed relationship imposed by the available transmission ratios, while the dotted line illustrates a representative engine-speed trajectory during acceleration. With a stepped gearbox (left), the engine repeatedly sweeps through a speed range and drops in speed at each gear change. With a CVT (right), engine speed can rise to a desired operating speed and remain approximately constant while the transmission continuously changes ratio, increasing again only after the high-ratio limit is reached. (Aaen, 2007)

1.2.2 Physical Layout and Single-Pulley Geometry

There are several forms of continuously variable transmission, including different mechanical layouts and different ways of producing the variable speed relationship. This work focuses on a belt-sheave CVT, shown conceptually in Figure 1.3. In this arrangement, two variable-radius pulleys are connected by a rubber V-belt. The pulley connected to the engine side is called the primary (or driving) pulley. The pulley connected to the downstream drivetrain is called the secondary (or driven) pulley. The belt transmits motion and torque between these two pulleys.

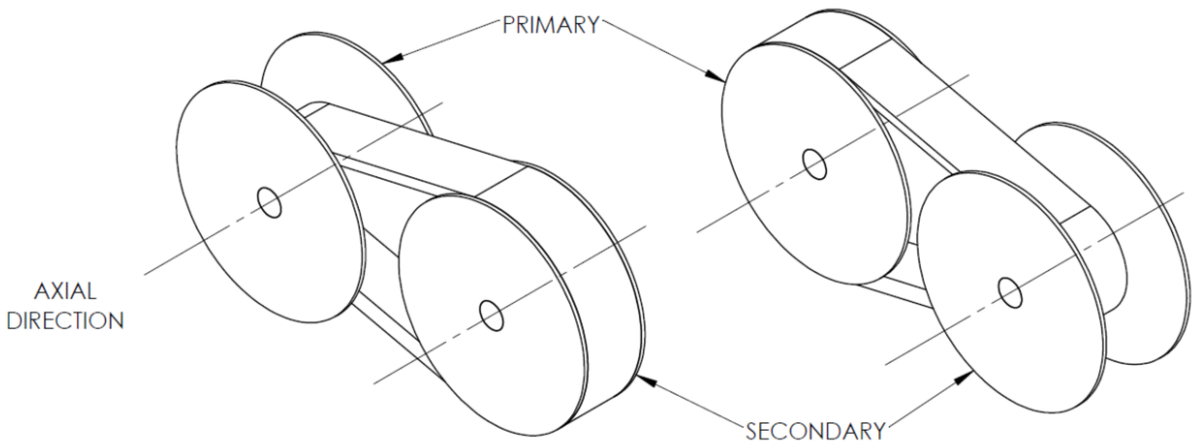


Figure 1.3: Two sample configurations of the belt-sheave CVT considered in this work. The primary pulley is connected to the engine side, the secondary pulley is connected to the downstream drivetrain, and the rubber V-belt transmits motion and torque between them. (Messick, 2018)

Before considering the full two-pulley system, it is helpful to look at the geometry of a single pulley. Both the primary and secondary pulleys share the same basic anatomy. Each pulley is made from two opposing conical sheaves that form a V-shaped groove. One sheave is fixed axially to the shaft, while the other can move along the shaft. The rubber V-belt sits between the sheave faces. This basic layout is shown in Figure 1.4.

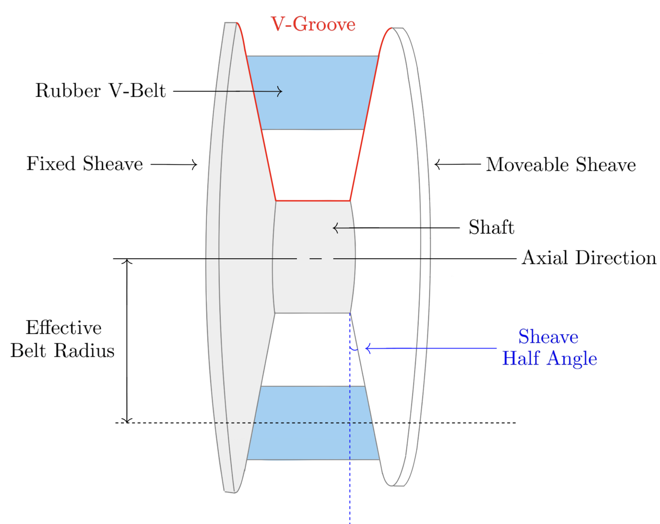


Figure 1.4: Generic anatomy of a belt-sheave CVT pulley. Each pulley consists of opposing conical sheaves that form a V-groove. Motion of the movable sheave changes the groove width and therefore changes the radius at which the belt rides.

The key geometric idea is simple: changing the axial separation between the sheaves changes the effective radius of the belt. If the movable sheave moves toward the fixed sheave, the groove becomes narrower. Since the belt has a finite width, it cannot occupy the same radial position in the narrower groove, so it is pushed outward along the conical sheave faces. The effective belt radius therefore increases.

If the movable sheave moves away from the fixed sheave, the groove becomes wider and the belt can fall deeper between the sheaves. This reduces the effective belt radius. At this level, the mechanism is just wedge geometry: axial motion of the sheave becomes radial motion of the belt. Closing the sheaves makes the belt radius larger, while opening the sheaves makes the belt radius smaller, as shown in Figure 1.5.

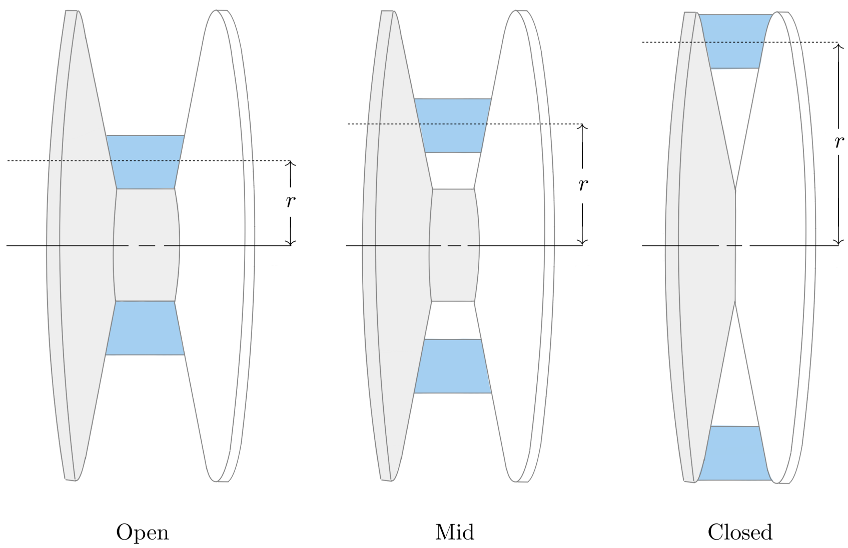


Figure 1.5: Single-pulley radius change. As the sheaves close, the V-groove narrows and the belt is pushed outward to a larger effective radius. As the sheaves open, the belt can fall deeper in the groove and the effective radius decreases.

1.2.3 Two-Pulley Coupling and Ratio Change

The full CVT contains two of the variable-radius pulleys described above. The important difference is that the two pulleys are connected by the same belt. Because the belt length is approximately fixed, the effective radii of the two pulleys cannot change independently.

If the belt rides outward on the primary pulley, it must ride inward on the secondary pulley. Likewise, if the primary radius decreases, the secondary radius must increase. This opposite motion is the core geometric behavior that allows the CVT ratio to change. In the physical mechanism, the actual motion of the sheaves occurs through contact forces, springs, actuator mechanisms, and the axial force balance of the pulleys.

The CVT begins launch in a low-speed, high-reduction configuration. In this state, the primary pulley has a small effective radius and the secondary pulley has a large effective radius. This behaves like a low gear in a conventional drivetrain: the engine-side pulley turns relatively quickly while the downstream side turns more slowly, amplifying torque through the drivetrain.

As the CVT upshifts, the primary sheaves close and the primary effective radius increases. At the same time, the secondary effective radius decreases. The transmission therefore moves from its launch configuration toward a higher-speed configuration. By the end of the shift, the primary radius is larger and the secondary radius is smaller. This coupled motion is shown in Figure 1.6.

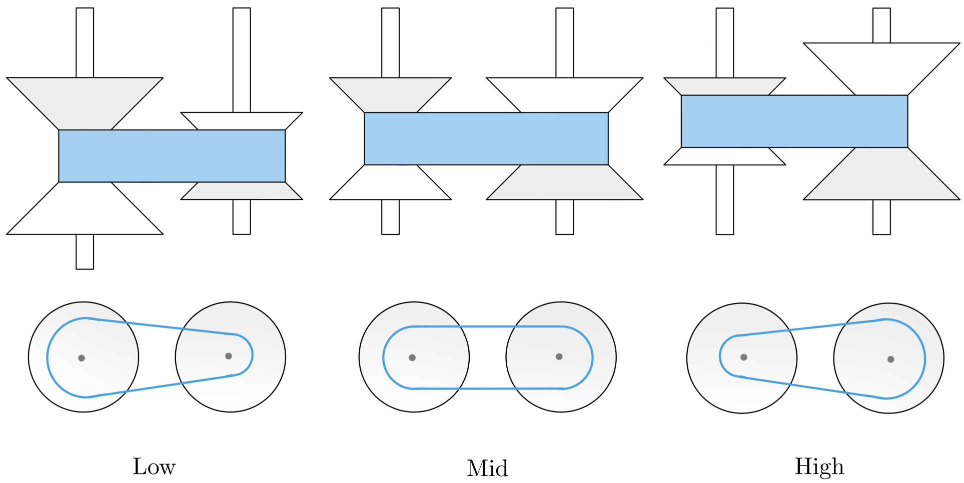


Figure 1.6: Low-, intermediate-, and high-ratio states of a belt–sheave CVT. As the belt moves outward on the primary pulley and inward on the secondary pulley, the effective radii vary in opposite directions while the belt length remains approximately constant. The lower views highlight the corresponding change in transmission ratio.

1.2.4 Torque Transfer, Clamping Force, and Slip

The geometry described so far determines the speed relationship between the primary and secondary pulleys. However, a speed relationship alone does not guarantee that torque can be transmitted. Unlike a gear pair, the belt and pulley are not mechanically locked together by teeth. Torque is transmitted through frictional traction between the belt and the sheave faces.

For traction to develop, the belt must be physically squeezed between the sheave faces. Figure 1.7 shows how this happens at a small portion of the belt–pulley contact. The circled region on the pulley wrap is enlarged to show the belt cross-section within the pulley V-groove. At that location, the movable sheave presses the belt toward the fixed sheave. This clamping action produces contact normal forces on the two sides of the belt, creating the contact pressure needed for frictional traction.

The same local interaction occurs continuously around the wrapped portion of the belt. At each position along the wrap, the normal forces between the belt and sheave faces allow a tangential friction force to act. Together, these small tangential traction forces transfer torque between the pulley and the belt. In practical terms, transferring more torque through the CVT requires enough clamping force to create enough available traction at the belt–pulley interface. More clamping force increases the normal contact force, which in turn increases the tangential force that friction can support.

If the available traction is not sufficient for the torque being transferred, the belt slips relative to the pulley. Slip reduces useful torque transfer and can produce heat, wear, and efficiency loss. A working CVT must therefore do two things at once: it must allow the pulley radii to change so the ratio can shift, and it must maintain enough clamping force for the belt to transmit torque without excessive slip.

This distinction is central to the rest of the model. The pulley geometry determines how the speed ratio changes. The contact mechanics determine how much torque can actually be carried through the belt–pulley contacts. A CVT is therefore not only a variable-geometry mechanism; it is also a clamping and traction problem.

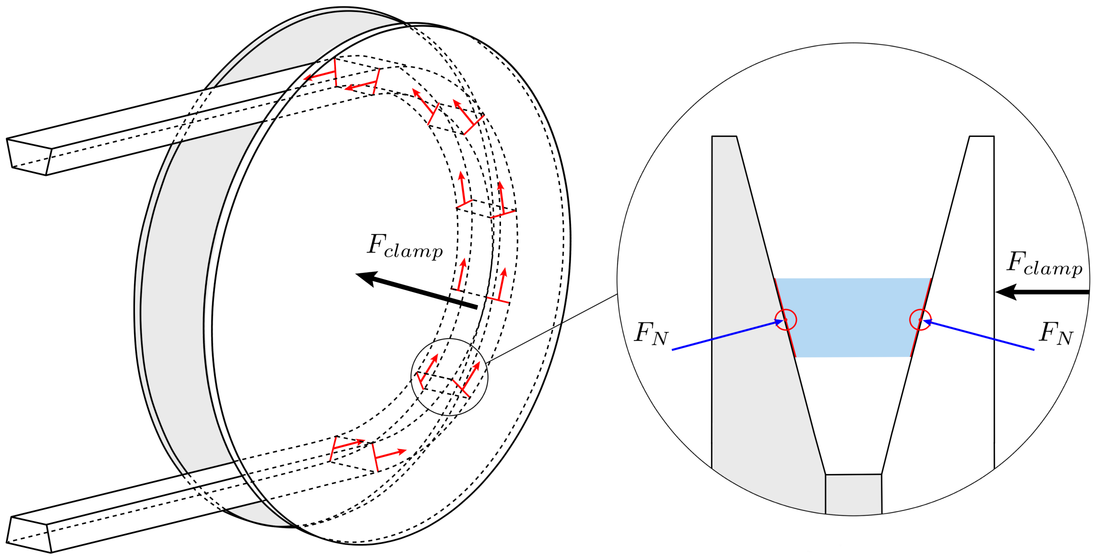


Figure 1.7: Clamping and traction in a rubber V-belt CVT. A representative location on the wrapped belt–pulley contact is circled and enlarged to show the local belt cross-section within the V-groove. Motion of the movable sheave toward the fixed sheave creates normal contact forces on the belt faces. These normal forces allow tangential frictional traction to act continuously along the wrap, collectively transmitting torque between the pulley and belt. The indicated tangential forces are representative local contributions to this distributed traction.

1.2.5 Mechanical Actuation in the Present CVT

Different CVTs generate radius change and clamping force in different ways. Some systems use electronic or hydraulic actuation to command pulley motion or clamping pressure directly. The system studied here is mechanically actuated: its pulley forces arise from mechanisms inside the primary and secondary pulley assemblies rather than from an externally commanded ratio.

These mechanisms are important because they determine both how the belt moves through the ratio range and how much clamping force is available for torque transfer. The primary pulley is mainly speed-sensitive, while the secondary pulley is mainly torque-reactive.

The primary pulley uses flyweights, ramps, and a spring. As engine speed increases, the flyweights move outward under centrifugal effects. The ramp surfaces convert this outward flyweight motion into a closing force on the primary sheaves. The spring resists this motion and helps set the speed at which the primary begins to close. As the primary closes, the belt is pushed outward, the primary effective radius increases, and the CVT shifts in the upshift direction introduced in Figure 1.6. The primary actuation mechanism is shown conceptually on the left side of Figure 1.8.

The secondary pulley uses a helix mechanism and spring loading. The helix can be understood like a screw-like interface: when torque is transmitted through the secondary, the angled helix surfaces convert part of that torque reaction into a clamping force between the sheaves. In this way, torque through the secondary helps create the clamping needed to carry torque through the belt. The spring provides preload and helps set the baseline clamping behavior.

The helix also means that secondary shift is not purely axial. As the secondary moves through its shift range, axial motion is coupled with relative rotation between secondary sheave components. This detail becomes important later when the secondary force model is developed, but the basic intuition is simply that the helix converts torque and relative rotation into axial clamping. The secondary actuation mechanism is shown conceptually on the right side of Figure 1.8.

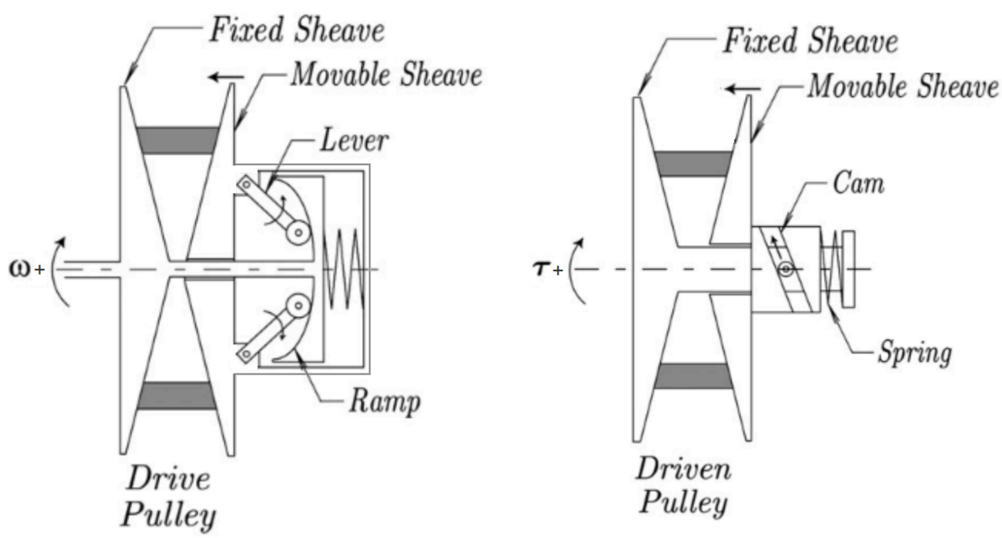


Figure 1.8: Conceptual illustration of the mechanical actuation mechanisms in the present CVT. Left: the primary (drive) pulley, where increasing rotational speed moves the flyweights outward and the ramp geometry converts that motion into a closing force on the sheaves, opposed by a spring. Right: the secondary (driven) pulley, where transmitted torque loads a cam or helix mechanism that converts part of the torque reaction into sheave clamping force, together with spring preload. In this way, the primary is mainly speed-sensitive and the secondary is mainly torque-reactive. (Julió and Plante, 2011)

Although the primary and secondary mechanisms respond most strongly to different physical quantities, they act on the same belt and the same shift process. Primary speed affects centrifugal actuation, transmitted torque affects the torque-reactive secondary, the resulting clamping forces determine the available belt traction, and the belt geometry couples motion of the two pulleys. As the operating conditions change, the balance among these effects can change as well, allowing the transmission to shift in either direction or to enter a different traction state. These coupled relationships form the physical foundation for the dynamic behaviour developed later in this work.

1.3 Motivation and Literature Review

CVT modelling is not a single, uniform problem. A model developed to size a variator, estimate belt efficiency, design a control strategy, or predict transient behaviour may involve the same physical components, but it will usually make different choices about which quantities are prescribed and which are solved. Some models are primarily design tools. Some focus on mechanical actuation and force balance. Some study belt forces, axial thrust, slip, or efficiency at fixed operating points. Others attempt to resolve transient shift behaviour or detailed belt–pulley contact mechanics.

The progression followed here is from design intuition, to component force balance, to mechanical actuation, to belt contact, to ratio change, and finally to general transient CVT behaviour. This ordering mirrors the way the modelling problem becomes difficult. A belt–sheave CVT can first be understood as a variable-radius transmission, then as a collection of mechanical actuators that generate axial force, then as a friction-limited belt contact problem, and finally as a coupled dynamic system in which pulley rotation, shift motion, belt behaviour, and contact state evolve together. The literature contains strong foundations for each of these pieces, but those foundations are distributed across several modelling traditions.

1.3.1 Framework for Classifying CVT Models

A useful way to compare CVT models is to ask what part of the system is being allowed to move, and what part is being held fixed. The same physical transmission can be treated as a geometry problem, a force-balance problem, a belt-contact problem, a shift-dynamics problem, or a coupled system-dynamics problem. Each viewpoint can be useful, but each one answers a different question. This is why the literature is best read by tracking the modelling choices that define each formulation, rather than by treating all CVT models as attempts at the same final system.

One of the first choices is how the ratio is represented. In the simplest case, the transmission is evaluated at a fixed ratio. A slightly broader approach computes the ratio from a quasi-static force balance at each operating point. Other models prescribe a ratio trajectory, or use an empirical relationship for the rate of ratio change. More detailed shift models treat ratio change as a mechanical process in its own right, involving radial belt motion, pulley motion, and contact forces (Kim et al., 2002; Sorge, 2008; Cammalleri and Sorge, 2009; Sorge and Cammalleri, 2011). These approaches are not interchangeable. A fixed-ratio model can explain forces at a known condition, while a shift model can explain how the system moves from one condition to another. Even then, a formulation that describes the mechanics of an imposed or already active shift does not necessarily determine when a shift begins, which direction it takes, or what operating condition follows it.

A second choice is how axial force is produced. Some models take pulley clamping forces as known inputs. This is natural when the goal is to study belt response under specified loading, or when the variator is externally controlled. Other models include the mechanism that generates those forces, such as centrifugal weights, springs, torque-sensitive cams, hydraulic pressure, or electronic actuation (Cammalleri, 2005; Kim et al., 2002; Skinner, 2020; Sorge and Cammalleri, 2011). The important distinction is not only the actuator type, but how completely its mechanical action is resolved. A clamping force may be prescribed directly, obtained from an instantaneous force balance or empirical relation, or produced by a mechanism whose own motion is treated dynamically. Each choice answers a different question about how the pulley force arises.

A third choice is how much of the belt–pulley contact is represented. At low fidelity, the belt may be treated through a capstan-style tension relation, often adjusted for the wedge action of the V-groove.

More detailed models include radial and tangential friction, sliding angle, belt penetration, bending stiffness, local normal pressure, tension variation, slip mechanisms, and efficiency losses (Gerbert, 1996; Kim and Kim, 1989; Sorge, 1996; Kong and Parker, 2005, 2006; Chen et al., 1998; Bertini et al., 2014; Messick, 2018). At still higher fidelity, the belt may be discretized into nodes or elements so that local belt motion and contact behaviour are solved directly (Julió and Plante, 2011; Ballew, 2015). Increasing contact fidelity can make the local mechanics more realistic, but it does not by itself determine how much of the complete transmission response is being solved. Local mechanical fidelity and global dynamic closure are therefore separate modelling choices.

A fourth choice is how the rotating system and its interaction with the surroundings are treated. Many models prescribe one or both pulley speeds, or impose the speed ratio algebraically. This is often appropriate for steady contact analysis, efficiency prediction, or controlled operating-point studies. Other models solve rotational equations of motion so that pulley speeds and transmitted torques evolve dynamically (Srivastava and Haque, 2007; Ballew, 2015; Duan et al., 2016). Any physical transmission must, of course, interact with whatever machinery is attached to it. The important distinction is whether those external systems supply only the mechanical loading associated with that interaction, or whether quantities belonging to the internal CVT mechanics—such as pulley-speed histories, clamping-force histories, axial pulley motion, or ratio trajectories—must also be prescribed. This distinction determines how much of the transmission operating state is actually predicted by the model.

A further choice is the operating region over which the same formulation remains valid. Some models are intended for steady or fully engaged operation. Others describe transient shifting once the variator is already engaged, focus specifically on clutching or engagement, or move through a predefined sequence of operating phases. A broader formulation must determine which mechanical regime is active as conditions change and remain valid as the transmission enters or leaves those regimes. A model may therefore reproduce behaviour accurately within one operating phase without necessarily providing the mechanics that determine how the transmission reaches, leaves, or reverses that phase.

These questions form a useful progression for understanding the modelling problem, but the associated modelling choices are partly independent. A formulation may resolve belt contact in great detail while prescribing global pulley motion. Another may solve pulley rotation dynamically while receiving clamping force as an external input. Another may model mechanical actuation carefully while treating ratio change through quasi-static or empirical relations. None of these choices is inherently wrong; they reflect the intended modelling objective. The important point is that model fidelity in one part of the CVT does not by itself imply closure of the complete transmission response.

The following review therefore considers both what each formulation was developed to predict and what quantities or operating conditions remain prescribed. It proceeds from practical and quasi-static design models, through mechanical actuation and belt-contact mechanics, to transient rubber-belt formulations and related dynamic models in metal-belt and chain CVTs. This provides a consistent basis for identifying which parts of the CVT dynamics are already well established and where the remaining modelling gaps lie.

1.3.2 Practical Design and Tuning Foundations

The practical understanding of mechanically actuated rubber V-belt CVTs developed largely through machine design and tuning practice. General machine design references such as Budynas and Nisbett

(2015) provide the background needed to treat the CVT as a real mechanical assembly involving belts, springs, clutches, shafts, friction interfaces, uncertainty, and design iteration. Not all of these effects are included in a compact model, but they provide useful context for deciding which behaviours are retained and which are idealized.

Practical tuning literature provides a more application-specific foundation. Aaen (2007) describes powersport CVTs using the language of engagement speed, shift speed, flyweights, primary springs, secondary springs, torque sensing, backshift, and speed variance. Its value is not that it provides a complete dynamic model, but that it builds component-level intuition from the perspective of a tuner or racer. It explains how changes to weights, springs, ramps, and helices are expected to affect behaviour, while leaving much of the coupled system response to experience and testing.

These sources are valuable because they describe the physical machine before it is reduced to equations. They also show why prediction is difficult: the same hardware change can affect engagement, shift rate, belt grip, backshift, engine speed, and vehicle acceleration at the same time. Qualitative tuning rules are therefore useful starting points, but they are not enough to determine the complete transient response of the CVT under changing operating conditions.

1.3.3 Design-Oriented and Quasi-Static CVT Models

Design-oriented CVT models commonly evaluate the transmission at a fixed operating condition or through a sequence of quasi-static operating points. Rather than asking how every transient state evolves in time, these models usually ask which geometry, actuator, belt, or tuning parameters should be selected to satisfy a desired operating requirement. In a fixed-operating-point analysis, quantities such as ratio and pulley speed may be specified while the corresponding forces, tensions, or capacities are calculated. In a quasi-static model, the operating condition may instead be obtained from an instantaneous force balance, while inertia and the time required to move between neighbouring equilibria are neglected. These approaches commonly relate vehicle requirements, pulley geometry, belt length, wrap angle, speed ratio, flyweight force, spring preload, helix force, belt tension, and torque capacity. Their main value is that they connect tunable hardware to expected operating behaviour.

A useful starting point is the numerical design method of Arango and Muñoz Alzate (2020), which begins from vehicle requirements, road specifications, and desired performance before working toward the CVT design. This is valuable because it places component selection in its proper context: the belt, pulley geometry, speed-ratio range, engine or motor selection, and mechanical centrifugal actuation are not independent choices, but responses to the requirements imposed by the vehicle and its operating route. In this sense, the work provides a design-level framework where system requirements drive CVT component choices.

Other design-oriented works focus more narrowly on particular parts of the variator design. Cammalleri (2005) develops a speed-torque-controlled rubber V-belt variator design using equilibrium relationships for a centrifugal driver and torque-sensitive driven pulley. La Battaglia et al. (2022) focuses on the kinematic design of motorcycle rubber V-belt CVTs, emphasizing the roller sliding profile of the front pulley as a critical design variable. Both works show how simplified mathematical models can be used to choose actuator or geometry parameters without requiring a complete transient simulation of the transmission.

Skinner (2020) provides a practical force-balance and tuning-oriented model. The work emphasizes quantifiable tuning objectives such as engagement speed, shift-out speed, static RPM shifting, and efficient power transfer, and relates those objectives to the primary and secondary subsystems. It is useful as an applied tuning reference because it connects the variables adjusted by teams to predicted operating behaviour, while remaining primarily an operating-point and force-balance model.

The common strength of these works is that they make CVT design less arbitrary. They identify the parameters that matter and show how geometry, springs, flyweights, helices, belt forces, and vehicle requirements can be related. They also highlight a recurring tension in CVT modelling: models that are simple enough to support early design may omit important coupled behaviour, while models that resolve more detailed mechanics can become difficult to apply directly as design tools. In this sense, design-oriented and quasi-static models occupy an important middle ground. They provide usable relationships for component selection and tuning, but they do not generally determine the complete time evolution of engagement, traction, shift motion, and pulley response.

1.3.4 Mechanical Actuation and Clamping Force Generation

After the basic design variables are identified, the next modelling question is how the CVT generates the axial forces that clamp the belt and drive ratio change. In externally controlled CVTs, these forces may be treated as prescribed hydraulic or electronic inputs. In mechanically actuated rubber V-belt CVTs, however, the forces are produced by the hardware itself. The primary force is typically speed-sensitive, while the secondary force is torque-sensitive.

Cammalleri (2005) provides an important design treatment of this problem through a speed-torque-controlled rubber V-belt variator. The centrifugal driver and torque-sensitive driven pulley are treated as part of the variator design, with equilibrium relationships connecting actuator geometry, spring loading, pulley torque, and the intended operating behaviour of the CVT. This establishes a quantitative link between the mechanical hardware and the clamping forces it is designed to produce.

Kim et al. (2002) also study a rubber belt CVT with mechanical actuators, including both steady-state and transient characteristics. Their driver and driven thrusts are obtained from the mechanical actuator geometry and instantaneous force balances, while the transient ratio response is represented separately by an experimentally identified first-order relationship. The work therefore connects mechanical primary and secondary actuation to transient CVT behaviour, but the motion of the actuator hardware itself is not what determines the shift acceleration.

Sorge and Cammalleri (2011) address the secondary side in substantially greater mechanical detail through the helical shift mechanics of rubber V-belt variators. Their formulation allows the fixed and movable driven faces to rotate differently and resolves how the belt-contact state determines the torque carried by the movable face. The resulting helix response therefore depends on spring loading, helix geometry, transmitted torque, and the internal relative motion of the driven pulley rather than on a simple assumed torque split. The instantaneous pulley radius and shift speed are nevertheless supplied to the contact calculation, so the resulting actuator relation determines the forces associated with a specified shift state rather than the acceleration of the movable secondary assembly itself.

Together, these works establish much of the mechanical foundation needed to model passive CVT actuation. They show how centrifugal primary mechanisms and torque-reactive secondaries generate configuration-dependent axial forces, and how detailed actuator geometry can materially change those forces. They also illustrate an important distinction introduced in Section 1.3.1:

including the actuator mechanism does not necessarily mean that the actuator is itself a dynamically propagated subsystem. In these formulations, the actuator force is generally closed configuration by configuration, while transient ratio motion is either outside the model or supplied through a separate shift description. This distinction becomes important when the actuator hardware has inertia and its motion is coupled directly to the shift and shaft dynamics.

1.3.5 Rubber V-Belt Contact, Slip, and Efficiency Models

Once axial force is generated by the actuation mechanisms, the belt-pulley interface determines what that force actually does. The sheaves may clamp the belt, but the resulting torque capacity, axial reaction, radial belt motion, slip, and loss all depend on how the rubber belt interacts with the pulley groove. A rubber V-belt therefore cannot be treated as an ideal kinematic constraint. It wedges into the sheaves, develops a tension distribution around the wrap, experiences radial and tangential friction, deforms under load, and dissipates energy through several mechanisms.

Classical belt theory gives the starting point: friction and wrap angle determine the maximum tension ratio that can be supported before gross sliding occurs. For V-belts, this picture is modified by the groove angle and the associated wedge effect. However, real belt slip is more complicated than a single capstan-style traction limit. Gerbert (1996) presents a unified treatment of belt slip in which creep, radial compliance, shear deflection, and seating/unseating effects all contribute to speed loss. The important point is not that every model must include all of these effects, but that "slip" is not one physical mechanism. A model should be clear about which slip and loss mechanisms it represents and which it leaves outside its scope.

The contact problem also determines the axial reaction felt by the sheaves. Kim and Kim (1989) developed a theoretical analysis of axial forces in a rubber V-belt CVT using simplified assumptions about the contact regions and belt force distribution. Sorge (1996) developed a compact model for axial thrust in V-belt drives, emphasizing radial penetration and the connection between belt forces and pulley thrust. These works are useful because they connect the belt contact state back to the sheave force balance: axial force is not only generated by the actuator, but also shaped by how the belt seats, slides, and transmits load in the groove.

A more detailed treatment is given by Kong and Parker (2005), who formulate the steady mechanics of belt-pulley systems using distributed belt equations. Kong and Parker (2006) extend this framework to pulley grooves and sliding friction. This line of work resolves local belt mechanics at much greater fidelity than simple force-balance models, including the distribution of forces and sliding behaviour around the contact arc.

Messick (2018) extends this contact-mechanics tradition for rubber V-belt CVTs by building on the Kong and Parker framework and validating axial-force and efficiency predictions experimentally. Messick's work is valuable for its contact-region formulation, belt property measurement, operating-range analysis, and dynamometer validation. It is also clear about its modelling scope: shifting is treated as a quasi-equilibrium process rather than as a transient response-time problem. In this sense, it is a strong reference for belt force and efficiency modelling, while leaving the time evolution of the complete CVT operating state to a different class of model.

Efficiency-focused studies provide another view of the same interface. Chen et al. (1998) experimentally investigated speed and torque losses in a rubber V-belt CVT, while Bertini et al. (2014) developed an analytical model for rubber V-belt CVT power losses including sliding, hysteresis, and entry/exit effects. These works show that transmission losses arise from several interacting mechanisms whose relative importance depends on the operating condition.

The contact and efficiency literature therefore provides the foundation for understanding how belt force, axial thrust, slip, and loss arise. It also reinforces the distinction between local fidelity and global dynamic closure. A high-fidelity contact model may be necessary for efficiency prediction, axial-force prediction, or detailed local belt mechanics. A lower-dimensional contact model may instead be appropriate when the modelling objective requires the belt to remain coupled transparently to actuator dynamics, pulley rotation, shift motion, and the broader transient operating state of the CVT.

1.3.6 Transient Rubber-Belt Models and Dynamic Closure

The contact and force-balance literature establishes how a rubber belt CVT behaves at a specified operating condition. Transient models extend this picture by asking how the transmission moves between operating conditions and which parts of that motion must be resolved in time. In the rubber-belt literature, this has been approached through models of shift mechanics, distributed belt motion, pulley rotational dynamics, and mechanically actuated vehicle response. A useful comparison among these works is therefore which parts of the CVT state are advanced dynamically and which quantities are supplied as operating conditions or inputs.

Sorge (2008) provides an important bridge from quasi-static analysis to transient shift mechanics. Rather than representing ratio change as a sequence of neighbouring equilibria, the formulation treats the radial migration of the belt through the pulley grooves as a time-dependent process coupled to circumferential belt motion, mass conservation, and sliding. Cammalleri and Sorge (2009) later develop approximate closed-form solutions for the same class of shift problem. These works establish a mechanical basis for transient ratio change, while taking the pulley loading and active shift condition as part of the problem definition.

A more detailed transient treatment is given by Julió and Plante (2011), who discretize the rubber belt and apply Newton's law to the individual belt elements. Their model predicts the time response of transmission ratio under specified drive-pulley axial force, drive-pulley speed, and load torque, while resolving belt elasticity, bending, damping, and frictional contact directly. The formulation is validated for steady and shifting conditions and provides substantially greater local belt detail than reduced force-balance models. The authors also report difficulty obtaining a stabilized prediction for the low-torque start-up condition in which belt clamping itself acts as the clutch, indicating that the operating range of the formulation is narrower than the full sequence of CVT engagement and shift behaviour.

Ballew (2015) extends this discretized-belt model family by removing the prescribed input-pulley speed used in earlier formulations. Applied torques, pulley inertias, and belt forces instead determine the pulley angular accelerations, allowing both shaft speeds to evolve dynamically. Engine and vehicle submodels are also included so that changes in external loading can feed back through the transmission rather than being represented by a prescribed pulley-speed history.

Axial actuation in Ballew (2015) remains a separate input to the belt–pulley model. The thesis notes that the axial-force inputs could be supplied either by a control system or by a model of a mechanically governed actuator. In the simulations presented, a PI controller with feed-forward gain generates the primary axial force, while the secondary axial force is held fixed. The formulation therefore extends the dynamic closure of the rotating system substantially, while leaving the source of the pulley clamping forces to a separate actuation model.

da Silva et al. (2023) present a reduced mechanically actuated rubber-belt CVT model coupled to a longitudinal vehicle simulation. Axial equations of motion are written for the movable sheaves, with flyweight loading on the primary and a torque-dependent secondary contribution. The operating sequence is divided into a clutching phase followed by a fully engaged phase, and the model is validated against a full-throttle acceleration. The work therefore provides an example of movable-sheave dynamics and mechanical actuation being retained in a transient application, while using a simplified, phase-specific representation of the CVT behaviour.

Taken together, these works show a clear progression in transient rubber-belt modelling. Ratio change has been treated as a mechanical process, belt motion and contact have been propagated directly, pulley rotational motion has been allowed to respond to applied torques, and movable-sheave inertia has been introduced in mechanically actuated applications. The formulations differ, however, in which parts of the transmission remain prescribed or are restricted to a particular operating phase. This makes the broader CVT literature useful for examining how other variator architectures have divided the boundary between prescribed actuation and dynamically resolved transmission behaviour.

1.3.7 Adjacent Metal-Belt and Chain CVT Dynamic Models

The broader CVT literature contains an extensive modelling tradition for metal push-belt and chain variators, particularly in automotive applications. These systems are not direct equivalents of dry rubber V-belt CVTs, since their load-carrying mechanisms, deformation, lubrication, and actuation systems differ. They nevertheless provide useful comparisons for transient ratio change, frictional contact, pulley dynamics, and the treatment of applied clamping forces.

Srivastava and Haque (2009) review the dynamics and control of belt and chain CVTs and summarize a wide range of steady and transient formulations, including slip and traction models, control-oriented shift laws, and detailed belt or chain mechanics. The review illustrates both the maturity of dynamic CVT modelling outside the rubber-belt literature and the importance of keeping the underlying transmission architecture in view when comparing models.

The Carbone–Mangialardi–Mantriota (CMM) family provides a well-developed example of mechanically derived shift dynamics for metal-belt and chain variators. Carbone et al. (2001, 2005) derive ratio-changing behaviour from belt or chain kinematics, mass conservation, friction, and pulley deformation, including creep-mode and slip-mode shifting. Experimental work by Yildiz et al. (2018) validates the resulting CMM shift dynamics under load. In these tests, the predicted quantity is the speed-ratio response to a prescribed change in clamping-force ratio, while drive-pulley angular speed, driven-side clamping force, and load torque are specified operating quantities. The formulation is therefore directed at the variator shift response under known actuation and loading.

Other metal-belt studies similarly treat transient ratio change and traction behaviour directly. Srivastava and Haque (2005, 2008) develop dynamic formulations that retain transient belt motion and examine the effects of friction characteristics and band-pack slip. Bonsen et al. (2003) study variator slip and its relationship to transmitted torque and clamping. These works extend the treatment of shift and traction beyond steady operating-point analysis while retaining the controlled clamping architecture typical of metal-belt automotive CVTs.

Schindler et al. (2012) develop a spatial transient multibody model of a pushbelt variator with detailed representation of the pulley sheaves, ring packages, friction, and unilateral contact. The model is used to study the internal transient response of the pushbelt system, including deformation and three-dimensional contact behaviour. In the simulations presented, the output pulley receives prescribed clamping force and load torque, while the input pulley is assigned angular-velocity and clamping-velocity histories. The detailed belt and pulley response is therefore resolved under a specified global variator excitation.

Chain CVTs provide another example of strongly coupled transient mechanics. Duan et al. (2016) include pulley rotational dynamics, movable-pulley axial dynamics, chain radial motion, chain and pulley deformation, and a speed-dependent friction model spanning micro-slip and macro-slip. The axial and rotational pulley dynamics are coupled directly through the chain forces. The movable-pulley equation is driven by an applied clamping force, however, so the generation of that clamping force remains outside the variator model.

These adjacent models show how transient shift mechanics, detailed contact behaviour, pulley rotation, and movable-pulley motion can be represented at substantially different levels of detail. They also illustrate a recurring modelling boundary: in hydraulically or otherwise externally actuated variators, clamping force and pulley actuation are naturally supplied as inputs to the transmission model. For a mechanically actuated rubber V-belt CVT, those same functions are produced by hardware contained within the transmission, making the treatment of the actuation mechanism itself an important part of how the model boundary is defined.

1.3.8 Synthesis of Prior Work

The literature does not point to a single missing ingredient in CVT modelling. Instead, several mature lines of work resolve different parts of the same machine. Practical and design references establish the component relationships used to size and tune rubber V-belt CVTs. Mechanical-actuation studies explain how centrifugal and torque-reactive hardware generate pulley forces. Contact and efficiency models resolve the belt–pulley interface at progressively greater local detail. Transient formulations add radial belt motion, ratio change, pulley rotation, movable-sheave dynamics, or distributed belt and chain motion. The resulting body of work provides strong foundations for the individual mechanisms required in a dynamic transmission model.

A recurring distinction among these formulations is the location of the model boundary. Any CVT must receive mechanical interaction from the systems attached to it, but the quantities treated as external inputs differ substantially among models. Depending on the modelling objective, pulley speed, axial force, clamping pressure, ratio motion, or actuator commands may be prescribed while the remaining transmission response is solved. These choices are appropriate for many component, control, and contact studies, but they also determine how much of the CVT itself is contained within the dynamic system.

For a mechanically actuated rubber V-belt CVT, the mechanism that generates clamping is itself part of the transmission. Its motion therefore cannot be separated completely from the response that it produces: the actuation, pulley, belt, and rotational mechanics form one coupled physical system. No formulation identified in the reviewed literature was found to close that complete mechanically actuated system dynamically from the loading applied at the primary and secondary shafts. Treating the clamping mechanism dynamically is one part of this gap, but the broader requirement is that the internal CVT response be solved as a coupled whole rather than assembled from separately prescribed or separately closed pieces.

A separate gap concerns the range of operating behaviour covered by the same formulation. Much of the literature is developed for an already engaged variator, a specified shift, or a prescribed sequence of operating phases. Individual works address clutching, upshift and backshift, ratio limits, or traction behaviour, but no formulation identified in the reviewed literature was found to remain mechanically valid across the complete operating sequence of disengagement and engagement, seated ratio limits, free shifting, and traction loss and recovery without imposing a predefined operating phase or trajectory. This is a different requirement from accurately modelling any one of those operating conditions: it requires the same physical formulation to remain valid as the active mechanical constraints change.

The remaining modelling opportunity is therefore defined by two independent forms of closure. The first is system-level closure of the complete mechanically actuated CVT, so that its internal state evolves as one coupled mechanical system in response to the loading applied at the primary and secondary shafts. The second is operating-domain closure, so that the same model remains valid as the transmission moves between its major mechanical regimes rather than being constructed around one specified phase or trajectory.

Table 1.1 summarizes selected formulations that most directly establish this progression. The table is not a ranking of model quality; each formulation reflects a different modelling objective. It instead shows which parts of the transmission are resolved within each model and which quantities remain prescribed as operating conditions or inputs.

Formulation	Principal behaviour resolved	Actuation / clamping	Quantities supplied or separately closed	Primary operating scope
Cammalleri (2005)	Mechanical variator design and equilibrium force relationships	Centrifugal primary and torque-sensitive secondary represented mechanically	Operating speed, torque, and equilibrium condition	Steady / quasi-static design
Kim et al. (2002)	Steady mechanical response with an empirical first-order ratio transient	Mechanical primary and secondary thrust relations	Operating conditions and experimentally identified transient shift relation	Engaged upshift and downshift response
Sorge (2008)	Transient radial belt migration, sliding, and ratio-change mechanics	Pulley loading forms part of the specified shift problem	Pulley loading and active shift condition	Engaged transient shift
Sorge and Cammalleri (2011)	Torque-reactive secondary helix mechanics with differential sheave rotation	Helix and spring represented mechanically	Instantaneous pulley radius and shift motion supplied to the secondary calculation	Engaged secondary shift mechanics
Messick (2018)	Distributed rubber-belt contact, axial thrust, and efficiency	Axial loading prescribed	Pulley speed, torque, clamping, and operating point	Steady / quasi-equilibrium contact and efficiency
Julió and Plante (2011)	Transient discretized rubber-belt motion and contact	Drive-pulley axial force prescribed	Drive-pulley speed, axial force, and load torque	Engaged steady and shifting operation
Ballew (2015)	Discretized belt dynamics with pulley rotational response to applied torques	Primary axial force controlled; secondary axial force fixed in reported cases	Axial-force inputs; actuator mechanism remains separate	Engaged transient operation with dynamic shaft response
Carbone et al. (2001, 2005); Yildiz et al. (2018)	Mechanically derived metal-variator ratio dynamics including creep and slip modes	Clamping-force ratio prescribed	Drive speed, clamping quantities, and load torque	Engaged underdrive / overdrive shifting
Schindler et al. (2012)	Detailed spatial pushbelt, pulley, friction, and unilateral-contact dynamics	Global pulley actuation prescribed	Input angular and clamping velocities; output clamping force and load torque	Detailed transient response of an engaged variator
Duan et al. (2016)	Coupled chain, pulley axial and rotational dynamics, deformation, and slip	Applied clamping force drives movable-pulley motion	Applied clamping and external loading	Engaged chain-CVT shift dynamics

Table 1.1: Selected CVT formulations illustrating the progression from operating-point and component models to increasingly coupled transient descriptions. The comparison emphasizes which behaviour is resolved within each formulation and which quantities remain prescribed according to its modelling objective.

1.4 Aim, Scope, and Contributions

The literature review identifies two remaining modelling needs for mechanically actuated rubber V-belt CVTs: closure of the transmission as one coupled dynamic system, and a formulation that remains valid as the CVT moves between its different operating regimes. The present work addresses these needs with a first-principles transient model of the CVT itself.

1.4.1 Aim

The aim of this work is to develop a general transient model of a mechanically actuated dry rubber V-belt CVT whose behaviour follows from its internal mechanics and from the mechanical loading applied at the primary and secondary shafts. The model is intended to determine its own ratio motion, clamping response, pulley dynamics, belt behaviour, and traction state rather than requiring those quantities to be prescribed as an operating history. External systems may therefore be coupled to either shaft without changing the internal CVT formulation.

The same physical model is also intended to remain applicable as the transmission moves between its different operating conditions. Engagement, operation at the ratio limits, free shifting, and changes in traction state are treated as different admissible states of the same CVT rather than as separate scenario-specific models. The direction and sequence of those transitions are determined by the evolving mechanical conditions.

A further aim is physical interpretability. The derivation is organized so that predicted behaviour can be traced back to the geometry, actuation, inertial response, belt mechanics, and active constraints that produced it. The assumptions, numerical implementation, and reported simulation cases are also documented so that the formulation can be reproduced and independently examined.

1.4.2 Scope

The modeled system is the mechanically actuated CVT itself. It includes the primary and secondary pulley assemblies, the mechanical hardware that produces their clamping forces, and the complete rubber V-belt loop. The model is deliberately independent of any particular engine, motor, dynamometer, vehicle driveline, or other surrounding system. Instead, those systems may be coupled to the CVT through the primary and secondary shafts without changing the internal transmission formulation.

The formulation retains the rotational motion of both pulley assemblies, the axial motions required for ratio change, the motion of the retained actuation hardware, and the transport motion of the belt. The retained component masses and inertias participate in the corresponding dynamic balances, while the actuation mechanisms and belt–pulley contacts provide the forces and reactions through which these motions influence one another.

The belt and contact mechanics are intentionally represented at lower order than in distributed deformable-belt formulations. The model does not attempt to resolve the complete spatial deformation of the rubber belt, local creep and shear fields, detailed seating and unseating behaviour, full actuator tribology, flexible pulley and shaft deformation, thermal evolution, wear, or a complete transmission-loss model. The principal consequences of these and the other idealizations are stated explicitly in Section 2.2. These choices define the physical resolution of the present formulation; they do not prevent more detailed component laws from being introduced in future work.

1.4.3 Contributions

The principal contribution of this work is a first-principles transient formulation of the mechanically actuated rubber V-belt CVT as a complete dynamic component. Its internal response is solved as one coupled mechanical system under the loading applied at the primary and secondary shafts, without requiring prescribed histories of ratio, clamping force, pulley speed, or shift direction. The same formulation is constructed to remain valid across the major operating regimes of the transmission rather than around one predefined shift or operating sequence.

Realizing this overall capability requires several specific mechanical and formulation developments. These are contributions in their own right, while also providing the physical closure needed by the complete CVT model:

1. **Dynamic primary centrifugal actuation.** The primary flyweight mechanism is treated as a moving inertial subsystem rather than as an instantaneous centrifugal-force relation. The derivation retains the effect of flyweight acceleration in the axial force balance and the configuration-dependent contribution of the flyweights to primary shaft angular momentum. The same actuator motion therefore participates consistently in both the shift and rotational dynamics.

2. **Dynamic axial–rotational coupling of the torque-reactive secondary.** The movable secondary sheave is allowed to rotate relative to the shaft as required by the helix geometry while translating axially. Its retained rotational inertia consequently enters both the axial dynamics of the movable assembly and the angular-momentum balance of the secondary shaft. This produces the reciprocal coupling required between shift motion, helix reaction, movable-sheave rotation, and shaft dynamics.

3. **Reduced transient belt, geometry, and traction closure.** The pulley geometry and fixed belt length are carried through position, velocity, and acceleration level so that the evolving pulley radii enter the dynamic balances consistently. The belt retains an independent transport motion, and the reduced wrap mechanics preserve the radial and tangential acceleration terms implied by the selected kinematics. Belt reactions and traction admissibility are then coupled directly to the pulley and shift dynamics.

4. **Common mechanics across CVT operating regimes.** The formulation includes the mechanical constraints associated with disengaged or deadzone motion, engagement, seated ratio limits, free shift, and traction-limited stick–slip operation within one regime-dependent model. Transitions are selected from the admissibility of the current mechanical state rather than from a scripted operating sequence. When activation of a rigid constraint changes the allowable velocity, the post-transition motion is obtained from the retained inertia and impulse–momentum balance rather than an arbitrary state reset.

Together, these developments provide the system-level and operating-domain closure identified as missing in Section 1.3.8. They also preserve a direct mechanical interpretation of the model: changes in the predicted CVT response can be traced to specific actuator, inertial, geometric, belt, or contact effects within the formulation.

1.5 Document Roadmap

The remainder of this work develops the model from system definition through simulation results. Chapter 2 formalizes the CVT as a simulation problem by defining the modeled transmission, its interaction with surrounding systems, the retained assumptions and coordinates, the state variables, and the mechanical regimes through which the model evolves. Chapter 3 then derives the governing mechanics, proceeding from pulley and belt geometry to axial actuation, pulley rotational dynamics, belt transport and contact, and finally the coupled transient model and its regime transitions.

Chapter 4 presents the simulation results and discussion using representative launch cases to examine the model behaviour and its response to targeted CVT tuning changes. It also documents the implementation and reproducibility artifacts and concludes the present study. The appendices collect supporting derivations and numerical details that are useful for reproducing or extending the formulation but would interrupt the main development.

Chapter 2

Model Definition and Simulation Problem

With the aim and scope established in Section 1.4, the next step is to define the CVT as a precise simulation problem. This requires a clear account of what belongs to the modeled transmission, how surrounding systems interact with it mechanically, and what information describes the CVT at a given instant.

The definition proceeds from the outside inward. Section 2.1 first establishes the modeled CVT and its shaft interfaces, after which Section 2.2 states the physical idealizations that set the resolution of the formulation. Section 2.3 then defines the coordinates and sign conventions used to describe the retained motions. With those choices fixed, Section 2.4 identifies the internal state carried from one instant to the next, while Section 2.5 introduces the active mechanical constraints that distinguish the admissible operating regimes. Finally, Section 2.6 separates the roles of fixed data, current state, shaft-interface loading, and instantaneous model results, completing the simulation problem used by the derivation in Chapter 3.

These definitions provide the common problem statement used by the derivation in Chapter 3. At any instant, the fixed CVT data, current internal state, and mechanical loading presented at the two shafts are known. The governing mechanics determine the corresponding instantaneous response and state derivatives, which may then be advanced through time by numerical integration.

2.1 Modeled CVT and Shaft Interfaces

The CVT is connected to the surrounding machinery through its primary and secondary shafts. These two connections form the external mechanical interfaces of the modeled transmission. Everything that physically belongs to the CVT between them is retained inside the model boundary.

The modeled CVT includes the primary and secondary pulley assemblies, the complete belt loop, and both belt–pulley contacts. It also includes the primary flyweights, ramp, and spring and the secondary spring and helix. Any masses and inertias retained for these components are included with them. Their forces and inertias therefore act together as parts of the same coupled mechanism.

Placing the model boundary at the two shafts keeps the CVT description independent of any particular surrounding system. An attached system supplies the mechanical loading that it presents at the corresponding shaft, so changing what is connected to the CVT changes the loading seen by the model rather than the definition of the transmission itself.

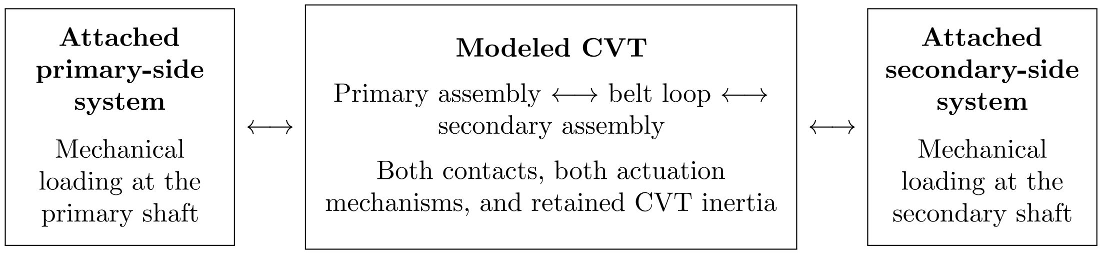


Figure 2.1: The modeled CVT and its external mechanical interfaces. Everything in the centre box is resolved as part of the transmission. An attached system acts on the CVT through the corresponding shaft, while the shaft motion remains part of the modeled response.

The same interface description applies at either shaft. Let $j \in \{p, s\}$ identify the primary or secondary side. The attached system contributes two mechanical quantities:

$$\tau_{\mathrm{B},j}, \qquad I_{\mathrm{B},j}.$$

The applied torque $\tau_{\mathrm{B},j}$ is the signed torque exerted on the CVT through shaft j. The equivalent inertia $I_{\mathrm{B},j}$ is the inertia of the attached system as seen at that same shaft. A rotor connected directly to the shaft contributes its own inertia; a wheel, drivetrain, or other mechanism connected through gearing contributes the corresponding inertia reflected to the shaft.

These quantities describe how the attached system loads the CVT, not how the shaft must move. The shaft speed remains part of the modeled motion. An attached system may use the current shaft speed, time, and its own internal state to determine the torque and equivalent inertia that it presents at the interface. Those quantities then act together with the inertia and mechanics inside the CVT to determine the shaft acceleration.

A positive applied torque acts in the positive rotational direction defined for the corresponding shaft, and the equivalent inertia is expressed about that same coordinate. The equivalent inertia contains only inertia belonging to the machinery attached beyond the shaft interface. Any pulley, sheave, flyweight, helix, or belt inertia retained inside the CVT is therefore excluded from $I_{\mathrm{B},j}$, so no component inertia is counted twice.

The terms *primary-side interface* and *secondary-side interface* identify the two shaft connections rather than a required direction of power flow. Torque may act through either interface in either rotational direction without changing which components belong to the CVT. Changing the attached driver or load therefore changes the loading at the interface, not the definition of the modeled CVT.

2.2 Modeling Assumptions

The assumptions below define the physical resolution of the formulation rather than a particular operating case. Each assumption is stated together with its principal consequence for the mechanics that can and cannot be represented. The consequence summaries are not intended to enumerate every secondary effect of an idealization; they identify the main physical resolution removed by that choice so that later results can be interpreted against a clear model boundary.

2.2.1 Material and Mechanism Idealizations

Assumption 1 (Time-Invariant Material and Surface State)

Model statement

The material properties and surface condition assigned to the CVT remain unchanged during a simulation. Intrinsic component dimensions are likewise not allowed to evolve through wear or thermal growth. The selected properties describe the hardware for the duration of the simulated event rather than evolving with its operating history.

Consequences for model fidelity

Temperature-dependent stiffness or friction, thermal state evolution, wear, glazing, conditioning, degradation, and other feedback from accumulated contact history are not represented. The formulation therefore describes short-term mechanical response for a specified material and surface state rather than the progressive evolution of that state.

Assumption 2 (Rigid Pulley and Actuation Hardware)

Model statement

The pulleys, shafts, actuation hardware, guides, and supporting CVT structure are represented as rigid bodies. Their prescribed geometry does not deform under the loads developed during operation.

Consequences for model fidelity

Pulley bending, shaft bending or torsion, guide and housing deflection, load-dependent groove geometry, and structural vibration generated by those compliances are outside the model. More detailed CVT formulations show that pulley deformation can alter local belt mechanics and axial loading; those effects would require compliant component models rather than changes to the rigid-body balances alone (Carbone et al., 2005; Duan et al., 2016).

Assumption 3 (Idealized Actuator Mechanics)

Model statement

The primary ramp and flyweight mechanism, secondary helix, guides, and springs follow their prescribed kinematic and force laws without additional contact friction, backlash, or local compliance unless such an effect is introduced explicitly by the selected component law.

Consequences for model fidelity

Ramp or roller friction, pin–helix friction, roller or pin spin, guide friction, backlash, helix-flank transitions, contact compliance, and unmodelled spring hysteresis or internal damping are omitted. Relaxing this assumption would add actuator-specific tribology and compliance to the same mechanism free bodies; for example, contact friction appears explicitly in more detailed centrifugal and helix treatments (Kim et al., 2002; Sorge and Cammalleri, 2011).

Assumption 4 (Fixed, Uniform Belt Cross-Section)

Model statement

The same prescribed cross-section is assigned everywhere along the belt and is carried unchanged as the belt runs and shifts. This description fixes the cross-sectional shape and dimensions and treats mass density as uniform through the section and along the belt. The belt therefore has a constant mass per unit length, and its cross-sectional centre of mass coincides with the geometric centroid. The cord layer is retained only as a prescribed tension-carrying line at the fixed depth d_{cord}; its distinct density and contribution to the through-section mass distribution are not resolved. As the section follows the curved modeled belt path, it does not compress, bulge, shear, warp, or grow under centrifugal or sheave loading.

Consequences for model fidelity

Accordingly, neither spatial variations caused by cogs, ribs, joints, local reinforcement, manufacturing variation, or nonuniform wear nor load-induced changes caused by lateral and radial compression, bulging, cross-sectional shear and warping, centrifugal growth, or local cog and rib deformation are represented. Concentrated cord mass and differences between the densities of the belt layers cannot shift the modeled centre of mass away from the geometric centroid. Changes in belt width, height, contact geometry, mass distribution, and internal strain energy caused by these omitted effects therefore lie outside the present formulation. Detailed belt models retain some of these spatial and compliance effects explicitly (Gerbert, 1996; Kong and Parker, 2005; Messick, 2018).

2.2.2 Belt Geometry and Kinematics

Assumption 5 (Aligned Pulley Axes and Fixed Center Distance)

Model statement

The primary and secondary pulley axes remain parallel at a fixed spacing. The pulley centers and shaft directions therefore do not change as the CVT shifts.

Consequences for model fidelity

Shaft misalignment, eccentricity, geometric runout, bearing-center motion, and the associated changes in belt path and face loading are not represented. Removing this assumption would require the belt geometry and contact loading to respond to the additional shaft and bearing motion.

Assumption 6 (Fixed Belt Length)

Model statement

The modeled belt has a fixed total length. Longitudinal extension is not treated as an independent deformation of the belt, so every admissible pulley configuration must be compatible with the same installed length.

Consequences for model fidelity

Tension-dependent elongation, longitudinal strain energy, extension waves, permanent elongation, and the longitudinal component of elastic creep are not represented. Detailed traction-belt models can retain these effects through a longitudinal compliance field; the present formulation instead assigns ratio change to the pulley and contact motions of a fixed-length belt (Gerbert, 1996; Kong and Parker, 2005; Messick, 2018).

Assumption 7 (Single Belt Transport Coordinate)

Model statement

The circumferential motion of the complete belt is represented by one global transport coordinate. Individual material points are not tracked around the loop, and the local tangential belt speed is not given an independent spatial field.

Consequences for model fidelity

The geometry-induced redistribution of local tangential material speed during shift, material-point migration through the wraps, and the additional momentum transport associated with that spatially varying velocity field are not resolved. The model still retains acceleration of the global belt transport and the motion of the prescribed wrap geometry. Shift-mechanics and discretized belt formulations resolve the omitted local transport in greater detail (Sorge, 2008; Cammalleri and Sorge, 2009; Julió and Plante, 2011).

Assumption 8 (Prescribed Planar Belt Path)

Model statement

At each instant, the belt is taken to wrap each pulley along a circular arc and to pass between the pulleys along two straight tangent spans in one common plane. The belt path is therefore set by the current pulley geometry rather than solved as a flexible dynamic curve.

While both pulleys are engaged, the belt is treated as taut and perfectly flexible. It carries load along its length through a single longitudinal tension, but it carries no compression, transverse shear force, or bending moment. The belt also remains in continuous contact with the sheave faces over each prescribed wrap. This engaged description is physically admissible only while the belt tension and distributed normal contact loading remain nonnegative.

Consequences for model fidelity

Slack or buckled spans within the engaged CVT, local belt lift-off, partial wrap, and contact reattachment are not represented. Bending stiffness, flexural strain energy, and bending hysteresis are also absent even though the prescribed belt path is curved. A calculated negative tension or normal loading therefore indicates that the assumed engaged path is no longer physically admissible; it does not represent a physical negative belt load.

Progressive or spiral radial migration within a wrap, seating and unseating trajectories, span sag, transverse belt vibration, and transverse inertia associated with straight-span reorientation are omitted. The same planar description also excludes the out-of-plane span skew, twist, and axial belt inertia that would arise when the two physical groove centerplanes move differently during shift. More detailed belt and shift-mechanics models resolve subsets of these path and contact effects explicitly (Sorge, 2008; Kong and Parker, 2005; Messick, 2018).

Assumption 9 (Reduced Belt Motion While the Primary Is Disengaged)

Model statement

While the primary sheaves are separated from the belt, no primary sheave normal resultant, tangential traction, or belt torque is retained. Any radial support provided at the minimum primary radius is not assigned a tangential contact law. Belt pretension and the changing shape of the slack spans are not retained as additional mechanical states.

The belt is instead taken to remain seated on the fully closed secondary and to be carried without slip at the secondary effective radius. The belt transport speed and secondary speed are therefore joined by one kinematic constraint throughout the disengaged-primary regime.

Consequences for model fidelity

The disengaged belt and secondary do not have independent transport coordinates. Slack-span motion, sag, belt whip, intermittent primary rubbing, secondary slip, and tension waves created as slack is taken up cannot be predicted. The model also cannot determine deadzone belt pretension or the transient contact loads produced by a detailed re-engagement history. Instead, the entire belt moves at the speed imposed by the secondary contact.

2.2.3 Belt–Pulley Contact Representation

Assumption 10 (Gross Tangential Coulomb Traction)

Model statement

During continuous motion, each belt–pulley interface is represented by a dry gross stick–slip law with fixed static and kinetic traction capacities for the selected surface state. The frictional contact action retained by the model acts in the circumferential belt-transport direction.

Consequences for model fidelity

The formulation can distinguish torque-carrying stick from gross kinetic slip, but it does not resolve radial friction, local creep or microslip, detailed presliding behaviour, lubrication or Stribeck effects, or friction variation with pressure, temperature, conditioning, or local sliding speed. Friction associated specifically with radial migration, seating, and unseating is also outside the retained contact direction. These mechanisms are treated separately in higher-fidelity rubber-belt contact models (Gerbert, 1996; Kong and Parker, 2006).

The static and kinetic traction limits are force-level laws for continuous motion. They do not supply a separate impulse-level friction law during an instantaneous change in mechanical constraint.

Assumption 11 (Wrap-Level Contact Resolution)

Model statement

The model may resolve how belt tension and normal loading vary around a pulley wrap, but each wrap is assigned one overall traction state at an instant. The contact is not resolved through the belt width or depth, and no separate local traction state is carried for different portions of the same wrap.

Consequences for model fidelity

Local adhesion and sliding subregions, spatial traction reversals, detailed pressure concentrations, edge loading, and cross-sectional contact-stress fields cannot be predicted. Resolving those effects requires a distributed or discretized contact description of the type used in higher-fidelity belt models (Kong and Parker, 2006; Messick, 2018; Julió and Plante, 2011).

Assumption 12 (Symmetric Sheave-Face Sharing)

Model statement

The two sheave faces at a pulley are assumed to share the total normal contact load equally. On the secondary pulley, the belt torque is also divided equally between the fixed and movable faces for the purpose of the movable-sheave balance. The primary sheave faces rotate together with their shaft and therefore have one common angular speed. On the secondary, helix motion gives the movable face a different instantaneous angular speed from the fixed face. Under Assumption 11 (Wrap-Level Contact Resolution), the wrap-level tangential law does not retain those two face contacts separately. It therefore uses one representative secondary contact speed, defined so that the aggregate contact retains the pulley-side tangential power implied by the two equal face-torque paths.

Consequences for model fidelity

Face-to-face differences in normal loading, transmitted torque, and local sliding state are not resolved. The power-equivalent representative speed preserves the pulley-side torque power implied by the equal-share reduction, but it does not make the two physical face speeds equal or recover their separate contact states. The helix-induced rotation of the movable face remains in the actuation and rotational balances and also enters the representative contact speed. Secondary sticking therefore denotes gross sticking of this aggregate contact, rather than simultaneous pointwise adhesion to both physical faces during helix motion. Face-resolved shift models show that the two secondary face torques and sliding conditions can depart substantially from this aggregate description during shifting (Sorge and Cammalleri, 2011).

2.2.4 Boundary and Transition Idealizations

Assumption 13 (Instantaneous, Perfectly Inelastic Constraint Activation)

Model statement

When a mechanical stop or a new kinematic constraint becomes active at finite relative velocity, its finite contact duration and local compliance are replaced by an instantaneous, perfectly inelastic event. This treatment applies to arrival at a mechanical travel stop, first belt engagement, and activation of the deadzone belt lock.

The generalized positions remain continuous. The generalized velocities may change, and their post-event values are determined by impulse–momentum balance over all retained inertial coordinates subject to the newly active kinematic constraints. Applied torques, spring forces, damping forces, and other bounded forces act over no resolved time during the event and therefore contribute no finite impulse. All masses and inertias retained by the continuous mechanics remain part of the momentum balance. A constraint release produces no impulse unless another constraint activates at the same instant.

Consequences for model fidelity

The event model does not resolve contact duration, local indentation, peak impact force, stress waves, rebound, chatter, or the high-frequency vibration created by compliant contact. Individual velocities are neither preserved nor set independently; incoming momentum is redistributed among the velocities that remain compatible with the new mechanical topology. Because the event is perfectly inelastic, kinetic energy need not be conserved. The decrease in kinetic energy is retained as the impact or topology-capture loss, but the model cannot divide that loss among local belt deformation, material damping, sound, heat, or structural vibration. Predicting those details would require a finite-duration compliant-contact model. An impulsive load applied through an external shaft boundary would likewise require an explicit boundary impulse rather than the continuous torque description.

Assumption 14 (Tangential Contact Through Boundary Events)

Model statement

A belt–pulley contact that is sticking immediately before an instantaneous boundary event remains constrained against tangential slip through the event. Contacts that are sliding, together with contacts created by the event, are not assigned a new impulsive no-slip constraint. Their relative tangential motion after the event, together with the ordinary continuous traction limits, determines the stick–slip state that follows.

Consequences for model fidelity

An existing sticking contact may transmit whatever tangential constraint impulse is required to preserve no slip; that impulse is not checked against a separate impulse-level friction bound. The model therefore cannot predict impact-induced loss of an existing stick, brief impulsive slip followed by readhesion, or a transition caused by exhausting an impulse-level friction capacity.

Conversely, normal contact or kinematic capture does not automatically glue a new belt–pulley interface tangentially. In particular, a newly formed primary contact may retain tangential relative motion and begin its continuous motion in slip. Resolving the omitted impulsive friction behaviour would require an impulse-level traction law coupled to the normal constraint impulse.

2.2.5 Other Omitted Losses

Assumption 15 (Negligible Unmodeled Parasitic Losses)

Model statement

Bearing, seal, windage, and auxiliary-hardware losses are neglected unless they are introduced explicitly by a component or external boundary model. No global empirical efficiency factor is inserted to represent these omitted mechanisms.

Consequences for model fidelity

Energy loss and damping arise only from mechanisms represented explicitly by the formulation, including gross kinetic slip and perfectly inelastic boundary or topology-capture events. Absolute efficiency may therefore be optimistic, and real hardware may exhibit additional damping or resisting torque when the omitted parasitic losses are significant. Adding such losses does not require changing the kinematic structure of the CVT; it requires supplying the corresponding physical torque or force laws.

Resulting fidelity Taken together, these assumptions retain the dominant rigid-body actuation, pulley rotation, sheave-shift motion, global belt transport, wrap tension and contact resultants along an admissible taut belt path, clamping, traction limitation, gross stick–slip mechanics, and momentum redistribution when a new mechanical constraint activates. They also retain the kinetic-energy loss associated with gross slip and perfectly inelastic capture.

The formulation does not reproduce the complete deformable belt and local contact problem, spatial variation within the belt construction, slack or buckled engaged spans, partial-wrap lift-off and reattachment, belt bending mechanics, slack-belt dynamics while the primary is disengaged, actuator tribology, flexible-structure response, finite-duration impact and rebound, impulse-level friction limits, thermal or wear evolution, or miscellaneous parasitic losses. The assumptions are separated by physical mechanism so that a higher-fidelity model can replace one closure without obscuring which additional behaviour that change is intended to recover.

2.3 Coordinate and Sign Conventions

This section defines the coordinates and sign conventions used throughout the formulation.

2.3.1 Rotational and Belt-Transport Directions

Figure 2.2 shows the two pulley assemblies face-on, with their parallel shafts extending away from the viewer. Let ω_p and ω_s denote the angular velocities of the primary and secondary pulley assemblies, respectively. In this view, counterclockwise is positive for both pulleys, as shown by the arrows in the figure. Thus, $\omega_p > 0$ and $\omega_s > 0$ denote counterclockwise rotation, while negative values denote clockwise rotation.

Let v_b denote the signed transport speed of belt material around the loop. The positive direction is the direction shown in Figure 2.2: from the secondary pulley to the primary pulley along the upper span, and from the primary pulley to the secondary pulley along the lower span. Thus, $v_b > 0$ denotes transport in this direction, while $v_b < 0$ denotes transport in the opposite direction.

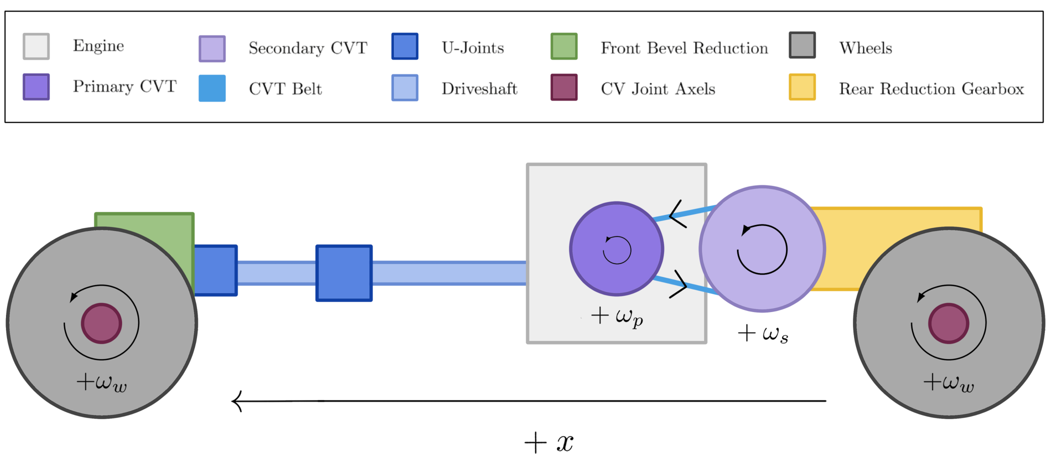


Figure 2.2: Rotational and belt-transport sign convention viewed from the outboard pulley faces, looking inward along the parallel shaft axes. Counterclockwise rotation is positive for both pulleys. With the primary pulley on the left and the secondary pulley on the right, positive belt transport runs from secondary to primary along the upper span and from primary to secondary along the lower span.

Torque follows the corresponding shaft convention. A torque acting on shaft $j \in \{p, s\}$ is positive when it acts in the rotational sense defined by $\omega_j > 0$. In particular, the interface torque $\tau_{B,j} > 0$ acts on the CVT in this positive direction. A torque that opposes positive shaft rotation is negative, but the torque coordinate itself does not change if the shaft reverses.

An attached engine, vehicle, dynamometer, or other system may use its own coordinates internally. The shaft speed and applied torque exchanged at its CVT interface are expressed using the convention defined here.

2.3.2 Local Axial Sheave Coordinates

Each movable sheave is assigned a local axial coordinate. Let x_p denote the position of the primary movable sheave and x_s the position of the secondary movable sheave. More generally, x_j denotes the corresponding coordinate for pulley $j \in \{p, s\}$.

For either pulley, $x_j = 0$ denotes the fully open position. Increasing x_j moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing x_j opens it. If $x_{j,\max}$ denotes the available travel, then

$$x_j = 0 \quad \text{is fully open}, \qquad x_j = x_{j,\max} \quad \text{is fully closed}.$$

The velocity follows the same convention: $\dot{x}_j > 0$ denotes closing motion and $\dot{x}_j < 0$ denotes opening motion. The fully open position is only the geometric origin of the coordinate; it does not prescribe the configuration from which a simulation must begin.

The two coordinates are local to their respective sheave pairs. Their positive directions may therefore point in opposite physical directions along a common global axis. What they share is their mechanical meaning: positive axial motion closes the corresponding pulley.

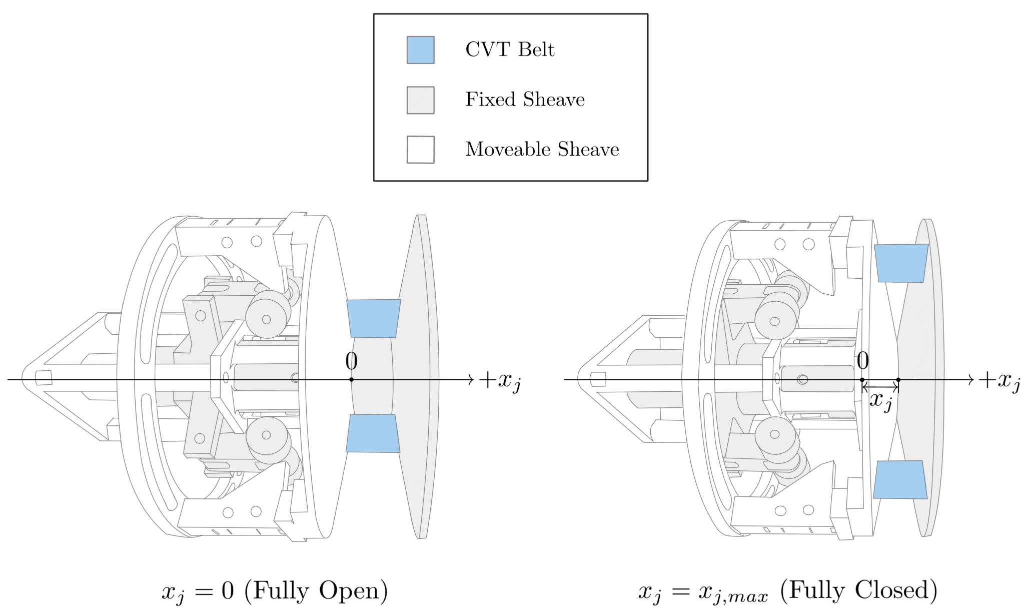


Figure 2.3: Local axial convention for the primary and secondary movable sheaves. For either pulley j, $x_j = 0$ denotes the fully open position and increasing x_j moves the movable sheave toward the fixed sheave. The fully closed position is $x_j = x_{j,\max}$. Because the coordinates are local, the positive primary and secondary directions need not point in the same global direction.

2.3.3 Local Radial, Tangential, and Axial Directions

At either pulley, the radial and tangential directions lie in the plane of pulley rotation. At any point around the pulley, the radial unit vector $\hat{\mathbf{e}}_r$ points outward from the pulley axis. The tangential unit vector $\hat{\mathbf{e}}_\theta$ is perpendicular to $\hat{\mathbf{e}}_r$ and points in the direction produced by the positive rotation defined in Section 2.3.1. A radial or tangential component is positive when it points along its corresponding unit vector.

These vectors form a local polar basis rather than a fixed Cartesian pair. Their directions therefore change with the point around the pulley, while $\hat{\mathbf{e}}_r$ remains outward and $\hat{\mathbf{e}}_\theta$ remains tangent in the positive rotational direction. The inset in Figure 2.4 shows this local basis.

The positive axial direction for pulley j is the direction of increasing x_j defined in Section 2.3.2. It lies along the pulley shaft and is perpendicular to the radial–tangential plane. In the face-on view of Figure 2.4, the shaft axis therefore passes through the page. That figure illustrates only the radial and tangential directions; the axial sign remains the local sheave-closing direction shown in Figure 2.3.

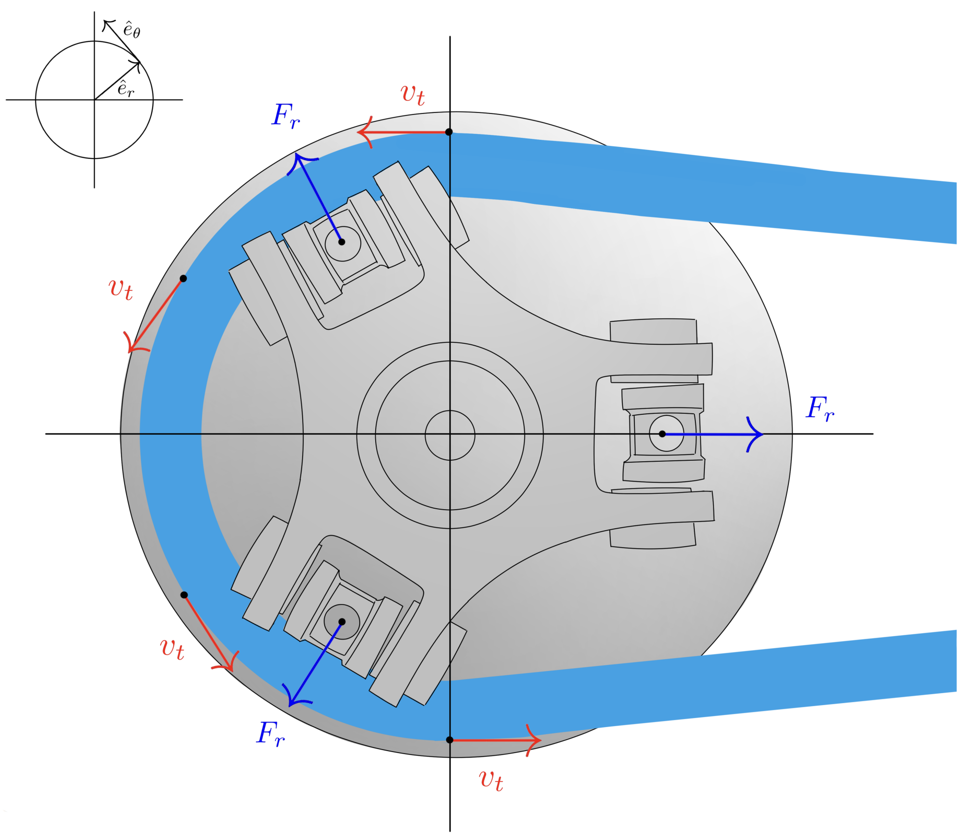


Figure 2.4: Radial and tangential directions in the plane of pulley rotation. The inset shows the local polar basis: $\hat{\mathbf{e}}_r$ points outward from the pulley axis, while $\hat{\mathbf{e}}_\theta$ is tangent in the positive rotational direction defined in Section 2.3.1. The basis changes orientation with the point around the pulley rather than remaining fixed like a Cartesian pair.

2.4 CVT State Variables

At any instant, the modeled CVT's motion and configuration are described by five internal numerical quantities. These quantities are collected in the state vector,

State Vector **(2.4.1)**

$$\boldsymbol{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \\ v_b \end{bmatrix}$$

Here, ω_p and ω_s are the primary and secondary shaft speeds. The shift coordinate s uniquely identifies the positions of both movable sheaves, while $\dot{s}$ is the corresponding shift velocity. The geometric relation between s and the local coordinates x_p and x_s is established in Section 3.1.3. The remaining entry, v_b, is the belt transport speed.

The primary and secondary pulley assemblies rotate about separate shafts. Although the belt couples their motion, neither shaft speed can generally be reconstructed from the other. Both ω_p and ω_s must therefore be retained. Absolute shaft angles are not required because the internal CVT mechanics depend on the current angular velocities, but not on the angular orientation of either pulley about its shaft.

The current pulley geometry and the configurations of the retained actuation mechanisms depend on the shift position, so s must also be retained. Shift motion has inertia and is governed by a second-order balance. Its present velocity $\dot{s}$ is therefore required alongside its position.

The belt transport speed is retained independently because belt motion need not equal the surface speed of either pulley when tangential slip is present. Under the uniform-belt and single-transport-coordinate assumptions, the mechanics do not depend on which material point occupies a particular location around the loop. No belt-phase or accumulated belt-displacement coordinate is therefore required.

Together, these five values contain all of the internal position and velocity information that must be carried from one instant to the next. For the selected CVT, these values, the current shaft-interface conditions, and the active mechanical constraints are sufficient for the formulation to determine

$$\dot{\boldsymbol{x}}(t) = \begin{bmatrix} \dot{\omega}_p \\ \dot{\omega}_s \\ \dot{s} \\ \ddot{s} \\ \dot{v}_b \end{bmatrix}. \tag{2.4.2}$$

This derivative gives the time rate of change of every state quantity. While the active constraints remain unchanged, advancing the state using these rates produces its value at the next instant, where the same evaluation is repeated. No other internal position or velocity variable must be retained.

The vector in Equation (2.4.1) describes only the modeled CVT. An engine, vehicle, dynamometer, or other attached system may carry additional states in its own model and use them when determining the torque and equivalent inertia presented at a shaft interface.

2.5 Mechanical Regimes and State Evolution

Consider the shift coordinate at one of its travel limits with $\dot{s} = 0$. The same position and velocity can describe two different mechanical conditions. If the surrounding forces press the movable sheave against the limit, a stop reaction is required and further motion into the stop is prevented. If the unconstrained tendency is away from the limit, the stop reaction vanishes and the sheave is free to accelerate away. The values stored in $\boldsymbol{x}$ are identical in both cases, but the motion that is allowed is not.

The missing information is which contacts and mechanical constraints are currently active. Their active set is called the *mechanical regime*. For the modeled CVT, a regime identifies which belt–pulley contacts exist, whether the shift coordinate is free or supported at a unilateral boundary, and whether each existing tangential contact is sticking or sliding. Let $q(t)$ denote the current regime. This label is not another position or velocity and does not introduce another moving part; it records the mechanical arrangement under which the state is evolving.

The complete internal condition needed to continue the model is therefore the pair $\big(\boldsymbol{x}(t), q(t)\big)$. The state vector records the current motion and configuration, while the regime identifies the constraints that determine which motion is admissible. For the selected CVT and a fixed regime, the current state and shaft-interface conditions determine the state derivative. At this level, that relation may be written as

$$\dot{\boldsymbol{x}} = \mathcal{F}_q(t, \boldsymbol{x}, \tau_{\mathrm{B},p}, I_{\mathrm{B},p}, \tau_{\mathrm{B},s}, I_{\mathrm{B},s}) \,, \tag{2.5.1}$$

The function $\mathcal{F}_q$ represents the mechanics of the selected CVT under the current mechanical arrangement. Its subscript indicates that the active contacts and constraints select the applicable form of the mechanical balances. While that regime remains admissible, the state advances continuously according to Equation (2.5.1).

Every regime is defined by physical conditions that must remain satisfied. An event occurs when the evolving motion reaches the boundary of those conditions: a contact forms or separates, a travel limit is reached or released, or an existing tangential contact can no longer maintain its current stick–slip condition. At that instant, the old mechanical arrangement no longer provides the admissible continuation of the motion and the regime must change.

Not every regime change alters the state instantaneously. he simplest case is a nonimpulsive regime change: the active force or constraint description changes without producing an impulse, so the positions and velocities pass through the event unchanged,

$$\boldsymbol{x}^{+} = \boldsymbol{x}^{-}, \tag{2.5.2}$$

where the superscripts $(-)$ and $(+)$ denote the values immediately before and after the event. The new regime changes the derivative obtained from Equation (2.5.1), even though the state itself is continuous.

A different transition is required when a rigid kinematic constraint becomes active with a nonzero velocity into that constraint. The configuration remains continuous, but one or more velocity components may change over the effectively instantaneous event. In general, the compatible post-event state may be written as

$$\boldsymbol{x}^{+} = \Delta_{q^{-}\to q^{+}}\big(\boldsymbol{x}^{-}\big)\,. \tag{2.5.3}$$

The transition law enforces the newly active kinematic constraint together with impulse–momentum balance for the retained inertia. It does not require each velocity or each coordinate momentum to remain unchanged. A constraint impulse can change the velocity components, and kinetic energy may decrease when the capture is inelastic. Once the transition is complete, continuous evolution resumes under the new regime.

The resulting motion has the repeated structure

$$\text{continuous evolution} \;\longrightarrow\; \text{event} \;\longrightarrow\; \text{constraint-consistent transition} \;\longrightarrow\; \text{new continuous evolution.} \tag{2.5.4}$$

Because the CVT combines continuous state evolution with discrete changes in its active mechanical arrangement, its motion forms a *hybrid initial-value problem.* A complete initial condition specifies both an initial state and an initial regime,

$$\boldsymbol{x}(t_0) = \boldsymbol{x}_0, \qquad q(t_0) = q_0, \tag{2.5.5}$$

with $\boldsymbol{x}_0$ satisfying the contacts and constraints represented by q_0. The regimes are therefore not separate CVT models. They are different admissible arrangements of the same pulley assemblies, belt, actuation mechanisms, and shaft interfaces. The state vector records how that mechanism is moving; the regime records which motion is presently allowed.

2.6 Model Quantities and Evaluation Flow

Advancing the CVT from one instant to the next requires more than the stored internal condition alone. The calculation also uses data that define the selected CVT and loading supplied through the two shaft interfaces, and it produces quantities that are needed only for the current evaluation. These quantities serve different roles in the model and should not all be treated as state.

Table 2.1 separates the four roles used throughout the formulation. The fixed data represented by its first row are collected in **??**, while the instantaneous quantities represented by its final row are listed by mechanical regime in **??**. Fixed CVT data may be specified directly or calculated once from the selected hardware during initialization. In either case, they define the CVT being modeled and remain unchanged while that simulation is advanced. The time-varying internal condition is instead carried by the state and regime introduced in the preceding sections.

Table 2.1: Roles of the quantities used during a CVT model evaluation.

Quantity role	**Contents**	**Use during evaluation**
Fixed CVT data	Physical and geometric data for the selected hardware, including any values calculated once during initialization	Define the particular CVT and the mechanical relations used to represent it; they are not advanced through time.
Current CVT condition	State vector $\boldsymbol{x}(t)$ and mechanical regime $q(t)$	Record the time-varying internal information retained from the preceding instant and identify the currently active constraints.
Shaft-interface data	Applied torque $\tau_{\mathrm{B},j}$ and equivalent inertia $I_{\mathrm{B},j}$ at each shaft	Supply the current mechanical effect of the systems attached to the primary and secondary shafts.
Instantaneous CVT results	State derivative, current internal mechanical quantities, and regime admissibility measures	Determine how the state can advance from the current instant and whether the active regime can continue; they are recomputed rather than retained as independent state.

The pair $(\boldsymbol{x}, q)$ is therefore the only time-varying internal CVT condition passed from one model evaluation to the next. The shaft-interface quantities are supplied for the current evaluation. An attached system may use time, shaft speed, and states belonging to its own model to determine those quantities, but those external states do not become states of the CVT.

Using these roles, one evaluation proceeds as follows:

1. The current time, state $\boldsymbol{x}(t)$, and regime $q(t)$ identify the internal CVT condition at the start of the evaluation.

2. The systems attached to the primary and secondary shafts supply the torque and equivalent inertia presented at their respective interfaces under the current shaft conditions.

3. The CVT mechanics use the fixed CVT data, current internal condition, and shaft-interface data to determine the instantaneous mechanical solution. This solution provides $\dot{\boldsymbol{x}}(t)$ together with the conditions used to test whether the current regime remains admissible.

4. If the regime remains admissible, the state is advanced using $\dot{\boldsymbol{x}}$ and the regime is retained. If an event is reached, the event state is established, the applicable transition is applied when required, and the new regime is selected.

5. The resulting state and regime become the current internal condition for the next evaluation, and the cycle repeats.

Chapter 3

Derivation

Chapter 2 established the CVT as a hybrid initial-value problem: at a given instant, the current state, mechanical regime, and shaft-interface loading are known, and the model must determine the corresponding instantaneous response. This chapter derives the mechanics required to perform that evaluation. The development begins with the most general continuous configuration, in which the belt is fully engaged and the CVT is free to shift, and then returns to the mechanical boundaries and regime changes that connect successive intervals of continuous motion.

Section 3.1 first develops the pulley and belt-loop kinematics, reducing the motion of both sheaves and both pulley radii to the shared shift coordinate. Using that kinematic structure, Section 3.2 derives the axial actuation and movable-sheave dynamics, while Section 3.3 develops the corresponding rotational dynamics of the two pulley assemblies. Section 3.4 then closes the mechanics of the belt itself, relating its transport motion, tension, pulley loading, and transmitted torque. These pieces are assembled in Section 3.5, where the admissible belt–pulley contact state completes the instantaneous dynamics of the freely shifting engaged CVT. Finally, **??** extends this result to the remaining mechanical regimes, showing how the continuous equations reduce when additional constraints become active and how the state is carried consistently through transitions between them.

3.1 Pulley Geometry

The conceptual overview introduced the CVT as two variable-radius pulleys connected by a single belt. Moving the pulley sheaves changes where the belt sits on each pulley and therefore changes the transmission ratio.

Both pulleys contain a movable sheave. Let x_p denote the axial position of the movable primary sheave and x_s the axial position of the movable secondary sheave. The geometric problem is therefore twofold: how does the axial position of each movable sheave set the belt radius on its own pulley, and how does the shared belt constrain the two pulley positions to move together?

The first question can be answered from the local geometry of each pulley. The second requires following the belt around the complete loop. Once these position relationships are known, they can be differentiated to obtain the corresponding radius rates and accelerations.

The natural place to begin is the geometry of a single pulley.

3.1.1 Local Pulley Geometry

The first part of the geometric question can be answered on a single pulley: how does the axial position of its movable sheave determine the pulley radius? The same conical geometry appears on both pulleys, so the relationship can be derived once and then applied to the primary and secondary.

The axial direction and primary-sheave convention were introduced in Section 2.3.2 and illustrated in Figure 2.3. The local coordinates x_p and x_s follow the same convention on their respective pulleys.

For either pulley $j \in \{p, s\}$, $x_j = 0$ denotes the fully open configuration. Increasing x_j moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing x_j opens it. The corresponding velocity follows the same sign convention: $\dot{x}_j > 0$ denotes closing motion and $\dot{x}_j < 0$ denotes opening motion.

The primary therefore begins at $x_p = 0$ and moves in the positive direction as it closes. The secondary begins in a closed position, with $x_s > 0$, and moves toward smaller values of x_s as it opens.

Consider a belt seated between two straight sheave faces with half-angle β, as shown in Figure 3.1. Let b_{in} denote the width of the belt at its inner surface. As assumed in Assumption 4 (Fixed, Uniform Belt Cross-Section), this width remains constant.

Suppose the movable sheave closes the pulley by a distance Δx_j. At the original belt radius, the available space between the sheaves is therefore reduced from b_{in} to

$$b_{\text{in}} - \Delta x_j.$$

The belt responds by moving outward through a radial distance Δr_j. At its new position, the angled sheave faces provide an additional axial distance d on each side.

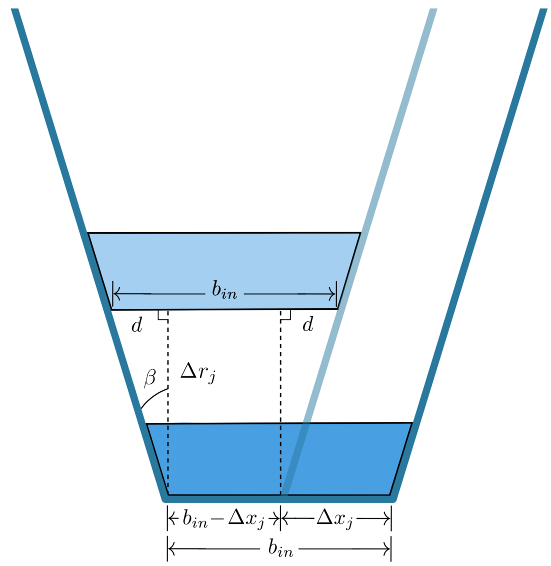


Figure 3.1: Local geometry of a belt moving outward between the sheave faces of pulley j. Closing the pulley by Δx reduces the available width at the original belt position to $b_{\text{in}} - \Delta x$. Moving outward by Δr provides an additional axial distance d at each sheave face.

At the new radius, the total space between the sheave faces must again equal the fixed belt width. Reading directly from Figure 3.1,

$$b_{\text{in}} = 2d + (b_{\text{in}} - \Delta x_j)\,. \tag{3.1.1}$$

The right triangle beside either sheave face relates d to the radial movement. For the sheave half-angle β,

$$\tan\beta = \frac{d}{\Delta r_j}, \tag{3.1.2}$$

and therefore

$$d = \Delta r_j \tan\beta. \tag{3.1.3}$$

Substituting Equation (3.1.3) into Equation (3.1.1) gives

$$b_{\text{in}} = 2\Delta r_j \tan\beta + (b_{\text{in}} - \Delta x_j)\,. \tag{3.1.4}$$

The constant belt-width terms cancel, leaving

$$\Delta r_j = \frac{\Delta x_j}{2\tan\beta}. \tag{3.1.5}$$

This is the basic axial-to-radial relationship of the pulley. Closing the pulley moves the belt outward, while opening it moves the belt inward. A smaller sheave angle produces more radial movement for the same axial motion, whereas a larger sheave angle produces less.

Taking the infinitesimal limit of Equation (3.1.5) gives

$$\frac{dr_j}{dx_j} = \frac{1}{2\tan\beta}. \tag{3.1.6}$$

Because the sheave faces are straight, this slope is constant.

Radius convention The derivation above follows the inner surface of the belt, where b_{in} is defined. The belt also has the finite radial height h_b, so its inner and outer surfaces occupy different radii. Let $r_{j,\text{inner}}$ and $r_{j,\text{outer}}$ denote those radii. They are related by

$$r_{j,\text{outer}} = r_{j,\text{inner}} + h_b. \tag{3.1.7}$$

The fixed-cross-section assumption in Assumption 4 (Fixed, Uniform Belt Cross-Section) makes h_b constant, so the axial-to-radial displacement derived above can be read from either surface. An absolute pulley radius must nevertheless identify which surface is being measured. The outer surface is used here because it provides a convenient and directly measurable reference. Throughout the geometric development,

$$r_j \equiv r_{j,\text{outer}}. \tag{3.1.8}$$

Using the outer surface gives a clear measure of where the belt is seated. It does not mean that every physical property of the belt acts at that radius.

The belt tension is carried along the cords embedded within the belt, so it acts through the centre of the cord layer. The distance from the pulley shaft to this line is therefore the lever arm through which belt tension transmits torque, making it the effective radius for gearing. This cord-line radius is commonly called the pitch radius (Gerbert, 1996).

Let d_{cord} denote the radial depth of the cord-layer centre from the outer surface, with $0 < d_{\text{cord}} < h_b$. Because the cord line lies this far inside the outer surface, its effective radius is

$$r_{j,\text{eff}} = r_{j,\text{outer}} - d_{\text{cord}}. \tag{3.1.9}$$

The belt mass is spread throughout the whole cross-section rather than concentrated at the outer surface or the cord line. To describe where this mass is with one radius, we use its centre of mass and denote the radius to that point by $r_{j,\text{cm}}$. Under the uniform-density approximation, the centre of mass is the geometric centroid of the trapezoid.

Let y measure radially inward from the outer surface, with $y = 0$ at the outer surface and $y = h_b$ at the inner surface, and let y_{cm} denote the depth of the centre of mass. These two depths and their corresponding radii are shown in Figure 3.2.

Draft figure placeholder: trapezoidal belt cross-section showing the outer and inner surfaces, b_{out}, b_{in}, and h_b. Measure y radially inward from the outer surface. Mark the cord-layer centre at d_{cord} and the mass centroid at y_{cm}, together with the radial lines $r_j = r_{j,\text{outer}}$, $r_{j,\text{eff}}$, and $r_{j,\text{cm}}$.

Figure 3.2: Radial reference lines within the belt cross-section. The outer-surface radius r_j locates the seated belt. The cord-line radius $r_{j,\text{eff}}$ is the lever arm through which belt tension transmits torque, while $r_{j,\text{cm}}$ locates the belt's centre of mass. The depths d_{cord} and y_{cm} are measured radially inward from the outer surface.

If b_{out} and b_{in} are the outer and inner widths, respectively, the straight sides of the trapezoid give

$$b(y) = b_{\text{out}} - \frac{b_{\text{out}} - b_{\text{in}}}{h_b}\, y. \tag{3.1.10}$$

Because the density is uniform, a thin strip at depth y contains mass in proportion to its area $b(y)\,dy$. The centroid depth is therefore the weighted average of y over the cross-section:

$$y_{\text{cm}} = \frac{\displaystyle\int_0^{h_b} y\, b(y)\, dy}{\displaystyle\int_0^{h_b} b(y)\, dy}. \tag{3.1.11}$$

The denominator is the belt cross-sectional area,

$$A_b = \int_0^{h_b} b(y)\, dy = \frac{h_b}{2}\left(b_{\text{out}} + b_{\text{in}}\right). \tag{3.1.12}$$

Using this area in the weighted average gives

$$\begin{aligned} y_{\text{cm}} &= \frac{1}{A_b}\int_0^{h_b} y\, b(y)\, dy \\ &= h_b \frac{b_{\text{out}} + 2b_{\text{in}}}{3\left(b_{\text{out}} + b_{\text{in}}\right)}. \end{aligned} \tag{3.1.13}$$

The result has the expected limiting behaviour. If $b_{\text{out}} = b_{\text{in}}$, the cross-section is rectangular and $y_{\text{cm}} = h_b/2$. For the usual trapezoid with $b_{\text{out}} > b_{\text{in}}$, the additional area near the outer surface places the centroid slightly closer to that surface, so $y_{\text{cm}} < h_b/2$. The belt mass therefore follows the radius

$$r_{j,\text{cm}} = r_{j,\text{outer}} - y_{\text{cm}}. \tag{3.1.14}$$

The outer-surface radius map still requires one known configuration to establish its absolute placement. Let $x_{j,\text{ref}}$ denote a known axial position at which the belt is seated, and let $r_{j,\text{ref}}$ denote the outer radius measured at that same position. The outer radius at any other seated position is then

$$r_j(x_j) = r_{j,\text{ref}} + \frac{x_j - x_{j,\text{ref}}}{2\tan\beta}. \tag{3.1.15}$$

The pair $(x_{j,\text{ref}}, r_{j,\text{ref}})$ simply anchors the linear relationship to a known physical configuration.

For the secondary pulley, a convenient reference is its known closed starting position. The secondary begins with the belt seated at a large radius and opens as the CVT shifts. Since opening corresponds to decreasing x_s, Equation (3.1.15) produces the corresponding decrease in r_s.

The primary introduces an important geometric asymmetry: it can open farther than the width of the belt. At $x_p = 0$, the distance between the primary sheaves can therefore exceed b_{in}. Initial positive motion of the movable primary sheave only removes this clearance, leaving the belt at the same radius. The primary radius begins to increase only once the movable sheave reaches the belt.

This interval is the *primary deadzone*. Let $x_{p,\text{dz}}$ denote the primary position at which the movable sheave first reaches the belt. Within the deadzone,

$$0 \le x_p < x_{p,\text{dz}},$$

the primary coordinate changes, but the belt remains at its minimum primary radius. At $x_p = x_{p,\text{dz}}$, the clearance has been removed and the belt becomes seated between the sheave faces. Further closure then moves the belt outward according to the local geometry derived above.

The first-contact position provides the natural reference for the seated primary branch:

$$x_{p,\text{ref}} = x_{p,\text{dz}}, \qquad r_{p,\text{ref}} = r_{p,\text{min}}.$$

The complete primary radius map is therefore

$$r_p(x_p) = \begin{cases} r_{p,\text{min}}, & 0 \le x_p < x_{p,\text{dz}}, \\ r_{p,\text{min}} + \dfrac{x_p - x_{p,\text{dz}}}{2\tan\beta}, & x_{p,\text{dz}} \le x_p \le x_{p,\text{max}}. \end{cases} \tag{3.1.16}$$

The linear axial-to-radial relationship in Equation (3.1.15) is standard in reduced CVT geometry models (Duan et al., 2016; Ballew, 2015). It applies to both pulleys whenever the belt is seated, while Equation (3.1.16) extends the primary description through the deadzone before contact.

The local geometry now tells us the outer-surface radius associated with either movable-sheave position. The two local descriptions belong to the same closed belt, so only certain primary and secondary positions are geometrically compatible.

3.1.2 Complete Belt-Loop Geometry

The local pulley geometry describes how axial motion changes the belt's outer-surface radius on each pulley. The belt, however, passes around both pulleys. To connect those radii, the belt must now be followed around the complete loop.

Under Assumption 8 (Prescribed Planar Belt Path), the modeled belt path consists of a circular wrap on each pulley joined by two straight tangent spans. The aligned, fixed-center geometry of Assumption 5 (Aligned Pulley Axes and Fixed Center Distance) makes the two spans equal. If each straight span has length l, and the primary and secondary wrap angles are ϕ_p and ϕ_s, the total geometric length of the loop is

$$L_{\text{geo}} = r_p\phi_p + r_s\phi_s + 2l. \tag{3.1.17}$$

The quantities l, ϕ_p, and ϕ_s follow directly from the geometry shown in Figure 3.3.

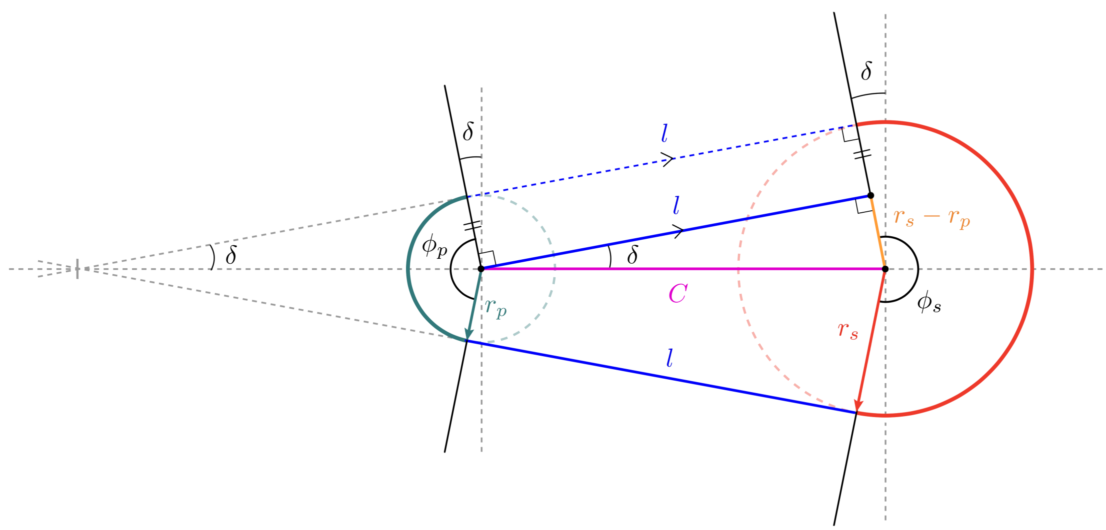


Figure 3.3: Geometry of the complete belt loop. The belt path consists of two wrapped arcs and two equal straight spans of length l. Unequal pulley radii incline the straight spans by the angle δ, producing the primary and secondary wrap angles ϕ_p and ϕ_s.

Consider the right triangle in the upper span in Figure 3.3. The line joining the pulley centres forms its hypotenuse and has length C. One side lies along the straight belt span and therefore has length l.

The other side is perpendicular to the span. The primary and secondary radii drawn to their respective contact points are also perpendicular to this span and are therefore parallel. As shown by the orange construction in Figure 3.3, the difference between their lengths gives the remaining side of the triangle, $r_s - r_p$. The triangle therefore satisfies

$$l^2 + (r_s - r_p)^2 = C^2, \tag{3.1.18}$$

and the length of either straight span is

$$l = \sqrt{C^2 - (r_s - r_p)^2}. \tag{3.1.19}$$

This expression requires

$$|r_s - r_p| < C. \tag{3.1.20}$$

This condition is already enforced by the physical separation of the pulleys. The pulley circles must remain apart, requiring $C > r_s + r_p$, and $r_s + r_p$ is necessarily greater than $|r_s - r_p|$.

The straight-span length accounts for only two parts of the belt path. The two wrapped portions still depend on how the straight spans meet the pulleys. If the pulley radii are equal, the spans are parallel to the line joining the pulley centres and the belt wraps halfway around each pulley. Both wrap angles are then equal to π.

When the radii differ, the straight spans rotate through the angle δ shown in Figure 3.3. The same right triangle gives

$$\sin\delta = \frac{r_s - r_p}{C}, \tag{3.1.21}$$

and therefore

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{3.1.22}$$

The angle δ is signed. For the configuration shown, $r_s > r_p$, so $\delta > 0$. The primary wrap loses an angle δ at each end of its contact arc, while the secondary wrap gains the same amount at each end. Their wrap angles are therefore

$$\phi_p = \pi - 2\delta, \qquad \phi_s = \pi + 2\delta. \tag{3.1.23}$$

If $r_p > r_s$, then $\delta < 0$, and the same equations automatically reverse which pulley has the larger wrap angle.

The complete belt length follows by adding the two wrapped arcs and the two straight spans. Substituting Equations (3.1.19), (3.1.22) and (3.1.23) into Equation (3.1.17) gives

Exact geometric belt length **(3.1.24)**

$$\begin{aligned} L_{\text{geo}}(r_p, r_s, C) = \pi\,(r_p + r_s) & \\ &+ 2\,(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \\ &+ 2\sqrt{C^2 - (r_s - r_p)^2} \end{aligned}$$

Each term still corresponds directly to part of the belt path. The first is the combined half-circle contribution from the two pulley wraps. The second accounts for the unequal wrap angles, and the third is the combined length of the two straight spans.

For any chosen values of r_p, r_s, and C, Equation (3.1.24) tells us how much belt is required to follow that path. The installed belt cannot change its total length as the pulleys shift. Under Assumption 6 (Fixed Belt Length), every possible pulley position must therefore satisfy

$$L_{\text{geo}}(r_p, r_s, C) = L_b, \tag{3.1.25}$$

where L_b is the fixed length of the same outer-surface path described by r_p and r_s, consistent with the convention in Equation (3.1.8). This is the geometric compatibility condition for the belt loop. Once any two of r_p, r_s, and C are known, the condition determines the third.

For a particular CVT, the first quantity to determine is the centre distance C. The pulley radii change as the sheaves move, but the pulley shafts do not move with them. Once the CVT is assembled, C remains fixed throughout operation.

The fixed centre distance can be found from any installed pulley position that can be measured. At that position, the belt length and the two pulley radii are known. Denoting the measured radii by $r_{p,\text{ref}}$ and $r_{s,\text{ref}}$, the only unknown in Equation (3.1.25) is C.

In the general unequal-radius case, C appears both inside the square root and inside the inverse sine. The exact relation therefore cannot be rearranged into a useful explicit expression for C. Instead, the centre distance is found with a one-dimensional root solve.

For a trial centre distance, the difference between the belt length required by the geometry and the belt length actually available is

$$\mathcal{R}_L(r_p, r_s, C) = L_{\text{geo}}(r_p, r_s, C) - L_b. \tag{3.1.26}$$

A positive residual means that the trial geometry requires more belt than is available. A negative residual means that it requires less. The correct centre distance is the value for which the required and available lengths match:

$$\mathcal{R}_L\left(r_{p,\text{ref}}, r_{s,\text{ref}}, C\right) = 0. \tag{3.1.27}$$

The root solver varies C until this condition is satisfied.

Remark (Simplified Belt-Length Relation)

The exact root solve is required because changing pulley radii alter both the straight-span lengths and the wrap angles. When the radius difference is small relative to the centre distance,

$$\frac{|r_s - r_p|}{C} \ll 1,$$

the straight-span angle is also small:

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

A common simplification is to neglect the resulting wrap-angle correction while retaining the exact straight-span length. The compatibility relation then becomes

$$L_b \approx \pi\left(r_p + r_s\right) + 2\sqrt{C^2 - \left(r_s - r_p\right)^2}.$$

Unlike the exact relation, this approximation can be rearranged directly. For example, the centre distance is

$$C \approx \sqrt{\left(r_s - r_p\right)^2 + \frac{1}{4}\left[L_b - \pi\left(r_p + r_s\right)\right]^2}.$$

The present model retains the exact relation. The effect of using the simplified form is examined in Appendix A.

Once the initial root solve has determined C, both the centre distance and the belt length are known constants. The compatibility condition can then be used to connect the two pulley radii at every subsequent position.

Suppose the primary radius r_p is known. Substituting r_p, C, and L_b into Equation (3.1.26) leaves the secondary radius as the only unknown. Its compatible value is found from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \qquad \text{solved for } r_s. \tag{3.1.28}$$

The resulting r_s is the secondary radius required for the same fixed belt to pass around both pulleys. The calculation also works in the opposite direction. If r_s is known instead, the compatible primary radius follows from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \qquad \text{solved for } r_p. \tag{3.1.29}$$

Thus, once C has been established, either pulley radius can be used to determine the other. The two radii remain local geometric quantities, but they cannot vary independently while belonging to the same fixed-length belt loop.

The arc-and-tangent construction used above is standard in belt-drive geometry. Budynas and Nisbett (2015) presents the same underlying construction for relating pulley radii, straight spans, wrap angles, and total belt length. The exact geometry is retained here because it provides the compatibility relation between the two shifting pulleys.

The local relations $r_p(x_p)$ and $r_s(x_s)$ now describe how each pulley produces its radius, while Equation (3.1.25) describes which pairs of radii can belong to the same belt loop. Combining these results gives the relationship between the two local axial coordinates.

3.1.3 From Local Pulley Coordinates to the Shift Coordinate

The preceding geometry gives a local relation for each pulley: x_p determines $r_p(x_p)$, and x_s determines $r_s(x_s)$. A complete CVT configuration, however, cannot be formed by choosing these positions independently. Both radii must belong to the same belt loop of length L_b. The remaining problem is therefore to identify the pairs (x_p, x_s) that satisfy this shared geometry.

For any proposed pair, the local pulley relations first give the two radii. Substituting those radii into the belt-length residual gives

$$\mathcal{R}_x(x_p, x_s) \equiv \mathcal{R}_L(r_p(x_p), r_s(x_s), C) = 0. \tag{3.1.30}$$

Every pair that makes Equation (3.1.30) equal to zero is a possible position of the two pulleys. Together, these pairs form the compatibility curve.

To reveal the shape of this curve, either local coordinate could be chosen as a starting point. Take x_p simply as an arbitrary choice. For each value of x_p, the primary relation gives $r_p(x_p)$. The shaft spacing fixes C, and L_b remains fixed by Assumption 6 (Fixed Belt Length), so the belt-length residual can then be solved for the required secondary radius r_s. The local secondary relation finally gives the corresponding position x_s:

$$x_p \longrightarrow r_p(x_p) \longrightarrow r_s \longrightarrow x_s.$$

Repeating these steps across the primary travel produces the representative compatibility curve in Figure 3.4.

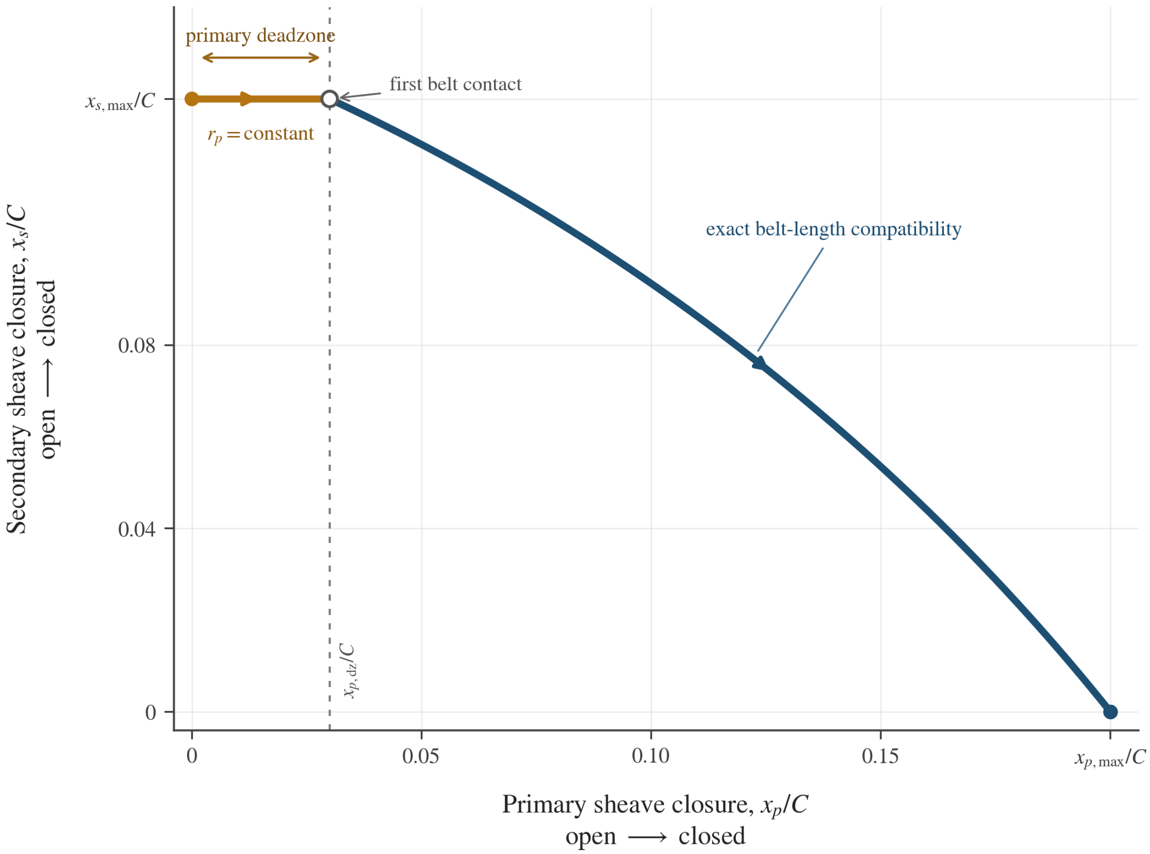


Figure 3.4: Representative compatibility curve between the local primary and secondary axial coordinates for an admissible belt–pulley geometry. Both coordinates are normalized by the same centre distance C, so their relative travel and the changing slope of the engaged branch are shown on a common scale. The dimensions are illustrative and do not represent a particular CVT.

The horizontal portion is created by the primary deadzone. While $0 \leq x_p < x_{p,\mathrm{dz}}$, the primary sheave closes only the clearance to the belt. The primary radius remains fixed, so the compatible secondary radius and position remain fixed as well.

Once the primary reaches the belt, further closure increases r_p. The fixed belt loop then requires r_s to decrease, which means that the secondary opens and x_s decreases. The two sheaves do not move by equal amounts, and the resulting branch is not linear. Changing the radii alters both wrap lengths and both straight spans, so the compatible secondary displacement changes throughout the shift.

The curve also reveals which local coordinate can represent the complete geometry. A vertical line at any x_p meets the curve once, giving one compatible x_s. At the deadzone height, however, a horizontal line meets every x_p between 0 and $x_{p,\mathrm{dz}}$. Thus x_s is a unique function of x_p over the full travel, whereas x_p is not a unique function of x_s.

The primary coordinate is therefore adopted as the system shift coordinate. This is the same s included in the state vector of **??**:

System shift coordinate **(3.1.31)**

$$s \equiv x_p, \qquad x_p(s) = s, \qquad x_s = x_s(s)$$

The secondary mapping is then

$$x_s(s) = \begin{cases} x_{s,\max}, & 0 \le s < x_{p,\mathrm{dz}}, \\ \underset{0 \le x_s \le x_{s,\max}}{\mathrm{root}} \left[\mathcal{R}_x(s, x_s) = 0\right], & x_{p,\mathrm{dz}} \le s \le x_{p,\max}. \end{cases} \tag{3.1.32}$$

The first branch holds the secondary fixed while the primary closes its clearance; the second returns the unique compatible position after belt contact. Both give $x_{s,\max}$ at $s = x_{p,\mathrm{dz}}$, so the mapping is continuous. The shift coordinate therefore determines both movable-sheave positions and, through the local pulley relations, both pulley radii. The position geometry is now complete. What remains is to determine how these positions and radii move when s changes with time.

3.1.4 Pulley-Radius Rates

The shift coordinate now determines the positions of both movable sheaves and the corresponding pulley radii. Its rate form asks the same geometric question over a small motion: if the primary closes by an amount ds, how far must the secondary open, and how quickly do the two pulley radii change?

We first find how each local coordinate and pulley radius changes per unit s. Throughout this subsection, a prime denotes differentiation with respect to s. For either pulley $j \in \{p, s\}$, the corresponding time rates are then

$$\dot{x}_j = x_j'(s)\dot{s}, \qquad \dot{r}_j = r_j'(s)\dot{s}. \tag{3.1.33}$$

Primary radius rate The primary side follows immediately from the definition $x_p(s) = s$:

$$x_p'(s) = 1, \qquad \dot{x}_p = \dot{s}. \tag{3.1.34}$$

The corresponding radius map is constant through the primary deadzone and linear after the movable sheave reaches the belt. Differentiating Equation (3.1.16) therefore gives

$$r_p'(s) = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ \dfrac{1}{2\tan\beta}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases} \tag{3.1.35}$$

Multiplying this slope by $\dot{s}$ gives

Primary Radius Rate **(3.1.36)**

$$\dot{r}_p = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ \dfrac{\dot{s}}{2\tan\beta}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

Thus, the primary sheave can move while crossing the deadzone even though the primary belt radius does not. After contact, the sheave angle fixes the radial change produced by each unit of primary closure.

Secondary coordinate and radius rates The secondary requires one additional step. Its local pulley relation tells us how r_s changes when x_s changes, but it does not tell us how far the secondary must move. That motion is set by the fixed belt loop.

At every compatible position,

$$L_{\mathrm{geo}}(r_p(s), r_s(s), C) = L_b. \tag{3.1.37}$$

The centre distance C is fixed by the pulley shafts, while L_b is constant under Assumption 6 (Fixed Belt Length). A small change in s therefore cannot change the geometric length of the belt path:

$$\frac{dL_{\mathrm{geo}}}{ds} = 0. \tag{3.1.38}$$

Both radii change during this motion. The change caused by r_p is the sensitivity of the belt length to r_p, multiplied by the actual rate r_p'. The secondary contributes in the same way. Applying the chain rule to Equation (3.1.37) gives

$$\frac{\partial L_{\mathrm{geo}}}{\partial r_p} r_p' + \frac{\partial L_{\mathrm{geo}}}{\partial r_s} r_s' = 0. \tag{3.1.39}$$

The two partial derivatives have a direct purpose: each tells us how much additional belt path would be required if one pulley radius increased while the other radius remained fixed. The actual pulley motion must make these two changes cancel. We therefore calculate the two sensitivities next.

The span angle and straight-span length entering the belt-length relation were derived in Equations (3.1.19) and (3.1.22). Writing them together in the form needed for differentiation,

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right), \qquad l = \sqrt{C^2 - (r_s - r_p)^2}, \tag{3.1.40}$$

Here, δ sets the wrap angles through Equation (3.1.23), while l is the length of either straight span. Using these quantities, the exact path length from Equation (3.1.24) is

$$L_{\mathrm{geo}} = \pi(r_p + r_s) + 2(r_s - r_p)\delta + 2l. \tag{3.1.41}$$

First, hold r_s and C fixed and increase r_p. The radius difference $r_s - r_p$ then decreases at the same rate:

$$\frac{\partial(r_s - r_p)}{\partial r_p} = -1. \tag{3.1.42}$$

Using this result in the definitions of δ and l gives

$$\begin{aligned}\frac{\partial \delta}{\partial r_p} &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(-\frac{1}{C}\right) \\ &= -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}} = -\frac{1}{l}, \end{aligned} \tag{3.1.43}$$

$$\begin{aligned}\frac{\partial l}{\partial r_p} &= \frac{1}{2\sqrt{C^2 - (r_s - r_p)^2}} \left[2(r_s - r_p)\right] \\ &= \frac{r_s - r_p}{l}. \end{aligned} \tag{3.1.44}$$

We can now differentiate the complete path length in Equation (3.1.41). The first term differentiates directly, the wrap correction requires the product rule, and the final term contains the changing straight spans:

$$\begin{aligned}\frac{\partial L_{\text{geo}}}{\partial r_p} &= \pi + 2\left[(-1)\delta + (r_s - r_p)\left(-\frac{1}{l}\right)\right] + 2\left(\frac{r_s - r_p}{l}\right) \\ &= \pi - 2\delta \\ &= \phi_p. \end{aligned} \tag{3.1.45}$$

The two terms containing $(r_s - r_p)/l$ cancel. The remaining sensitivity is the primary wrap angle.

The secondary calculation follows the same path, but increasing r_s increases the radius difference:

$$\frac{\partial(r_s - r_p)}{\partial r_s} = 1. \tag{3.1.46}$$

The corresponding derivatives of δ and l are

$$\begin{aligned}\frac{\partial \delta}{\partial r_s} &= \frac{1}{\sqrt{1-\left(\dfrac{r_s-r_p}{C}\right)^2}}\left(\frac{1}{C}\right)\\ &= \frac{1}{l},\end{aligned} \tag{3.1.47}$$

$$\begin{aligned}\frac{\partial l}{\partial r_s} &= \frac{1}{2\sqrt{C^2-(r_s-r_p)^2}}\left[-2(r_s-r_p)\right]\\ &= -\frac{r_s-r_p}{l}.\end{aligned} \tag{3.1.48}$$

Differentiating the full path length with these signs gives

$$\begin{aligned}\frac{\partial L_{\text{geo}}}{\partial r_s} &= \pi + 2\left[(1)\delta + (r_s-r_p)\left(\frac{1}{l}\right)\right] + 2\left(-\frac{r_s-r_p}{l}\right)\\ &= \pi + 2\delta\\ &= \phi_s.\end{aligned} \tag{3.1.49}$$

Again, the terms involving the changing straight-span length cancel. The sensitivity to the secondary radius is the secondary wrap angle. Together,

$$\frac{\partial L_{\text{geo}}}{\partial r_p} = \phi_p, \qquad \frac{\partial L_{\text{geo}}}{\partial r_s} = \phi_s. \tag{3.1.50}$$

This result also has a simple geometric interpretation. If one pulley radius increases by a small amount while the other is held fixed, the required belt path increases by that radial change multiplied by the angle over which the belt wraps the pulley. The remaining changes in the tangent geometry cancel one another.

Returning to the fixed-length condition Equation (3.1.39), the two radial changes must therefore satisfy

$$\phi_p r_p' + \phi_s r_s' = 0. \tag{3.1.51}$$

After primary contact, the local pulley relation gives the same axial-to-radial slope on both pulleys. Since $x_p' = 1$,

$$r_p' = \frac{1}{2\tan\beta}, \qquad r_s' = \frac{1}{2\tan\beta}x_s'. \tag{3.1.52}$$

Substituting these two slopes into Equation (3.1.51) gives

$$\phi_p\left(\frac{1}{2\tan\beta}\right) + \phi_s\left(\frac{1}{2\tan\beta}x_s'\right) = 0. \tag{3.1.53}$$

The common local pulley slope cancels, leaving

$$\phi_p + \phi_s x_s' = 0. \tag{3.1.54}$$

Solving for the slope of the secondary mapping gives

Secondary Coordinate Slope **(3.1.55)**

$$x_s'(s) = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

Within the deadzone, the primary radius does not change, so neither the compatible secondary radius nor x_s changes. After contact, $x_s' < 0$: primary closure requires the secondary to open. Its magnitude is not generally one and changes with the two wrap angles, which is why the engaged branch in Figure 3.4 is curved rather than straight.

The time rate of the secondary coordinate is now obtained by multiplying this slope by $\dot{s}$:

$$\dot{x}_s = x_s'(s)\dot{s} = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}\dot{s}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases} \tag{3.1.56}$$

Applying the local axial-to-radial slope once more gives the secondary radius rate:

Secondary Radius Rate **(3.1.57)**

$$\dot{r}_s = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}\dfrac{\dot{s}}{2\tan\beta}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

The rates above were written for r_p and r_s, which denote the outer-surface radii. The inner surface, cord line, and mass centre remain at fixed depths within the belt cross-section. When the belt moves radially on either pulley, all four reference radii therefore change by exactly the same amount. For either pulley $j \in \{p, s\}$,

$$r_j' = r_{j,\mathrm{inner}}' = r_{j,\mathrm{eff}}' = r_{j,\mathrm{cm}}', \qquad \dot{r}_j = \dot{r}_{j,\mathrm{inner}} = \dot{r}_{j,\mathrm{eff}} = \dot{r}_{j,\mathrm{cm}}. \tag{3.1.58}$$

The choice of reference changes the absolute radius, but not its rate. The two pulley-radius rates are now determined by s and $\dot{s}$. However, the secondary multiplier $-\phi_p/\phi_s$ changes as the radii and wrap angles change. Even if $\dot{s}$ is constant, the secondary rate is not. Its acceleration must therefore include the changing slope of the compatibility curve.

3.1.5 Pulley-Radius Accelerations

The pulley-radius rates in Equations (3.1.36) and (3.1.57) were obtained by multiplying the slope of each geometric mapping by the shift velocity $\dot{s}$. Their accelerations follow by asking how those rates change. On the primary, the mapping slope is constant within each branch. On the secondary, the slope in Equation (3.1.55) changes as the pulley radii and wrap angles change. The secondary acceleration must therefore account for both the imposed shift acceleration and the changing geometric slope.

Primary radius acceleration The primary coordinate requires no additional geometry. The system-coordinate definition in Equation (3.1.31) gives $x_p(s) = s$, so

$$x_p'(s) = 1, \qquad x_p''(s) = 0, \tag{3.1.59}$$

and therefore

$$\dot{x}_p = \dot{s}, \qquad \ddot{x}_p = \ddot{s}. \tag{3.1.60}$$

The general shift-to-time relation in Equation (3.1.33) gives the primary radius rate in the form

$$\dot{r}_p = r_p'(s)\dot{s}. \tag{3.1.61}$$

Differentiating this product with respect to time gives

$$\begin{aligned} \ddot{r}_p &= \frac{d}{dt}\left[r_p'(s)\dot{s}\right] \\ &= \frac{dr_p'}{dt}\dot{s} + r_p'(s)\ddot{s}. \end{aligned} \tag{3.1.62}$$

Because r_p' is itself a function of s,

$$\frac{dr_p'}{dt} = r_p''(s)\dot{s}. \tag{3.1.63}$$

Substituting Equation (3.1.63) into Equation (3.1.62) gives

$$\ddot{r}_p = r_p''(s)\dot{s}^2 + r_p'(s)\ddot{s}. \tag{3.1.64}$$

Within either smooth branch of the primary radius map, $r_p'(s)$ is constant. Hence

$$r_p''(s) = 0. \tag{3.1.65}$$

Using the primary slopes from Equation (3.1.35), the primary radius acceleration is

Primary Radius Acceleration **(3.1.66)**

$$\ddot{r}_p = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ \dfrac{\ddot{s}}{2\tan\beta}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

The result is direct: within the deadzone the primary radius remains fixed, while after contact the shift acceleration is converted into radial acceleration by the same constant sheave slope found in Equation (3.1.35).

Secondary coordinate curvature The secondary mapping is not linear after contact. Its first derivative was found from the fixed-length rate balance in Equation (3.1.54):

$$\phi_p + \phi_s x_s' = 0.$$

The balance in Equation (3.1.54) contains the information needed for the second derivative. We will differentiate it once more with respect to s, which first requires the rates at which the two wrap angles change with s.

The required geometric quantities were defined in Equations (3.1.22) and (3.1.23). Written together, they are

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right), \qquad \phi_p = \pi - 2\delta, \qquad \phi_s = \pi + 2\delta. \tag{3.1.67}$$

As s changes, both r_p and r_s change, so the radius difference inside δ changes at the rate

$$\frac{d}{ds}(r_s - r_p) = r_s' - r_p'. \tag{3.1.68}$$

Applying the chain rule to δ, and using the straight-span length from Equation (3.1.19) in the final line, gives

$$\begin{aligned} \delta' &= \frac{1}{\sqrt{1 - \left(\dfrac{r_s - r_p}{C}\right)^2}} \left(\frac{r_s' - r_p'}{C}\right) \\ &= \frac{r_s' - r_p'}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \frac{r_s' - r_p'}{l}. \end{aligned} \tag{3.1.69}$$

After primary contact, the local pulley slopes are given by Equation (3.1.52):

$$r_p' = \frac{1}{2\tan\beta}, \qquad r_s' = \frac{x_s'}{2\tan\beta}.$$

Substituting these slopes into Equation (3.1.69) gives

$$\begin{aligned} \delta' &= \frac{\dfrac{x_s'}{2\tan\beta} - \dfrac{1}{2\tan\beta}}{l} \\ &= \frac{x_s' - 1}{2l\tan\beta}. \end{aligned} \tag{3.1.70}$$

The wrap-angle derivatives now follow directly from Equation (3.1.67):

$$\phi_p' = -2\delta', \qquad \phi_s' = 2\delta'. \tag{3.1.71}$$

We can now differentiate the secondary slope balance in Equation (3.1.54). The primary term contributes ϕ_p', while the secondary term requires the product rule because both ϕ_s and x_s' change with s:

$$\begin{aligned} 0 &= \frac{d}{ds}\left(\phi_p + \phi_s x_s'\right) \\ &= \phi_p' + \phi_s' x_s' + \phi_s x_s''. \end{aligned} \tag{3.1.72}$$

Substituting the wrap-angle derivatives from Equation (3.1.71) gives

$$-2\delta' + 2\delta' x_s' + \phi_s x_s'' = 0. \tag{3.1.73}$$

Solving first for x_s'',

$$\begin{aligned} \phi_s x_s'' &= 2\delta'(1 - x_s'), \\ x_s'' &= \frac{2\delta'(1 - x_s')}{\phi_s}. \end{aligned} \tag{3.1.74}$$

Using the expression for δ' from Equation (3.1.70),

$$\begin{aligned} x_s'' &= \frac{2}{\phi_s}\left(\frac{x_s' - 1}{2l\tan\beta}\right)(1 - x_s') \\ &= -\frac{(1 - x_s')^2}{\phi_s l\tan\beta}. \end{aligned} \tag{3.1.75}$$

On the engaged branch, the first-derivative result in Equation (3.1.55) gives $x_s' = -\phi_p/\phi_s$. The remaining factor can therefore be simplified:

$$\begin{aligned} 1 - x_s' &= 1 + \frac{\phi_p}{\phi_s} \\ &= \frac{\phi_s + \phi_p}{\phi_s} \\ &= \frac{2\pi}{\phi_s}, \end{aligned} \tag{3.1.76}$$

where $\phi_p + \phi_s = 2\pi$ follows from Equation (3.1.23). Substituting this result into Equation (3.1.75) gives the curvature of the secondary coordinate mapping:

Secondary Coordinate Curvature **(3.1.77)**

$$x_s''(s) = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{4\pi^2}{\phi_s^3 l \tan\beta}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

The deadzone branch in Equation (3.1.32) is horizontal, so both x_s' and x_s'' are zero there. On the engaged branch, $x_s'' < 0$: as the primary closes, the secondary mapping becomes progressively steeper in the opening direction.

Secondary axial and radial accelerations The curvature in Equation (3.1.77) describes how the slope of the compatibility curve changes with position. To convert it into acceleration, begin with the secondary axial velocity from Equation (3.1.56):

$$\dot{x}_s = x_s'(s)\dot{s}.$$

Differentiating with respect to time gives

$$\begin{aligned} \ddot{x}_s &= \frac{d}{dt}\left[x_s'(s)\dot{s}\right] \\ &= \frac{dx_s'}{dt}\dot{s} + x_s'(s)\ddot{s}. \end{aligned} \tag{3.1.78}$$

Because x_s' changes only through s,

$$\frac{dx_s'}{dt} = x_s''(s)\dot{s}. \tag{3.1.79}$$

The secondary axial acceleration is therefore

$$\ddot{x}_s = x_s'(s)\ddot{s} + x_s''(s)\dot{s}^2. \tag{3.1.80}$$

Substituting the slope from Equation (3.1.55) and the curvature from Equation (3.1.77) gives

Secondary Axial Acceleration **(3.1.81)**

$$\ddot{x}_s = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}\ddot{s} - \dfrac{4\pi^2}{\phi_s^3 l \tan\beta}\dot{s}^2, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

The local radius derivative in Equation (3.1.6) is the constant $1/(2\tan\beta)$. Applying it to the secondary gives

$$\ddot{r}_s = \frac{1}{2\tan\beta}\ddot{x}_s. \tag{3.1.82}$$

Substituting Equation (3.1.81) into Equation (3.1.82) gives

Secondary Radius Acceleration **(3.1.83)**

$$\ddot{r}_s = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}\dfrac{\ddot{s}}{2\tan\beta} - \dfrac{2\pi^2}{\phi_s^3 l \tan^2\beta}\dot{s}^2, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

The two terms have different physical origins. The first comes from the imposed shift acceleration, converted through the coordinate slope in Equation (3.1.55) and the local pulley slope in Equation (3.1.6). The second comes from the curvature in Equation (3.1.77): even when $\ddot{s} = 0$, moving through the engaged branch changes $-\phi_p/\phi_s$, so the secondary velocity continues to change.

As with the radius rates, the fixed offsets between the four radial references give, for either pulley $j \in \{p, s\}$,

$$\ddot{r}_j = \ddot{r}_{j,\mathrm{inner}} = \ddot{r}_{j,\mathrm{eff}} = \ddot{r}_{j,\mathrm{cm}}. \tag{3.1.84}$$

Remark (Deadzone Boundary)

The idealized maps in Equations (3.1.16) and (3.1.32) change slope when the primary first reaches the belt at $s = x_{p,\mathrm{dz}}$. Their first derivatives therefore change discontinuously, and their ordinary second derivatives are not defined at that single point. The acceleration relations in Equations (3.1.66) and (3.1.83) apply within the deadzone and engaged branches, while first contact is treated as the transition between them.

Remark (Practical Consequences of the Pulley Geometry)

The radius, rate, and acceleration relations derived in this section describe how the pulley geometry evolves as the CVT shifts. Appendix C explores the practical consequences of these results in more detail. In particular, it examines the resulting geometric CVT ratio and ratio-change rate across the shift travel, together with how the current pulley geometry and sheave angle shape the available ratio range and ratio-change sensitivity.

3.1.6 Geometric Closure and Model Scope

Specifying the shift state now determines the complete modeled pulley geometry. The definition in Equation (3.1.31) sets the primary position directly, while Equation (3.1.32) gives the unique compatible secondary position. The local pulley maps in Equations (3.1.15) and (3.1.16) convert those two positions into the outer-surface radii used to describe the belt path. Finally, the rate and acceleration relations in Equations (3.1.36), (3.1.57), (3.1.66) and (3.1.83) determine how those radii move. Thus, for any admissible shift state,

$$(s, \dot{s}, \ddot{s}) \quad \longmapsto \quad (r_p, \dot{r}_p, \ddot{r}_p, r_s, \dot{r}_s, \ddot{r}_s)\,.$$

Here r_p and r_s retain the outer-surface convention used throughout the geometric development. The corresponding inner-surface, effective, and centre-of-mass radii follow directly from their fixed positions within the belt cross-section. Nothing else has to be solved for.

These results follow from the geometry derived above under the idealizations stated in Section 2.2. The belt cross-section and total length remain fixed under Assumption 4 (Fixed, Uniform Belt Cross-Section) and Assumption 6 (Fixed Belt Length), while Assumption 8 (Prescribed Planar Belt Path) constrains the belt to circular wraps and straight tangent spans. Together with the aligned pulley axes and fixed centre distance of Assumption 5 (Aligned Pulley Axes and Fixed Center Distance), these conditions leave one modeled belt path for each admissible value of s.

Relation to prior transient models These choices also define the level of detail represented by the present geometry. It determines the complete belt path that is globally compatible with the imposed shift position, but it does not track the trajectory of each belt element through the pulley wraps.

Duan et al. (2016) begin from closely related open-belt arc-and-tangent geometry and the same wedge relationship between movable-sheave motion and nominal radius motion. Their chain-CVT formulation retains additional internal deformation. Chain extension and its rate contribute to the span tension, and pulley deformation allows the local radius followed by a chain element to differ from the nominal pulley radius. The present formulation instead fixes the belt length through Assumption 6 (Fixed Belt Length) and prescribes the arc-and-tangent belt path through Assumption 8 (Prescribed Planar Belt Path). Consequently, elastic path-length change and deformation-driven departure from that prescribed path are not represented in this geometric closure.

Rubber-belt transient models resolve the same shifting process at several levels of detail. Analytical shift-mechanics models, such as those of Sorge (2008) and Cammalleri and Sorge (2009), describe local radial belt migration through a pulley wrap and couple it to circumferential transport, sliding, and the mechanics of the contact region. Discretized rubber-belt models, such as those of Julió and Plante (2011) and Ballew (2015), instead obtain the evolving belt geometry from the motion and contact state of individual belt elements. These approaches can resolve local deformation, tension redistribution, and contact migration in greater detail, but require additional spatial or nodal states and the contact laws that govern them.

The distinction is therefore one of model resolution. The present formulation uses the global compatibility condition in Equation (3.1.25) to find the representative pulley-radius pair and its derivatives directly from s, $\dot{s}$, and $\ddot{s}$. The cited transient models additionally ask how the belt or chain deforms and migrates within that path. Those internal motions lie outside the present geometric layer; within the assumptions stated above, the global pulley geometry and its motion are fully determined.

This closure determines which sheave and radius motions are compatible with the modeled belt path, but it does not determine which compatible acceleration occurs. In particular, every acceleration relation above still contains the shift acceleration $\ddot{s}$. Its value must be consistent with the forces and inertias acting on the two movable pulley assemblies.

3.2 Axial Actuation and Sheave Balances

The preceding section established how the two movable sheaves must move together. It did not determine the motion itself. That requires the forces acting on each movable assembly and the inertia that those forces must accelerate.

The primary and secondary remain separate mechanisms even though the belt geometry couples their positions. Each has its own actuator, spring, moving hardware, and belt reaction, so each must satisfy its own axial equation of motion. We therefore complete the primary balance in its local coordinate x_p, then do the same for the secondary in x_s. Once each mechanism has been interpreted and every local force has been derived, the geometric maps developed above in Section 3.1 express both balances through the shared shift coordinate s. This final step shows how the two local free bodies act on one shift motion.

3.2.1 Primary Movable-Sheave Balance

The primary is easiest to understand by separating rotation about the shaft from motion along it. The entire primary rotates with the shaft at ω_p, but its parts do not all move axially. The fixed sheave remains at one axial position. Opposite it, the movable sheave, ramp carrier, and the hardware attached rigidly to them slide together along the shaft. Motion in the closing direction narrows the pulley groove and clamps the belt; motion in the opening direction widens the groove.

The flyweights provide the closing drive. They are attached to the fixed sheave and carried around the shaft with it, but each can also pivot about its attachment to the fixed sheave. As the primary rotates, centrifugal loading tends to swing each flyweight radially outward until it presses against a ramp carried by the movable sheave. Because the ramp surface is inclined, its contact reaction has an axial component that drives the movable sheave in the closing direction. The primary spring pushes the movable sheave in the opening direction, and the belt pushes back through its normal reaction on the movable face.

This leaves two distinct moving bodies in the model. The movable sheave, ramp carrier, and attached hardware form one translating assembly with axial coordinate x_p. The flyweights rotate with the primary and pivot separately, but do not translate rigidly with x_p; they deliver their closing action to the movable sheave through the ramp contacts. During a shift, the net axial force must accelerate the translating assembly, while the flyweights require their own rotational balance. We therefore begin by isolating the translating assembly, then determine the force transmitted to it by the flyweights. This mechanism and the resulting axial free body are shown together in Figure 3.5.

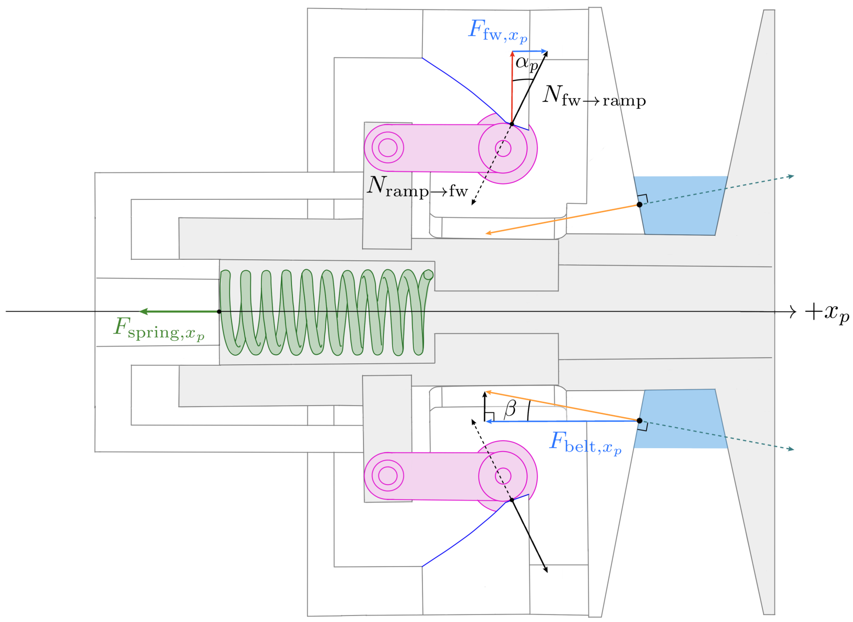


Figure 3.5: Axial free-body diagram of the primary movable assembly (unshaded). The symmetric sides of the cross-section are used to show the flyweight–ramp and belt–sheave force resolutions separately. Dotted arrows show the equal-and-opposite forces on the flyweight and belt, with F_{clamp} indicating the movable sheave's squeeze on the belt.

As established in Section 3.1, positive x_p closes the primary. Let m_p be the total mass of the movable sheave, ramp carrier, and every other component that translates rigidly with them. The three axial forces shown in Figure 3.5 are:

- F_{fw,x_p}, the closing force delivered to the ramp carrier by the flyweight contacts;
- F_{spring,x_p}, the opening force exerted by the primary spring; and
- F_{belt,x_p}, the opening force exerted by the belt on the movable face.

Newton's second law along the shaft is therefore

$$m_p \ddot{x}_p = F_{\mathrm{fw},x_p} - F_{\mathrm{spring},x_p} - F_{\mathrm{belt},x_p}. \tag{3.2.1}$$

This free body tells us exactly what remains to be found. The spring force will follow from its compression, and the belt force from the normal reaction on the conical face. The flyweight force is harder. It is the axial result of a contact on a body that pivots rather than translating rigidly with the sheave. We therefore determine F_{fw,x_p} first.

Force Transmitted by the Flyweights

Flyweight free body Figure 3.6 isolates one flyweight in the radial–axial plane fixed to the primary. The flyweight is pinned at P and pivots about the circumferential axis through that point. Its pose is described by q_f, measured from $+x_p$ to the body-fixed reference line shown in the figure. At $q_f = 0$, this line is aligned with $+x_p$; at $q_f = \pi/2$, it points radially outward. Increasing q_f therefore describes the outward-pivoting direction.

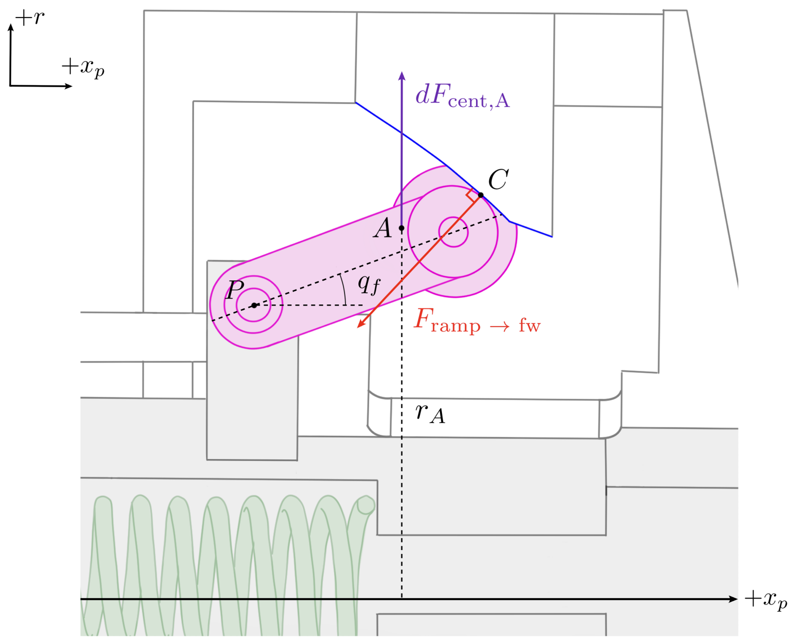


Figure 3.6: Free body of one primary flyweight in the frame rotating with the primary. Centrifugal loading acts throughout the flyweight mass; the marked element A identifies one infinitesimal mass dm at shaft radius r_A and its outward load. The ramp-contact normal at C and the pivot reaction at P complete the force system.

As the primary rotates at ω_p, the flyweight experiences centrifugal loading throughout its mass. Each infinitesimal element is loaded radially outward according to its distance from the primary shaft. The marked element A is one example: for mass dm at shaft radius r_A, its outward load is

$$dF_{\mathrm{cent},A} = \omega_p^2 r_A \, dm. \tag{3.2.2}$$

Every other element is loaded in the same way at its own shaft radius. Together, these forces form the distributed centrifugal loading of the flyweight.

Because the flyweight's relative motion is confined to the radial–axial plane, the Euler and Coriolis inertial forces are circumferential and have no moment about the circumferential pivot axis. The q_f balance therefore depends on ω_p, but not on $\dot{\omega}_p$.

Taking moments about P separates the roles of these loads. The pivot reaction passes through P and therefore contributes no moment. The radial lines of action of the centrifugal loads generally do not pass through P, so together they produce a moment in the direction of increasing q_f. The ramp normal produces the opposing moment.

The figure shows one flyweight, whereas the primary contains a symmetric set whose members follow the same q_f. Let $\tau_{f,\text{cent}}$ and $\tau_{f,\text{contact}}$ denote the respective magnitudes of the centrifugal and contact moments summed over this complete set. Since their shared coordinate q_f is a rotation about the flyweight pivots, let I_f denote the sum of each flyweight's mass moment of inertia about its own pivot axis. With positive moment defined in the direction of increasing q_f, rotational Newton's second law gives

$$\tau_{f,\text{cent}} - \tau_{f,\text{contact}} = I_f \ddot{q}_f. \tag{3.2.3}$$

Mechanism geometry The moment balance is written in the flyweight coordinate q_f, whereas the force required by the movable-sheave balance must be expressed through the axial coordinate x_p. The ramp geometry connects the two: each sheave position fixes the corresponding flyweight pose.

The centrifugal moment in Equation (3.2.3) comes from the distributed loading shown in the figure, so its value depends on how the complete flyweight mass is distributed about the primary shaft. The marked element A contributes $r_A^2\, dm$ to the mass moment of inertia about that shaft. Adding the corresponding contributions from every element of every flyweight gives the complete shaft-axis inertia $J_{f,p}$. Since x_p fixes the flyweight pose, it fixes every element's shaft radius and therefore fixes $J_{f,p}$. This shaft-axis inertia should not be confused with the pivot-axis inertia I_f in Equation (3.2.3), which is constant for rigid flyweights.

The mechanism therefore supplies two configuration maps:

$$q_f = q_f(x_p), \qquad J_{f,p} = J_{f,p}(x_p). \tag{3.2.4}$$

Together with the constant I_f, these maps abstract the physical geometry of the flyweight mechanism into the quantities required by the general force balance.

Remark (Constructing the flyweight maps)

The mechanism-specific construction is not carried out here. Appendix D.1 works through two common flyweight arrangements, showing how their geometry and mass distribution produce $q_f(x_p)$, $J_{f,p}(x_p)$, and I_f.

With these mechanism properties specified, the remaining task is to express the two moments and $\ddot{q}_f$ in Equation (3.2.3) through the primary motion. We begin with the moments.

Both could be found by resolving the forces and constructing their moment arms for the specific flyweight geometry. A more convenient approach is equivalent virtual work. Hold the mechanism at its current position and imagine a small compatible motion. During this motion, a moment τ_f acting through dq_f performs the work

$$\tau_f\, dq_f = \delta W_f. \tag{3.2.5}$$

We can therefore find each moment by calculating the work of its underlying load during the same motion.

For the centrifugal loading, at fixed instantaneous angular speed, the work associated with a change dJ in shaft-axis inertia is $\frac{1}{2}\omega^2\, dJ$. Applying this relation to the flyweights gives

$$\tau_{f,\text{cent}}\, dq_f = \delta W_{f,\text{cent}} = \frac{1}{2}\omega_p^2\, dJ_{f,p}. \tag{3.2.6}$$

For the contact moment, the corresponding work is the force–displacement work $F\, d\ell$. The movable sheave moves axially through dx_p, so only the axial component of the ramp-contact force performs work. This component is F_{fw,x_p}, the closing force introduced by the movable-sheave free body, and its work is therefore $F_{\text{fw},x_p}\, dx_p$. Under Assumption 3 (Idealized Actuator Mechanics), the ramp contact is treated as an ideal frictionless constraint, so no work is dissipated at the contact. The work of the contact moment therefore appears entirely as axial work on the movable assembly. Using the positive magnitudes defined above,

$$\tau_{f,\text{contact}}\, dq_f = \delta W_{f,\text{contact}} = F_{\text{fw},x_p}\, dx_p. \tag{3.2.7}$$

Flyweight acceleration The two moment terms are now written in forms that can be returned to the rotational balance. Its remaining term is the flyweight angular acceleration. Ramp contact makes this acceleration a consequence of the sheave motion rather than an independent state. Since the contact geometry already gives q_f as a function of x_p, its angular velocity is

$$\dot{q}_f = \frac{dq_f}{dx_p}\dot{x}_p, \tag{3.2.8}$$

and one more derivative gives

$$\ddot{q}_f = \frac{dq_f}{dx_p}\ddot{x}_p + \frac{d^2q_f}{dx_p^2}\dot{x}_p^2. \tag{3.2.9}$$

The second term appears when the amount of flyweight rotation required per unit sheave travel changes along the ramp. Even at constant $\dot{x}_p$, the flyweight then changes angular speed as it moves.

Axial closing force There is now an expression for every term in the rotational balance. Multiplying Equation (3.2.3) by dq_f and substituting Equations (3.2.6), (3.2.7) and (3.2.9) brings them together,

$$\frac{1}{2}\omega_p^2 dJ_{f,p} - F_{\text{fw},x_p} dx_p = I_f\left(\frac{dq_f}{dx_p}\ddot{x}_p + \frac{d^2q_f}{dx_p^2}\dot{x}_p^2\right) dq_f. \tag{3.2.10}$$

Rearranging for the force transmitted to the movable sheave gives

$$\boxed{\begin{aligned} F_{\mathrm{fw},x_p} = {} & \frac{1}{2}\omega_p^2 \frac{dJ_{f,p}}{dx_p} - I_f \left(\frac{dq_f}{dx_p}\right)^2 \ddot{x}_p \\ & - I_f \frac{dq_f}{dx_p}\frac{d^2 q_f}{dx_p^2}\dot{x}_p^2. \end{aligned}} \tag{3.2.11}$$

The result now has the form required by the movable-sheave free body. The first term is the centrifugal drive and therefore uses the shaft-axis property $J_{f,p}$. The final two terms arise from acceleration about the flyweight pivots and therefore use the pivot-axis property I_f: the first reflects $\ddot{x}_p$ through the motion ratio, while the second appears when that motion ratio changes with position.

Remark (Admissible flyweight maps)

The completed force law uses $q_f''(x_p)$ and $J_{f,p}'(x_p)$, while the selected coordinate directions require primary closure to correspond to outward flyweight rotation. Over the admissible shift range, the supplied mechanism maps must therefore satisfy

$$q_f \in C^2, \qquad J_{f,p} \in C^1, \qquad 0 < \frac{dq_f}{dx_p} < \infty.$$

These are requirements on the fitted or constructed maps, not additional equations of motion.

The one-sided flyweight contact can transmit compression but not tension, so the force predicted by Equation (3.2.11) must satisfy $F_{\mathrm{fw},x_p} \geq 0$.

Spring and Belt Forces

Returning to the axial force balance, the two remaining forces in Equation (3.2.1) act directly on the translating assembly.

Primary spring Let k_p be the spring stiffness and $x_{p,\mathrm{pre}}$ its installed compression when the primary is open at $x_p = 0$. Closing the sheave by x_p adds the same amount of compression, so the spring opening force is

$$F_{\mathrm{spring},x_p} = k_p \left(x_{p,\mathrm{pre}} + x_p\right). \tag{3.2.12}$$

Belt reaction Let N_p be the total integrated normal load over both primary sheave faces. This load is coupled to the belt and the secondary balance, so it remains an unknown at this stage. Under Assumption 12 (Symmetric Sheave-Face Sharing), the movable face carries one half of this total load. Its normal is inclined by the sheave half-angle β from the axial direction, so the opening force on the movable face is

$$F_{\mathrm{belt},x_p} = \frac{N_p}{2}\cos\beta. \tag{3.2.13}$$

Completed Primary Balance

Each force in the original free body now has a mechanism-level expression. Substitute Equations (3.2.11) to (3.2.13) into Equation (3.2.1) and collect the terms multiplying $\ddot{x}_p$:

$$\boxed{\begin{aligned}\left[m_p + I_f\left(\frac{dq_f}{dx_p}\right)^2\right]\ddot{x}_p = {} & \frac{1}{2}\omega_p^2\frac{dJ_{f,p}}{dx_p} \\ & - I_f\frac{dq_f}{dx_p}\frac{d^2q_f}{dx_p^2}\dot{x}_p^2 \\ & - k_p\left(x_{p,\mathrm{pre}} + x_p\right) - \frac{N_p}{2}\cos\beta.\end{aligned}} \tag{3.2.14}$$

The flyweights now appear through the same two axes introduced by their free body. Rotation about the primary shaft, described by $J_{f,p}$, provides the speed-dependent closing drive. Rotation about the flyweight pivots, described by I_f, adds to the inertia that the axial motion must accelerate. If the motion ratio dq_f/dx_p changes along the ramp, the flyweights also accelerate even when $\dot{x}_p$ is constant, producing the term proportional to $\dot{x}_p^2$. The spring and belt forces act directly on the movable assembly and oppose closure exactly as shown in the first free body.

This is the primary axial equation of motion. It states how the flyweight action is divided among spring compression, belt loading, and acceleration of the translating and pivoting hardware.

Point-Mass Limit and Relation to Existing Models

The general force law becomes the familiar classical actuator model when the distributed flyweight mass is collapsed to its centre of mass and the flyweight's pivot dynamics are neglected. Let m_f be the total mass of the symmetric flyweight set and let $r_f(x_p)$ be the common instantaneous centre-of-mass radius. Under the point-mass assumption, the shaft-axis inertia reduces to

$$J_{f,p} = m_f r_f^2. \tag{3.2.15}$$

Substituting this point-mass inertia into the centrifugal term of Equation (3.2.11), while omitting the pivot-acceleration terms, gives

$$F_{\mathrm{fw},x_p}^{\mathrm{pm}} = m_f r_f \omega_p^2 \frac{dr_f}{dx_p}. \tag{3.2.16}$$

This is the familiar point-mass result. The factor $m_f r_f \omega_p^2$ is the total centrifugal loading, while dr_f/dx_p is the local radial-to-axial motion ratio that converts that loading into sheave force. The path need not be straight, and its motion ratio need not be constant. A sliding mass on a straight ramp is one simple case; for a pivoted flyweight, $r_f(x_p)$ instead follows from the flyweight pose and its mass geometry.

If the movable sheave is also in axial equilibrium, the point-mass actuator force balances the opening forces from the spring and belt:

$$m_f r_f \omega_p^2 \frac{dr_f}{dx_p} = k_p\left(x_{p,\mathrm{pre}} + x_p\right) + \frac{N_p}{2}\cos\beta. \tag{3.2.17}$$

This reduction appears directly in the quasi-static model of Messick (2018). Their three wedge-shaped flyweights slide along linear ramps without changing orientation. Measured CAD geometry supplies the centre-of-mass radius and the radial-to-axial displacement ratio, giving a force of the same form as Equation (3.2.16). Friction is neglected, and the resulting primary force is supplied to the belt model as a boundary condition; shifting is treated as a sequence of equilibrium configurations rather than a time-dependent response. Arango and Muñoz Alzate (2020) use the same instantaneous force conversion to construct both sliding-weight ramps and pivoted flyweight-cam profiles. Their contact geometry is designed so that the available centrifugal force supplies the required clamping force through the local mechanical advantage.

Other treatments retain more of the contact geometry while keeping the same instantaneous-equilibrium closure. Kim et al. (2002) resolve the two contacts of a centrifugal roller and include contact friction when converting roller centrifugal force into driver thrust. The current roller radius and contact angles vary with ratio, but no roller acceleration enters the thrust equation. Skinner (2020) resolve a flyweight, its supporting link, and the ramp reactions in a complete free body, including the centrifugal loading of both the flyweight and the link. In the present notation, those separate mass contributions could be collected in $J_{f,p}(x_p)$. Skinner's derivation therefore retains a substantially richer equilibrium force path than a single fixed ramp factor, but still closes the flyweight system by force equilibrium at each configuration.

Despite their different levels of geometric detail, none of these actuator relations includes the acceleration of the flyweight mechanism. Messick does not model shift time, Skinner solves algebraically for the speed required at each configuration, and Kim places the time dependence in a separate experimentally identified first-order ratio law, using different fitted coefficients for upshift and downshift. Conversely, the transient belt model of Ballew (2015) imposes primary axial force through a controller rather than resolving the mechanical actuator. Across these examples, transient shift motion is therefore obtained without returning the flyweights' own acceleration to the primary axial balance.

The distinction of the present formulation is therefore not that it can describe a curved ramp or a detailed flyweight shape; the equilibrium models above can already retain substantial geometric detail. Here, the complete mass distribution enters through $J_{f,p}(x_p)$, so the centrifugal drive is not restricted to a centre-of-mass point model. More importantly, the flyweight moment balance is not closed at instantaneous equilibrium. Mapping $I_f \ddot{q}_f$ into x_p produces both the reflected pivot inertia and the changing-motion-ratio term in Equation (3.2.14). Consequently, the flyweight drive, spring force, belt reaction, translating mass, and pivoting mass all determine the same shift acceleration rather than being divided between an equilibrium actuator law and a separate shift-response model. No equivalent explicit reflection of the flyweights' pivoting inertia was identified in the reviewed mechanically actuated rubber-belt CVT formulations.

This extension does not retain every effect represented in the earlier models. Structural deformation is excluded by Assumption 2 (Rigid Pulley and Actuation Hardware), while Assumption 3 (Idealized Actuator Mechanics) treats the ramp contact as an ideal constraint rather than resolving its tribology or local compliance. The roller-contact friction retained by Kim et al. (2002), for example, therefore lies outside the present force law; roller spin and contact compliance are likewise not resolved. The added fidelity is in the coupled mechanism dynamics, not in frictional or compliant contact detail.

3.2.2 Secondary Movable-Sheave Balance

While the primary closes mainly in response to rotational speed, the secondary is torque reactive: torque transmitted through the pulley can also increase the axial force clamping the belt.

To see how, begin with the two secondary sheaves rotationally disconnected. The fixed sheave rotates with the secondary shaft. The movable sheave can slide along that shaft and rotate relative to the fixed sheave. Because the belt contacts the two faces separately, it applies torque to each body. With no connection between them, the two bodies can accelerate differently and rotate relative to one another.

Now place a pin on the movable sheave in a straight slot running along the shaft. The movable sheave remains free to slide, but the pin prevents relative rotation. Whenever the two sheaves tend to rotate differently, the pin presses against the side of the slot and a normal reaction develops. Because the slot is axial, this normal acts entirely in the circumferential direction. It transfers torque between the sheaves but produces no axial force.

A helix is the same slot tilted around the shaft. The contact normal remains perpendicular to the slot surface, but now has both circumferential and axial components. The circumferential component continues to couple the rotational motion of the sheaves, while the axial component acts on the movable sheave. Relative rotation is therefore allowed only when accompanied by axial travel along the helix.

These three cases and their contact reactions are compared in Figure 3.7.

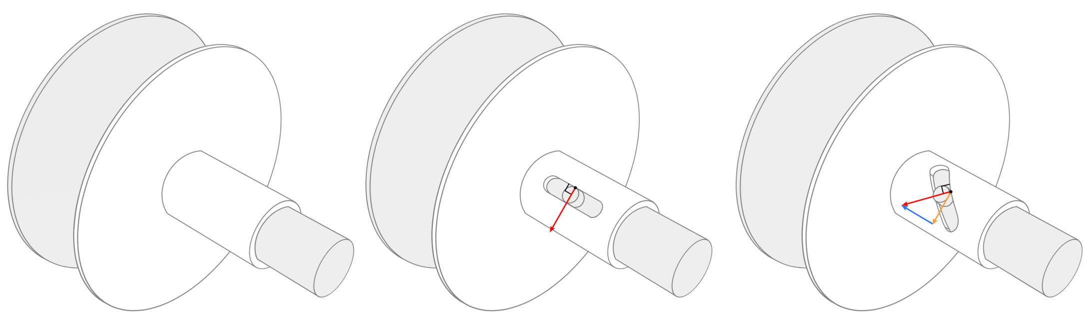


Figure 3.7: How the helix creates torque-reactive axial loading. With no connection, the two sheaves may rotate independently. A straight axial slot couples their rotation through a purely circumferential reaction; inclining the slot adds an axial component to that same reaction, coupling relative rotation to axial travel.

If the axial component of the helix reaction is balanced by the other axial loads, the movable sheave remains at its current position. Its relative angle then remains fixed, and the two sheaves rotate together with the shaft. If the axial loads do not balance, their resultant accelerates the movable sheave along the helix, and it rotates by the amount required by the slot geometry.

In the actual secondary, the belt and spring load this mechanism. The belt applies torque to the two sheave faces separately. The secondary spring is installed both compressed and twisted: its torsional preload tends to rotate the movable sheave relative to the fixed sheave, while its axial compression pushes the movable sheave toward closure. The torsional action loads the helix; the axial action enters the movable-sheave balance directly.

The axial free body gives the target for the derivation.

Axial Free Body

Figure 3.8 isolates the movable assembly in the axial direction.

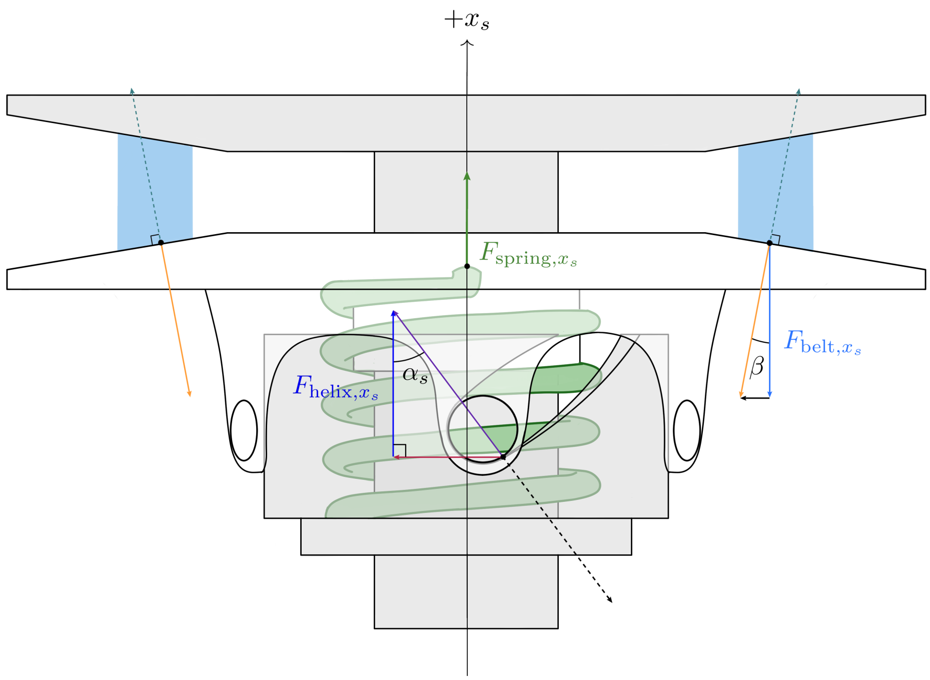


Figure 3.8: Axial free body of the complete movable secondary assembly. The axial spring force and belt reaction act directly on the translating assembly, while the secondary's torque-reactive behaviour enters through the pin–helix force F_{helix,x_s}.

Let m_s be the mass of the movable sheave and every component that translates rigidly with it. Three axial loads act on this assembly. The pin–helix contact contributes the signed force F_{helix,x_s}. The compressed spring pushes the movable sheave toward the fixed sheave in $+x_s$. The belt pushes the two sheave faces apart, so its reaction on the movable face acts in $-x_s$, just as on the primary. Newton's second law therefore gives

$$m_s \ddot{x}_s = F_{\text{helix},x_s} + F_{\text{spring},x_s} - F_{\text{belt},x_s}. \tag{3.2.18}$$

This free body makes the remaining task clear. The direct spring and belt forces follow from compression and contact geometry. The helix force does not: the tilted slot has established that an axial reaction exists, but not its magnitude or sign. Because the same contact reaction must also carry the torque that constrains relative rotation, its rotational effect is resolved first. This sets the order of the derivation:

1. use the rotational balance to find the torque carried by the helix;
2. use the helix geometry to convert that torque into axial force; and
3. add this force to the direct spring and belt loads in the axial balance.

Rotational Loading of the Movable Sheave

As established in Section 3.1, x_s is measured from the open secondary toward the fixed sheave, so $+x_s$ closes the pulley. Let θ_s measure the rotation of the movable sheave relative to the fixed sheave in the same positive rotational sense already assigned to ω_s (see Section 2.3.1). Only the conventional torque-reactive helix orientation is considered: as the secondary closes in $+x_s$, the movable sheave turns relative to the fixed sheave in $+\theta_s$.

With these directions set, the first question is what loads the helix circumferentially. The belt applies torque to the movable face, and the twisted spring applies torque between the two sheaves. The pin–helix contact supplies the torque that constrains the resulting relative motion to follow the slot.

Our goal is to find the torque reacted at the pin–helix contact. To do so, isolate the movable assembly and apply Newton's second law for rotation. Use the following quantities:

- $I_{s,M}$: rotational inertia of the movable sheave and every component that rotates rigidly with it;
- $\omega_{s,M}$: absolute angular velocity of that assembly;
- $\tau_{s,M}$: signed belt torque applied to the movable face;
- $\tau_{s,\text{spring}}$: signed spring torque applied to the movable assembly; and
- $\tau_{s,\text{helix}}$: the contact torque at the pin–helix connection, produced by the circumferential component of the contact normal. A positive $\tau_{s,\text{helix}}$ acts on the movable assembly in $-\theta_s$.

The figures show one representative pin and track. The quantities listed above and later in this section are the totals over the complete set of loaded pin–track contacts, so no additional track multiplier is required.

The rotational balance is therefore

$$\tau_{s,M} + \tau_{s,\text{spring}} - \tau_{s,\text{helix}} = I_{s,M}\dot{\omega}_{s,M}. \tag{3.2.19}$$

Solving for the helix torque gives

$$\tau_{s,\text{helix}} = \tau_{s,M} + \tau_{s,\text{spring}} - I_{s,M}\dot{\omega}_{s,M}. \tag{3.2.20}$$

Equation (3.2.20) gives the contact torque we want, but its right-hand side still needs three relations:

- the share of belt torque applied to the movable face, $\tau_{s,M}$;
- the torque supplied by the wound spring, $\tau_{s,\text{spring}}$; and
- the absolute angular acceleration of the movable assembly, $\dot{\omega}_{s,M}$.

The belt term follows immediately from the face-torque assumption.

Movable-face belt torque Let τ_s be the total signed torque exerted by the belt on the secondary, so $\tau_s = \tau_{s,F} + \tau_{s,M}$. Under Assumption 12 (Symmetric Sheave-Face Sharing), the two faces receive equal torque shares. Only the movable-face share appears in its free body:

$$\tau_{s,M} = \frac{\tau_s}{2}. \tag{3.2.21}$$

This equal split is a reduced-contact assumption. During shift, the movable face rotates slightly faster or slower than the fixed face. The two belt–face contacts can therefore have different sliding conditions, so their torques need not divide equally.

Remark (Why the face-torque split can matter)

Sorge and Cammalleri (2011) resolve the two contacts separately and show that the face torques can depart substantially from an equal split; distributed tangential forces on the two faces can even act in opposite directions over parts of the wrap. Since $\tau_{s,M}$ enters Equation (3.2.20) directly, this is also a potential error in the torque reacted by the helix.

The remaining terms both depend on the helix kinematics: $\dot{\omega}_{s,M}$ depends on how the movable sheave rotates as it shifts, and the spring torque depends on how far that motion unwinds the spring. We therefore derive the relation between x_s and θ_s next.

Motion Imposed by the Helix

The helix geometry tells us how axial shift changes the relative angle of the movable sheave. Unwrap one track from its effective cylindrical radius r_h, as seen in Figure 3.9. A small relative rotation $d\theta_s$ moves the pin by $r_h\, d\theta_s$ circumferentially while the sheave moves by dx_s axially. Let α_s be the local angle of the unwrapped track, measured from the positive circumferential direction toward the positive axial direction. For the conventional helix orientation considered here,

$$0 < \alpha_s < \frac{\pi}{2}.$$

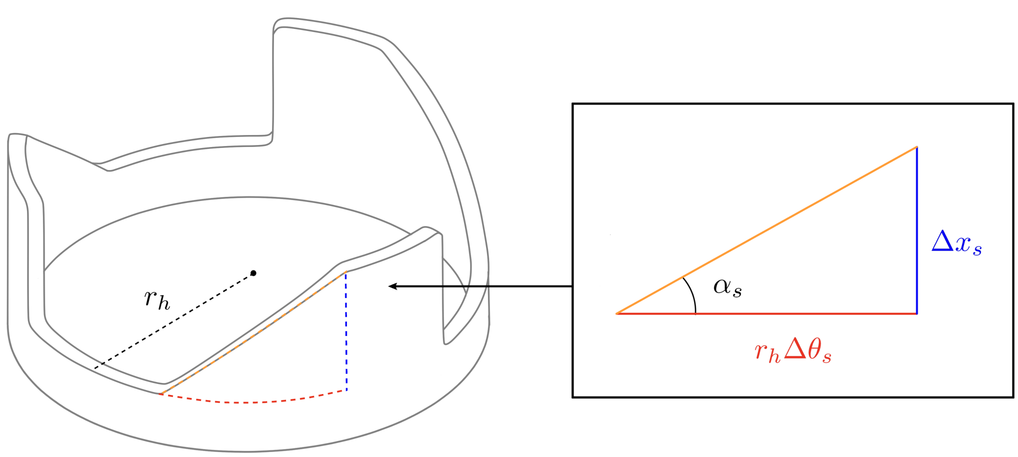


Figure 3.9: Unwrapped secondary-helix geometry. The same track angle relates relative rotation to axial travel and reaction torque to axial force.

Taking the infinitesimal limit of the triangle in Figure 3.9,

$$\tan\alpha_s = \frac{dx_s}{r_h\, d\theta_s}, \qquad \frac{d\theta_s}{dx_s} = \frac{1}{r_h \tan\alpha_s}. \tag{3.2.22}$$

Thus, $d\theta_s/dx_s$ is the relative rotation required per unit axial travel. It is positive for the conventional helix orientation assumed here. Returning to the straight axial slot from the opening example, $\alpha_s \to \pi/2$ and $d\theta_s/dx_s \to 0$, as expected.

Taking $\theta_s(0) = 0$ at the open secondary, integrate the local motion ratio to obtain

$$\theta_s(x_s) = \int_0^{x_s} \frac{1}{r_h \tan\alpha_s(\xi)}\, d\xi. \tag{3.2.23}$$

Equation (3.2.23) defines the helix as a function $\theta_s(x_s)$: for each axial position, it gives the relative angle enforced by the track. The function need not come from a constant helix angle. Any supplied function may be used provided it is single-valued and strictly increasing over the admissible shift range.

Rotation of the movable sheave We can now use this function to determine the two remaining quantities in the rotational balance. First consider the absolute motion of the movable sheave. The fixed sheave rotates with the secondary shaft at ω_s, while the movable sheave also has the relative rotation θ_s:

$$\omega_{s,M} = \omega_s + \dot{\theta}_s = \omega_s + \frac{d\theta_s}{dx_s}\dot{x}_s. \tag{3.2.24}$$

Because θ_s was defined in the same positive rotational sense as ω_s, the relative speed enters with the plus sign shown here. Differentiating gives the angular acceleration required by the rotational balance:

$$\dot{\omega}_{s,M} = \dot{\omega}_s + \frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2. \tag{3.2.25}$$

The shaft acceleration $\dot{\omega}_s$ remains unknown here because ω_s is a state of the coupled model. It is solved together with the axial motion rather than by the helix relation alone. The term proportional to $\ddot{x}_s$ is the angular acceleration caused by axial acceleration. The term proportional to $\dot{x}_s^2$ appears only when the motion ratio varies with position.

Torsional Spring Loading The same helix function tells us how much the spring has unwound. Let $k_{s,\theta}$ be the spring's torsional stiffness and $\theta_{s,\mathrm{pre}}$ its signed twist at $x_s = 0$. Increasing θ_s reduces this twist, leaving $\theta_{s,\mathrm{pre}} - \theta_s(x_s)$. The resulting torque on the movable assembly is

$$\tau_{s,\mathrm{spring}} = k_{s,\theta}\left[\theta_{s,\mathrm{pre}} - \theta_s(x_s)\right]. \tag{3.2.26}$$

For the installed winding assumed here, this quantity is positive while the spring remains preloaded. The opposite winding direction can be represented by a signed $\theta_{s,\mathrm{pre}}$.

We now have relations for all three terms on the right-hand side of Equation (3.2.20). Substituting them gives the pin–helix contact torque:

$$\begin{aligned}\tau_{s,\text{helix}} = {} & \frac{\tau_s}{2} + k_{s,\theta}\left[\theta_{s,\text{pre}} - \theta_s(x_s)\right] \\ & - I_{s,M}\left[\dot{\omega}_s + \frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2\right].\end{aligned} \tag{3.2.27}$$

This completes the rotational part of the derivation. The next step is to use the pin–helix contact geometry to find the axial force associated with this torque.

Axial Component of the Helix Reaction

The torque above is the moment produced by the circumferential component of the contact normal. Let F_{helix,x_s} denote the signed axial force produced by the helix reaction, positive in $+x_s$.

Under Assumption 3 (Idealized Actuator Mechanics), the pin–helix contact is treated as an ideal frictionless constraint. The same compatible-work argument used for the primary therefore converts the rotational contact load into its axial effect without contact dissipation. Consider the motion of the movable sheave relative to the helix carrier. The contact normal does no net work in this relative motion. Its axial component contributes $F_{\text{helix},x_s}\,dx_s$, while its circumferential component contributes $-\tau_{s,\text{helix}}\,d\theta_s$. Their sum is zero, so

$$F_{\text{helix},x_s}\,dx_s = \tau_{s,\text{helix}}\,d\theta_s. \tag{3.2.28}$$

Using the motion ratio gives

$$F_{\text{helix},x_s} = \tau_{s,\text{helix}}\frac{d\theta_s}{dx_s}. \tag{3.2.29}$$

The motion ratio $d\theta_s/dx_s$ is therefore the mathematical origin of the axial effect: it converts the contact torque into axial force. In the straight-slot limit used in the opening example, this ratio is zero, so the slot can react torque while the axial conversion vanishes.

Substituting Equation (3.2.27) gives the helix force in terms of the model variables:

$$\boxed{\begin{aligned}F_{\text{helix},x_s} = \frac{d\theta_s}{dx_s}\Bigg\{ & \frac{\tau_s}{2} + k_{s,\theta}\left[\theta_{s,\text{pre}} - \theta_s(x_s)\right] \\ & - I_{s,M}\left[\dot{\omega}_s + \frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2\right]\Bigg\}.\end{aligned}} \tag{3.2.30}$$

This gives F_{helix,x_s}. The belt and wound spring provide the driving torques inside the braces; part of that torque may be required to accelerate the movable assembly, and only the remainder is converted into axial thrust.

Remark (Admissible helix maps)

The completed force uses the first two derivatives of $\theta_s(x_s)$. Over the admissible shift range, the supplied helix map must therefore be single-valued and twice continuously differentiable, with a finite, nonzero motion ratio:

$$\theta_s \in C^2, \qquad 0 < \frac{d\theta_s}{dx_s} < \infty.$$

A manufactured multi-angle helix may still be represented, but nominal segments must be joined smoothly rather than by a mathematical corner.

The difficult force in the preliminary axial balance is now known. The two remaining forces act directly on the translating assembly.

Direct Spring and Belt Forces

Direct axial spring force The spring force follows directly from its compression. Let $k_{s,x}$ be the axial stiffness and $x_{s,\text{pre}}$ the compression at $x_s = 0$. Closing the secondary by x_s lets the spring extend by the same amount, so its remaining compression is $x_{s,\text{pre}} - x_s$. Hence,

$$F_{\text{spring},x_s} = k_{s,x}\left(x_{s,\text{pre}} - x_s\right). \tag{3.2.31}$$

This force and the torque in Equation (3.2.26) come from the same installed spring. Following the reduced decomposition of Kim et al. (2002), the model represents its axial compression and torsional twist with separate linear laws. This use of prescribed, uncoupled spring laws is part of Assumption 3 (Idealized Actuator Mechanics); compression–twist coupling, hysteresis, and end-contact friction are not retained.

Belt opening reaction The belt opening force follows the same projection already used on the primary. Let N_s be the integrated normal load over both secondary faces. Under Assumption 12 (Symmetric Sheave-Face Sharing), the movable face carries one half of this total load, giving

$$F_{\text{belt},x_s} = \frac{N_s}{2}\cos\beta. \tag{3.2.32}$$

As on the primary, N_s is not determined by this local free body. It remains a coupled unknown supplied by the belt-contact and remaining system equations.

Completed Local Secondary Balance

We now have a relation for every term in the axial free body. Substitute them into Equation (3.2.18) and collect the terms multiplying $\ddot{x}_s$:

$$
\boxed{\begin{aligned}\left[m_s + I_{s,M}\left(\frac{d\theta_s}{dx_s}\right)^2\right]\ddot{x}_s = {} & \left[\frac{\tau_s}{2} + k_{s,\theta}\left(\theta_{s,\text{pre}} - \theta_s\right) - I_{s,M}\dot{\omega}_s\right]\frac{d\theta_s}{dx_s} \\ & - I_{s,M}\frac{d\theta_s}{dx_s}\frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2 \\ & + k_{s,x}\left(x_{s,\text{pre}} - x_s\right) - \frac{N_s}{2}\cos\beta.\end{aligned}} \tag{3.2.33}
$$

The completed equation separates into four physical pieces:

- $m_s + I_{s,M}(d\theta_s/dx_s)^2$ is the local axial inertia. Just as the flyweight pivot inertia appeared in the primary balance, the helix makes the secondary's relative-rotational inertia part of the load that axial motion must accelerate;
- the first bracket contains the belt and torsional-spring torques, less the portion required to accelerate the movable assembly with the secondary shaft;
- the term proportional to $\dot{x}_s^2$ accounts for a helix motion ratio that varies with position; and
- the last two terms are the direct spring closing force and belt opening force.

This is the axial equation of motion we set out to obtain. If its right-hand side remains zero and the sheave is initially stationary, x_s and θ_s remain fixed while the two sheaves rotate together. A nonzero right-hand side accelerates the sheave axially, and $\theta_s(x_s)$ fixes the accompanying relative rotation.

Classical Limit and Relation to Existing Models

As on the primary, the full dynamic balance should recover the established actuator equilibrium when the shift mechanism is stationary. Consider a fixed secondary configuration at constant shaft speed, $\dot{x}_s = \ddot{x}_s = \dot{\omega}_s = 0$. Under the equal face-torque split adopted here, the helix force reduces to

$$
F_{\text{helix},x_s} = \left[\frac{\tau_s}{2} + k_{s,\theta}\left(\theta_{s,\text{pre}} - \theta_s\right)\right]\frac{d\theta_s}{dx_s}. \tag{3.2.34}
$$

Adding the direct spring and belt loads gives the corresponding equilibrium condition for the complete movable assembly:

$$
\left[\frac{\tau_s}{2} + k_{s,\theta}\left(\theta_{s,\text{pre}} - \theta_s\right)\right]\frac{d\theta_s}{dx_s} + k_{s,x}\left(x_{s,\text{pre}} - x_s\right) = \frac{N_s}{2}\cos\beta. \tag{3.2.35}
$$

These equations expose the classical mechanism directly. The axial spring supplies one closing force, while the belt torque and torsional-spring torque are converted into a second closing force by the helix motion ratio. The factor 1/2 is not a property of the helix itself; it follows from the equal face-torque closure used here. A face-resolved contact model would instead use the torque $\tau_{s,M}$ actually applied to the movable face.

The important comparison is how this equilibrium relation is used. In the present formulation, Equations (3.2.34) and (3.2.35) are stationary limits of a dynamic balance. In much of the mechanically actuated CVT literature, an equivalent algebraic relation remains the actuator closure while the transmission is shifting.

The formulation of Kim et al. (2002) is a direct example. Their driven thrust is the sum of a spring force and a torque-cam force obtained from instantaneous equilibrium. The cam term retains the applied torque, torsional-spring load, cam geometry, and cam friction. Suppressing that friction and adopting the equal face-torque split used here gives the same spring-plus-torque-cam structure as Equation (3.2.35). Their shift transient is not obtained from the movable secondary's equation of motion, but from a separately identified first-order ratio law. Kim et al. therefore retain contact friction that is absent here, but do not use the secondary assembly's inertia to determine its shift acceleration.

The design-oriented sources examined here apply the same configuration-by-configuration logic at different levels of detail. Messick (2018) calculates the secondary thrust required by a steady-state belt solution and uses it as a target for helix design, explicitly excluding the shift response time. Skinner (2020) calculates the compression-spring, torsional-spring, and torque-feedback contributions to secondary clamping algebraically. Arango and Muñoz Alzate (2020) construct the torque-sensing cam profile from the transmitted torque, spring force, required axial force, and axial displacement over the operating range. These treatments are useful for sizing and tuning the mechanism, but they do not determine its motion from a movable-sheave equation of motion.

Among the reviewed sources, Sorge and Cammalleri (2011) provide the most detailed treatment of the belt–face–helix interaction during shifting. Their distributed contact model allows the fixed and movable faces to rotate at different speeds and determines the torque $M_{N,s}$ carried by the movable face. The resulting driven-side actuator relation combines the spring load with a helix contribution having the same torque-times-motion-ratio structure as Equation (3.2.34). It therefore replaces the assumed $\tau_s/2$ with a face-resolved torque and can capture substantial departures from an equal split.

That contact calculation is shift-aware, but the secondary actuator itself is still closed quasi-statically. Sorge and Cammalleri prescribe the instantaneous radius and the dimensionless shift speed $\rho = \dot{r}/(\omega_f r)$, then solve for the corresponding belt-contact state. A nonzero ρ changes the relative face speeds, sliding directions, contact tractions, and face-torque split, but it is not produced by the forces on the movable secondary. Because no axial or relative-rotational equation of motion is written for the movable half-pulley, its translational inertia and helix-reflected rotational inertia are absent by construction.

The models therefore retain complementary parts of the physics. Kim et al. retain cam friction, while Sorge and Cammalleri resolve the two belt–face contacts and their unequal torques. The present reduced model retains neither capability. It instead carries the movable assembly's rotational balance through the general constraint $\theta_s(x_s)$ and solves its free axial motion. This introduces the shaft-acceleration coupling, the reflected inertia $I_{s,M}(d\theta_s/dx_s)^2$, and the velocity-squared term generated when the helix motion ratio varies with position. Among the reviewed mechanically actuated rubber-belt formulations, no equivalent explicit projection of the movable side's rotational inertia through a mechanical secondary helix was identified.

This distinction is deliberately narrow. The rigid-hardware, ideal-actuator, and symmetric-face idealizations are stated explicitly in Assumption 2 (Rigid Pulley and Actuation Hardware), Assumption 3 (Idealized Actuator Mechanics), and Assumption 12 (Symmetric Sheave-Face Sharing). They leave the compliant and frictional actuator details, and the face-resolved load redistribution of the models above, outside the present balance. The comparison is therefore not a claim of greater local contact fidelity; it isolates the added treatment of the movable assembly's inertia and helix-constrained dynamics.

3.2.3 Coupling Through the Shared Shift Coordinate

The primary and secondary balances have been derived in their natural local coordinates, x_p and x_s. The fixed-length belt geometry makes those motions compatible. Choosing the primary displacement as the shared shift coordinate leaves the primary balance unchanged and supplies the secondary motion through the map $x_s = x_s(s)$.

Local-to-Shared Kinematics

Because the shared coordinate is the primary displacement,

$$x_p = s, \qquad \dot{x}_p = \dot{s}, \qquad \ddot{x}_p = \ddot{s}. \tag{3.2.36}$$

The secondary displacement follows from the fixed-length belt map. Its first two time derivatives are

$$x_s = x_s(s), \qquad \dot{x}_s = \frac{dx_s}{ds}\dot{s}, \qquad \ddot{x}_s = \frac{dx_s}{ds}\ddot{s} + \frac{d^2x_s}{ds^2}\dot{s}^2. \tag{3.2.37}$$

Positive x_p and x_s both denote sheave closing. Increasing s closes the primary but opens the secondary, so $dx_s/ds < 0$. If the belt geometry is nonlinear, this motion ratio changes through the shift and $d^2x_s/ds^2 \neq 0$. Both derivatives are determined by the belt-geometry relation established earlier.

Two Conditions on One Shift Motion

Substituting Equation (3.2.36) into Equation (3.2.14) gives the primary condition directly:

Primary Condition on the Shared Shift Motion **(3.2.38)**

$$\begin{aligned}\left[m_p + I_f\left(\frac{dq_f}{ds}\right)^2\right]\ddot{s} = {} & \frac{1}{2}\omega_p^2\frac{dJ_{f,p}}{ds} \\ & - I_f\frac{dq_f}{ds}\frac{d^2q_f}{ds^2}\dot{s}^2 \\ & - k_p\left(x_{p,\text{pre}} + s\right) - \frac{N_p}{2}\cos\beta.\end{aligned}$$

Because $x_p = s$, this is simply the local primary balance written using the shared coordinate.

The secondary condition follows just as directly. Substituting Equation (3.2.37) into Equation (3.2.33) and collecting terms gives

Secondary Condition on the Shared Shift Motion **(3.2.39)**

$$
\begin{aligned}
&\left[m_s + I_{s,M}\left(\frac{d\theta_s}{dx_s}\right)^2\right]\frac{dx_s}{ds}\,\ddot{s} \\
&= \frac{d\theta_s}{dx_s}\left[\frac{\tau_s}{2} + k_{s,\theta}\left(\theta_{s,\text{pre}} - \theta_s\right) - I_{s,M}\dot{\omega}_s\right] \\
&\quad + k_{s,x}\left(x_{s,\text{pre}} - x_s\right) - \frac{N_s}{2}\cos\beta \\
&\quad - \Bigg\{\left[m_s + I_{s,M}\left(\frac{d\theta_s}{dx_s}\right)^2\right]\frac{d^2x_s}{ds^2} \\
&\qquad\qquad + I_{s,M}\frac{d\theta_s}{dx_s}\frac{d^2\theta_s}{dx_s^2}\left(\frac{dx_s}{ds}\right)^2\Bigg\}\dot{s}^2.
\end{aligned}
$$

Here x_s, θ_s, and their local derivatives are evaluated at $x_s = x_s(s)$. Since $dx_s/ds < 0$, a force that closes the secondary drives the shift toward decreasing s, thereby opening the primary. Conversely, the secondary belt force acts in the opening direction and drives the shift toward increasing s. The kinematic map therefore carries the inverse coupling between the two pulleys directly into the acceleration balance.

The two terms proportional to $\dot{s}^2$ have different geometric origins. The term containing d^2x_s/ds^2 appears because the belt geometry can require a changing amount of secondary travel per unit primary travel. The term containing $d^2\theta_s/dx_s^2$ is inherited from the local secondary balance and appears when the helix motion ratio varies with position; substituting $\dot{x}_s = (dx_s/ds)\dot{s}$ supplies its squared local-to-shared motion ratio.

The two balances are not added into one axial free body. They remain separate conditions on the same shift acceleration: the same $\ddot{s}$ must satisfy both movable assemblies. The belt reactions N_p and N_s, together with the coupled shaft and contact quantities, allow both conditions to hold. Those unknowns are supplied by the rotational, belt, and contact relations developed later.

Relation to Existing Dynamic Mappings

Transient belt and pulley models provide broad precedent for dynamic ratio prediction. Julió and Plante (2011) discretize the rubber belt and integrate its motion, while Ballew (2015) allow both pulley speeds to evolve from the applied torques and inertias. In that model family, however, pulley axial force is prescribed and the radius follows from belt compression; a movable-sheave equation $m_j\ddot{x}_j = \sum F_{j,x}$ is not integrated.

Movable-sheave dynamics and shared-coordinate reductions also predate the present model. Duan et al. (2016) retain movable-pulley mass in a chain CVT and couple the pulley motions through chain deformation and tension. Most directly, da Silva et al. (2023) write axial equations for both movable sheaves of a mechanically actuated rubber-belt CVT, impose a linear relation between their displacements, and reduce the two translations to one shift equation. Thus transient shifting, axial sheave inertia, and a shared shift coordinate are not individually new.

The distinction here lies in how these ingredients are combined. The fixed-length belt geometry supplies the nonlinear map $x_s(s)$, so both dx_s/ds and d^2x_s/ds^2 remain in the secondary acceleration.

At the same time, the flyweight pivot inertia and the movable secondary's relative-rotational inertia remain tied to their own mechanism maps, $q_f(x_p)$ and $\theta_s(x_s)$. The result is a pair of local free-body balances expressed as simultaneous conditions on one shift acceleration.

This reduction also defines the tradeoff. It does not retain the spatial belt deformation fields of Julió and Plante (2011) and Ballew (2015), or the prescribed-shift, face-resolved torque distribution of Sorge and Cammalleri (2011). It instead keeps the actuator and movable-sheave inertias explicit while closing the nominal belt geometry without spatial discretization. Among the reviewed mechanically actuated rubber-belt formulations, no equivalent combination of these features was identified.

3.2.4 What the Axial Balances Establish

The primary and secondary now have complete axial equations of motion in their natural coordinates. On the primary, shaft speed produces a closing action because the flyweight centres of mass move outward as the movable sheave closes. Part of that action accelerates the flyweights about their pivots before the remainder is transmitted to the movable assembly. On the secondary, the belt and torsional-spring torques act through the helix. Part of their action changes the rotation of the movable sheave relative to the shaft-side assembly before the remainder appears as axial force.

Although the two actuators are physically different, their constrained motions produce the same underlying structure. The map $q_f(x_p)$ reflects the flyweights' pivot inertia into the primary axial coordinate, while the helix map $\theta_s(x_s)$ reflects the movable-sheave rotational inertia into the secondary coordinate. In each case, the local motion ratio contributes an effective axial inertia, and variation of that ratio with position produces a velocity-squared term. The primary closing action is additionally set by dr_f/dx_p, while the secondary balance couples to the shaft acceleration because its movable sheave rotates relative to an assembly that is itself rotating.

The shared belt geometry adds a second layer of constrained motion. The primary map $x_p(s) = s$ carries its local balance directly into the shift coordinate, whereas the secondary map $x_s(s)$ reflects its local motion into the same $\ddot{s}$ and contributes its own velocity-squared curvature term. Both free-body relations must therefore be satisfied by one shift motion.

They do not, however, determine that motion in isolation. At this stage, the two axial equations contain the five unknown quantities

$$\ddot{s}, \qquad N_p, \qquad N_s, \qquad \tau_s, \qquad \dot{\omega}_s.$$

The belt normal loads, transmitted secondary torque, and secondary shaft acceleration follow only when the rotational and belt-contact mechanics are added. The two balances derived here therefore form the axial part of one simultaneous dynamic solution for the engaged CVT.

This completes the axial force description. Attention now turns from translation along the pulley shafts to rotation about them.

3.3 Rotational Dynamics of the Pulley Assemblies

The preceding section established how forces along the pulley shafts move the sheaves, but its equations still contain quantities belonging to rotation about those shafts. The primary flyweight force depends on the primary speed ω_p, while the secondary balance contains the transmitted torque τ_s and the secondary acceleration $\dot{\omega}_s$. The axial equations can use these quantities, but cannot supply them.

The primary pulley is carried around the primary shaft at ω_p, while the secondary rotates about its own shaft at ω_s. These are separate rotations, so their speeds need not change together. The mechanisms inside each pulley also move relative to these underlying shaft rotations. The primary flyweights pivot as the primary rotates, while the secondary movable sheave translates and rotates relative to its shaft through the helix. These internal motions are already tied to the shift coordinate, but they also change how angular momentum is distributed within each pulley as the CVT shifts.

The remaining question is therefore what governs the evolution of ω_p and ω_s, and consequently how shifting affects that evolution.

3.3.1 Following the Torque

To determine how ω_p and ω_s evolve, each pulley side requires a rotational equation of motion. That equation must account for the angular momentum carried about the corresponding shaft and the torques capable of changing it. The first step is therefore to identify the rotating systems represented by the two shaft speeds.

The primary-side system contains the primary shaft and the complete primary pulley assembly. The movable assembly and flyweights remain part of this system even while they move relative to the fixed sheave, since they continue to carry angular momentum about the primary axis. The secondary-side system is defined in the same way: it contains the secondary shaft and the complete secondary pulley assembly, including the helix-guided motion of the movable sheave. The belt passes between these two systems and is not included in either one.

The resulting division is shown in Figure 3.10.

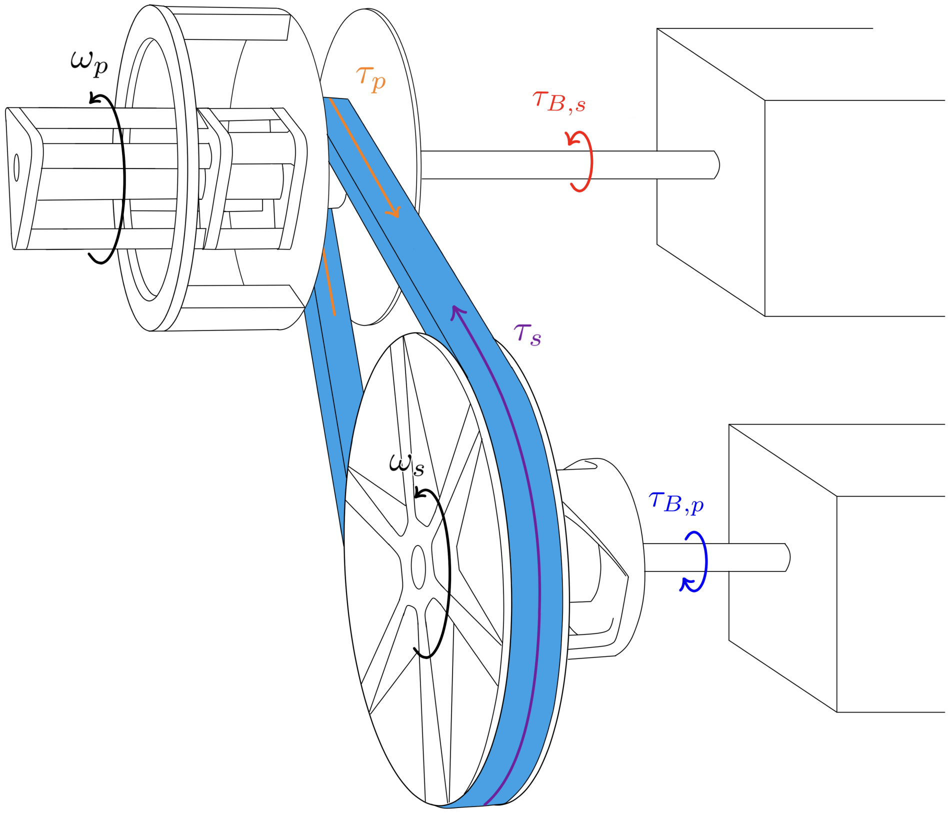


Figure 3.10: Rotating systems and torque path of the engaged CVT. The shaft-side boundaries represent the torque and equivalent inertia presented by the attached systems. The belt remains outside the two pulley-side systems and interacts with them through the contact torques τ_p and τ_s. The arrows show the signed directions associated with forward power transfer.

Begin with the primary. When its speed changes, the primary shaft must accelerate not only the pulley itself, but also the rotating hardware attached beyond the CVT boundary. As established in Section 2.1, that external hardware is represented at the shaft by the equivalent inertia $I_{\mathrm{B},p}$, while $\tau_{\mathrm{B},p}$ is the torque it applies to the primary side. The remaining external interaction is the belt, which exerts the torque τ_p on the primary pulley.

The secondary follows the same construction. Its attached hardware contributes the equivalent inertia $I_{\mathrm{B},s}$ and applies the boundary torque $\tau_{\mathrm{B},s}$, while the belt exerts τ_s on the secondary pulley.

The signs follow directly from the convention established in Section 2.3.1. During forward power transfer, the primary boundary drives the primary pulley and the belt resists it. The belt then drives the secondary pulley against the resisting secondary boundary. Thus,

$$\tau_{\mathrm{B},p} > 0, \qquad \tau_p < 0, \qquad \tau_s > 0, \qquad \tau_{\mathrm{B},s} < 0.$$

Although the primary and secondary contain different mechanisms, their rotational balances now have the same outer structure. Let $j \in \{p, s\}$ denote the pulley side under consideration.

The angular momentum on side j has two contributions. The fixed equivalent boundary inertia rotates at ω_j, so it carries angular momentum $I_{\mathrm{B},j}\omega_j$. Let $H_{\mathrm{CVT},j}$ denote the angular momentum carried by the corresponding shaft and pulley assembly, including the internal motions of its components. The total is therefore

$$H_j = I_{\mathrm{B},j}\omega_j + H_{\mathrm{CVT},j}. \tag{3.3.1}$$

Bearing, seal, windage, and other unmodeled resisting torques are omitted under Assumption 15 (Negligible Unmodeled Parasitic Losses). Within the resulting pulley-side free body, the applied boundary torque and belt torque are therefore the only external torques retained. Newton's second law for rotation gives

$$\tau_{\mathrm{B},j} + \tau_j = \frac{dH_j}{dt}.$$

Substituting Equation (3.3.1) writes this as

$$\tau_{\mathrm{B},j} + \tau_j = \frac{d}{dt}\left(I_{\mathrm{B},j}\omega_j + H_{\mathrm{CVT},j}\right).$$

The equivalent boundary inertia is fixed, so differentiating yields the common pulley-side equation of motion:

Common Pulley-Side Rotational Balance **(3.3.2)**

$$\tau_{\mathrm{B},j} + \tau_j = I_{\mathrm{B},j}\dot{\omega}_j + \frac{dH_{\mathrm{CVT},j}}{dt}, \qquad j \in \{p, s\}.$$

The way shifting changes the angular momentum carried within a pulley is contained in $H_{\mathrm{CVT},j}$. On the primary, the flyweights redistribute mass relative to the shaft as they pivot. On the secondary, the movable sheave rotates relative to the shaft-side assembly as it follows the helix. The two pulley sides therefore share the balance in Equation (3.3.2), but require separate expressions for their internal angular momentum. Beginning with the primary, each assembly can now be examined in turn.

3.3.2 Primary Pulley Assembly

The primary is the simpler of the two pulley assemblies because nearly all of its hardware shares the shaft speed ω_p. The shaft, sheaves, ramp carrier, and other coaxial components may translate during a shift, but that translation is along the shaft and therefore does not change their distance from its axis. Their combined shaft-axis inertia can consequently be collected into one constant quantity, $I_{p,\mathrm{const}}$.

The flyweights are different. They are carried around the primary shaft at ω_p, but their mass moves radially as they pivot. The same motion was already used in Section 3.2.1 to determine the centrifugal action that closes the primary. There, the flyweight geometry was summarized by the shaft-axis inertia map $J_{f,p}(x_p)$. Since $x_p = s$, that map is now $J_{f,p}(s)$.

It is useful to recall what this inertia represents. The quantity $J_{f,p}$ measures the flyweight mass distribution about the *primary shaft*. It is therefore distinct from the constant pivot-axis inertia I_f that appeared in the flyweight moment balance of Equation (3.2.3). The two quantities describe the same flyweights about different axes and enter different equations of motion.

At a given shift position, the angular momentum carried inside the primary CVT boundary is therefore

$$H_{\mathrm{CVT},p} = [I_{p,\mathrm{const}} + J_{f,p}(s)]\,\omega_p. \tag{3.3.3}$$

If the shift position were fixed, differentiating this expression would simply multiply the current inertia by $\dot{\omega}_p$. During a shift, however, $J_{f,p}$ changes as well. Because its only configuration variable is s, differentiating Equation (3.3.3) gives

$$\frac{dH_{\mathrm{CVT},p}}{dt} = [I_{p,\mathrm{const}} + J_{f,p}(s)]\,\dot{\omega}_p + \frac{dJ_{f,p}}{ds}\,\dot{s}\,\omega_p. \tag{3.3.4}$$

The two terms describe different ways of changing the same angular momentum. The first accelerates the primary with the flyweights held at their current configuration. The second changes the angular momentum because the flyweight mass itself moves toward or away from the shaft while already rotating.

Substituting this result into the common pulley-side balance from Equation (3.3.2) gives the primary equation of motion:

Primary Pulley Rotational Balance **(3.3.5)**

$$\begin{aligned}\tau_{\mathrm{B},p} + \tau_p = {}& [I_{\mathrm{B},p} + I_{p,\mathrm{const}} + J_{f,p}(s)]\,\dot{\omega}_p \\ & + \frac{dJ_{f,p}}{ds}\,\dot{s}\,\omega_p.\end{aligned}$$

The connection to the axial flyweight balance is now explicit. In Equation (3.2.38), the derivative $dJ_{f,p}/ds$ determined how shaft rotation produces a centrifugal closing action. Here, the same derivative determines how that closing motion changes the shaft-axis angular momentum. The two appearances are not separate flyweight corrections: they are the axial and rotational consequences of the same radial redistribution of flyweight mass.

For the conventional primary mechanism, outward flyweight motion increases $J_{f,p}$. During an upshift, $\dot{s} > 0$, so the rotating primary must supply the angular momentum required by that outward motion in addition to accelerating the shaft. During a backshift the redistribution reverses, and the flyweights return angular momentum to the shaft-side motion.

If the shift is held fixed, $\dot{s} = 0$, and the flyweight mass distribution no longer changes. The additional term vanishes and the familiar fixed-inertia form is recovered:

$$\tau_{\mathrm{B},p} + \tau_p = [I_{\mathrm{B},p} + I_{p,\mathrm{const}} + J_{f,p}(s)]\,\dot{\omega}_p, \qquad \dot{s} = 0. \tag{3.3.6}$$

3.3.3 Secondary Pulley Assembly

The secondary differs from the primary in one important way. During a shift, the complete secondary remains one mechanical assembly, but not every part of that assembly has the same angular speed.

The secondary shaft, fixed sheave, and hardware locked to them rotate at ω_s. Let their combined shaft-axis inertia inside the CVT boundary be $I_{s,F}$. The movable sheave and the components rigidly attached to it have inertia $I_{s,M}$. As established while deriving the secondary axial balance, the helix makes this movable assembly rotate relative to the shaft-side assembly as it translates. Its absolute angular speed is therefore

$$\omega_{s,M} = \omega_s + \frac{d\theta_s}{dx_s}\dot{x}_s, \tag{3.3.7}$$

which is the relation already obtained in Equation (3.2.24). The angular momentum carried by the secondary must therefore retain the two angular speeds separately:

$$H_{\mathrm{CVT},s} = I_{s,F}\omega_s + I_{s,M}\omega_{s,M}. \tag{3.3.8}$$

The free body here is larger than the movable-sheave free body used in Section 3.2.2. There, only the movable assembly was isolated, so the spring and helix torques crossed the boundary and had to appear explicitly. Here, the shaft-side and movable-side hardware are enclosed together. The torque exerted by the helix on the movable side is accompanied by an equal-and-opposite torque on the shaft-side assembly, and both actions lie inside the same free body. The torsional spring produces the same kind of internal pair. These torques therefore cancel from the moment balance of the complete secondary. Their mechanical effect has not disappeared: the helix still allows the movable side to rotate at $\omega_{s,M}$ rather than ω_s, and that difference remains in Equation (3.3.8).

The common pulley-side balance in Equation (3.3.2) requires the time rate of change of the angular momentum inside this boundary. Because Equation (3.3.8) contains two angular velocities, differentiating it requires the corresponding two angular accelerations:

$$\frac{dH_{\mathrm{CVT},s}}{dt} = I_{s,F}\dot{\omega}_s + I_{s,M}\dot{\omega}_{s,M}. \tag{3.3.9}$$

The first acceleration, $\dot{\omega}_s$, is the shaft acceleration governed by the present rotational balance. The second is not another independent degree of freedom. The helix already tied the movable-side rotation to the axial sheave motion in Section 3.2.2. In particular, differentiating the helix kinematics there gave Equation (3.2.25):

$$\dot{\omega}_{s,M} = \dot{\omega}_s + \frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2. \tag{3.3.10}$$

The rotational balance has therefore led back to a kinematic relation already established by the axial mechanism. Substituting Equation (3.3.10) into Equation (3.3.9) gives

$$\frac{dH_{\mathrm{CVT},s}}{dt} = (I_{s,F} + I_{s,M})\,\dot{\omega}_s + I_{s,M}\left[\frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2\right]. \tag{3.3.11}$$

Substituting this rate into the common pulley-side rotational balance in Equation (3.3.2) produces the local secondary equation of motion:

Secondary Pulley Rotational Balance **(3.3.12)**

$$\tau_{\mathrm{B},s} + \tau_s = (I_{\mathrm{B},s} + I_{s,F} + I_{s,M})\,\dot{\omega}_s + I_{s,M}\left[\frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2\right].$$

The first term accelerates the rotation shared by the complete secondary. The bracketed terms change only the movable side's rotation relative to the shaft. They are the rotational counterpart of the helix-reflected inertia already encountered in the secondary axial balance.

The complete CVT uses the shared shift coordinate rather than x_s as an independent coordinate. The required mapping was already established in Equation (3.2.37). Substituting

$$\dot{x}_s = \frac{dx_s}{ds}\dot{s}, \qquad \ddot{x}_s = \frac{dx_s}{ds}\ddot{s} + \frac{d^2x_s}{ds^2}\dot{s}^2$$

into Equation (3.3.12) gives the same rotational balance in the coordinate used by the complete model:

$$\begin{aligned}\tau_{\mathrm{B},s} + \tau_s = {} & (I_{\mathrm{B},s} + I_{s,F} + I_{s,M})\,\dot{\omega}_s \\ & + I_{s,M}\frac{d\theta_s}{dx_s}\frac{dx_s}{ds}\ddot{s} \\ & + I_{s,M}\left[\frac{d\theta_s}{dx_s}\frac{d^2x_s}{ds^2} + \frac{d^2\theta_s}{dx_s^2}\left(\frac{dx_s}{ds}\right)^2\right]\dot{s}^2.\end{aligned} \tag{3.3.13}$$

All local helix quantities in this expression are evaluated at $x_s = x_s(s)$. Its signs follow directly from the kinematics already chosen. For the conventional helix, $d\theta_s/dx_s > 0$, while the belt geometry gives $dx_s/ds < 0$ over the engaged shift range. Hence

$$\frac{d\theta_s}{ds} = \frac{d\theta_s}{dx_s}\frac{dx_s}{ds} < 0.$$

During an upshift, $\dot{s} > 0$, so $\dot{\theta}_s < 0$: the secondary opens and the movable side rotates more slowly than the shaft-side assembly. The sign of the corresponding angular-momentum term is therefore carried by the mechanism maps rather than assigned separately in the rotational balance.

If the shift is held fixed, $\dot{s} = \ddot{s} = 0$. The helix then produces no relative motion, and the fixed and movable sides rotate together. The balance reduces to

$$\tau_{\mathrm{B},s} + \tau_s = (I_{\mathrm{B},s} + I_{s,F} + I_{s,M})\,\dot{\omega}_s. \tag{3.3.14}$$

3.3.4 What the Rotational Balances Establish

The primary and secondary now have complete shaft-axis equations of motion. Both remain direct statements of $\sum \tau = dH/dt$. The additional terms appear because the hardware that actuates the pulleys continues to move while the pulley assemblies rotate, so shifting changes the angular momentum stored inside the assemblies themselves.

On the primary, this connection is carried by the same flyweight inertia map $J_{f,p}(s)$ that appeared in the axial derivation. There, $dJ_{f,p}/ds$ contributes to the centrifugal closing action because the flyweights move outward as the primary closes. Here, $J_{f,p}(s)$ and its derivative determine how much shaft-axis angular momentum those same flyweights carry as their radial mass distribution changes. The axial and rotational balances are therefore two views of the same flyweight motion: one asks how rotation drives the movable sheave, while the other asks how that moving flyweight mass changes the rotational response of the pulley.

The secondary shows the same reciprocity through $I_{s,M}$ and the helix map $\theta_s(x_s)$. In the axial balance, the helix reflects the movable assembly's rotational inertia into the sheave motion, so part of the available torque is required to change the movable side's rotation before the remainder appears as axial force. In the rotational balance, that same relative motion means the movable side carries angular momentum at a speed different from the shaft-side assembly, so the axial shift acceleration enters the shaft-axis balance. Again, the two equations are describing one constrained piece of hardware from different free bodies.

Mechanical actuation therefore creates a direct coupling between the axial and rotational dynamics inside each pulley assembly. This is distinct from the coupling carried by the belt reaction. Belt contact can connect the two kinds of balance because its normal action loads the sheaves axially while its tangential action transmits torque about the shafts. The present terms add a second path: the flyweight and helix mechanisms that generate the clamping response also carry inertia, so their own motion appears directly in the rotational equations. The actuator is therefore part of both balances, rather than acting only as a source of clamping force whose influence reaches shaft rotation through the belt.

This distinction gives a precise comparison with earlier transient models. In the rubber-belt formulation of Ballew (2015), belt-contact torque changes pulley speed through a fixed lumped pulley inertia, while pulley axial force is supplied to the belt model. Axial and rotational behaviour can still interact through the belt solution, but motion of the clamping hardware does not alter the pulley angular momentum itself. Duan et al. (2016) provide a stronger adjacent precedent in a chain CVT: rotational dynamics, movable-pulley axial dynamics, chain motion, and contact forces are solved as a coupled system. Their shaft balance nevertheless retains a fixed pulley inertia, while the movable-pulley mass appears in a separate axial equation. The coupling is therefore carried through the chain, contact forces, and changing radius rather than through actuator motion entering both balances.

A mechanically actuated rubber-belt comparison is provided by da Silva et al. (2023), who retain axial equations for both movable sheaves and reduce their translations to a shared shift coordinate. Their rotational treatment uses a fixed primary-pulley inertia. This establishes direct precedent for mechanical clamping, movable-sheave inertia, and coupled shift motion, but not for the additional shaft-axis angular-momentum terms produced by the moving actuator hardware.

The helical model of Sorge and Cammalleri (2011) provides a complementary secondary comparison. Their formulation resolves the relative motion of the driven sheaves and the generally unequal belt torques acting on the two faces in substantially greater local detail. That work therefore captures secondary helix and face-contact mechanics that are reduced here. The distinction in the present balance is different: the helix-constrained movable assembly is retained as part of the complete secondary angular momentum, so the same relative motion that appears in its axial dynamics also appears in the shaft-axis equation.

Taken together, these comparisons show that axial–rotational coupling itself is not the new ingredient. Existing formulations can produce that coupling through belt or chain reactions, changing radii, and shared shift motion. Among the reviewed reduced-order formulations for mechanically actuated rubber-belt CVTs, however, no equivalent pair of pulley balances was identified in which the moving clamping hardware itself provides this additional inertial coupling on both sides: the primary flyweights change their shaft-axis mass distribution as they generate closing action, while the secondary movable assembly changes its shaft-axis angular momentum through the same helix motion that produces its axial response.

The result is a single mechanical picture of each pulley rather than separate axial and rotational descriptions joined only through transmitted load. The same mechanism maps that determine how the pulley clamps also determine how its internal motion changes the shaft dynamics. When those actuator motions cease, the additional terms vanish and the familiar fixed-configuration inertia balances are recovered.

3.4 Belt Transport and Pulley Contact Mechanics

The preceding sections established the axial and rotational equations of motion for the two pulley assemblies, but both stop at the same remaining mechanical body: the belt. The axial balances still contain the belt normal loads N_p and N_s, while the rotational balances contain the belt torques τ_p and τ_s. The belt transport speed v_b is also retained as a state, but its acceleration has not yet been established.

So far, each of these quantities has appeared only through the action of the belt on one of the pulley assemblies. Newton's third law gives the same interactions from the opposite side. The force and torque exerted by the belt on a pulley are accompanied by equal-and-opposite actions on the belt itself. The remaining mechanics can therefore be viewed from the belt rather than from either pulley.

This change of free body is important because the belt is not two independent contacts. It is one closed body connecting both pulley assemblies. Whatever forces and torques are exchanged at the primary must coexist with those at the secondary and with the motion of the belt between them. The belt mechanics must therefore be self-consistent around the complete loop.

3.4.1 Complete-Belt Transport

The belt transport speed provides the natural starting point. The belt speed v_b can change only if something pushes or pulls the belt along its path. Those tangential actions come from the primary and secondary pulley contacts. The belt's transport acceleration must therefore come from the two pulley wraps.

At either wrap, the mechanism is simple. The pulley contact has both normal and tangential components. Normal contact acts perpendicular to the local belt path, so it cannot push or pull the belt in the direction that defines v_b. Frictional traction, by contrast, is spread over the contact and acts tangent to the belt at every point. Traction acting in the direction of belt travel does positive work on the belt, while traction acting against belt travel does negative work and removes energy from it. Moreover, when the complete closed belt is considered, the tensions between adjacent belt sections are internal and cancel. The primary and secondary tractions are therefore the only external actions that can change the belt's transport speed.

Each small traction force also acts at a radius from the shaft and therefore produces a small moment on the pulley. Adding these moments around the wrap gives the contact torque τ_j already used in the pulley rotational balance. Thus, τ_j is simply the accumulated turning effect of the same tangential contact forces that can change v_b.

Newton's third law places equal-and-opposite traction forces on the belt. Their total moment is therefore $-\tau_j$, and at the effective contact radius $r_{j,\text{eff}}$, they are equivalent to the net tangential pull $-\tau_j/r_{j,\text{eff}}$. If the pull at one wrap supplies exactly the power removed by the pull at the other, the belt simply carries that power between the pulleys. If the two actions do not balance, the resulting energy excess or deficit must be taken up by the belt.

The belt assumptions leave that imbalance only one place to go. The fixed length of Assumption 6 (Fixed Belt Length) excludes longitudinal extension and the strain energy it would store. The single transport coordinate of Assumption 7 (Single Belt Transport Coordinate) gives every belt region the same speed v_b, so one region cannot speed up independently of the rest. The belt's only retained longitudinal energy is therefore $\frac{1}{2}m_b v_b^2$. Net power into the belt increases v_b; net power out decreases it.

The energy argument now explains why an imbalance between the pulley actions must change v_b. Newton's second law determines the corresponding signed acceleration directly, where the whole belt loop is treated as a single body. Place one boundary around the entire loop. From the force accounting above, the only external tangential actions on this free body are the two equivalent pulley pulls, acting on the complete belt mass m_b. The complete-belt transport balance is therefore

Complete-Belt Transport Balance **(3.4.1)**

$$\begin{aligned}\sum F_{b,t}^{\text{ext}} &= -\frac{\tau_p}{r_{p,\text{eff}}} - \frac{\tau_s}{r_{s,\text{eff}}} \\ &= \frac{d}{dt}\left(m_b v_b\right) = m_b \dot{v}_b.\end{aligned}$$

This balance gives the basic longitudinal motion of the belt. If the two pulley contacts exert a net force in the positive belt direction, the belt speeds up; if they exert a net force in the opposite direction, it slows down. If the two actions balance, the belt can carry power between the pulleys without changing v_b.

Interpreting the Transport Balance

As far as speeding up or slowing down the complete belt is concerned, this result is sufficient. The quantity v_b, however, describes only how quickly belt material travels along its prescribed path. Consider one marked piece of belt. It approaches a pulley along one straight span and leaves along the other. Its direction of travel has changed. The complete-belt balance says how its speed changes, but contains nothing that explains this change in direction. Left to itself, the marked piece would continue straight past the pulley. Its motion around the wrap is therefore possible only because forces act on it throughout that turn.

The clearest way to separate these forces from the transport motion is to remove the motion altogether. Consider first the familiar case of a rope wrapped around a classical pulley. Pull equally on the two ends of the rope. If both pulls are increased together, the tension throughout the rope rises and the pulley becomes more heavily loaded, even though no tension difference has been introduced. The taut rope is redirected around the pulley, and in doing so it exerts a resultant load on it. The pulley supplies the corresponding equal-and-opposite contact reaction.

The same basic effect occurs when the CVT belt is wrapped around a sheave. Suppose the belt and pulley are stationary and the entry and exit tensions are both $500\,\mathrm{N}$. Their difference is zero, so there is no transmitted torque or longitudinal acceleration, yet the tensioned belt still loads the pulley through the wrap. Raising both the entry and exit tensions to $1000\,\mathrm{N}$ changes neither the tension difference nor the belt's motion, but the belt now loads the pulley more strongly and the corresponding normal contact increases. The geometry and motion have not changed; the pulley loading has.

The loading described above is carried through the belt–sheave contact around the wrap. In the model, the resultant normal contact loads on the primary and secondary pulleys are denoted by N_p and N_s. Their axial components already appear in the primary and secondary movable-sheave balances; see Equations (3.2.14) and (3.2.33). The static example therefore has an important consequence: even with no transmitted torque or belt acceleration, changing the belt tension changes N_p and N_s, and can therefore change the shift balance. Determining these normal loads requires knowing the tension carried around each pulley. The remaining difficulty is that this tension is not generally constant around the pulley.

As established above, a pulley transfers torque through tangential traction distributed around its wrap. As the belt passes through the contact, this traction changes how strongly the belt is being pulled. The tension leaving the wrap therefore generally differs from the tension entering it. In the simplest steady, inertia-free picture, the transmitted torque is equal to the pulley radius multiplied by this tension difference. Retaining the inertia of the wrapped belt modifies that balance, but not the basic interpretation: torque transfer tells us how the tension changes through the contact.

The quantity needed to determine the normal loading is therefore itself changing as the belt moves through the wrap. Knowing the entry and exit tensions tells us only how that process begins and ends, not what tension is carried at each point between them. The normal contact must therefore be determined locally as well.

3.4.2 Local Equations of Motion on a Pulley Wrap

The previous subsection leaves one local question: how is the tension carried at a particular point in the wrap related to the contact force acting there? The next step is therefore to build the free-body diagram of an infinitesimal piece of belt within the wrap of either pulley $j \in \{p, s\}$.

Let θ increase through the wrap in the positive belt-transport direction defined in Section 2.3.1, and isolate the piece bounded by cuts at θ and $\theta + d\theta$. Let $T_j(\theta)$ denote the tension carried by the belt at angular position θ. The tensions at the two cuts are therefore $T_j(\theta)$ and $T_j(\theta + d\theta)$. To keep the labels in the free-body diagram compact, define

$$T_j \equiv T_j(\theta), \qquad dT_j \equiv T_j(\theta + d\theta) - T_j(\theta). \tag{3.4.2}$$

The tension at the second cut can then be written exactly as $T_j + dT_j$. Thus, dT_j is the signed change in tension across the element: it is positive if the tension increases with θ and negative if the tension decreases. No direction of change has yet been assumed.

Once the piece is cut free, the surrounding belt pulls on its two exposed ends. Under Assumption 8 (Prescribed Planar Belt Path), the belt is perfectly flexible and follows the prescribed planar path. Each cut therefore carries only a longitudinal tension force directed along the local belt tangent; there is no transverse shear force or bending moment. At both cuts, the surrounding belt pulls away from the isolated piece.

Because the belt curves around the pulley, those two tension forces do not point in the same direction. The belt tangent turns through $d\theta$ from one cut to the other. It is therefore convenient to draw the local directions at the midpoint of the element: $\hat{\mathbf{e}}_r$ points radially outward from the pulley axis, while $\hat{\mathbf{e}}_\theta$ points with increasing θ. Each end tangent is inclined by $d\theta/2$ from the midpoint tangent. The left-hand view in Figure 3.11 shows this in-plane geometry.

The pulley also acts on the element through both sheave faces. To complete the free-body diagram, these contact forces must be added to the cut-end tensions. Their geometry cannot be represented clearly in the pulley plane, so the right-hand view in Figure 3.11 shows the cross-section of the same element. Under Assumption 4 (Fixed, Uniform Belt Cross-Section), its shape and mass centroid remain fixed. Under Assumption 12 (Symmetric Sheave-Face Sharing), the two faces carry equal shares of the normal contact. Let dN_j denote the sum of the magnitudes of the two face-normal forces; each face therefore exerts a force of magnitude $dN_j/2$. Their axial components oppose one another and cancel, whereas their outward radial components add to $dN_j \sin\beta$.

The remaining contact action is tangential. Under Assumption 10 (Gross Tangential Coulomb Traction), let $dF_{t,j}$ denote the resultant tangential force exerted by both sheave faces on the element. It is defined as positive when it acts in the direction of increasing θ; its actual sign will be determined by the contact state. Because this force acts in the pulley plane, it appears in the left-hand view of Figure 3.11. The two views therefore complete a single free-body diagram: the left shows the forces acting around the wrap, while the right shows how the two face-normal forces produce the radial contact force appearing on the left.

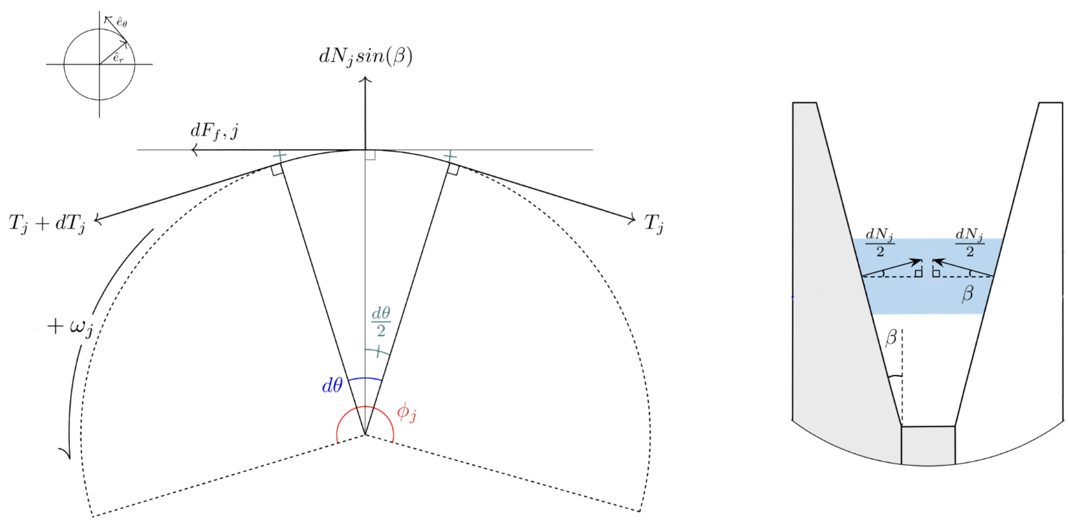


Figure 3.11: Two views of the same infinitesimal belt element, bounded by θ and $\theta + d\theta$ within the wrap of pulley j. The in-plane free-body diagram on the left shows the cut-end tensions $T_j = T_j(\theta)$ and $T_j + dT_j = T_j(\theta + d\theta)$, the midpoint directions, the half-angle $d\theta/2$, the signed tangential contact force $dF_{t,j}$, and the outward radial contact force $dN_j \sin\beta$. The cross-section on the right shows the two face-normal forces of magnitude $dN_j/2$, whose axial components cancel and whose radial components add.

Equations of motion for the infinitesimal element With the free-body diagram complete, Newton's second law can now be assembled in the two local directions. The radial direction comes first because it contains the normal contact that motivated the local analysis.

In the radial direction, the two sheave-face forces supply the combined outward component $dN_j \sin\beta$. The surrounding belt pulls inward at both cuts. Since each end tangent is inclined by $d\theta/2$ from the midpoint tangent, the corresponding inward components are $T_j \sin(d\theta/2)$ and $(T_j + dT_j)\sin(d\theta/2)$. Taking outward as positive, Newton's second law therefore gives

$$dN_j \sin\beta - T_j \sin\left(\frac{d\theta}{2}\right) - (T_j + dT_j)\sin\left(\frac{d\theta}{2}\right) = dm_j a_{r,j}, \tag{3.4.3}$$

where dm_j is the mass of the element and $a_{r,j}$ is its signed acceleration in the outward radial direction.

The same element must also satisfy Newton's second law in the tangential direction. The face-normal forces have no tangential component. At the cut at θ, the tension pulls opposite increasing θ, giving the component $-T_j \cos(d\theta/2)$. At the cut at $\theta + d\theta$, the tension pulls with increasing θ, giving $(T_j + dT_j)\cos(d\theta/2)$. Including the signed tangential contact force $dF_{t,j}$, the tangential equation is therefore

$$(T_j + dT_j)\cos\left(\frac{d\theta}{2}\right) - T_j \cos\left(\frac{d\theta}{2}\right) + dF_{t,j} = dm_j a_{\theta,j}, \tag{3.4.4}$$

where $a_{\theta,j}$ is the signed acceleration in the direction of increasing θ.

The force side of each equation is now fully determined by the free-body diagram. Their right-hand sides still express the belt element's response only in the general form of mass times acceleration. Reducing them requires knowing how much belt mass lies within the angle $d\theta$, and how that material accelerates while it travels around a pulley whose wrap radius can simultaneously move as the CVT shifts. These quantities follow from the belt cross-section and the radius kinematics already established in Section 3.1.

Mass and motion of the belt element The mass and acceleration of the element must be evaluated at the belt's mass centroid. As established in Equations (3.1.8) and (3.1.14), r_j locates the outer surface used to describe where the belt is seated, whereas $r_{j,\mathrm{cm}}$ locates the mass centroid. Under Assumption 4 (Fixed, Uniform Belt Cross-Section), the centroid remains a fixed radial distance from the outer surface. Its radial velocity and acceleration are therefore the radius rate and acceleration already derived for the seated belt:

$$\dot{r}_{j,\mathrm{cm}} = \dot{r}_j, \qquad \ddot{r}_{j,\mathrm{cm}} = \ddot{r}_j.$$

The element spans the angle $d\theta$, so its arc length at the mass centroid is $r_{j,\mathrm{cm}}d\theta$. Multiplying this length by the belt cross-sectional area A_b from Equation (3.1.12) and the belt density ρ_b gives the element mass:

$$dm_j = \rho_b A_b r_{j,\mathrm{cm}}\, d\theta. \tag{3.4.5}$$

The element can move in two directions at once. It moves radially with the shifting wrap at speed $\dot{r}_j$, while the belt's transport carries it tangentially at speed v_b. Its velocity is therefore

$$\mathbf{v}_{b,j} = \dot{r}_j\,\hat{\mathbf{e}}_r + v_b\,\hat{\mathbf{e}}_\theta. \tag{3.4.6}$$

Acceleration is the time derivative of this velocity. Both scalar speeds can change, but the local directions also change as the element travels around the pulley. Differentiating while retaining both effects gives

$$\begin{aligned}\mathbf{a}_{b,j} &= \frac{d}{dt}\left(\dot{r}_j\hat{\mathbf{e}}_r + v_b\hat{\mathbf{e}}_\theta\right)\\ &= \ddot{r}_j\hat{\mathbf{e}}_r + \dot{r}_j\dot{\hat{\mathbf{e}}}_r + \dot{v}_b\hat{\mathbf{e}}_\theta + v_b\dot{\hat{\mathbf{e}}}_\theta.\end{aligned} \tag{3.4.7}$$

The two basis-vector derivatives follow from how quickly the element moves around the wrap. At the centroidal radius, its tangential speed and angular rate are related by

$$v_b = r_{j,\mathrm{cm}}\dot{\theta}, \qquad \dot{\theta} = \frac{v_b}{r_{j,\mathrm{cm}}}. \tag{3.4.8}$$

As θ increases, the outward radial direction turns toward $\hat{\mathbf{e}}_\theta$, while the tangential direction turns toward $-\hat{\mathbf{e}}_r$. Multiplying these changes per unit angle by the angular rate in Equation (3.4.8) gives

$$\dot{\hat{\mathbf{e}}}_r = \frac{v_b}{r_{j,\text{cm}}}\hat{\mathbf{e}}_\theta, \qquad \dot{\hat{\mathbf{e}}}_\theta = -\frac{v_b}{r_{j,\text{cm}}}\hat{\mathbf{e}}_r. \tag{3.4.9}$$

Substituting these basis rates into Equation (3.4.7) and collecting the radial and tangential terms gives

$$\mathbf{a}_{b,j} = \left(\ddot{r}_j - \frac{v_b^2}{r_{j,\text{cm}}}\right)\hat{\mathbf{e}}_r + \left(\dot{v}_b + \frac{\dot{r}_j v_b}{r_{j,\text{cm}}}\right)\hat{\mathbf{e}}_\theta. \tag{3.4.10}$$

The acceleration components required by the two equations of motion are therefore

$$a_{r,j} = \ddot{r}_j - \frac{v_b^2}{r_{j,\text{cm}}}, \tag{3.4.11}$$

$$a_{\theta,j} = \dot{v}_b + \frac{\dot{r}_j v_b}{r_{j,\text{cm}}}. \tag{3.4.12}$$

The radial component contains the outward acceleration of the shifting wrap and the familiar inward centripetal acceleration produced as the belt turns around the pulley. The tangential component contains the change in belt speed and an additional contribution from the radial velocity changing direction as the element moves through the wrap. Together with Equation (3.4.5), these two components now provide the complete inertial sides of the radial and tangential balances.

Radial equation of motion We first return to the radial balance in Equation (3.4.3). Substituting the element mass from Equation (3.4.5) and the radial acceleration from Equation (3.4.11), then collecting the two inward tension components, gives

$$dN_j \sin\beta - (2T_j + dT_j)\sin\left(\frac{d\theta}{2}\right) = \rho_b A_b \left(r_{j,\text{cm}}\ddot{r}_j - v_b^2\right) d\theta. \tag{3.4.13}$$

The desired local quantity is the normal contact carried per unit wrap angle, $dN_j/d\theta$. Dividing Equation (3.4.13) by $d\theta$ gives

$$\sin\beta\,\frac{dN_j}{d\theta} - (2T_j + dT_j)\,\frac{\sin(d\theta/2)}{d\theta} = \rho_b A_b \left(r_{j,\text{cm}}\ddot{r}_j - v_b^2\right).$$

Only the tension term still depends on the finite angular width of the element. To put its sine factor into the standard small-angle form, write

$$\frac{\sin(d\theta/2)}{d\theta} = \frac{1}{2}\frac{\sin(d\theta/2)}{d\theta/2}.$$

Redistributing this factor of $1/2$ through $2T_j + dT_j$, the radial balance can be written directly as

$$\sin\beta\,\frac{dN_j}{d\theta} - \left(T_j + \frac{dT_j}{2}\right)\frac{\sin(d\theta/2)}{d\theta/2} = \rho_b A_b \left(r_{j,\text{cm}}\ddot{r}_j - v_b^2\right).$$

The infinitesimal element is obtained by taking $d\theta \to 0$. Because $dT_j = T_j(\theta + d\theta) - T_j(\theta)$, continuity of the tension field gives $dT_j \to 0$. At the same time, the small-angle ratio tends to one:

$$T_j + \frac{dT_j}{2} \longrightarrow T_j, \qquad \frac{\sin(d\theta/2)}{d\theta/2} \longrightarrow 1.$$

The complete local radial balance is consequently

$$\boxed{\sin\beta \, \frac{dN_j}{d\theta} = T_j - \rho_b A_b \left(v_b^2 - r_{j,\mathrm{cm}}\ddot{r}_j\right).} \tag{3.4.14}$$

To see the physical meaning of this result most clearly, it is useful first to consider a stationary belt on a fixed wrap. Setting $v_b = 0$ and $\ddot{r}_j = 0$ gives

$$\sin\beta \, \frac{dN_j}{d\theta} = T_j. \tag{3.4.15}$$

We now recover the result anticipated from the physical picture: increasing the belt tension increases the normal loading carried per unit wrap angle. This remains true even when the tension is uniform through the contact, so there is no tension difference and no torque is transmitted. The normal loading therefore depends on the absolute tension level, not only on the tension difference across the wrap.

Returning to the full balance, motion modifies this stationary relationship. As the belt turns around the pulley, part of the inward tension is required to provide its centripetal acceleration, reducing the outward sheave reaction at a given tension. An outward wrap acceleration has the opposite effect and increases that reaction.

Tangential equation of motion Finally, we complete the tangential equation of motion introduced in Equation (3.4.4). Substituting the element mass from Equation (3.4.5) and the tangential acceleration from Equation (3.4.12) gives the complete balance

$$(T_j + dT_j)\cos\left(\frac{d\theta}{2}\right) - T_j \cos\left(\frac{d\theta}{2}\right) + dF_{t,j} = \rho_b A_b r_{j,\mathrm{cm}} d\theta \left(\dot{v}_b + \frac{\dot{r}_j v_b}{r_{j,\mathrm{cm}}}\right). \tag{3.4.16}$$

The desired result is again a local relation per unit wrap angle. Cancelling the common T_j contribution from the two end tensions, simplifying the acceleration term, and dividing through by $d\theta$ gives

$$\frac{dT_j}{d\theta}\cos\left(\frac{d\theta}{2}\right) + \frac{dF_{t,j}}{d\theta} = \rho_b A_b \left(r_{j,\mathrm{cm}}\dot{v}_b + \dot{r}_j v_b\right).$$

The remaining cosine records that the end tensions are not quite parallel on a finite element. In the infinitesimal limit, the two end tangents coincide and

$$\lim_{d\theta \to 0} \cos\left(\frac{d\theta}{2}\right) = \cos(0) = 1.$$

The cosine therefore becomes a factor of one; the tension-gradient term remains. The local tangential equation of motion is

$$\frac{dT_j}{d\theta} + \frac{dF_{t,j}}{d\theta} = \rho_b A_b \left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b \right). \tag{3.4.17}$$

This balance identifies the two tangential forces acting per unit wrap angle: the change in belt tension and the traction supplied by the pulley. Their sum produces the tangential acceleration of the belt. What remains is to relate that tangential contact force to the normal contact already present in the radial balance.

Under Assumption 10 (Gross Tangential Coulomb Traction), the sheave contact obeys Coulomb friction. While the belt sticks, the tangential force is a reaction that can act in either direction and take any magnitude up to the static limit. Once the belt slides, its magnitude is fixed by kinetic friction and its direction opposes the relative sliding. Both cases relate the tangential force on the element to the same normal-force increment dN_j, but with different rules for the allowable proportionality.

Rather than carrying a separate magnitude condition and direction sign through the remaining equations, define the signed traction utilization λ_j by

Signed Traction Utilization **(3.4.18)**

$$dF_{t,j} = \lambda_j \, dN_j, \qquad \lambda_j = \begin{cases} \lambda_{j,\mathrm{st}}, & |\lambda_{j,\mathrm{st}}| \le \mu_s, \quad \text{sticking}, \\ \sigma_j \mu_k, & \sigma_j \in \{-1, +1\}, \quad \text{sliding}. \end{cases}$$

The magnitude of λ_j gives the tangential force used per unit normal load, while its sign gives the direction of that force. Thus, $\lambda_j > 0$ acts with increasing θ, $\lambda_j < 0$ acts against it, and $\lambda_j = 0$ represents normal contact with no tangential force. In sliding, σ_j selects the direction that opposes the relative motion, while μ_k fixes the magnitude. Under Assumption 11 (Wrap-Level Contact Resolution), one value of λ_j represents the complete wrap at a given instant.

Dividing $dF_{t,j} = \lambda_j dN_j$ by $d\theta$ and substituting it into Equation (3.4.17) gives the complete local tangential balance:

$$\boxed{\frac{dT_j}{d\theta} + \lambda_j \frac{dN_j}{d\theta} = \rho_b A_b \left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b \right).} \tag{3.4.19}$$

To expose the physical meaning of this result, it is useful again to remove the effects of motion. For a fixed wrap and steady belt speed, $\dot{r}_j = 0$ and $\dot{v}_b = 0$, so the tangential inertia vanishes and the balance becomes

$$\frac{dT_j}{d\theta} = -\lambda_j \frac{dN_j}{d\theta}. \tag{3.4.20}$$

We now recover the expected local relationship: the pulley traction must be balanced by an equal and opposite tension gradient. Traction therefore forces the tension to evolve around the wrap, and integrating this local evolution produces the tension difference associated with transmitted torque.

Since the normal contact is compressive, $dN_j/d\theta \geq 0$: positive traction makes the tension fall as θ increases, while traction in the opposite direction makes it rise. Retaining the full balance, some of the traction may be used to accelerate the belt, but the same underlying link between traction and tension evolution remains.

3.4.3 Tension Evolution Through a Pulley Wrap

The balances derived above describe the forces on one infinitesimal belt element at an arbitrary position within a pulley wrap. The next step is to carry that local description across the complete contact, so that the effect of every element from the wrap entrance to the wrap exit can be accumulated.

For pulley j, let θ increase from 0 at the entrance to ϕ_j at the exit. The belt enters with tension $T_{j,\text{in}}$, carries the tension $T_j(\theta)$ at an intermediate position, and leaves with tension $T_{j,\text{out}}$:

$$0 \leq \theta \leq \phi_j, \qquad T_j(0) = T_{j,\text{in}}, \qquad T_j(\phi_j) = T_{j,\text{out}}.$$

Both $T_j(\theta)$ and the normal loading $dN_j/d\theta$ are needed throughout this interval. Tension provides the natural starting point: $T_{j,\text{in}}$ is the entrance boundary value, and following its evolution through the wrap produces $T_{j,\text{out}}$. Once that tension field is known, the radial balance recovers the normal loading at every position.

The tangential balance, Equation (3.4.19), contains the desired tension gradient but remains coupled to $dN_j/d\theta$. Rearranging the radial balance, Equation (3.4.14), gives

$$\frac{dN_j}{d\theta} = \frac{1}{\sin\beta}\left[T_j - \rho_b A_b\left(v_b^2 - r_{j,\text{cm}}\ddot{r}_j\right)\right]. \tag{3.4.21}$$

Substituting this expression into the tangential balance, then expanding and rearranging, gives

$$\begin{aligned}
&\frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin\beta}\left[T_j - \rho_b A_b\left(v_b^2 - r_{j,\text{cm}}\ddot{r}_j\right)\right] \\
&\qquad = \rho_b A_b\left(r_{j,\text{cm}}\dot{v}_b + \dot{r}_j v_b\right) \\
&\frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin\beta}T_j = \rho_b A_b\left(r_{j,\text{cm}}\dot{v}_b + \dot{r}_j v_b\right) + \rho_b A_b\frac{\lambda_j}{\sin\beta}\left(v_b^2 - r_{j,\text{cm}}\ddot{r}_j\right).
\end{aligned}$$

Factoring the common $\rho_b A_b$ on the right gives the local equation that governs tension evolution through either pulley wrap:

Pulley-Wrap Tension Evolution **(3.4.22)**

$$\begin{aligned}
\frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin\beta}T_j = \rho_b A_b\Big[r_{j,\text{cm}}\dot{v}_b + \dot{r}_j v_b& \\
+\frac{\lambda_j}{\sin\beta}\left(v_b^2 - r_{j,\text{cm}}\ddot{r}_j\right)\Big]&, \qquad j \in \{p, s\}.
\end{aligned}$$

The two local balances have now been reduced to one first-order equation for $T_j(\theta)$.

Solving the tension equation Before Equation (3.4.22) can be integrated through the wrap, its coefficients must be considered along the spatial coordinate θ. At a fixed instant, the circular wrap prescribed by Assumption 8 (Prescribed Planar Belt Path) gives one $r_{j,\mathrm{cm}}$, $\dot{r}_j$, and $\ddot{r}_j$ for pulley j. Likewise, Assumption 7 (Single Belt Transport Coordinate) makes v_b and $\dot{v}_b$ common throughout the belt, while Assumption 11 (Wrap-Level Contact Resolution) assigns one λ_j to the complete wrap. None of these quantities therefore varies with θ during the integration.

To expose the structure of the equation, define the two constant coefficients

$$\begin{aligned} a_j &= \frac{\lambda_j}{\sin\beta}, \\ c_j &= \rho_b A_b \left[r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b + a_j \left(v_b^2 - r_{j,\mathrm{cm}} \ddot{r}_j \right) \right]. \end{aligned} \tag{3.4.23}$$

Then Equation (3.4.22) becomes

$$\frac{dT_j}{d\theta} + a_j T_j = c_j. \tag{3.4.24}$$

To integrate Equation (3.4.24), it is useful to combine the two terms on the left into a single derivative. The exponential $e^{a_j\theta}$ is chosen because its derivative is $a_j e^{a_j\theta}$, matching the coefficient multiplying T_j. The product rule then gives

$$\frac{d}{d\theta}\left(e^{a_j\theta} T_j\right) = e^{a_j\theta}\left(\frac{dT_j}{d\theta} + a_j T_j\right). \tag{3.4.25}$$

Multiplying Equation (3.4.24) by $e^{a_j\theta}$ therefore gives

$$\frac{d}{d\theta}\left(e^{a_j\theta} T_j\right) = c_j e^{a_j\theta}. \tag{3.4.26}$$

The entrance condition $T_j(0) = T_{j,\mathrm{in}}$ now supplies the lower boundary. Integrating from the wrap entrance to an arbitrary position θ gives

$$\int_0^\theta \frac{d}{d\vartheta}\left(e^{a_j\vartheta} T_j(\vartheta)\right) d\vartheta = c_j \int_0^\theta e^{a_j\vartheta}\, d\vartheta. \tag{3.4.27}$$

For $a_j \neq 0$, evaluating the integrals yields

$$e^{a_j\theta} T_j(\theta) - T_{j,\mathrm{in}} = \frac{c_j}{a_j}\left(e^{a_j\theta} - 1\right), \tag{3.4.28}$$

and isolating $T_j(\theta)$ gives

$$T_j(\theta) = T_{j,\mathrm{in}} e^{-a_j\theta} + \frac{c_j}{a_j}\left(1 - e^{-a_j\theta}\right). \tag{3.4.29}$$

Substituting the constant coefficients back into Equation (3.4.29) and rearranging gives the tension carried at every position through the wrap:

$$
\begin{aligned}
T_j(\theta) &= T_{j,\text{in}} e^{-\frac{\lambda_j}{\sin\beta}\theta} \\
&\quad + \rho_b A_b \Big(v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin\beta}{\lambda_j} \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b \right) \Big) \Big(1 - e^{-\frac{\lambda_j}{\sin\beta}\theta} \Big), \qquad \lambda_j \neq 0.
\end{aligned} \tag{3.4.30}
$$

At the wrap exit, $\theta = \phi_j$ and $T_j(\phi_j) = T_{j,\text{out}}$. Evaluating Equation (3.4.30) at that endpoint gives

$$
\begin{aligned}
T_{j,\text{out}} &= T_{j,\text{in}} e^{-\frac{\lambda_j}{\sin\beta}\phi_j} \\
&\quad + \rho_b A_b \Big(v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin\beta}{\lambda_j} \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b \right) \Big) \Big(1 - e^{-\frac{\lambda_j}{\sin\beta}\phi_j} \Big), \qquad \lambda_j \neq 0.
\end{aligned} \tag{3.4.31}
$$

The closed form is written for $\lambda_j \neq 0$ because its derivation divided by a_j, but zero utilization presents no difficulty. For completeness, setting $\lambda_j = 0$ directly in the tangential balance of Equation (3.4.19) gives

$$
\frac{dT_j}{d\theta} = \rho_b A_b \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b \right). \tag{3.4.32}
$$

Integrating from the wrap entrance to θ, and then evaluating the result at $\theta = \phi_j$, gives

$$
\begin{aligned}
T_j(\theta) &= T_{j,\text{in}} + \rho_b A_b \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b \right) \theta, \\
T_{j,\text{out}} &= T_{j,\text{in}} + \rho_b A_b \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b \right) \phi_j.
\end{aligned} \tag{3.4.33}
$$

Interpreting the wrap solution The local balances have now been carried across the complete wrap. For a specified entrance tension and instantaneous wrap state, Equation (3.4.30) gives the tension at every position, while Equation (3.4.31) gives the resulting exit tension. Within the constant-wrap description established by Assumption 11 (Wrap-Level Contact Resolution), the solution has therefore retained every transient term admitted by the model: belt acceleration, radial migration, radial acceleration, and centripetal inertia. These instantaneous quantities are uniform with respect to θ, even though the resulting tension and normal loading vary through the wrap. Improving the contact resolution further would therefore mean allowing local belt speed, radial motion, and traction state to vary within the wrap itself, for which the constant-coefficient solution would be replaced by a spatially varying model (Kong and Parker, 2006; Messick, 2018; Julió and Plante, 2011).

The full expression describes the wrap even while the belt speed and pulley radius are changing. When those changes stop, the transient terms disappear, leaving a direct connection to the familiar steady belt relation. Holding the pulley radius fixed and the belt transport speed steady gives $\dot{r}_j = \ddot{r}_j = \dot{v}_b = 0$. Substituting these conditions directly into the entry-to-exit result, Equation (3.4.31), and rearranging gives

$$
\frac{T_{j,\text{out}} - \rho_b A_b v_b^2}{T_{j,\text{in}} - \rho_b A_b v_b^2} = e^{-\frac{\lambda_j \phi_j}{\sin\beta}}. \tag{3.4.34}
$$

This is the inertia-corrected structure used by Gerbert (Gerbert, 1996), written in the same in/out convention as the full wrap solution. The correction has a direct physical meaning: $\rho_b A_b v_b^2$ is the portion of the tension needed simply to turn the moving belt, whereas the remainder produces normal contact and is therefore called the tractive tension. The ratio shows that it is this tractive portion, rather than the total tension, that friction causes to evolve across the steady wrap. At the traction limit, $|\lambda_j| = \mu$, while the sign of λ_j determines whether the tension rises or falls from the wrap entrance to the wrap exit.

The same comparison leads directly to the familiar classical limit. If belt inertia is neglected and the V-groove is removed by taking $\beta = 90°$, then the centrifugal correction vanishes and $\sin\beta = 1$, so Equation (3.4.34) becomes

$$\frac{T_{j,\text{out}}}{T_{j,\text{in}}} = e^{-\lambda_j \phi_j}. \tag{3.4.35}$$

This is the Euler–Eytelwein equation in the same signed in/out convention (Euler, 1769; Eytelwein, 1808). No new law has been introduced: the classical result is simply what remains of the entry-to-exit balance after transient acceleration, centrifugal inertia, and V-groove amplification are removed. Read in reverse, the full solution restores each of those effects while preserving the same underlying idea that distributed traction changes the tension through the wrap. With that tension now known at every position, Equation (3.4.21) can recover the corresponding normal loading, allowing the pulley load and transmitted torque to be assembled from the same contact distribution.

3.4.4 Pulley Contact Loads and Torque

With $T_j(\theta)$ now known, the same distributed contact can supply the two belt actions left in the pulley equations: the normal resultant N_j and the contact torque τ_j. Begin with the radial balance. Equation (3.4.21) gives the normal loading per unit wrap angle, so accumulating that distribution from the entrance at $\theta = 0$ to the exit at $\theta = \phi_j$ gives

$$N_j = \frac{1}{\sin\beta} \int_0^{\phi_j} \left[T_j(\theta) - \rho_b A_b \left(v_b^2 - r_{j,\text{cm}} \ddot{r}_j \right) \right] d\theta. \tag{3.4.36}$$

Here, N_j is the total normal load integrated over both sheave faces. Since $T_j(\theta)$ is supplied by Equation (3.4.30), this integral now determines the normal resultant required in Equations (3.2.14) and (3.2.33).

It is important to note that, under Assumption 8 (Prescribed Planar Belt Path), this mathematical solution is physically admissible only while

$$T_j(\theta) \geq 0, \qquad \frac{dN_j}{d\theta} \geq 0, \qquad 0 \leq \theta \leq \phi_j. \tag{3.4.37}$$

A negative tension indicates slack, while a negative normal-loading gradient indicates local lift-off; neither represents a physical negative load.

Although Equation (3.4.36) determines N_j, evaluating it requires the complete tension field. The tangential balance gives the same resultant directly from the wrap boundary values. Under Assumption 11 (Wrap-Level Contact Resolution), λ_j is constant throughout the wrap, so integrating Equation (3.4.19) from $\theta = 0$ to $\theta = \phi_j$ gives

$$T_{j,\mathrm{out}} - T_{j,\mathrm{in}} + \lambda_j N_j = \rho_b A_b \phi_j \left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b \right). \tag{3.4.38}$$

For $\lambda_j \neq 0$, rearranging gives the normal resultant in its compact entry-to-exit form:

Pulley-Wrap Normal Resultant **(3.4.39)**

$$N_j = \frac{T_{j,\mathrm{in}} - T_{j,\mathrm{out}} + \rho_b A_b \phi_j \left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b \right)}{\lambda_j}, \qquad \lambda_j \neq 0.$$

The division by λ_j only excludes zero traction from this compact form. When $\lambda_j = 0$, the tangential balance no longer determines N_j, but the radial integral remains valid. Substituting the zero-traction tension field from Equation (3.4.33) into Equation (3.4.36) gives

$$\begin{aligned} N_j = {} & \frac{\phi_j}{\sin\beta} \left[T_{j,\mathrm{in}} - \rho_b A_b \left(v_b^2 - r_{j,\mathrm{cm}} \ddot{r}_j \right) \right] \\ & + \frac{\rho_b A_b \phi_j^2}{2 \sin\beta} \left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b \right). \end{aligned} \tag{3.4.40}$$

Thus $\lambda_j = 0$ removes the tangential contact force, not the normal contact: the pulley can remain clamped even though the modeled contact transmits no torque.

The axial derivation already established how the normal contact enters each movable-sheave free body. Under Assumption 12 (Symmetric Sheave-Face Sharing), the movable face carries $N_j/2$; projecting that load onto the local shaft axis gives the previously introduced belt-opening term

$$F_{\mathrm{belt},x_j} = \frac{N_j}{2} \cos\beta, \qquad j \in \{p, s\}.$$

With N_j now known, this determines the belt-opening term left unknown in Equations (3.2.14) and (3.2.33).

The remaining unresolved belt term is the contact torque τ_j in Equations (3.3.5) and (3.3.13). Recall from Section 3.4.1 that $dF_{t,j} = \lambda_j \, dN_j$ acts on the belt, while the equal-and-opposite force acts on the pulley. Using the cord-line lever arm $r_{j,\mathrm{eff}}$ from Equation (3.1.9), and then using the integrated tangential balance in Equation (3.4.38), gives

$$\begin{aligned} \tau_j &= -r_{j,\mathrm{eff}} \int_0^{\phi_j} \lambda_j \frac{dN_j}{d\theta} \, d\theta \\ &= -r_{j,\mathrm{eff}} \lambda_j N_j \\ &= r_{j,\mathrm{eff}} \left[T_{j,\mathrm{out}} - T_{j,\mathrm{in}} - \rho_b A_b \phi_j \left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b \right) \right], \qquad j \in \{p, s\}. \end{aligned} \tag{3.4.41}$$

The final line makes the transient correction explicit: the pulley torque is set by the entry-to-exit tension change only when the tangential acceleration of the wrapped belt vanishes.

For a specified wrap state, the local mechanics are now complete, but the absolute tension level is not. Equation (3.4.30) maps the entrance tension $T_{j,\mathrm{in}}$ to the exit tension $T_{j,\mathrm{out}}$, but an isolated wrap cannot determine either boundary value on its own. Those tensions continue into the adjoining straight spans and, ultimately, around the same closed belt. The next step is therefore to reconnect the two pulley wraps through the straight spans and require the tension to be compatible after one complete circuit of the belt loop.

3.4.5 Reassembling the Closed Belt

Dividing the complete belt at the four in–out-of-wrap points in Figure 3.12 produces four regions: the primary wrap, the secondary wrap, and the upper and lower straight spans. Newton's second law has already been applied throughout both wrapped regions. The two straight spans remain. They carry no distributed pulley contact, but they still contain belt mass, so their end tensions need not be equal while the belt accelerates.

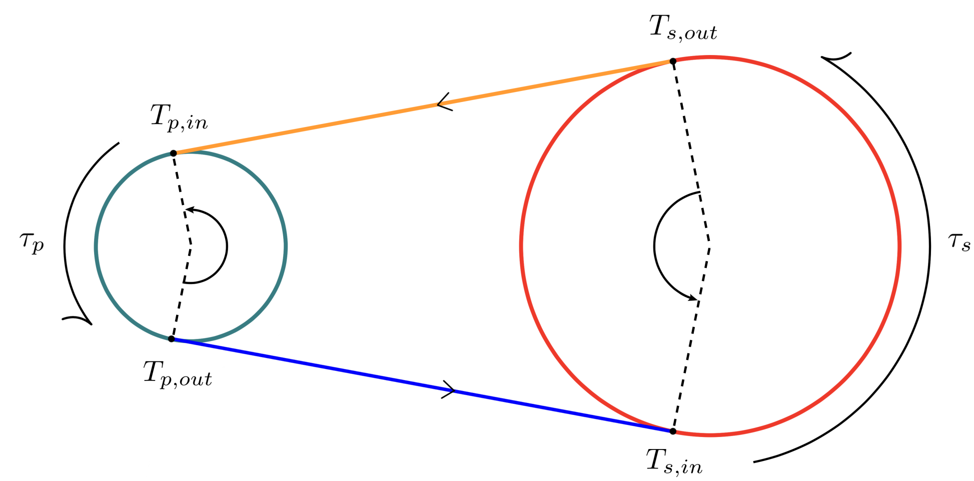


Figure 3.12: Complete belt divided into two wrapped regions and two straight spans. The arrows show the positive belt-transport direction. The upper span runs from the secondary wrap exit, $T_{s,\text{out}}$, to the primary wrap entrance, $T_{p,\text{in}}$; the lower span runs from $T_{p,\text{out}}$ to $T_{s,\text{in}}$. The torque arrows identify the contact actions on the wrapped regions already treated above.

As the in–out-of-wrap points move during a shift, each straight span also rotates. Assumption 8 (Prescribed Planar Belt Path) prescribes this changing path but explicitly omits the transverse inertia associated with its reorientation. Accordingly, Newton's second law is applied only along the instantaneous straight span.

Take this direction as positive with belt travel, and let p_u and p_l be the tangential momenta of the belt particles occupying the two spans at the instant considered; Newton's second law follows those same particles. The downstream tension acts positively and the upstream tension negatively, so

$$\begin{aligned} T_{p,\text{in}} - T_{s,\text{out}} &= \frac{dp_u}{dt}, \\ T_{s,\text{in}} - T_{p,\text{out}} &= \frac{dp_l}{dt}. \end{aligned} \tag{3.4.42}$$

At that instant, a straight span $i \in \{u, l\}$ has mass $\rho_b A_b l_i$. Under Assumption 7 (Single Belt Transport Coordinate) and Assumption 4 (Fixed, Uniform Belt Cross-Section), its stored tangential momentum is $p_i = \rho_b A_b l_i v_b$, so the obvious first step is to differentiate:

$$\frac{d}{dt}\left(\rho_b A_b l_i v_b\right) = \rho_b A_b \left(l_i \dot{v}_b + \dot{l}_i v_b\right).$$

The term $\rho_b A_b \dot{l}_i v_b$ exposes the complication. Each station marks where the belt enters or leaves the current pulley geometry; it is not fixed to a particular belt particle. As the stations move, belt particles pass between the wraps and straight span, so the region contains a changing mass. The derivative above therefore describes the momentum stored within a moving region, not yet the material momentum rate required by Newton's second law. The momentum crossing its boundaries must also be included.

Let v_i^- and v_i^+ denote the velocities of the upstream and downstream stations, respectively, projected along the straight span. The belt crosses those boundaries at the relative speeds $v_b - v_i^-$ and $v_b - v_i^+$. Adding the outgoing momentum flux and subtracting the incoming flux gives

$$\begin{aligned}\frac{dp_i}{dt} = &\frac{d}{dt}\left(\rho_b A_b l_i v_b\right)\\ &+ \rho_b A_b v_b \left[\left(v_b - v_i^+\right) - \left(v_b - v_i^-\right)\right].\end{aligned} \tag{3.4.43}$$

These same station velocities determine the changing distance between them:

$$\dot{l}_i = v_i^+ - v_i^-. \tag{3.4.44}$$

Expanding the stored momentum and substituting Equation (3.4.44) gives

$$\begin{aligned}\frac{dp_i}{dt} &= \rho_b A_b \left(l_i \dot{v}_b + \dot{l}_i v_b\right) + \rho_b A_b v_b \left(v_i^- - v_i^+\right)\\ &= \rho_b A_b \left(l_i \dot{v}_b + \dot{l}_i v_b - \dot{l}_i v_b\right)\\ &= \rho_b A_b l_i \dot{v}_b.\end{aligned} \tag{3.4.45}$$

The cancellation is exact. The station motion changes the momentum stored within the geometric span by $\rho_b A_b \dot{l}_i v_b$, but the same motion produces an equal and opposite momentum flux through its boundaries. Once that flux is included, the moving stations leave no additional term in the tangential balance; only the acceleration of the belt transport remains.

The aligned-pulley geometry gives both straight spans the same length l, as shown in Equation (3.1.19). Substituting Equation (3.4.45) into the two Newton balances therefore yields

$$\begin{aligned}T_{p,\text{in}} - T_{s,\text{out}} &= \rho_b A_b l \dot{v}_b,\\ T_{s,\text{in}} - T_{p,\text{out}} &= \rho_b A_b l \dot{v}_b.\end{aligned} \tag{3.4.46}$$

Because the right-hand sides of the two straight-span balances are identical, their tension differences must be equal:

$$T_{p,\text{in}} - T_{s,\text{out}} = T_{s,\text{in}} - T_{p,\text{out}} \tag{3.4.47}$$

Rearranging this result places the two tensions belonging to each pulley on the same side:

$$T_{p,\text{in}} + T_{p,\text{out}} = T_{s,\text{in}} + T_{s,\text{out}}. \tag{3.4.48}$$

The closed belt therefore requires the sum of the primary boundary tensions to match the sum of the secondary boundary tensions. The remaining task is to evaluate that sum from the mechanics already obtained for a generic pulley wrap.

The integrated tangential balance in Equation (3.4.38) gives the boundary-tension difference as

$$T_{j,\text{out}} - T_{j,\text{in}} = \rho_b A_b \phi_j \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b\right) - \lambda_j N_j. \tag{3.4.49}$$

This and the entry-to-exit solution in Equation (3.4.31) are two relations for the same two boundary tensions. Solving them for the entrance tension gives

$$\begin{aligned} T_{j,\text{in}} = \rho_b A_b &\left[v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin\beta}{\lambda_j}\left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b\right)\right] \\ &+ \frac{\lambda_j N_j - \rho_b A_b \phi_j \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b\right)}{1 - e^{-\frac{\lambda_j}{\sin\beta}\phi_j}}, \qquad \lambda_j \neq 0. \end{aligned} \tag{3.4.50}$$

The required sum can now be written as $2T_{j,\text{in}} + (T_{j,\text{out}} - T_{j,\text{in}})$. Substituting Equations (3.4.49) and (3.4.50) and simplifying gives

$$\begin{aligned} T_{j,\text{in}} + T_{j,\text{out}} = 2\rho_b A_b &\left[v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin\beta}{\lambda_j}\left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b\right)\right] \\ &+ \frac{1 + e^{-\frac{\lambda_j}{\sin\beta}\phi_j}}{1 - e^{-\frac{\lambda_j}{\sin\beta}\phi_j}} \left[\lambda_j N_j - \rho_b A_b \phi_j \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b\right)\right], \qquad \lambda_j \neq 0. \end{aligned} \tag{3.4.51}$$

Applying this expression to the primary and secondary sides of Equation (3.4.48) eliminates all four temporary boundary tensions:

Closed-Loop Tension Compatibility **(3.4.52)**

$$
\begin{aligned}
&2\rho_b A_b \left[v_b^2 - r_{p,\mathrm{cm}}\ddot{r}_p + \frac{\sin\beta}{\lambda_p}\left(r_{p,\mathrm{cm}}\dot{v}_b + \dot{r}_p v_b\right)\right] \\
&\quad + \frac{1 + e^{-\frac{\lambda_p}{\sin\beta}\phi_p}}{1 - e^{-\frac{\lambda_p}{\sin\beta}\phi_p}} \left[\lambda_p N_p - \rho_b A_b \phi_p \left(r_{p,\mathrm{cm}}\dot{v}_b + \dot{r}_p v_b\right)\right] \\
&= 2\rho_b A_b \left[v_b^2 - r_{s,\mathrm{cm}}\ddot{r}_s + \frac{\sin\beta}{\lambda_s}\left(r_{s,\mathrm{cm}}\dot{v}_b + \dot{r}_s v_b\right)\right] \\
&\quad + \frac{1 + e^{-\frac{\lambda_s}{\sin\beta}\phi_s}}{1 - e^{-\frac{\lambda_s}{\sin\beta}\phi_s}} \left[\lambda_s N_s - \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}}\dot{v}_b + \dot{r}_s v_b\right)\right], \qquad \lambda_p\lambda_s \neq 0.
\end{aligned}
$$

The displayed form assumes nonzero utilization only because the analytical wrap solution contains factors of $1/\lambda_j$. The zero-utilization limit remains regular. Using the zero-traction relations from Equations (3.4.33) and (3.4.40) gives

$$
T_{j,\mathrm{in}} + T_{j,\mathrm{out}} = 2\rho_b A_b \left(v_b^2 - r_{j,\mathrm{cm}}\ddot{r}_j\right) + \frac{2N_j \sin\beta}{\phi_j}, \qquad \lambda_j = 0. \tag{3.4.53}
$$

This expression replaces the corresponding side of Equation (3.4.52) whenever $\lambda_p = 0$ or $\lambda_s = 0$. The four boundary tensions have therefore disappeared from the closed-loop equation: the straight spans require the two boundary-tension sums to match, and the local wrap mechanics express each sum in terms of the pulley normal load, traction utilization, and retained belt kinematics. Once the coupled mechanical equations determine those quantities, Equations (3.4.31) and (3.4.50) recover the boundary tensions and Equation (3.4.30) recovers the tension through each wrap.

3.4.6 Belt-Mechanics Closure and Scope

The closed-loop compatibility in Equation (3.4.52) is the final relation required from the belt itself. Rather than fixing the absolute tension level through a prescribed tight- or slack-side boundary condition, it makes that level emerge from the two pulley-wrap solutions together with the momentum balances of the two straight spans. This way of closing the loop also defines the level of belt resolution retained by the present model.

This treatment combines ideas that more commonly appear at different levels of resolution. Classical belt relations describe the tension change through a single wrap once an endpoint tension and contact state are supplied (Gerbert, 1996). Steady whole-belt boundary-value models couple the pulley contacts and free spans, but typically require operating-point boundary data such as a prescribed slack-side tension (Kong and Parker, 2006; Messick, 2018). Transient discretized rubber-belt models instead enforce whole-loop compatibility through the states of many belt elements (Julió and Plante, 2011; Ballew, 2015). The present formulation occupies a reduced middle ground: it retains one global belt-transport coordinate and an analytical closed-loop compatibility without introducing spatial or nodal belt states.

The chain-CVT model of Duan et al. (2016) provides the closest comparison for the retained transient physics. Both formulations account for centripetal inertia, tangential belt or chain acceleration, and the additional tangential acceleration produced as the contact radius changes. Duan also couples the pulley axial and rotational dynamics through the contact solution, as is done here. One difference is that Duan neglects radial acceleration when reducing the final spatial wrap equations, whereas the present radial balance retains it.

Duan resolves considerably more physics within each pulley wrap. The chain trajectory may depart locally from the nominal pitch circle; radial and circumferential relative motion determine the sliding direction; friction varies continuously with sliding speed; and pulley deformation changes the contact geometry with angular position. Tension and normal loading are therefore obtained from coupled spatial equations. In the present model, Assumption 8 (Prescribed Planar Belt Path) fixes a circular contact path on rigid pulley geometry, Assumption 7 (Single Belt Transport Coordinate) assigns one transport speed to the belt, and Assumption 11 (Wrap-Level Contact Resolution) assigns one traction utilization λ_j to each wrap. The present formulation consequently retains the mechanism-level transient dynamics while sacrificing within-wrap variation of migration, microslip, traction reversal, and pulley deformation.

The complete belt loop is also closed differently. Duan concentrates the chain's longitudinal deformation in the tight span, obtains its absolute tension from a lumped spring–damper law, and assumes that tension is equal at the driving-pulley entrance and driven-pulley exit. The tension difference across the slack span then accelerates the mass of the complete chain. Under Assumption 6 (Fixed Belt Length), the present formulation has no longitudinal stretch law from which to obtain an absolute tension. Instead, both straight spans are treated as massive belt regions, including the momentum carried through their moving in–out-of-wrap boundaries. Combining those balances with the two wrap solutions closes the loop without prescribing a tight- or slack-side tension.

The comparison is therefore best understood as a difference in where each model places its resolution. Duan resolves the local contact, friction, chain elasticity, and pulley deformation in greater detail. The present formulation uses a coarser contact description, but carries radial acceleration, both straight-span inertias, moving-boundary momentum transport, movable-sheave dynamics, and pulley rotation through the complete belt loop. The distributed tangential action within each wrap is reduced to the single signed utilization λ_j, allowing these mechanism-level transients to be retained without introducing a spatially discretized belt state.

3.5 Tangential Contact-State Closure

The fully engaged CVT has now been followed through every retained motion and free body. Section 3.1 relates the sheave positions and pulley radii, Section 3.2 balances the two movable assemblies, Section 3.3 balances the two rotating assemblies, and Section 3.4 balances the belt around both wraps and straight spans. No additional free body or degree of freedom remains to be introduced.

Looking across those relations, only two quantities remain without relations that fix their values at the current instant: λ_p and λ_s. The reason lies in how λ_j was introduced in Equation (3.4.18). There, the separate Coulomb conditions on traction magnitude, direction, and whether the contact is sticking or sliding were collected into the single signed utilization λ_j. That compression let the remainder of the belt derivation carry one contact quantity through the wrap tension, normal loading, pulley torque, and closed-loop belt relations without reopening the piecewise contact law at every step.

Importantly, this is different from adding one more balance. The equations derived so far keep the same form as the CVT moves. Changing a belt–pulley contact between sticking and sliding, however, is a change of mechanical regime: the actual relation used to determine λ_j changes with the contact state. This is the type of regime change introduced in Section 2.5, where a change in the active contact or constraint state changes the equations governing the mechanism. What remains is therefore not another motion or free-body balance, but the choice of which tangential contact relation governs each wrap.

3.5.1 Relative Motion at the Pulley Contacts

With the remaining ambiguity reduced to the two Coulomb branches, the first useful distinction is the contact motion itself. Sticking and sliding differ in whether the belt has relative tangential motion with respect to the pulley contact represented by a wrap. Before that relative motion can be written, the surface speed represented by each reduced contact must therefore be identified.

For the primary, this comparison is direct. Both sheave faces rotate with the primary shaft, so their angular speed is ω_p. The secondary is different. Under Assumption 11 (Wrap-Level Contact Resolution), the complete wrap carries one traction state rather than separate states for the fixed and movable faces, while the helix kinematics already established that the two physical faces do not generally have the same speed during a shift. The fixed face rotates at ω_s, whereas Equation (3.2.24) gives the movable-face speed $\omega_{s,M} = \omega_s + \dot{\theta}_s$.

One representative secondary surface speed is therefore required. Its value is fixed by the same face-torque reduction already used in Assumption 12 (Symmetric Sheave-Face Sharing). In particular, Equation (3.2.21) gives $\tau_{s,M} = \tau_s/2$, and therefore $\tau_{s,F} = \tau_s/2$. The instantaneous power delivered by the belt torque to the two faces is

$$
\begin{aligned}
P_{s,\text{faces}} &= \tau_{s,F}\omega_s + \tau_{s,M}\omega_{s,M} \\
&= \frac{\tau_s}{2}\omega_s + \frac{\tau_s}{2}\left(\omega_s + \dot{\theta}_s\right) \\
&= \tau_s\left(\omega_s + \frac{1}{2}\dot{\theta}_s\right).
\end{aligned}
\tag{3.5.1}
$$

Let $\bar{\omega}_s$ be the angular speed assigned to the single aggregate secondary contact. The same total torque represented by that contact carries power $\tau_s\bar{\omega}_s$. Equating this power with Equation (3.5.1) fixes the representative speed. The primary and secondary contact speeds can therefore be written together as

$$
\boxed{\bar{\omega}_p = \omega_p, \qquad \bar{\omega}_s = \omega_s + \frac{1}{2}\dot{\theta}_s = \frac{\omega_s + \omega_{s,M}}{2}.}
\tag{3.5.2}
$$

Thus the average in Equation (3.5.2) is not an additional empirical approximation. Once one secondary traction state and the equal face-torque split have been adopted, power consistency fixes this representative speed. The arithmetic mean appears specifically because the two torque shares are equal. It remains an aggregate quantity: the model still does not determine the separate sliding states of the fixed and movable faces.

Returning now to relative motion, look at pulley j. The belt moves through the wrap with signed transport speed v_b, while the represented pulley surface moves tangentially at $r_{j,\text{eff}}\bar{\omega}_j$. Their difference defines the relative tangential speed,

$$v_{\mathrm{rel},j} = v_b - r_{j,\mathrm{eff}}\bar{\omega}_j, \qquad j \in \{p, s\}. \tag{3.5.3}$$

The sign follows the common belt-transport and rotational directions defined in Section 2.3.1. Positive $v_{\mathrm{rel},j}$ means the belt moves ahead of the represented pulley contact in the positive belt direction, while a negative value means it moves the other way.

If $v_{\mathrm{rel},j} \neq 0$, the contact is already sliding. Its relative motion fixes the direction of friction, and kinetic Coulomb friction fixes the magnitude. The sliding branch of Equation (3.4.18) therefore gives

$$\begin{aligned} \sigma_j &= -\operatorname{sgn}(v_{\mathrm{rel},j}), \\ \lambda_j &= \sigma_j \mu_k, \qquad v_{\mathrm{rel},j} \neq 0. \end{aligned} \tag{3.5.4}$$

Sliding therefore closes λ_j directly from the current contact motion. Sticking begins at the opposite kinematic condition,

$$v_{\mathrm{rel},j} = 0 \qquad \Longleftrightarrow \qquad v_b = r_{j,\mathrm{eff}}\bar{\omega}_j. \tag{3.5.5}$$

The important point is that zero relative speed does not imply zero tangential force. It only says that the belt and represented pulley contact move together at that instant. Static friction may still supply a nonzero reaction, with λ_j taking whatever signed value is required to maintain sticking so long as the static limit in Equation (3.4.18) is not exceeded.

That speed match must also persist. If the belt speed and represented pulley-surface speed begin changing at different rates, relative motion appears immediately. Differentiating Equation (3.5.2) gives

$$\begin{aligned} \dot{\bar{\omega}}_p &= \dot{\omega}_p, \\ \dot{\bar{\omega}}_s &= \dot{\omega}_s + \frac{1}{2}\ddot{\theta}_s \\ &= \dot{\omega}_s + \frac{1}{2}\left[\frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2\right], \end{aligned} \tag{3.5.6}$$

where the final line follows directly from Equation (3.2.25). Differentiating Equation (3.5.3) then gives the relative tangential acceleration,

$$\begin{aligned} a_{\mathrm{rel},j} &\equiv \dot{v}_{\mathrm{rel},j} \\ &= \dot{v}_b - r_{j,\mathrm{eff}}\dot{\bar{\omega}}_j - \dot{r}_{j,\mathrm{eff}}\bar{\omega}_j. \end{aligned} \tag{3.5.7}$$

The first two terms are the direct comparison between belt acceleration and the acceleration of a represented pulley surface at fixed radius. The final term appears because the contact radius is not fixed during a shift. As established in Equation (3.1.58), the effective radius moves with the pulley radius, so $r_{j,\mathrm{eff}}\omega_j$ can change even when ω_j is momentarily constant. A sticking belt must follow that change as well.

Established sticking therefore requires

$$a_{\text{rel},j} = 0. \tag{3.5.8}$$

The two Coulomb branches now leave different things to determine. Sliding already supplies λ_j. Sticking instead supplies the condition that the mechanics must satisfy, $a_{\text{rel},j} = 0$, while λ_j remains the static reaction required to make that condition hold within the available friction limit.

3.5.2 Determining the Sticking Traction

For a sticking contact, the previous subsection leaves a simple-looking pair: λ_j is unknown, while the contact must satisfy $a_{\text{rel},j} = 0$. At first, this seems no different from any other unknown and equation already added to the model. If λ_j entered the mechanics linearly, it could simply be solved algebraically alongside the other instantaneous unknowns.

Whether that is possible can be read directly from the equations already derived. The pulley torque in Equation (3.4.41), for example, contains

$$\tau_j = -r_{j,\text{eff}} \lambda_j N_j, \tag{3.5.9}$$

so an unknown λ_j multiplies the unknown normal resultant N_j. The equation is therefore no longer linear in the unknowns. If this product were the only complication, however, another rearrangement might still isolate λ_j. The stronger obstacle comes from the wrap relation carried into the closed-loop belt compatibility. Through Equation (3.4.51), the same λ_j also appears reciprocally and inside terms such as

$$\frac{\sin\beta}{\lambda_j}, \qquad \frac{1 + e^{-\frac{\lambda_j \phi_j}{\sin\beta}}}{1 - e^{-\frac{\lambda_j \phi_j}{\sin\beta}}}. \tag{3.5.10}$$

With the same unknown appearing both in products with other unknowns and inside reciprocal and exponential terms, there is no useful algebraic rearrangement that isolates the required static utilization.

There is one important simplification. This nonlinear dependence comes from the unknown static λ_j itself; the other remaining accelerations and contact loads appear linearly. If the static utilization were known, those quantities could be obtained directly from the balances already derived.

That suggests turning the problem around. Rather than trying to isolate λ_j explicitly, choose a candidate static utilization and determine the response it would produce. The resulting accelerations give $a_{\text{rel},j}$ through Equation (3.5.7). Since sticking requires the relative acceleration to vanish, $a_{\text{rel},j}$ can be used directly as the residual for the static-traction solve,

$$a_{\text{rel},j} = 0. \tag{3.5.11}$$

This is the same basic idea used for the geometric compatibility roots in Section 3.1. A nonzero residual means that the candidate traction would make the belt and pulley surface begin accelerating relative to one another. The required static utilization is the value for which the residual vanishes. Finding that value is therefore a root solve in the unknown static traction.

If only one wrap is sticking, the other utilization is already fixed by its sliding branch. One static λ_j and one sticking residual remain, so the root is one-dimensional. If both wraps are sticking, the two static utilizations influence the same mechanical response and must be varied together until both residuals vanish,

$$a_{\mathrm{rel},p}(\lambda_p, \lambda_s) = 0, \qquad a_{\mathrm{rel},s}(\lambda_p, \lambda_s) = 0. \tag{3.5.12}$$

This leaves only a one- or two-dimensional nonlinear search in the static traction utilizations. For each candidate value or pair, everything else is obtained from the linear balances already derived.

3.5.3 The Common Mechanical Solve

The two contact branches now reduce to the same mechanical calculation. Sliding supplies λ_j directly, while sticking tests candidate values as its root is found. In either case, evaluating the instantaneous response begins with one traction utilization for each wrap. The remaining step is to solve the mechanics for that pair.

With λ_p and λ_s specified, the equations already derived can now be read simply for the quantities they still have to determine. The axial balances expose the first of these unknowns. The primary balance in Equation (3.2.38) relates the shared shift acceleration $\ddot{s}$ to the primary normal resultant N_p. The secondary balance in Equation (3.2.39) contains the same $\ddot{s}$, the secondary normal resultant N_s, and also the secondary shaft acceleration $\dot{\omega}_s$ and belt torque τ_s.

The rotational balances add the shaft dynamics. The primary and secondary relations in Equations (3.3.5) and (3.3.13) contain $\dot{\omega}_p$ and $\dot{\omega}_s$, together with the belt torques τ_p and τ_s; the secondary relation also contains the same $\ddot{s}$ through the helix kinematics. The torques may look like two more unknowns, but the specified traction pair removes them immediately. From Equation (3.4.41),

$$\tau_j = -r_{j,\mathrm{eff}} \lambda_j N_j, \qquad j \in \{p, s\}, \tag{3.5.13}$$

so each pulley torque is set by the same normal resultant already appearing in the axial balances.

The complete-belt transport balance in Equation (3.4.1) brings in the belt acceleration $\dot{v}_b$. Its two tangential pulley actions are again the belt torques, so after the relation above is substituted the transport equation involves $\dot{v}_b$, N_p, and N_s for the specified traction pair.

Finally, the closed-loop tension compatibility in Equation (3.4.52) relates the same belt acceleration and normal resultants to the radial accelerations of the two wraps. Those radial accelerations introduce no new degree of freedom: Equations (3.1.66) and (3.1.83) already express them in terms of the current geometry, $\dot{s}$, and the same shift acceleration $\ddot{s}$. The four temporary boundary tensions have likewise already disappeared from the closed-loop relation.

No new unknown appears as these equations are brought together. The quantities left to determine are exactly

$$\mathbf{z} = \begin{bmatrix} \dot{\omega}_p & \dot{\omega}_s & \dot{v}_b & \ddot{s} & N_p & N_s \end{bmatrix}^{\mathsf{T}}. \tag{3.5.14}$$

At this point, every other term in the six equations is fixed by the current state, the specified traction pair, and the component relations already derived. Each equation is also linear in the six entries of $\mathbf{z}$. Collecting them therefore gives

$$\mathbf{A}(\lambda_p, \lambda_s)\mathbf{z} = \mathbf{b}(\lambda_p, \lambda_s). \tag{3.5.15}$$

To write $\mathbf{A}$ and $\mathbf{b}$ out cleanly, a small amount of temporary notation is useful. For this matrix only, primes shorten derivatives: x_s', r_j', q_f', and $J_{f,p}'$ denote derivatives with respect to s, while θ_s' denotes differentiation with respect to x_s; double primes follow the same convention. The only additional symbols are the two analytical wrap factors repeated in the closed-loop row,

$$\begin{aligned}
\mathcal{C}_{v,j} &\equiv \begin{cases} \dfrac{2\sin\beta}{\lambda_j} - \phi_j \dfrac{1+e^{-\frac{\lambda_j\phi_j}{\sin\beta}}}{1-e^{-\frac{\lambda_j\phi_j}{\sin\beta}}}, & \lambda_j \neq 0, \\ 0, & \lambda_j = 0, \end{cases} \\
\mathcal{C}_{N,j} &\equiv \begin{cases} \lambda_j \dfrac{1+e^{-\frac{\lambda_j\phi_j}{\sin\beta}}}{1-e^{-\frac{\lambda_j\phi_j}{\sin\beta}}}, & \lambda_j \neq 0, \\ \dfrac{2\sin\beta}{\phi_j}, & \lambda_j = 0. \end{cases}
\end{aligned} \tag{3.5.16}$$

These two symbols are temporary assembly notation only. Their $\lambda_j = 0$ branches are the finite limits already established in Equation (3.4.53).

The coefficient matrix is then

$$\mathbf{A} = \begin{bmatrix}
I_{\mathrm{B},p} + I_{p,\mathrm{const}} + J_{f,p} & 0 & 0 & 0 & r_{p,\mathrm{eff}}\lambda_p & 0 \\
0 & I_{\mathrm{B},s} + I_{s,F} + I_{s,M} & 0 & I_{s,M}\theta_s' x_s' & 0 & r_{s,\mathrm{eff}}\lambda_s \\
0 & 0 & 0 & m_p + I_f (q_f')^2 & \frac{1}{2}\cos\beta & 0 \\
0 & I_{s,M}\theta_s' & 0 & \big[m_s + I_{s,M}(\theta_s')^2\big] x_s' & 0 & \frac{1}{2}\big(\cos\beta + r_{s,\mathrm{eff}}\lambda_s\theta_s'\big) \\
0 & 0 & m_b & 0 & -\lambda_p & -\lambda_s \\
0 & 0 & \rho_b A_b\big(r_{p,\mathrm{cm}}\mathcal{C}_{v,p} - r_{s,\mathrm{cm}}\mathcal{C}_{v,s}\big) & 2\rho_b A_b\big(r_{s,\mathrm{cm}} r_s' - r_{p,\mathrm{cm}} r_p'\big) & \mathcal{C}_{N,p} & -\mathcal{C}_{N,s}
\end{bmatrix}. \tag{3.5.17}$$

The corresponding known-side vector is

$$\mathbf{b} = \begin{bmatrix}
\tau_{\mathrm{B},p} - J_{f,p}' \dot{s}\,\omega_p \\
\tau_{\mathrm{B},s} - I_{s,M}\big[\theta_s' x_s'' + \theta_s''(x_s')^2\big]\dot{s}^2 \\
\dfrac{1}{2}\omega_p^2 J_{f,p}' - I_f q_f' q_f'' \dot{s}^2 - k_p(x_{p,\mathrm{pre}} + s) \\
\theta_s' k_{s,\theta}(\theta_{s,\mathrm{pre}} - \theta_s) + k_{s,x}(x_{s,\mathrm{pre}} - x_s) - \Big\{\big[m_s + I_{s,M}(\theta_s')^2\big]x_s'' + I_{s,M}\theta_s'\theta_s''(x_s')^2\Big\}\dot{s}^2 \\
0 \\
2\rho_b A_b\big(r_{p,\mathrm{cm}} r_p'' - r_{s,\mathrm{cm}} r_s''\big)\dot{s}^2 + \rho_b A_b v_b\big(\dot{r}_s \mathcal{C}_{v,s} - \dot{r}_p \mathcal{C}_{v,p}\big)
\end{bmatrix}. \tag{3.5.18}$$

The row order follows the derivation: the first two rows are the pulley rotational balances, the next two are the axial balances, the fifth is the complete-belt transport balance, and the sixth is the closed-loop tension compatibility.

For a specified traction pair, every entry of $\mathbf{A}$ and $\mathbf{b}$ is fixed by the current state and the component relations already derived. Solving Equation (3.5.15) then gives the two shaft accelerations, belt acceleration, shift acceleration, and the two normal resultants. A sliding λ_j reaches this solve directly from kinetic friction; a sticking λ_j reaches the same solve as a candidate value whose resulting $a_{\mathrm{rel},j}$ is returned to the root described in Section 3.5.2.

The four contact combinations can now be read off directly:

	Secondary sliding	**Secondary sticking**
Primary sliding	Both λ_p and λ_s are fixed by kinetic friction. One 6×6 solve gives the instantaneous response.	λ_p is fixed by kinetic friction, while λ_s is found from the one-dimensional root $a_{\mathrm{rel},s} = 0$. Each candidate λ_s is evaluated through the same 6×6 solve.
Primary sticking	λ_s is fixed by kinetic friction, while λ_p is found from the one-dimensional root $a_{\mathrm{rel},p} = 0$. Each candidate λ_p is evaluated through the same 6×6 solve.	λ_p and λ_s are found together from $a_{\mathrm{rel},p} = 0$ and $a_{\mathrm{rel},s} = 0$. Each candidate pair is evaluated through the same 6×6 solve.

The six mechanical balances are therefore common to all four cases. What changes with the contact regime is only how the traction pair supplied to those balances is obtained: directly from kinetic friction, from one static-traction root, or from the two sticking conditions together.

3.5.4 Contact Admissibility and Regime Transitions

The four contact combinations above are enough to obtain an instantaneous response for any assumed tangential regime. They do not yet tell us whether that regime can actually be maintained, or what should happen when it cannot.

For a sticking contact, λ_j is the static reaction required to keep the belt and pulley moving together, as introduced in Equation (3.4.18). The root in Section 3.5.2 finds the value of that reaction that satisfies $a_{\mathrm{rel},j} = 0$. Sticking is physically available only while the required reaction lies within the static Coulomb limit,

$$|\lambda_j| \leq \mu_s. \tag{3.5.19}$$

If this condition is satisfied, the contact can supply the traction needed to remain stuck. If the required $|\lambda_j|$ would exceed μ_s, no admissible static reaction can preserve zero relative motion and the sticking regime must end.

Sliding is simpler away from zero relative speed. While $v_{\mathrm{rel},j} \neq 0$, its sign identifies the sliding direction and Equation (3.5.4) fixes the corresponding kinetic traction. The next ambiguity appears only when $v_{\mathrm{rel},j}$ reaches zero. At that instant the belt and pulley are momentarily moving together, so the first possibility to test is sticking. The required static λ_j is found from $a_{\mathrm{rel},j} = 0$ and checked against Equation (3.5.19). If it is admissible, the contact captures into sticking.

If sticking is not admissible, the contact must leave $v_{\mathrm{rel},j} = 0$ on a sliding branch. For a contact that arrived at zero while already sliding, two continuations are possible: the relative velocity may grow again in the same direction as before, or the sliding direction may reverse and grow in the opposite direction. At the event itself, $v_{\mathrm{rel},j} = 0$, so the relative velocity cannot distinguish between those two possibilities. The relative acceleration does: its sign tells which way $v_{\mathrm{rel},j}$ begins to move away from zero.

For a candidate kinetic sign σ_j, consistency therefore requires

$$\sigma_j \, a_{\mathrm{rel},j} < 0, \qquad v_{\mathrm{rel},j} = 0, \tag{3.5.20}$$

because σ_j is defined opposite to the direction of sliding. The valid kinetic branch is the one whose solved acceleration produces the relative-motion direction assumed by that branch. The same check selects the initial sliding direction when a sticking contact first releases.

Together, these checks also make the order of a regime update clear. Consider a sliding–sliding solution in which the primary relative velocity reaches zero. The secondary contact can initially be left on its current kinetic branch while the primary is tested as sticking. If the required static λ_p is admissible, the candidate regime becomes sticking–sliding. If it is not, the primary remains sliding and the kinetic branch consistent with Equation (3.5.20) is selected instead. The resulting traction pair changes the common mechanical response, so that response is solved again and the contact conditions at both wraps are checked once more. If both contacts reach a regime boundary at the same instant, the same logic is applied to the combined candidate regime rather than updating the two wraps independently.

These tangential contact changes are nonimpulsive regime transitions in the sense introduced in Section 2.5. The belt and pulley reach the event with continuous position and velocity; changing the active contact law changes the acceleration that follows rather than resetting the state instantaneously. The remaining structural regime changes associated with engagement and shift boundaries require a separate treatment.

3.6 3.6

The freely shifting engaged CVT is now mechanically closed. For any admissible tangential contact regime, Section 3.5 determines how the pulley speeds, belt transport, and shift coordinate evolve from the current state. But that solution does more than describe the mechanism at one instant: as it is integrated forward in time, it moves the hardware.

Consider what that motion can do. Continued backshift closes the secondary until it can move no farther; continued primary opening can then separate the primary sheave from the belt. In the opposite direction, continued upshift eventually carries the primary to the end of its closing travel. The equations developed so far can bring the CVT to each of these configurations, but once one is reached, the physical problem has changed. A sheave that was free to move may now be supported by a stop, or a belt–sheave interaction that previously transmitted force may no longer exist.

Reaching one of these boundaries therefore changes the shift regime of the CVT, in the same sense that a change between sticking and sliding changes a tangential contact regime in Section 3.5. That structural change has two separate consequences. First, the active contacts, supports, and permitted motions are different, so the continuous governing equations must change with the regime. Second, the state arriving at the boundary may contain a velocity that the new regime cannot support, so the incoming motion must be transferred into a compatible outgoing state.

These are different mechanical problems. The first determines how the CVT accelerates once the new regime is active. The second determines how its velocity changes at the instant the regime is entered. Both are required if the model is to pass through a shift-regime boundary without replacing the mechanics by an imposed reset.

3.6.1 Shift Domain and Structural Regimes

Before the governing equations can change with the shift regime, the boundaries between those regimes must first be located along the shift motion. The shift coordinate already traces the primary through its full travel, so the geometry derived in Section 3.1 identifies where contact is gained or lost, where hardware begins to support the motion, and which arrangement exists on either side of each boundary. Following the primary from fully open to fully closed makes those changes explicit.

Consider first the primary closing during engaged upshift. Its groove narrows, moving the belt outward. The fixed belt length requires the secondary radius to decrease, so the secondary opens. This motion can continue until the primary reaches its closing stop at $x_p = x_{p,\mathrm{max}}$. Further upshift would require the primary to close farther, into the stop. The belt remains between both pairs of sheave faces, so the mechanism can be held at this ratio with both pulley contacts present.

Opening the primary reverses these motions. The belt moves inward on the primary and outward on the secondary, bringing the secondary toward its fully closed stop. The secondary reaches that stop when the primary is at $x_p = x_{p,\mathrm{dz}}$, as given by Equation (3.1.32). Now try to continue the same engaged backshift: the belt would have to move farther outward on the secondary, which would require its sheave to close beyond the stop. With shift motion at rest and primary belt contact maintained, the secondary stop can therefore hold the mechanism at its minimum ratio. This is the *engaged low-ratio seat.*

Yet the primary has not reached its own opening stop. Its groove can open wider than the belt, leaving the clearance introduced in Equation (3.1.16). The primary can therefore open farther if it separates from the belt. That opening no longer moves the belt inward or requires the secondary to close farther. Under the deadzone mapping, the secondary stays fully closed and both prescribed belt radii remain fixed while the primary crosses its clearance. Only when the primary reaches $x_p = 0$ does its own opening stop prevent further travel.

The secondary stop thus limits the engaged backshift before the primary exhausts its opening travel. Closing the primary through the same clearance first brings it back to the belt; further closure then moves the belt outward and opens the secondary away from its stop.

Since $s \equiv x_p$ by Equation (3.1.31), the same travel and boundary positions in the generalized shift coordinate are

$$0 \le s \le s_{\mathrm{max}}, \qquad s_{\mathrm{dz}} \equiv x_{p,\mathrm{dz}}, \qquad s_{\mathrm{max}} \equiv x_{p,\mathrm{max}}. \tag{3.6.1}$$

These motions divide the full primary travel into the five structural arrangements summarized in Table 3.1.

Structural regime	Primary belt contact	Axial motion or support
Fully open primary stop	Absent	Primary held at $s = 0$; secondary remains fully closed.
Free deadzone	Absent	Primary free for $0 < s < s_{\mathrm{dz}}$; secondary remains fully closed.
Engaged low-ratio seat	Present	Fully closed secondary stop holds the engaged mechanism at $s = s_{\mathrm{dz}}$.
Free engaged shift	Present	Free shift for $s_{\mathrm{dz}} < s < s_{\max}$; both sheaves follow the belt geometry.
Engaged upper primary stop	Present	Primary closing stop holds the engaged mechanism at $s = s_{\max}$.

Table 3.1: Structural arrangements over the primary travel. The secondary retains belt contact in all five arrangements.

A boundary position does not, by itself, determine which shift regime is active. At $s = s_{\mathrm{dz}}$, the secondary may be held against its fully closed stop while the primary remains in contact with the belt. The primary may instead begin closing, opening the secondary away from that stop, or it may begin opening and separate from the belt into the clearance. All three continuations pass through the same value of s. What changes is which contacts and supports are active and which motions they permit. The shift regime must therefore be carried as part of the mechanical description; it cannot be inferred from the shift coordinate alone. This is the same regime-dependent logic introduced in Section 2.5, now applied to the changing mechanical arrangement of the CVT.

The same boundary also shows that shift regime and tangential contact state describe different parts of the mechanics. While the secondary stop holds the engaged mechanism at $s = s_{\mathrm{dz}}$, the two shafts and the belt can continue moving. The stop determines whether further axial motion is available, but it does not determine whether the belt sticks to either pulley or slides relative to it. Each engaged shift regime must therefore still be combined with the corresponding tangential contact regimes from Section 3.5.

The three boundary positions and the five arrangements in Table 3.1 therefore define where structural regime changes may occur and which arrangements are physically admissible there. They do not assign a unique regime to each value of s; at a shared boundary, the active contacts, supports, and direction of continuation still matter. The active shift regime and tangential contact regimes together specify the mechanical regime that the model must track as the CVT evolves.

3.6.2 Governing Equations in Each Shift Regime

Knowing which shift regime is active tells us which contacts and supports exist, but it does not yet determine how the CVT moves within that regime. The continuous equations must reflect those physical conditions. The useful starting point is the mechanical system already assembled for free engaged motion in Equation (3.5.15). Its six rows are the primary and secondary rotational balances, the primary and secondary axial balances, the complete-belt transport balance, and the closed-loop tension compatibility. A different shift regime does not require these mechanics to be derived again. It changes which of those balances remain active, which contact terms disappear, and which motions are prescribed by the hardware.

Disengaged Primary and Deadzone Motion

At the fully open end of travel, the primary sheaves do not touch the belt, and the same remains true while the primary crosses the clearance toward $s = s_{\mathrm{dz}}$. The immediate consequence is

$$N_p = 0, \qquad \tau_p = 0. \tag{3.6.2}$$

Putting this contact loss back into the assembled mechanics shows exactly what changes on the primary side. The rotational balance in Equation (3.3.5) and the axial balance in Equation (3.2.38) still describe the same primary hardware, but their belt terms vanish. The primary rotational row therefore becomes

$$\tau_{\mathrm{B},p} = \left[I_{\mathrm{B},p} + I_{p,\mathrm{const}} + J_{f,p}(s)\right]\dot{\omega}_p + \frac{dJ_{f,p}}{ds}\,\dot{s}\,\omega_p, \tag{3.6.3}$$

while the primary axial row becomes

$$\begin{aligned}\left[m_p + I_f\left(\frac{dq_f}{ds}\right)^2\right]\ddot{s} = {} & \frac{1}{2}\omega_p^2\frac{dJ_{f,p}}{ds} \\ & - I_f\frac{dq_f}{ds}\frac{d^2q_f}{ds^2}\dot{s}^2 - k_p\left(x_{p,\mathrm{pre}} + s\right).\end{aligned} \tag{3.6.4}$$

The belt side changes more substantially. During engaged shift, the secondary axial balance in Equation (3.2.39) participates in the shared shift motion, while Equation (3.4.52) closes the taut belt path through both pulley contacts. Neither role remains in the deadzone. The secondary stays fully closed while s moves only the primary, so its axial balance no longer supplies an equation for $\ddot{s}$. Primary separation also breaks the two-contact tension loop, so the engaged closed-loop compatibility cannot be retained.

This is where the reduced deadzone belt model is needed. The fully closed secondary cannot close farther to take up the remaining belt slack and establish the taut path used by the engaged model. The belt model also has no longitudinal strain or pretension state from which a residual tension could be recovered while that slack remains. Although the belt may still rest on the minimum-radius primary support, that contact is not represented as a clamped belt contact. Resolving the loads produced by a slack or pretensioned belt as it is taken up would therefore require additional belt elasticity and slack-span geometry beyond the present reduced description.

Under Assumption 9 (Reduced Belt Motion While the Primary Is Disengaged), the disengaged belt is instead kept seated on the fully closed secondary and carried with its represented contact surface. Because the secondary has no axial or helix-relative motion, $\dot{\theta}_s = 0$, and Equation (3.5.2) reduces to $\bar{\omega}_s = \omega_s$. Hence,

$$v_b = r_{s,\mathrm{eff,dz}}\bar{\omega}_s = r_{s,\mathrm{eff,dz}}\omega_s, \qquad r_{s,\mathrm{eff,dz}} \equiv r_{s,\mathrm{eff}}(s_{\mathrm{dz}}). \tag{3.6.5}$$

The radius is fixed, so

$$\dot{v}_b = r_{s,\mathrm{eff,dz}}\dot{\omega}_s. \tag{3.6.6}$$

The secondary rotational balance remains active, but at fixed secondary geometry it takes the form given by Equation (3.3.14). The complete-belt transport balance in Equation (3.4.1) also remains active. The deadzone lock ties these two motions together, so their internal secondary contact torque can be eliminated by combining the balances. This gives

$$\tau_{\mathrm{B},s} = \left[I_{\mathrm{B},s} + I_{s,F} + I_{s,M} + m_b r_{s,\mathrm{eff,dz}}^2\right] \dot{\omega}_s. \tag{3.6.7}$$

The final term is the complete belt transport inertia reflected through the fixed secondary radius. Primary disengagement removes the primary belt load without removing the belt mass that the secondary must still accelerate.

The active free-deadzone system is now visible. Its unknown accelerations are

$$\mathbf{z}_{\mathrm{dz}} = \begin{bmatrix}\dot{\omega}_p & \dot{\omega}_s & \dot{v}_b & \ddot{s}\end{bmatrix}^\mathsf{T}, \tag{3.6.8}$$

and the four active relations are Equations (3.6.3), (3.6.4), (3.6.6) and (3.6.7). They may therefore be assembled, just as in the engaged case, as

$$\mathbf{A}_{\mathrm{dz}}\mathbf{z}_{\mathrm{dz}} = \mathbf{b}_{\mathrm{dz}}. \tag{3.6.9}$$

For $0 < s < s_{\mathrm{dz}}$, this reduced system determines $\dot{\omega}_p$, $\dot{\omega}_s$, $\dot{v}_b$, and $\ddot{s}$ while the primary moves freely through the clearance.

Fully open primary stop. The boundary at $s = 0$ introduces a different kind of regime change. Primary belt contact is already absent there, so no additional interaction disappears. Instead, the hardware prevents the primary from moving beyond the end of its travel. A sustained stop condition requires

$$s = 0, \qquad \dot{s} = 0, \qquad \ddot{s} = 0. \tag{3.6.10}$$

Inside the free deadzone, the primary axial balance in Equation (3.6.4) determines $\ddot{s}$. At the stop, that acceleration is no longer unknown because the hardware prescribes it to be zero. The force balance must still hold, so a new quantity appears instead: the reaction exerted by the stop on the primary.

Let $F_{\mathrm{stop},0} \geq 0$ denote the magnitude of this reaction in the $+s$ direction, the only direction in which the stop can push the primary. The primary axial balance becomes

$$0 = F_{\mathrm{fw},x_p} - F_{\mathrm{spring},x_p} + F_{\mathrm{stop},0}, \tag{3.6.11}$$

and therefore

$$F_{\mathrm{stop},0} = F_{\mathrm{spring},x_p} - F_{\mathrm{fw},x_p}. \tag{3.6.12}$$

The axial balance has not been removed; its role has changed. During free deadzone motion it determines the primary acceleration. At the stop the acceleration is known, so the same balance determines the support force. Equivalently, the free-deadzone system replaces its primary axial row by $\ddot{s} = 0$, and the original axial balance is evaluated afterward to recover $F_{\text{stop},0}$.

The fully open stop can sustain the shift regime only while

$$F_{\text{stop},0} \geq 0. \tag{3.6.13}$$

A negative value would require the stop to pull the primary in the $-s$ direction, which a unilateral contact cannot do. Once the required reaction reaches zero and the unconstrained primary tendency points into the deadzone, the stop no longer supports the motion.

Engagement and the Low-Ratio Seat

Closing through the deadzone leaves the belt radii unchanged until the primary reaches $s = s_{\text{dz}}$. At that point the movable primary sheave reaches the belt, so the primary normal load and contact torque can reappear and the engaged belt mechanics become active again. The secondary is still at its fully closed position, so the first engaged shift regime is the low-ratio seat rather than free shift.

For a sustained low-ratio seat,

$$s = s_{\text{dz}}, \qquad \dot{s} = 0, \qquad \ddot{s} = 0. \tag{3.6.14}$$

The fixed shift also gives $\dot{\theta}_s = 0$, so Equation (3.5.2) again reduces to $\bar{\omega}_s = \omega_s$. The shafts and belt may nevertheless continue to rotate and accelerate, and the two pulley contacts retain the tangential contact regimes developed in Section 3.5.

The stop mechanics are the same type just introduced at $s = 0$, but the supported member is now the secondary. The secondary axial balance in Equation (3.2.39) would determine the shared shift acceleration during free engagement. At the low-ratio seat, $\ddot{s} = 0$ is prescribed instead, and the fully closed secondary stop supplies the reaction needed to satisfy the original secondary free body in Figure 3.8.

Let $F_{\text{stop,LR}} \geq 0$ denote the magnitude of that reaction in the secondary opening direction. Using Equations (3.2.30) and (3.2.32), the held secondary balance is

$$0 = F_{\text{helix},x_s} + F_{\text{spring},x_s} - F_{\text{belt},x_s} - F_{\text{stop,LR}}, \tag{3.6.15}$$

so

$$F_{\text{stop,LR}} = F_{\text{helix},x_s} + F_{\text{spring},x_s} - F_{\text{belt},x_s}. \tag{3.6.16}$$

As before, the stopped axial balance changes role rather than disappearing. The only additional complication is that the reaction cannot be evaluated before the coupled engaged response is known. The belt term contains the unknown N_s, while the helix term contains the transmitted secondary torque and $\dot{\omega}_s$.

For a trial traction pair (λ_p, λ_s), the mechanical core in Equation (3.5.15) normally determines $\ddot{s}$ together with $\dot{\omega}_p$, $\dot{\omega}_s$, $\dot{v}_b$, N_p, and N_s. At the seat, the secondary axial row is replaced by $\ddot{s} = 0$. The primary axial row in Equation (3.2.38) remains active because the primary itself is not supported by a stop. The rotational balances in Equations (3.3.5) and (3.3.13), the complete-belt transport balance in Equation (3.4.1), and the closed-loop tension compatibility in Equation (3.4.52) retain their engaged forms. Together these active rows determine the fixed-geometry response for the chosen tangential contact regimes.

The original secondary axial balance is then evaluated with the solved quantities to recover $F_{\mathrm{stop,LR}}$ from Equation (3.6.16). The low-ratio seat is supportable only while

$$F_{\mathrm{stop,LR}} \geq 0. \tag{3.6.17}$$

The stop reaction is a hardware load path, not an additional belt normal force. Any spring or helix loading carried directly by the stop does not add to N_s and therefore does not increase the secondary tangential traction capacity.

Once the secondary moves away from its closing stop while primary belt contact is retained, its axial row becomes active again. All six rows of Equation (3.5.15) are then the free engaged system already developed in Section 3.5. No additional continuous equation is required in the interior shift regime.

Upper Primary Stop

Free upshift continues until the primary reaches $s = s_{\mathrm{max}}$. Belt contact remains present, but the primary can no longer move farther in the $+s$ direction. This is the same unilateral support mechanism as the previous two stops, now applied to the primary axial row of the engaged system. A sustained upper-stop state satisfies

$$s = s_{\mathrm{max}}, \qquad \dot{s} = 0, \qquad \ddot{s} = 0. \tag{3.6.18}$$

Let $F_{\mathrm{stop,max}} \geq 0$ denote the magnitude of the primary stop reaction in the $-s$ direction. Using the primary axial balance in Equation (3.2.38),

$$0 = F_{\mathrm{fw},x_p} - F_{\mathrm{spring},x_p} - F_{\mathrm{belt},x_p} - F_{\mathrm{stop,max}}, \tag{3.6.19}$$

so

$$F_{\mathrm{stop,max}} = F_{\mathrm{fw},x_p} - F_{\mathrm{spring},x_p} - F_{\mathrm{belt},x_p}. \tag{3.6.20}$$

The constrained solve follows the same pattern as the low-ratio seat. The primary axial row is replaced by $\ddot{s} = 0$, while the secondary axial, rotational, belt-transport, and closed-loop compatibility rows remain active. The solved engaged response is then substituted into Equation (3.6.20) to recover $F_{\mathrm{stop,max}}$. The upper stop can sustain the shift regime only while

$$F_{\mathrm{stop,max}} \geq 0. \tag{3.6.21}$$

As at low ratio, the recovered reaction is carried by the hardware rather than added to the belt normal load.

The deadzone and stop-supported regimes therefore modify the assembled mechanics in two distinct ways. Primary separation removes the primary belt interaction and replaces the engaged belt closure with the reduced secondary-carried belt dynamics of Equation (3.6.9). A travel stop instead preserves the remaining contacts while prescribing the stopped axial motion.

The three stop-supported shift regimes then follow the same rule: impose $\ddot{s} = 0$ in place of the axial row belonging to the supported member, solve the remaining active mechanics, and evaluate the original axial balance to recover the unilateral stop reaction. The resulting cases are summarized in Table 3.2.

Supported shift regime	Stopped member	Reaction direction	Balance used to recover the reaction
Fully open primary stop	Primary	$+s$	Primary axial balance
Engaged low-ratio seat	Secondary	Opening secondary	Secondary axial balance
Engaged upper primary stop	Primary	$-s$	Primary axial balance

Table 3.2: Hardware reactions for the three stop-supported shift regimes. In each case, the stopped member's original axial balance is evaluated after the corresponding constrained continuous response has been solved.

Each stop is a unilateral contact, so all three supported shift regimes require

$$F_{\text{stop},0} \geq 0, \qquad F_{\text{stop,LR}} \geq 0, \qquad F_{\text{stop,max}} \geq 0. \tag{3.6.22}$$

The continuous mechanics are now defined for the fully open stop, free deadzone, low-ratio seat, free engaged shift, and upper primary stop. Within each shift regime, the active balances determine the instantaneous accelerations and any support reaction required by the hardware. They do not, by themselves, determine an instantaneous change in velocity when a new kinematic constraint is reached.

3.6.3 Shift-Regime Transitions and Momentum Transfer

The continuous equations determine how the CVT evolves while a shift regime remains active. A stop-supported regime immediately exposes the remaining problem. At the fully open stop, low-ratio seat, and upper primary stop, $\ddot{s} = 0$ is imposed by the active constraint. Once that condition has replaced the free axial equation, the constrained system cannot itself produce an acceleration away from the boundary. It needs another quantity to tell it when the support is no longer physically required.

That quantity is the stop reaction already recovered in Section 3.6.2. A positive reaction means the stop is carrying load and must remain active. As the surrounding loading changes, the required reaction can fall to zero. Continuing the constrained solution past that point may then require a negative reaction, which would make the stop pull rather than push. The common unilateral condition is therefore

$$F_{\text{stop}} \geq 0, \qquad F_{\text{stop}} = 0 \text{ at the release surface.} \tag{3.6.23}$$

The stop is released when this surface is crossed from the admissible side toward a negative required reaction. The constraint is then removed and the corresponding free continuous system is evaluated at the same state. The free deadzone equations resume after release from $s = 0$, while the free engaged equations resume after release from the low-ratio or upper stop. The newly active free balance again determines $\ddot{s}$.

No velocity reset is needed for this departure. A sustained stop already has $\dot{s} = 0$, so its velocity is compatible with the free regime that replaces it. Position and velocity therefore pass through unchanged, as in the nonimpulsive regime transition of Equation (2.5.2); only the active acceleration law changes. If a tangential contact-regime change at the same instant changes a recovered stop reaction from admissible to negative, the same rule applies immediately and no constrained segment is started with a stop that would have to pull.

The low-ratio seat has a second possible release because two unilateral interactions coexist there. The secondary stop can release while primary belt contact remains, sending the CVT into free engaged shift. The primary can also separate from the belt while the secondary remains at its fully closed position, sending the CVT into the deadzone. A vanishing primary normal resultant is necessary for this separation, but is not sufficient by itself. If the belt load were removed and the free primary immediately accelerated closed again, the surfaces would simply re-establish contact. Let $\ddot{s}_{\mathrm{cf}}$ denote the acceleration obtained from the contact-free primary balance in Equation (3.6.4), evaluated at the seated state. Primary separation therefore requires

$$N_p = 0, \qquad \ddot{s}_{\mathrm{cf}} < 0. \tag{3.6.24}$$

The first condition removes the compressive support; the second confirms that the released primary actually tends to move into the deadzone. These conditions determine when an established shift regime can end without an instantaneous change in velocity. Arrival at a new boundary can be different.

Finite-Speed Arrival at a Boundary

Consider free engaged upshift reaching the primary closing stop at $s = s_{\max}$ with $\dot{s}^- > 0$. Immediately before contact, the primary and secondary hardware and the belt all carry motion associated with the shift. A sustained upper-stop regime, however, requires

$$\dot{s}^+ = 0. \tag{3.6.25}$$

The upper-stop equations cannot determine how this finite velocity change occurs. They begin with the stop already active, and their imposed $\ddot{s} = 0$ only describes the acceleration after the incoming motion has been arrested.

The same problem occurs when free deadzone motion reaches the fully open primary stop. At $s = s_{\mathrm{dz}}$, it is even more revealing because the kinematic mapping itself changes. In the deadzone, s moves only the primary while the secondary axial position, secondary helix-relative angle, and both represented belt radii remain fixed. In engaged shift, the same $\dot{s}$ moves both sheaves, changes both belt radii, and rotates the secondary movable sheave through its helix. Simply carrying $\dot{s}$ across that boundary would therefore assign new velocities to physical hardware that was not moving that way immediately before the event.

The coordinate s is not a material body. It describes several linked motions, but it has no momentum of its own. Momentum belongs to the actual masses and inertias whose velocities are represented through ω_p, ω_s, v_b, and $\dot{s}$. When a new stop or a new shift-regime kinematic mapping makes some incoming motions incompatible with the outgoing regime, those physical momenta must be transferred together.

Under Assumption 13 (Instantaneous, Perfectly Inelastic Constraint Activation), the unresolved finite-duration interaction is represented as an instantaneous, perfectly inelastic event. Configuration remains continuous through the event, while the retained velocities are allowed to change according to impulse and momentum balance subject to the kinematic conditions of the outgoing regime.

Physical Velocities at the Event

The momentum calculation must contain the same inertial motions retained by the continuous model. Begin by collecting the four velocity entries already carried by the state,

$$\boldsymbol{\nu} = \begin{bmatrix} \omega_p & \omega_s & v_b & \dot{s} \end{bmatrix}^{\mathsf{T}}. \tag{3.6.26}$$

This vector introduces no new degrees of freedom. It is only a compact way to reconstruct the velocity of every retained physical component at the event configuration.

On the primary, the shaft-side rotation and the configuration-dependent flyweight orbital motion are carried by ω_p. The movable primary assembly translates at $\dot{s}$, while the flyweights also pivot at the rate already derived in Equation (3.2.8),

$$\dot{q}_f = \frac{dq_f}{ds}\dot{s}. \tag{3.6.27}$$

The secondary contains an additional coupling. Its shaft-side hardware rotates at ω_s, while the movable assembly has the absolute angular speed from Equation (3.2.24),

$$\omega_{s,M} = \omega_s + \frac{d\theta_s}{ds}\dot{s}. \tag{3.6.28}$$

Its axial speed is $\dot{x}_s = (dx_s/ds)\dot{s}$. Arresting shift motion can therefore change more than an axial velocity. Removing the helix-relative term changes the angular momentum carried by the secondary movable assembly and can change ω_s during the event. That angular momentum is not assumed to remain within the secondary alone because belt constraint impulses may act at the same instant.

The belt also carries shift-associated motion that must not disappear from the event model. For an infinitesimal element on pulley wrap j, Equation (3.4.6) gave

$$\mathbf{v}_{b,j} = \dot{r}_j\hat{\mathbf{e}}_r + v_b\hat{\mathbf{e}}_\theta. \tag{3.6.29}$$

The transport component v_b is already carried by the complete belt mass. During engaged shift, however, the belt material on each wrap also moves radially as its running radius changes. Integrating the element mass from Equation (3.4.5) around the wrap gives

$$m_{\text{wrap},j} = \rho_b A_b r_{j,\text{cm}}\phi_j, \qquad j \in \{p, s\}, \tag{3.6.30}$$

and the corresponding radial kinetic energy is

$$\mathcal{T}_{b,r} = \frac{1}{2} \sum_{j \in \{p,s\}} m_{\text{wrap},j} \left(\frac{dr_j}{ds} \dot{s} \right)^2 . \tag{3.6.31}$$

This does not duplicate the complete-belt transport energy $m_b v_b^2/2$. The transport and radial velocities are orthogonal components of Equation (3.6.29). The full belt participates in transport, while only the material currently carried on the pulley wraps participates in this retained radial motion. In the deadzone, Equations (3.1.36) and (3.1.57) gives $dr_p/ds = dr_s/ds = 0$, so the radial wrap velocities vanish even though the belt mass itself remains.

The physical velocities and their associated inertial weights are collected in Table 3.3.

Retained motion	Physical velocity	Mass or inertia
Primary shaft rotation and flyweight orbital motion	ω_p	$I_{\mathrm{B},p} + I_{p,\text{const}} + J_{f,p}(s)$
Primary movable axial motion	$\dot{s}$	m_p
Flyweight pivot motion	$(dq_f/ds)\dot{s}$	I_f
Secondary shaft-side rotation	ω_s	$I_{\mathrm{B},s} + I_{s,F}$
Secondary movable-side rotation	$\omega_s + (d\theta_s/ds)\dot{s}$	$I_{s,M}$
Secondary movable axial motion	$(dx_s/ds)\dot{s}$	m_s
Complete-belt transport	v_b	m_b
Primary-wrap radial motion	$(dr_p/ds)\dot{s}$	$m_{\text{wrap},p}$
Secondary-wrap radial motion	$(dr_s/ds)\dot{s}$	$m_{\text{wrap},s}$

Table 3.3: Physical velocity modes retained during a shift-regime transition. The kinematic derivatives are evaluated using the active one-sided mapping at the event configuration.

Every row in the table comes from a motion already retained in the continuous derivation. The flyweight orbital and pivot contributions are the two kinetic terms identified in Equation (D.4.5); the secondary movable-side row is the same helix-induced rotation retained throughout the secondary mechanics. Any equivalent inertia supplied through a shaft boundary is included with the corresponding shaft-side rotational inertia. The event model therefore introduces no additional stored inertia. It only collects the existing physical motions before asking how they can continue across a change of kinematics.

Impulse-Momentum Projection

For mechanical regime q, collect the physical velocities in Table 3.3 into $\mathbf{y}_q$. At a fixed configuration each is linear in the entries of $\boldsymbol{\nu}$, so

$$\mathbf{y}_q = \mathbf{J}_q \boldsymbol{\nu}. \tag{3.6.32}$$

The rows of $\mathbf{J}_q$ are the velocity coefficients already displayed in the table. The matrix changes when the shift regime changes because the kinematic derivatives change. In the deadzone, $dx_s/ds =$

$d\theta_s/ds = dr_p/ds = dr_s/ds = 0$; in engaged shift those same derivatives describe the secondary axial motion, helix rotation, and radial motion of the belt wraps. The material components remain the same even though their permitted velocities change.

Let $\mathbf{W}$ be the diagonal matrix of the corresponding masses and inertias. The kinetic energy represented by mechanical regime q is then

$$\mathcal{T}_q = \frac{1}{2}\boldsymbol{\nu}^\top \mathbf{J}_q^\top \mathbf{W} \mathbf{J}_q \boldsymbol{\nu}. \tag{3.6.33}$$

The remaining information comes from the velocity constraints that must hold immediately after the event. Let

$$\mathbf{C}_+ \boldsymbol{\nu}^+ = \mathbf{0} \tag{3.6.34}$$

collect those conditions. A mechanical stop contributes $\dot{s}^+ = 0$. Under Assumption 14 (Tangential Contact Through Boundary Events), a pulley contact that was already sticking before the event retains its no-slip condition through the event, whereas a sliding contact or a newly created contact is not impulsively forced to stick.

The no-slip rows must use the same representative contact velocities as the continuous model. From Equations (3.5.2) and (3.5.5), the useful constraint rows for the ordering in Equation (3.6.26) are

$$\begin{aligned}
\mathbf{c}_{\text{stop}} &= \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}, \\
\mathbf{c}_{p,\text{stick}} &= \begin{bmatrix} -r_{p,\text{eff}} & 0 & 1 & 0 \end{bmatrix}, \\
\mathbf{c}_{s,\text{stick}} &= \begin{bmatrix} 0 & -r_{s,\text{eff}} & 1 & -\dfrac{r_{s,\text{eff}}}{2}\dfrac{d\theta_s}{ds} \end{bmatrix}, \\
\mathbf{c}_{s,\text{dz}} &= \begin{bmatrix} 0 & -r_{s,\text{eff,dz}} & 1 & 0 \end{bmatrix}.
\end{aligned} \tag{3.6.35}$$

The $\dot{s}$ coefficient in $\mathbf{c}_{s,\text{stick}}$ follows directly from $\bar{\omega}_s = \omega_s + \dot{\theta}_s/2$. It vanishes when the shift is fixed, but must be retained during a finite-speed engaged event so that the impulsive no-slip condition uses the same power-consistent aggregate contact speed as the continuous equations. The final row is different in character. It is the reduced secondary-belt lock of Equation (3.6.5), not an ordinary tangential contact regime, and is active whenever the outgoing shift regime is deadzone.

Over the unresolved event duration, the bounded spring, actuator, and shaft loads produce no finite impulse. The constraint interactions can. Momentum balance over the retained physical modes gives

$$\mathbf{J}_+^\top \mathbf{W} \left(\mathbf{J}_+ \boldsymbol{\nu}^+ - \mathbf{J}_- \boldsymbol{\nu}^- \right) + \mathbf{C}_+^\top \boldsymbol{\Gamma} = \mathbf{0}, \tag{3.6.36}$$

where $\boldsymbol{\Gamma}$ contains the impulses associated with the outgoing constraints. Combining this balance with Equation (3.6.34) gives

$$\begin{bmatrix} \mathbf{J}_+^\top \mathbf{W} \mathbf{J}_+ & \mathbf{C}_+^\top \\ \mathbf{C}_+ & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\nu}^+ \\ \boldsymbol{\Gamma} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_+^\top \mathbf{W} \mathbf{J}_- \boldsymbol{\nu}^- \\ \mathbf{0} \end{bmatrix}. \tag{3.6.37}$$

The configuration is continuous through the instantaneous event, so $s^+ = s^-$. The velocity entries are not reset independently. They are solved together so that the outgoing physical motions satisfy the new kinematics while accounting for the momentum carried into the event. This is why an axial stop can change ω_s, and why the radial motion of the belt wraps participates when $\dot{s}$ is removed.

Kinetic energy is not imposed as another conservation law. Under the perfectly inelastic idealization,

$$\mathcal{T}^+ \leq \mathcal{T}^-. \tag{3.6.38}$$

The difference is the kinetic energy removed by the unresolved impact or kinematic capture. The present model does not divide that loss among local deformation, material damping, sound, heat, or structural vibration. Energy therefore checks the physical consistency of the projection, while Equations (3.6.34) and (3.6.36) determine the velocity transfer.

Shift-Regime Event Rules

The three exact geometric event surfaces are

$$s = 0, \qquad s = s_{\mathrm{dz}}, \qquad s = s_{\max}. \tag{3.6.39}$$

Their meaning depends on the active shift regime and the direction in which the surface is reached.

Free deadzone opening reaches $s = 0$ with $\dot{s}^- < 0$. The outgoing state remains in the deadzone kinematic mapping but acquires the stop condition $\dot{s}^+ = 0$, while the secondary-belt lock remains active. The projection in Equation (3.6.37) determines the compatible post-impact velocities. The lower-stop closure then recovers $F_{\mathrm{stop},0}$ from Equation (3.6.12). A nonnegative reaction admits the supported regime. If the projected state would already require a negative reaction, the collision has still occurred, but the stop cannot support the following continuous motion and releases immediately into free deadzone motion.

Closing through the deadzone reaches $s = s_{\mathrm{dz}}$ with $\dot{s}^- > 0$. The primary sheave reaches the belt and the engaged kinematic mapping becomes active. Before contact, the secondary axial and helix-relative velocities and both radial wrap velocities are zero; after contact, those motions are tied to the outgoing $\dot{s}^+$. Copying the incoming shift speed would therefore create those physical velocities without accounting for the momentum required to accelerate them. The event projection instead distributes the incoming momentum over the newly coupled motions.

The pre-existing deadzone secondary-belt lock is retained through this capture, expressed on the engaged side using the representative secondary contact speed. No primary tangential stick row is added simply because normal contact has appeared. After the projection, the engaged closure evaluates the new primary relative speed and selects the admissible tangential contact regime in the ordinary way.

Engaged backshift reaches the same geometric position with $\dot{s}^- < 0$, but the physical event is different. The secondary movable sheave now arrives at its fully closed stop with axial and helix-relative motion, and the two belt wraps carry radial shift motion. The secondary stop can arrest those motions. The primary, however, has not reached an opening stop. If it retains finite opening motion, continuing that motion requires primary separation and therefore the deadzone mapping rather than a blanket reset of the shared coordinate to $\dot{s}^+ = 0$.

For a finite-speed return with a real deadzone, the event projection therefore maps the engaged motion into the deadzone while retaining whatever primary opening motion is compatible with the outgoing physical momentum. The secondary axial and helix-relative velocities and both radial wrap velocities become zero through the same momentum-consistent transfer. Entering the deadzone also activates $\mathbf{c}_{s,\mathrm{dz}}$. This is the one intentional exception to the ordinary tangential impact rule: even if the secondary contact was sliding immediately before separation, the outgoing deadzone model is defined by its reduced secondary-belt lock, so that lock is imposed during the capture.

If repeated rigid captures reduce the remaining shift motion to the zero-velocity limit while primary contact and the secondary stop remain admissible, the sustained low-ratio seat becomes the continuous successor.

Remark (Rigid low-ratio capture)

With perfectly rigid geometry and perfectly inelastic events, return to the low-ratio boundary can produce a shrinking sequence of secondary-stop impacts and primary make–break events as the removable shift kinetic energy approaches zero. This sequence does not introduce another physical shift regime. A numerical implementation may complete the limiting capture once the additional kinetic energy removed by seating is indistinguishable from zero at its working precision. That threshold is a numerical completion criterion, not a new mechanical parameter.

Once the low-ratio seat is established, its departures use the nonimpulsive conditions derived at the beginning of this subsection. A zero crossing of $F_{\mathrm{stop,LR}}$ toward a negative required reaction releases the secondary stop into free engaged shift. Primary separation instead requires Equation (3.6.24); if those conditions are met, the deadzone mapping and its secondary-belt lock become active.

Free engaged upshift reaches $s = s_{\max}$ with $\dot{s}^- > 0$. The shift regime remains engaged, but the upper stop adds $\mathbf{c}_{\mathrm{stop}}$, so $\dot{s}^+ = 0$. Any pulley contact that was already sticking retains its corresponding tangential constraint under Assumption 14 (Tangential Contact Through Boundary Events); a sliding contact does not acquire one. The full projection matters because eliminating $\dot{s}$ also removes secondary helix-relative rotation and both belt-wrap radial velocities. Their incoming momentum is redistributed among the remaining compatible motions rather than discarded by setting $\dot{s}$ to zero alone.

The upper-stop closure is then evaluated at the projected state and $F_{\mathrm{stop,max}}$ is recovered from Equation (3.6.20). A nonnegative reaction admits the supported regime. A negative required reaction causes immediate release into free engaged motion. During an established upper-stop interval, the later zero crossing of that reaction toward inadmissibility produces the same nonimpulsive release described above.

An arrival projection can also change the tangential contact conditions or the normal loads that support the candidate successor. The outgoing continuous closure is therefore always evaluated at the projected state before integration resumes. Stop reactions must remain nonnegative, primary contact at the low-ratio boundary must satisfy Equation (3.6.24), and every existing pulley contact must satisfy the tangential admissibility conditions of Section 3.5.4. If one condition fails, the corresponding support or tangential regime changes at the same event time.

The resulting shift-regime transitions are summarized in Table 3.4.

Transition	Event condition	State transfer
Free deadzone to fully open stop	$s = 0$, $\dot{s}^- < 0$	Inelastic projection with $\dot{s}^+ = 0$; retain the deadzone secondary-belt lock.
Fully open stop to free deadzone	$F_{\text{stop},0} = 0$, crossing toward a negative required reaction	Nonimpulsive release; free deadzone equations resume at the same position and velocities.
Deadzone to engaged motion	$s = s_{\text{dz}}$, $\dot{s}^- > 0$	Project from the deadzone to the engaged kinematic mapping; retain the secondary-belt lock through capture; do not impose primary stick.
Engaged backshift to low-ratio boundary	$s = s_{\text{dz}}$, $\dot{s}^- < 0$	Secondary-stop capture; finite opening motion maps into the deadzone, while the zero-motion limit can settle on the engaged seat.
Low-ratio seat to free engaged shift	$F_{\text{stop,LR}} = 0$, crossing toward a negative required reaction	Nonimpulsive release of the secondary stop.
Low-ratio seat to deadzone	$N_p = 0$ and $\ddot{s}_{\text{cf}} < 0$	Primary contact separates; activate the deadzone mapping and its secondary-belt lock.
Free engaged shift to upper stop	$s = s_{\max}$, $\dot{s}^- > 0$	Inelastic projection with $\dot{s}^+ = 0$; retain only tangential stick constraints already active before impact.
Upper stop to free engaged shift	$F_{\text{stop,max}} = 0$, crossing toward a negative required reaction	Nonimpulsive release; free engaged equations resume at the same position and velocities.

Table 3.4: Shift-regime event conditions and state-transfer rules. Tangential contact-regime changes remain governed by Section 3.5.4.

The event logic now separates two different operations. An active unilateral constraint is retained only while its reaction remains admissible; when that reaction reaches zero and would otherwise become negative, the constraint is released without changing the velocity. A newly reached constraint or kinematic mapping that is incompatible with the incoming velocity instead requires the momentum projection in Equation (3.6.37). The reaction condition decides when existing support is no longer needed, while the impulse calculation determines how the momentum already carried into a new boundary is transferred.

Chapter 4

Results and Discussion

4.1 Simulation Results

This section compares four launch simulations: a nominal baseline case, a lighter-flyweight case, a steeper-helix case with reduced secondary torque reactivity, and a combined case with both a shallower initial primary ramp and a steeper helix. The purpose of these results is not to match experimental vehicle data directly, but to show that the model produces mechanically interpretable launch behaviour and responds to tuning changes in physically sensible ways.

The results are organized around the launch shift curves of these four cases. These curves first provide an overall picture of the launch, including engagement, low-ratio operation, active upshift, and high-ratio behaviour. The ratio terminology follows **??**: low ratio means larger R (greater speed reduction), and high ratio means smaller R. following subsections then explain those trends by examining the axial force balance across the CVT and the resulting torque-transfer and slip behaviour. In particular, the combined case is included to show a clearly slip-dominated launch, in which the model predicts loss of the low-ratio hold and sustained slip through the early portion of the launch.

Taken together, the four cases are intended to show three things: first, that the baseline tuning produces a coherent and interpretable launch trajectory; second, that changes to primary and secondary tuning shift the launch behaviour in the expected direction; and third, that when the tuning is made sufficiently unfavorable, the model captures the transition to prolonged

slipping rather than continuing to predict an unrealistic no-slip response. The implementation used to generate these simulations, along with repository, release, and data-access details, is documented in Section 4.2.

4.1.1 Simulation Cases and Phase Convention

The results section compares four full-throttle launch simulations from rest. One case is treated as the nominal baseline, while the remaining three are targeted tuning variations chosen to isolate specific changes in CVT behaviour. All cases use the same vehicle, gearbox, tire, and engine model parameters; they differ only in the clutch tuning parameters noted below.

Table 4.1: Launch cases considered in the results section. All parameters not listed remain equal to the nominal baseline.

Case	Description	Purpose in results section
D	Default tuning	Baseline reference case
F	Lighter flyweights	Shows the effect of reduced primary centrifugal clamping
H	Steeper helix	Shows the effect of reduced secondary torque reactivity
C	Shallower initial primary ramp + steeper helix	Produces a strongly slip-dominated launch

To keep the discussion consistent across the different plots, the launch is described in terms of a small number of recurring operating phases. These phases are not introduced as separate events for each case, but as common regions of behaviour used to interpret the launch trajectories and their underlying force and torque balances.

Throughout the following subsections, these phases will be indicated directly on the plots using low-opacity background shading. The same color convention is used wherever possible so that the reader can track the evolution of each case across the shift, clamping-force, and torque-transfer results.

The phase colors are

- **engagement** — light blue,
- **low-ratio operation** — light green,
- **active upshift** — light yellow, and
- **high-ratio operation** — light red.

Not every case is expected to exhibit all four phases equally clearly. The phase shading is therefore used as a visual guide to the dominant operating regime, rather than as a claim that every boundary is perfectly sharp.

4.1.2 Launch Shift Curves

The overall launch behaviour of the four tuning cases is first compared in Figure 4.1. Each trajectory is plotted in engine-speed–vehicle-speed coordinates, so that engagement, ratio change, and the final high-ratio condition can all be viewed on the same axes. All four simulations begin from the same launch initial conditions,

$$\omega_p(0) = 1800\,\text{rpm}, \qquad \omega_s(0) = 0,$$

where ω_p is the primary angular speed and ω_s is the driven angular speed corresponding to zero vehicle speed. From this state, the system is released into a full-throttle forward simulation using the complete coupled model developed in the preceding sections.

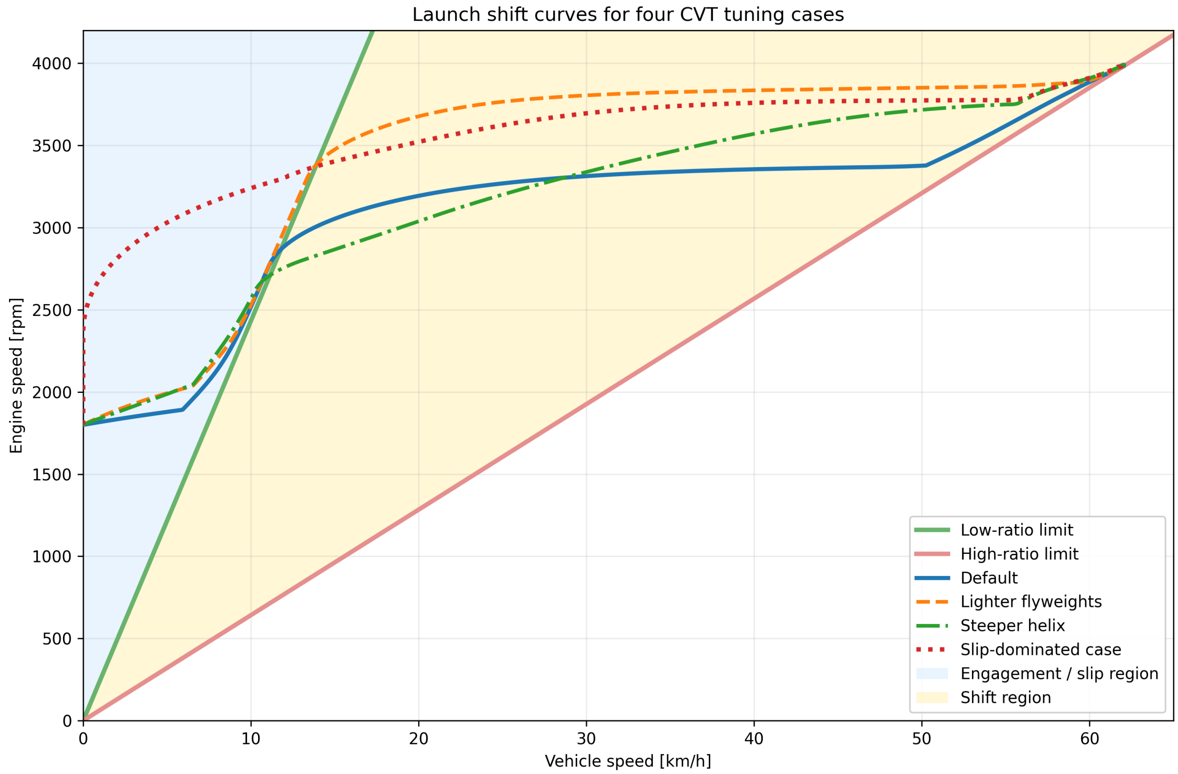


Figure 4.1: Engine-speed–vehicle-speed launch trajectories for the four tuning cases. The green line is the low-ratio limit, corresponding to a fully unshifted CVT. The red line is the high-ratio limit, corresponding to a fully shifted CVT. The blue shaded region to the left of the low-ratio line corresponds to launch engagement, where slip must be present. The yellow region between the two limit lines is the no-slip shift region.

The two reference lines are the kinematic no-slip limits of the CVT. The **low-ratio line** represents the fully unshifted transmission, where the CVT is in its torque-multiplying launch condition. The **high-ratio line** represents the fully shifted transmission, where the CVT has reached its final speed-increasing condition. Between these two limits lies the admissible shift region. A trajectory cannot move to the right of the high-ratio line, and any portion of a trajectory lying to the left of the low-ratio line must involve slip, since even the fully unshifted CVT cannot support such a low vehicle speed for the corresponding engine speed.

Read from left to right, the figure therefore shows the full launch sequence. The launch begins in the blue engagement region, where the vehicle is still too slow for no-slip operation. After crossing the low-ratio boundary, the trajectory enters the shift region. In a well-behaved launch, the CVT then tends to move approximately horizontally across this region, increasing vehicle speed while holding engine speed near a preferred operating value. This nearly horizontal portion will be referred to here as the *flat-shift region*. As the shift completes, the trajectory approaches the high-ratio line, which forms the final no-slip boundary of the launch.

With that interpretation in place, the differences between the four tuning cases are already visible. The **default** case shows the clearest baseline behaviour: a short engagement region, a brief low-ratio hold, and then a relatively flat shift through the middle of the plot before approaching the high-ratio boundary. This is the most balanced of the four trajectories, since it neither leaves low ratio too early nor rises excessively in engine speed during the main shift.

The **lighter-flyweight** case retains low ratio for longer before crossing into the shift region. Its flat-shift portion is also noticeably higher in engine speed than the default case, so the launch spends more of the main shift at elevated primary speed. The overall shape is still recognizable as a conventional launch trajectory, but the shift is delayed and carried out at a higher engine-speed level.

The **steeper-helix** case behaves differently. It leaves the low-ratio boundary much earlier, so the launch shows little sustained low-ratio hold. Its shift path is also less flat: instead of moving nearly horizontally across the shift region, it climbs upward more strongly as vehicle speed increases. This indicates that the launch does not settle into the same nearly constant engine-speed shift seen in the default case.

The **combined slip-dominated** case is the most extreme. It remains in the engagement region for much longer, showing no clear low-ratio hold before entering the shift region. Once it does cross into the no-slip range, its shift still occurs at a comparatively high engine speed, and the early launch is much more dominated by slip than by clean low-ratio operation.

At this stage, Figure 4.1 is intended to establish the overall shape of the four launches and the main qualitative differences between them. The key observations are the amount of low-ratio retention, the height and flatness of the main shift segment, and the extent of the initial slip-dominated portion of the launch. The following subsections explain these differences in terms of the underlying axial force balance and torque-transfer behaviour of the CVT.

4.1.3 Axial Force Balance Through the Launch

The shift-curve trajectories of Figure 4.1 indicate when each case remains near low ratio, when it begins to upshift, and how quickly it reaches high ratio. To explain those differences mechanically, Figure 4.2 compares the total axial force acting on the primary and secondary pulleys for each tuning case, with the same launch phases shaded in the background.

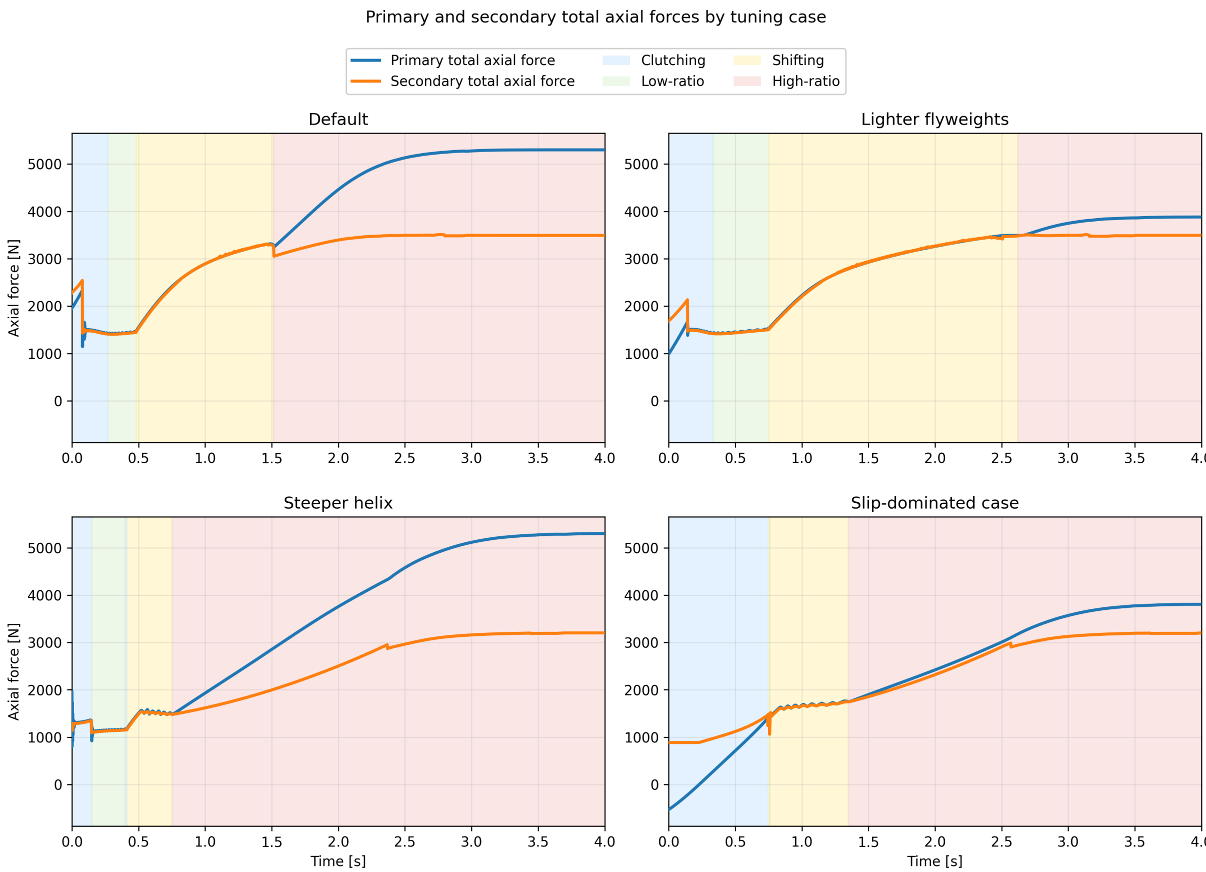


Figure 4.2: Primary and secondary total axial forces during launch for the four tuning cases. Blue curves show total primary axial force, orange curves show total secondary axial force. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as force-balance diagrams for the shift. While the secondary axial force exceeds the primary axial force, the transmission is held toward low ratio. Once the primary force overtakes the secondary, the pulley pair is driven into upshift. The timing of that crossover therefore determines how long the launch remains near low ratio and how early the transmission begins moving through its ratio range.

The **default** case shows the most balanced behaviour. During the initial clutching phase, the secondary force begins noticeably above the primary force, so the transmission is held near its low-ratio condition rather than shifting immediately. This produces the short but clear low-ratio hold seen previously in Figure 4.1. As the launch progresses, the primary axial force rises and eventually overtakes the secondary, initiating the main upshift. Importantly, the secondary force does not remain fixed while this happens. It also rises throughout the shift, so the system continues to resist ratio change even after upshift begins. This is mechanically important: if the secondary force stayed nearly constant while the primary force rose rapidly, the transmission would sweep through the shift range much more abruptly. Instead, both sides grow together, giving the default case its smoother and more controlled shift.

The **lighter-flyweight** case makes this point especially clear. Relative to the default case, the primary force is reduced in the early launch, while the secondary remains comparatively strong.

The initial force gap is therefore larger, meaning the low-ratio condition is held more firmly and for longer. This matches the longer low-ratio segment seen in the shift curves. Later in the launch, the primary still rises enough to drive upshift, but it does so later and at a higher engine-speed level than in the default case. In other words, this case does not fundamentally change the character of the shift; it mainly delays it by weakening the primary closing tendency during engagement and early launch.

The **steeper-helix** case behaves differently. Here, the initial difference between secondary and primary force is very small, and the primary overtakes the secondary much earlier. This explains why the corresponding shift curve showed little sustained low-ratio hold and entered the shifting region so quickly. We also see a differently shaped force evolution throughout the shift itself. In the default and lighter-flyweight cases, both forces continue rising together in a gradual way through the main shift region. In the steeper-helix case, both forces rise sharply early on and then level off much sooner. The two forces still remain relatively close to one another, but the overall shape they trace is different. This helps explain why the earlier shift curve did not display the same flat-shift behaviour as the default case. Rather than maintaining a long, nearly constant-speed shift, the transmission shows a more rising shift trajectory, consistent with this less gradual force evolution.

The **slip-dominated** case is the most extreme. Its early launch is characterized by a prolonged clutching phase and no meaningful low-ratio hold. This is consistent with the earlier shift-curve figure, where the trajectory spent much longer in the slip-dominated region before entering the no-slip shift region. Importantly, the beginning of geometric shift does not imply that the system has already stopped slipping. It only indicates that the shift coordinate has begun to move away from its initial value. The more important feature in this case is the low magnitude of both axial forces in the early launch. Compared with the other cases, neither pulley develops a strong axial loading during the initial portion of the event. This weak overall clamping is a key reason why this case behaves poorly, and it will be seen more directly in the following torque-transfer plots where the associated slip remains elevated for much longer.

One additional feature is visible near the beginning of each case. The early spike in secondary axial force is the torque-reactive contribution of the secondary. It appears most strongly during the initial launch when transmitted torque rises rapidly, and it provides an immediate resisting effect against premature upshift. This feature is especially useful because it shows that the secondary is responding dynamically to transmitted torque rather than acting as a static spring-only restraint. The influence of this torque reactivity will be seen more directly in the following torque-transfer results, but its signature is already visible here in the early force histories.

Taken together, Figure 4.2 provides the mechanical explanation for the differences observed in Figure 4.1. A larger initial gap between secondary and primary force corresponds to stronger low-ratio retention, as seen in the lighter-flyweight case. A very small gap corresponds to premature upshift, as seen in the steeper-helix case. The default case lies between these extremes and therefore produces the most balanced launch. Perhaps most importantly, the figure shows that the shift is not driven by the primary alone. The secondary rises with it throughout the launch, and the actual shift behaviour emerges from the evolving balance between the two rather than from either force in isolation. At the same time, this figure only explains the shift tendency and overall clamping levels; it has not yet quantified the resulting slip behaviour, which is addressed next.

4.1.4 Torque Bounds, No-Slip Demand, and Coupling Torque

The force-balance results of Figure 4.2 explain when each case tends to remain at low ratio and when it begins to shift. However, force balance alone does not determine whether the belt is actually able to transmit the torque required for sticking. To examine that question, Figure 4.3 compares four quantities during the launch: the primary and secondary traction bounds, the no-slip coupling torque τ_{ns}, and the actual transmitted coupling torque τ_c.

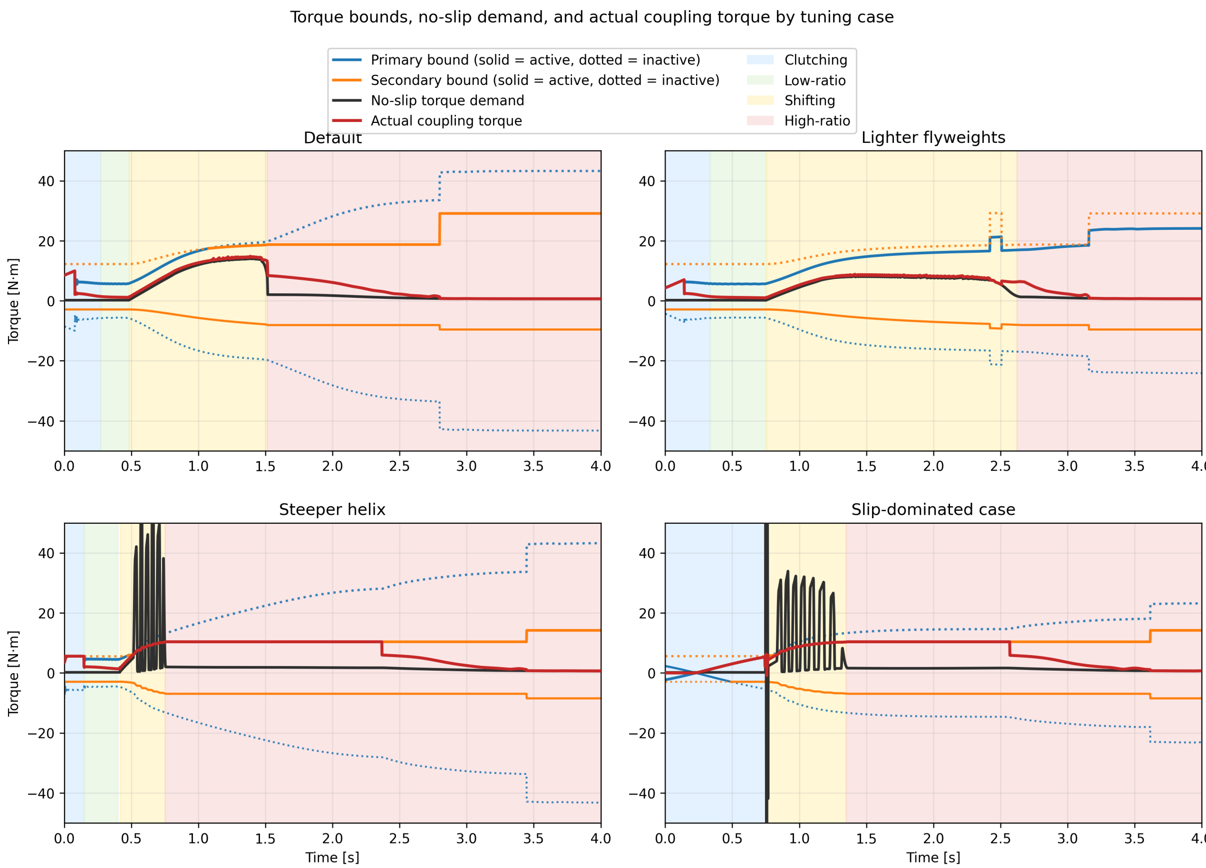


Figure 4.3: Torque bounds, no-slip demand, and actual coupling torque for the four tuning cases. Blue and orange curves show the primary and secondary bounds, with the active bound drawn solid and the inactive bound dotted. The black curve shows the no-slip torque demand τ_{ns}, and the red curve shows the actual transmitted coupling torque τ_c. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as traction-capacity diagrams. At any instant, the belt can only remain in stick if the required no-slip torque lies inside the admissible interval defined by the active upper and lower bounds. When τ_{ns} lies inside that interval, the actual coupling torque can follow the no-slip demand closely. When τ_{ns} exceeds the available bounds, sticking is no longer admissible and the transmitted torque must saturate at the corresponding active limit.

A first broad trend is immediately visible. The cases that settle out of slip early in the launch are precisely the cases with larger admissible positive torque capacity during the initial clutching and low-ratio phases. In the **default** and **lighter-flyweight** cases, the upper torque bound rises to a relatively large value early in the launch. This gives the system enough admissible coupling torque to reduce the belt-speed mismatch and recover toward sticking. By contrast, the **slip-dominated** case begins with essentially no admissible positive torque and only develops usable capacity more gradually. That lower early torque capacity directly explains why the slip-dominated case spends much longer in the clutching region.

This same distinction appears in the evolution of τ_c. In the cases that recover stick early, the actual coupling torque relaxes toward and then tracks the no-slip demand through the end of clutching and into low-ratio operation. Once low-ratio launch is established, the red and black curves remain close, indicating that the required torque stays admissible and sustained slip is not reintroduced immediately.

A second important pattern appears once the transmission begins to shift. In all four cases, the no-slip torque rises during the shift phase. This is consistent with the form of τ_{ns}, in which the ratio-rate term becomes active once the CVT begins moving away from its initial ratio. In the **default** and **lighter-flyweight** cases, this rise remains below the admissible upper torque bound, so the actual coupling torque continues to follow the no-slip demand and large renewed slip does not occur. In the **steeper-helix** and **slip-dominated** cases, however, τ_{ns} rises above the admissible upper bound during the shift. At that point, sticking is no longer feasible, the actual coupling torque saturates at the active maximum, and renewed slip appears until the state returns closer to the sticking condition.

The **steeper-helix** and **slip-dominated** cases also show visibly oscillatory behaviour in the no-slip torque during the shift onset. This is consistent with the oscillations already seen in the preceding axial-force plots. The most direct source is the ratio-rate contribution to τ_{ns}: if the shift dynamics are not smooth, then the implied no-slip torque need not evolve smoothly either. In these two poorer tuning cases, the shift motion appears more weakly damped, so the no-slip torque exhibits stronger excursions during the transition into active shifting. A plausible reason is that the present model neglects parasitic losses and other non-critical frictional effects, which in a real system would provide additional damping and suppress some of this oscillatory behaviour. This is therefore a useful limitation to note: the model captures the overall stick–slip transitions and the active limiting mechanism, but the transient oscillations in the most aggressive cases are likely amplified by the intentionally idealized loss assumptions.

Another common feature is the gradual relaxation of the actual coupling torque toward the no-slip demand rather than an instantaneous snap to it. This comes from the regularized transition law used in the simulation, discussed in Appendix B.3, which replaces a discontinuous switch with a smooth tanh-based approximation. Its purpose is numerical: it reduces chattering and improves stability near the stick–slip transition. The effect is visible here as a slight lag in the recovery of perfect sticking, but it does not change the main qualitative behaviour of the four cases. The same limiting patterns are still present, and the cases that lose admissibility remain clearly distinguishable from those that do not.

Finally, each case shows a noticeable jump in the admissible torque bound once full sticking is re-established near the end of the launch. This reflects the change in limiting behaviour used by the model after slip is removed, together with the corresponding increase in effective friction behaviour assumed in the sticking regime. The important point for the present discussion is that this jump occurs only after the system has already relaxed back toward the no-slip state; it is therefore a consequence of stick recovery rather than its cause.

Taken together, Figure 4.3 closes the interpretation begun in the previous two subsections. The shift curves showed the overall launch behaviour, and the axial-force plots explained the tendency to hold or leave low ratio. The present figure then shows whether the required coupling torque is actually admissible throughout those phases. The default and lighter-flyweight cases maintain sufficient positive torque capacity to recover stick early and remain near the no-slip demand through the main shift. The steeper-helix and slip-dominated cases do not: in both, the no-slip torque exceeds the available upper bound during the shift, forcing the actual coupling torque to saturate and reintroducing slip. In this way, the torque plots both confirm the solid patterns seen earlier and reveal one of the model's clearest shortcomings: the most weakly damped cases display exaggerated oscillatory transients due to the idealized loss assumptions.

4.1.5 Launch Performance Comparison

The preceding subsections established the mechanical differences between the four tuning cases. The shift-curve plots showed the overall launch trajectory, the axial-force plots explained the tendency to hold or leave low ratio, and the torque-bound plots showed whether the required coupling torque remained admissible throughout the launch. The final step is to compare how those differences affect overall launch performance.

Two simple metrics are used here: the time required to travel 50 m, and the time required to reach 60 km/h. These are useful because they summarize the launch in two complementary ways. The 50 m time reflects the integrated quality of the launch as a whole, while the 60 km/h time emphasizes how quickly the drivetrain can build vehicle speed through the main acceleration phase.

Table 4.2: Launch performance metrics for the four tuning cases.

Case	**Time to** 50 m [s]	**Time to** 60 km/h [s]
Default	3.93	2.27
Lighter flyweights	4.26	2.78
Steeper helix	4.19	2.82
Slip-dominated case	4.41	2.98

The default case performs best on both metrics, which is consistent with the earlier mechanism-level results. It develops enough admissible coupling torque to reduce slip early, retains low ratio long enough to launch effectively, and then transitions into a controlled shift without reintroducing large sustained slip. This is the most balanced use of the CVT: the transmission first allows a strong initial torque transfer into the driveline, and then continues accelerating the vehicle while keeping the shift progression smooth and admissible.

This point is worth emphasizing. The coupling torque plays two roles at once. It is the torque that loads the engine through the transmission, limiting how freely the primary (and therefore the engine) can accelerate, and it is also the torque delivered through the CVT that accelerates the vehicle from rest. A strong early coupling torque is therefore beneficial when it is admissible, because it quickly pulls the belt-speed mismatch down and simultaneously applies useful drive torque to the vehicle. The default case benefits from exactly this behaviour: it uses the initial slip region to establish a relatively large transmitted torque early in the launch, which helps "kick-start" vehicle acceleration before the main shift begins.

The lighter-flyweight case performs worse on both metrics, even though its shift remains comparatively well behaved. This matches the earlier interpretation that its main drawback is delayed primary loading during the early launch. By holding low ratio longer but developing less aggressive early coupling, it sacrifices initial acceleration and does not recover that loss later.

The steeper-helix case shows a different trade-off. It leaves low ratio too early and re-enters slip during the shift, which degrades the quality of the launch. However, because it still develops appreciable coupling torque early in the event, its 50 m time remains slightly better than the lighter-flyweight case. Its slower time to 60 km/h, on the other hand, is consistent with the less stable shift behaviour and the renewed slip seen in the torque-bound plots.

The slip-dominated case performs worst overall. This is the expected result from the earlier figures: its early clamping forces are weakest, its admissible torque rises too slowly, and its launch remains slip-dominated for too long. As a result, it spends too much of the start of the event establishing torque transfer rather than converting that torque cleanly into forward acceleration.

Overall, these performance metrics support the mechanism-level interpretation developed throughout the section. The best launch is not produced by simply holding low ratio as long as possible, nor by shifting as early as possible. Instead, good performance requires a balance: sufficient early admissible coupling torque to accelerate the vehicle strongly from rest, enough low-ratio retention to avoid premature upshift, and a smooth enough shift that the required no-slip torque does not repeatedly exceed the available traction limits.

4.2 Implementation and Reproducibility

The source code for the CVT simulator used in this work is available at `https://github.com/gr812b/CVT-Simulator`. This implementation includes the mathematical model developed in the main derivation sections along with the practical numerical regularization strategy described in Appendix B.3.

To support reproducibility, the exact version used to generate the figures and tables in this manuscript will be archived as a tagged release:

- repository release (to be finalized): `https://github.com/gr812b/CVT-Simulator/releases/tag/vX.Y.Z`,
- archival DOI landing page (placeholder): `https://doi.org/XX.XXXX/zenodo.XXXXXXX`.

Unless otherwise stated, simulation cases in Section 4.1 correspond to that archived release. The repository README and release notes describe the execution workflow, configuration files, and plotting scripts used to produce the reported results.

An interactive web deployment of the simulator is also available for demonstration purposes: `https://cvt.kaiarseneau.dev`. This live endpoint is provided for accessibility and does not replace the archived release as the canonical reproducibility artifact.

4.3 Conclusion

This work developed a unified first-principles dynamic model for a mechanically actuated dry rubber V-belt CVT and its coupling to the vehicle drivetrain. Starting from pulley geometry, belt-length closure, actuator mechanics, and rotational dynamics, the derivation produced a closed nonlinear state-space formulation in which ratio evolution, torque transmission, slip admissibility, and vehicle response are solved consistently in time.

The completed formulation brings together several mechanisms that strongly govern mechanically actuated CVT behaviour: primary flyweight actuation, secondary torque-reactive helix clamping, belt geometric closure, axial shift dynamics, traction-limited torque transfer, and longitudinal drivetrain coupling. The main outcome is a single dynamic framework in which these effects interact directly, while remaining mechanically interpretable in terms of the underlying pulley forces, ratio change, and torque limits.

The simulation results showed coherent launch behaviour across the tuning cases considered. The shift-curve results captured distinct differences in engagement, low-ratio retention, main-shift shape, and approach to high ratio. The axial-force results showed that these differences could be traced back to the evolving balance between primary and secondary clamping. The torque-bound results then showed when the no-slip coupling demand remained admissible and when renewed slip was forced by the available traction limits. Together, these results formed a consistent mechanism chain from tuning changes, to force balance, to torque admissibility, to overall launch behaviour.

An important outcome of the study is that the model preserved physical intuition across multiple levels. Changes in tuning did not simply produce different output curves; they produced changes that remained understandable in mechanical terms. Stronger or weaker low-ratio retention appeared through the initial clamping-force gap, earlier or later upshift appeared through the timing of force crossover, and reintroduced slip appeared when the no-slip torque rose beyond the admissible coupling bounds. This consistency gives confidence that the formulation is capturing the dominant interactions it was intended to represent.

The results also highlighted useful limitations of the present model. In the more aggressive cases, the no-slip torque demand exhibited visibly underdamped oscillatory transients during shift onset. These were consistent with irregular shift-rate evolution in the model and suggest that the idealized loss assumptions, particularly the neglect of parasitic losses and other non-critical frictional effects, leave the system with less damping than would be expected in real hardware. The formulation also remains limited to one-dimensional vehicle motion and does not yet include thermal effects, wear progression, or more detailed tribological contact evolution.

Overall, the present work provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behaviour during transient launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behaviour remains traceable back to clear mechanical causes. With further parameter identification, experimental validation, and extension to include more realistic damping and loss mechanisms, the framework developed here can serve as a useful basis for both engineering interpretation and future design-oriented CVT simulation.

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Appendix A

Belt Length Small-Angle Approximation

This appendix quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models. Using parameter values representative of the physical system considered in this work, we demonstrate that the approximation introduces substantial error in the computed center-to-center distance, then further in the ratio, which is critical for accurate dynamic modeling.

A.1 Standard Approximation

Many simplified treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this approximation, the belt-length constraint **??** reduces to

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{A.1.1}$$

A.2 Derivation of the Approximate Center Distance

To determine the center-to-center distance at the initial configuration, we substitute $r_p = r_{p,\text{outer,min}}$ and $r_s = r_{s,\text{outer,max}}$ into Equation (A.1.1) and solve for C.

Step 1: Isolate the square root. Rearranging Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) = 2\sqrt{C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}. \tag{A.2.1}$$

Step 2: Square both sides.

$$\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2 = 4\left[C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2\right]. \tag{A.2.2}$$

Step 3: Solve for C^2.

$$C^2 = \frac{\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2. \tag{A.2.3}$$

Step 4: Take the square root. Expanding the squared term in the numerator:

$$\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2 = L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2,$$

the approximate center distance becomes

$$\boxed{C_\text{approx} = \sqrt{\frac{L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}.} \tag{A.2.4}$$

This closed-form expression avoids the need for numerical root finding but neglects the $\sin^{-1}$ term in the exact constraint.

A.3 Error Analysis

We now evaluate the approximation error using parameter values from the physical CVT system considered in this work.

System Parameters

Parameter	Value	SI Value
Belt length L_b	37.538 in	0.9535 m
Belt height h_b	0.613 in	0.015 57 m
Groove half-angle β	11.5°	0.2007 rad
Inner primary radius	0.8125 in	0.020 637 5 m
Primary outer radius $r_{p,\text{outer,min}}$	1.4255 in	0.036 21 m
Secondary outer radius $r_{s,\text{outer,max}}$	4.0 in	0.1016 m

The primary outer radius is computed as the inner radius plus belt height: $r_{p,\text{outer,min}} = 0.8125 + 0.613 = 1.4255\,\text{in}$.

Computed Center Distances Solving the exact belt-length constraint **??** numerically yields

$$C_\text{exact} = 0.251\,74\,\text{m} \approx 9.911\,\text{in}.$$

Evaluating the approximate formula Equation (A.2.4) yields

$$C_\text{approx} = 0.268\,37\,\text{m} \approx 10.566\,\text{in}.$$

Error Metrics The absolute and relative errors are:

$$\Delta C = C_\text{approx} - C_\text{exact} = 0.016\,63\,\text{m} = 0.655\,\text{in}, \tag{A.3.1}$$

$$\varepsilon_\text{rel} = \frac{\Delta C}{C_\text{exact}} = \frac{0.01663}{0.25174} = 6.61\%. \tag{A.3.2}$$

Assessment The small-angle approximation introduces a relative error of nearly 7% in the center-to-center distance for this system. This error arises because the ratio

$$\frac{r_{s,\text{outer,max}} - r_{p,\text{outer,min}}}{C} = \frac{0.1016 - 0.03621}{0.25174} = 0.260$$

is not small. The assumption $\lambda \approx 0$ is therefore not justified for this CVT geometry.

Since the center distance C propagates into all subsequent geometric calculations—wrap angles, tangent span lengths, and derived quantities such as the transmission ratio—a 7% error in C can compound into larger errors in predicted system behavior.

A.4 Propagation to Secondary Radius and Transmission Ratio

To demonstrate how the center distance error propagates into operationally relevant quantities, we now examine the secondary radius and CVT ratio at two shift positions: zero shift and maximum shift.

A.4.1 Derivation of the Approximate Secondary Radius

Given a known primary outer radius $r_{p,\text{outer}}$ and center distance C, the secondary outer radius can be obtained by solving the belt-length constraint. Under the small-angle approximation Equation (A.1.1), this yields a closed-form expression.

Step 1: Isolate the square root. From Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer}} + r_{s,\text{outer}}) = 2\sqrt{C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2}. \tag{A.4.1}$$

Step 2: Square both sides.

$$\left[L_b - \pi r_{p,\text{outer}} - \pi r_{s,\text{outer}}\right]^2 = 4\left[C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2\right]. \tag{A.4.2}$$

Step 3: Expand and collect terms. Expanding the left side:

$$(L_b - \pi r_{p,\text{outer}})^2 - 2\pi(L_b - \pi r_{p,\text{outer}})r_{s,\text{outer}} + \pi^2 r_{s,\text{outer}}^2.$$

Expanding the right side:

$$4C^2 - 4r_{s,\text{outer}}^2 + 8r_{p,\text{outer}}r_{s,\text{outer}} - 4r_{p,\text{outer}}^2.$$

Collecting powers of $r_{s,\text{outer}}$ yields a quadratic equation:

$$(\pi^2+4)r_{s,\text{outer}}^2 - 2\left[\pi L_b - \pi^2 r_{p,\text{outer}} + 4r_{p,\text{outer}}\right]r_{s,\text{outer}} + \left[(L_b - \pi r_{p,\text{outer}})^2 + 4r_{p,\text{outer}}^2 - 4C^2\right] = 0. \tag{A.4.3}$$

Step 4: Apply the quadratic formula. The physically meaningful root (smaller secondary radius for overdrive configuration) is

$$\boxed{r_{s,\text{approx}} = \frac{\pi L_b + (4-\pi^2)r_{p,\text{outer}} - 2\sqrt{(\pi^2+4)C^2 - L_b^2 + 4\pi L_b r_{p,\text{outer}} - 4\pi^2 r_{p,\text{outer}}^2}}{\pi^2+4}.} \tag{A.4.4}$$

This closed-form expression enables direct computation of the approximate secondary radius without numerical root finding.

A.4.2 Zero Shift Configuration

At zero shift, the CVT is in its lowest ratio (maximum reduction) configuration. This is also the geometry used to compute the approximate center distance C_{approx}, so much of that initial center-distance error cancels when the same configuration is used again to recover the secondary radius.

Primary Radius at Zero Shift At $s = 0$, the primary is within the deadzone, so from **??**:

$$r_{p,\text{outer}}(0) = r_{p,\text{outer,min}} = 0.036\,21\,\text{m}.$$

Secondary Radius Comparison Solving the exact belt-length constraint with C_{exact} yields

$$r_{s,\text{exact}} = 0.101\,60\,\text{m}.$$

Evaluating the approximate formula Equation (A.4.4) with $C_{\text{approx}} = 0.268\,37\,\text{m}$ yields

$$r_{s,\text{approx}} = 0.101\,60\,\text{m}.$$

Transmission Ratio Error Computing effective radii with $h_b/2 = 0.007\,79\,\text{m}$:

Quantity	**Exact**	**Approximate**
$r_{p,\text{eff}}$	0.0284 m	0.0284 m
$r_{s,\text{eff}}$	0.0938 m	0.0938 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	3.300	3.300

The ratio error at zero shift is

$$\varepsilon_R = \frac{|3.300 - 3.300|}{3.300} \approx 0.000\%.$$

Thus, although C_{approx} is already 6.61% too large, the induced error has little visible effect at minimum shift because the radius calculation is being evaluated at the same operating point used to construct that approximate center distance in the first place.

Figure A.1: Scaled diagram of the CVT at zero shift ($s = 0$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The two solutions nearly overlap at this operating point, indicating negligible ratio error in this case.

A.4.3 Maximum Shift Configuration

At maximum shift, the primary and secondary pulleys are closest in size, representing a near-crossover configuration.

Primary Radius at Maximum Shift Using **??** with $s = s_{\max} = 0.75\,\text{in} = 0.019\,05\,\text{m}$ and deadzone $s_{\text{dz}} = 0.002\,489\,2\,\text{m}$:

$$\begin{aligned}
r_{p,\text{outer}}(s_{\max}) &= r_{p,\text{outer,min}} + \frac{s_{\max} - s_{\text{dz}}}{2\tan\beta} \\
&= 0.03621 + \frac{0.01905 - 0.0024892}{2 \times \tan(11.5°)} \\
&= 0.03621 + \frac{0.01656}{0.4070} \\
&= 0.0769\,\text{m}.
\end{aligned} \tag{A.4.5}$$

Secondary Radius Comparison Solving the exact belt-length constraint **??** numerically with $C_{\text{exact}} = 0.251\,74\,\text{m}$ yields

$$r_{s,\text{exact}} = 0.066\,73\,\text{m}.$$

Evaluating the approximate formula Equation (A.4.4) with $C_{\text{approx}} = 0.268\,37\,\text{m}$ yields

$$r_{s,\text{approx}} = 0.056\,26\,\text{m}.$$

Transmission Ratio Error The CVT ratio is defined using effective radii, which account for belt seating depth (**??**):

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}.$$

With $h_b/2 = 0.007\,79\,\text{m}$:

Quantity	**Exact**	**Approximate**
$r_{p,\text{eff}}$	0.0691 m	0.0691 m
$r_{s,\text{eff}}$	0.0589 m	0.0485 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	0.853	0.701

Both models predict overdrive ($R < 1$), but the approximation substantially underestimates the ratio. The ratio error is

$$\varepsilon_R = \frac{|0.701 - 0.853|}{0.853} = 17.8\%.$$

Unlike the minimum-shift case, this operating point does not benefit from the same cancellation, so the center-distance error propagates into a much larger error in secondary radius and transmission ratio.

Figure A.2: Scaled diagram of the CVT at maximum shift ($s = s_{\max}$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The approximation underestimates the secondary radius, leading to a lower predicted ratio than the exact model.

A.5 Summary of Approximation Errors

Configuration	R_{exact}	R_{approx}	**Ratio Error**
Zero shift ($s = 0$)	3.300	3.300	0.000%
Maximum shift ($s = s_{\text{max}}$)	0.853	0.701	17.8%

The small-angle approximation error is strongly operating-point dependent. At zero shift, most of the center-distance error cancels because the approximate center distance was itself constructed from that same geometry. At maximum shift, that cancellation is lost and the ratio error becomes substantial.

These errors arise from the initial 6.61% error in center distance, which propagates nonlinearly through the belt-length geometry in a state-dependent way. Since the transmission ratio directly enters the rotational equations of motion (**??**), this approximation can still produce materially inaccurate dynamics near maximum shift.

This analysis justifies the use of exact numerical solutions for all geometric computations in this work.

Appendix B

Numerical Methods Summary

This section documents the numerical algorithms used for belt-geometry root finding and dynamic simulation, including the specific tolerances, bracketing strategies, and solver configurations employed in the implementation.

B.1 Root Finding for Belt-Length Constraint

Given a known primary outer radius r_1 and fixed center-to-center distance C, the secondary outer radius r_2 is the unique root of the belt-length constraint. This scalar root problem is obtained by rearranging the exact belt length closure from **??** into the residual form:

$$g(r_2;\, r_1) = \pi(r_1 + r_2) + 2(r_2 - r_1)\arcsin\left(\frac{r_2 - r_1}{C}\right) + 2\sqrt{C^2 - (r_2 - r_1)^2} - L_b = 0, \qquad \text{(B.1.1)}$$

where L_b is the fixed belt length. A separate invocation solves for the center-to-center distance C itself at initialization, using the same equation evaluated at the boundary radii $(r_{1,\min},\, r_{2,\max})$ with C as the unknown; this is exactly the initialization step implied by **??** before the ratio law **??** can be evaluated over the shift range.

Bracketing strategy. The arcsin term in Equation (B.1.1) is real only when $|r_2 - r_1| \le C$, so the natural domain for r_2 is the open interval $(r_1 - C,\, r_1 + C)$. Within this interval a sign change is guaranteed: at the lower boundary $r_2 \to r_1 - C$, the arcsin saturates to $-\pi/2$ and the square root vanishes, giving $g \to 2\pi r_1 - 2\pi C - L_b < 0$ for all physically valid parameters. At the upper boundary $r_2 \to r_1 + C$, by symmetry $g \to 2\pi r_1 + 2\pi C - L_b > 0$, satisfied whenever the belt length is less than the circumference subtended by the wider pulley pair — a requirement any real CVT must satisfy. A small safety margin $\varepsilon = 10^{-9}$ m is applied at both ends to keep arcsin away from its domain boundary where the function becomes numerically steep.

For center-to-center distance solving, the bracket is instead the interval $(\Delta r + \varepsilon,\ L_b/2)$, where $\Delta r = |r_{2,\text{eff}} - r_{1,\text{eff}}|$ is the effective-radius difference at the boundary configuration. The lower bound ensures arcsin is defined; the upper bound is a conservative geometric limit (a straight two-span belt cannot exceed L_b across a gap of $L_b/2$). Both brackets are validated by an explicit sign-change check before invoking the solver, so any failure to bracket immediately identifies an infeasible geometry rather than producing a silent wrong answer.

Brent's method. Both roots are found using `scipy.optimize.brentq`, an implementation of Brent's algorithm Brent (1973). The method combines three strategies: bisection (guaranteed bracket-halving), secant interpolation (superlinear convergence near smooth roots), and inverse quadratic interpolation (order-≈ 1.84 convergence when three previous iterates are well-conditioned). Whenever the faster interpolation steps would place the next iterate outside the current bracket, the algorithm falls back to bisection, preserving the worst-case $O(\log_2(1/\varepsilon))$ guarantee of pure bisection. Because the sign-change brackets described above are guaranteed for all valid CVT geometries, `brentq` is certain to converge. Both root solves use an absolute tolerance of 10^{-9} m, well below any physically meaningful length precision.

B.2 ODE Integration for Dynamic Simulation

The full CVT state vector $\mathbf{x} = (\omega_p, \omega_s, s, \dot{s}, v_b)$ is integrated using `scipy.integrate.solve_ivp`. The governing system is written as a first-order ODE system in **??**, with rotational sub-equations **??** and **??**, axial dynamics **??**, belt-speed dynamics for v_b, and algebraic geometry closures **??** and **??**. Although **??** is second-order in s, introducing $(s, \dot{s})$ as separate states converts the complete model to first-order form for time integration. The time integrator is used with the `RK45` method — the Dormand–Prince explicit embedded Runge–Kutta pair Dormand and Prince (1980). The method propagates the state with a 5th-order formula and simultaneously computes a 4th-order companion estimate for local error control, using the same six function evaluations. The tableau is FSAL (first-same-as-last): the final stage of each accepted step is reused as the first stage of the next, so only five new evaluations are needed per step in steady operation.

Adaptive step size control. At each candidate step the per-component local error estimate $\hat{e}_i$ is compared to a mixed tolerance threshold. A step is accepted when

$$\max_i \frac{|\hat{e}_i|}{\mathrm{atol} + \mathrm{rtol} \cdot |y_i|} \leq 1, \tag{B.2.1}$$

and the next step size is scaled proportionally to $(\mathrm{tol}/\hat{e})^{1/5}$. The tolerances used in this work are $\mathrm{atol} = 10^{-6}$ and $\mathrm{rtol} = 10^{-4}$ (Table B.1). The absolute floor prevents spurious step rejection when a state variable passes through zero (e.g. $\dot{s} \approx 0$ at shift equilibrium or $\omega_p \approx 0$ at engine start), where a purely relative criterion would demand unreachable accuracy.

Parameter	Value	Interpretation
`atol`	10^{-6}	absolute floor: 1 µm on s, 1 µrad/s on ω
`rtol`	10^{-4}	relative tolerance: 0.01% of current state magnitude
`t_eval`	10 000 pts	uniform output grid over 30 s

Table B.1: ODE solver tolerance parameters.

Event detection. Phase transitions (shift engagement, boundary contact, back-shift onset) are located using `solve_ivp`'s event mechanism. These events correspond to qualitative changes in the dynamics around the bounded shift coordinate used in **??** and the full state evolution in **??**. After each accepted step the solver evaluates the event functions on a degree-4 dense-output interpolant and applies a root-finding pass to detect any sign change within the step. When a crossing is found it is bracketed at sub-tolerance precision before being reported, so transitions are located with the same accuracy as the integration tolerance rather than being snapped to the coarse output grid.

Phase splitting at hard constraints. The shift bounds $s \in [0, s_{\max}]$ with $\dot{s} = 0$ at both endpoints are handled by partitioning the trajectory into separate `solve_ivp` calls rather than imposing inequality constraints inside the ODE. Each phase — normal shifting, clamped-at-maximum, and back-shifting — presents a smooth right-hand side to the integrator, allowing RK45 to operate at full 5th-order accuracy throughout. The terminal event of each phase supplies the initial condition of the next. This is numerically cleaner than directly clamping **??** inside a single solve because it preserves a smooth right-hand side within each phase. Up to ten alternating full-shift / back-shift cycles are supported before the simulation terminates, preventing unbounded phase loops for unusual parameter regimes.

B.3 Slip-Law Regularization and Blend Parameter Selection

The traction law in **??** is physically clear but discontinuous at $v_{\text{rel}} = 0$, which can induce step-size chatter near stick–slip transitions. A practical regularized formulation uses three ideas: a smooth saturation law, a clamped no-slip torque request, and a blending rule between those two limits.

First define a smooth saturation torque

$$\tau_{\text{sat}} = \tau_{\max} \tanh\left(\frac{v_{\text{rel}}}{v_{\text{smooth}}}\right), \tag{B.3.1}$$

where $v_{\text{smooth}} > 0$ sets the transition width in slip-speed units. For $|v_{\text{rel}}| \gg v_{\text{smooth}}$, this approaches the usual friction-limited value $\pm\tau_{\max}$, while near zero it removes the sharp sign discontinuity.

Next, clamp the no-slip torque request:

$$\tau_{\text{req}} = \text{clip}(\tau_{\text{ns}},\, -\tau_{\max},\, \tau_{\max})\,. \tag{B.3.2}$$

This is the largest admissible torque consistent with the no-slip request, without allowing the requested value to exceed traction capacity.

Then define the blend factor

$$\alpha = \text{clip}\left(\frac{|v_{\text{rel}}|}{v_{\text{blend}}},\, 0,\, 1\right), \tag{B.3.3}$$

with $v_{\text{blend}} > 0$ defining the width of the blending region, and use the transmitted torque

$$\tau_c = (1 - \alpha)\,\tau_{\text{req}} + \alpha\,\tau_{\text{sat}}. \tag{B.3.4}$$

This has a useful interpretation. For very small mismatch, $\alpha \approx 0$, so the model stays close to the clamped no-slip demand. For larger mismatch, $\alpha \to 1$, so the law transitions to the smooth Coulomb limit. In other words, the regularized model behaves like a requested no-slip torque near stick and like a friction-limited sliding law once the mismatch becomes appreciable.

How to choose v_{smooth}. Choose the smoothing scale small enough that $\tanh(v_{\text{rel}}/v_{\text{smooth}})$ is close to a sign law through most of the slip regime, but not so small that the solver again sees an effectively discontinuous transition. A practical workflow is:

1. Start with v_{smooth} around 0.5–2% of a representative relative-speed magnitude seen during actual slip.
2. Decrease it until predicted macro outputs (ratio trajectory, launch time, peak speeds) stop changing materially.
3. Increase it slightly if event localization or adaptive step-size behavior becomes noisy near $v_{\text{rel}} \approx 0$.

The blend scale v_{blend} should generally be chosen greater than or equal to v_{smooth}, so that the handoff from clamped demand to the smooth Coulomb limit is not sharper than the smoothing law itself.

Appendix C

Rate and Acceleration of the Geometric CVT Ratio

The main derivation uses the individual pulley-radius rates and accelerations from Section 3.1.4. Those same results can be combined to show how quickly the available geometric CVT ratio changes as the sheaves move.

Recall the geometric CVT ratio from **??**:

$$R_g = \frac{r_s}{r_p}, \tag{C.0.1}$$

The individual primary and secondary radius rates and accelerations have already been established in Section 3.1.4. This appendix only combines those results to show how the available geometric ratio itself changes as the sheaves move.

C.0.1 Rate of Change of the Geometric CVT Ratio

Both the numerator and denominator in Equation (C.0.1) can change with time. Using the quotient rule,

$$\begin{aligned} \dot{R}_g &= \frac{d}{dt}\left(\frac{r_s}{r_p}\right) \\ &= \frac{\dot{r}_s r_p - r_s \dot{r}_p}{r_p^2} \\ &= \frac{\dot{r}_s}{r_p} - \frac{r_s}{r_p^2}\dot{r}_p. \end{aligned} \tag{C.0.2}$$

The secondary-radius result in Equation (3.1.57) supplies the coupled secondary motion. Within the active-shift branch,

$$\dot{r}_s = -\frac{\phi_p}{\phi_s}\dot{r}_p. \tag{C.0.3}$$

Substituting this into Equation (C.0.2) gives

$$\begin{aligned}
\dot{R}_g &= \frac{1}{r_p}\left(-\frac{\phi_p}{\phi_s}\dot{r}_p\right) - \frac{r_s}{r_p^2}\dot{r}_p \\
&= -\frac{\dot{r}_p}{r_p}\left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p}\right) \\
&= -\frac{\dot{r}_p}{r_p}\left(\frac{\phi_p}{\phi_s} + R_g\right).
\end{aligned} \tag{C.0.4}$$

Finally, Equation (3.1.36) gives $\dot{r}_p = \dot{s}/(2\tan\beta)$ in the active-shift branch. Therefore,

Rate of Change of Geometric CVT Ratio **(C.0.5)**

$$\dot{R}_g = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ -\dfrac{\dot{s}}{2\tan\beta\, r_p}\left(\dfrac{\phi_p}{\phi_s} + R_g\right), & s_{\mathrm{dz}} < s \le s_{\max}. \end{cases}$$

For an upshift, $\dot{s} > 0$, so the available geometric reduction decreases as the belt moves outward on the primary pulley.

C.0.2 Geometry Families in the Radius Plane

The preceding result gives the local rate at which the geometric ratio changes once the pulley geometry is known. To interpret that result in terms of actual CVT design choices, it is useful to first ask which pulley-radius combinations are available in the first place.

Recall once more that the geometric CVT ratio is

$$R_g = \frac{r_s}{r_p}. \tag{C.0.6}$$

Thus, every point in the (r_p, r_s) plane corresponds to one geometric ratio, and every constant-ratio contour is a line of geometrically similar pulley states. This view is helpful because it separates three different design questions that are easy to blur together:

1. What geometric ratio does a given radius pair provide?
2. Which radius pairs are compatible with a selected belt and installed geometry?
3. How does the chosen compatible path influence the way ratio changes as the CVT shifts?

The first question is answered by the constant-R_g contours. The other two come from the belt-length compatibility relation. Once one of the global geometric quantities—belt length L_b or center-to-center distance C—is fixed, the compatibility relation defines a family of allowable radius pairs in the same plane.

Figure C.1 compares these two views directly. The left panel fixes the center distance and varies the belt length, while the right panel fixes the belt length and varies the center distance. In both panels, the solid contours are constant geometric ratio. The dashed families are the geometrically compatible radius pairs selected by the remaining design choice.

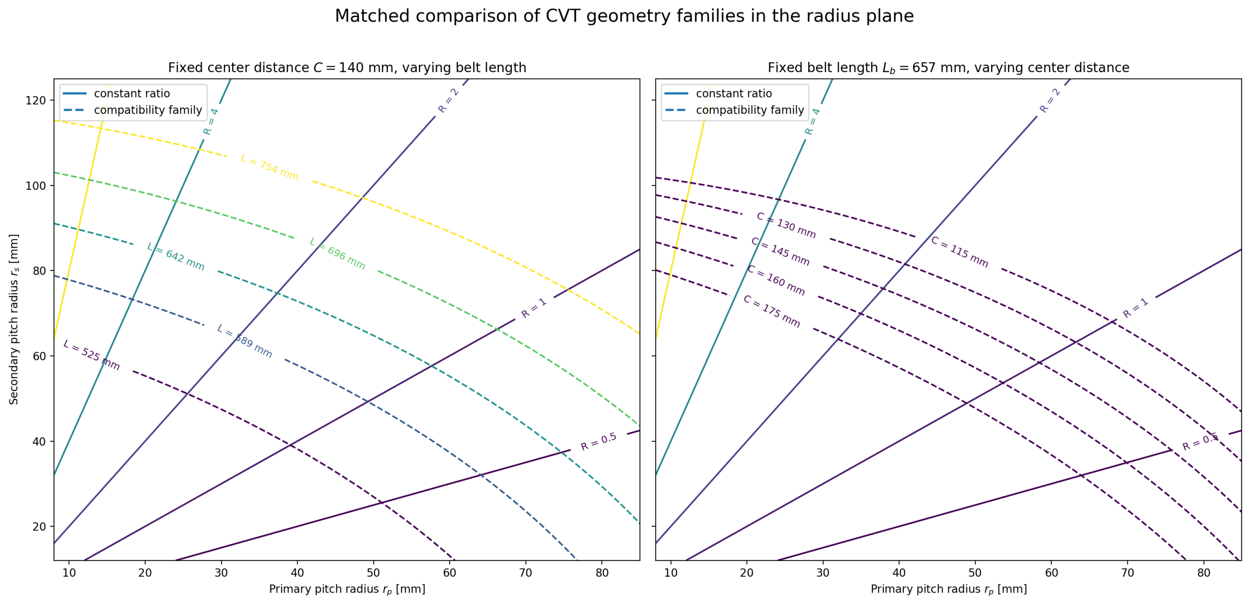


Figure C.1: Matched comparison of geometry families in the pulley-radius plane. In both panels, the solid contours are constant geometric ratio $R_g = r_s/r_p$. In (a), the center-to-center distance is fixed and the dashed families correspond to different belt lengths. In (b), the belt length is fixed and the dashed families correspond to different center-to-center distances.

This figure makes two points especially clear.

First, meeting a target ratio aperture is not only a matter of choosing two endpoint ratios. The selected belt and installed geometry determine which radius pairs are even reachable. A longer or shorter belt shifts the compatible family through the plane, and so does a change in center distance. Since the constant-ratio lines are not parallel to those compatibility families, changing L_b or C changes not only the reachable endpoints, but also the geometric route between them.

Second, the figure makes the roles of belt selection and packaging easier to distinguish. The belt length is largely a component-selection choice, while the center distance is largely a packaging choice. Both appear in the same compatibility relation, but they move the admissible family in different ways. This is why a target ratio range cannot be discussed cleanly without also specifying either the chosen belt or the available center-to-center distance.

In this sense, Figure C.1 is the static geometric view of the design problem. It shows what ratio states are available. The next question is dynamic in a geometric sense: once the CVT moves along one of those compatible paths, how much ratio change does a small increment of axial travel produce?

C.0.3 Exploring the Local Ratio-Change Map

The expression for $\dot{R}_g$ gives the instantaneous rate of change of the geometric ratio. To compare different parts of the geometry without committing to a particular time history, it is useful to divide out the shift speed and look instead at the local ratio change produced by a small increment of axial travel:

$$\frac{dR_g}{ds} = -\frac{1}{2\tan\beta\, r_p}\left(\frac{\phi_p}{\phi_s} + R_g\right). \tag{C.0.7}$$

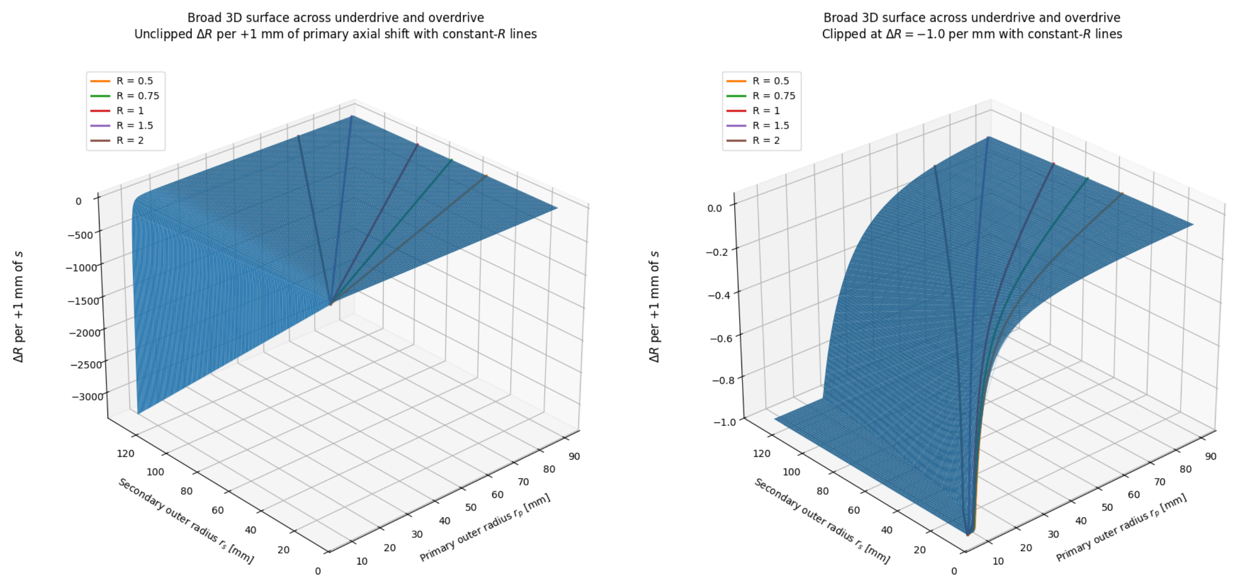
For a small positive increment Δs,

$$\Delta R_g \approx \frac{dR_g}{ds}\Delta s. \tag{C.0.8}$$

The plots below take $\Delta s = 1\,\mathrm{mm}$, so the displayed quantity may be read as the local change in geometric ratio caused by an additional 1 mm of primary closure. The negative sign simply reflects the usual upshift direction: increasing s reduces the geometric reduction ratio.

A sensitivity field over the radius plane The first view is a broad surface over the radius plane. Here, r_p and r_s are treated as independent coordinates so that every part of the geometry can be inspected. A real CVT does not wander arbitrarily across this surface; its fixed belt length and center distance constrain it to one belt-length-compatible path. Even so, the full surface is valuable because it shows where the geometry itself makes ratio change especially strong or weak.

Figure C.2 shows two versions of this surface. The unclipped view exposes the dominant trend: the local ratio change becomes much larger in magnitude as the primary radius becomes small. This follows directly from the explicit $1/r_p$ factor in Equation (C.0.7), and is reinforced by the fact that the geometric ratio $R_g = r_s/r_p$ is also largest in that same region. The clipped view hides the most extreme values so that the otherwise compressed mid-range curvature can be seen more clearly. The orange curve marks the $R_g = 1$ locus. It is a convenient reference, but not a special boundary in the local ratio-change law itself.



(a) Broad radius domain. The full scale exposes the rapidly increasing magnitude of local ratio change near small r_p.

(b) The same domain with values below -1 per 1 mm clipped, exposing the curvature through the mid-range.

Figure C.2: Local geometric-ratio change for a 1 mm increase in primary axial closure across a broad pulley-radius domain. The orange curve marks the $R_g = 1$ locus.

How belt choice and installed geometry select a path The broad surface is useful, but a particular CVT design only experiences the part of that field lying along its own compatible geometry path. This is where the earlier radius-plane view becomes directly relevant. Once a belt length and center distance are fixed, the CVT is constrained to one compatibility curve in the (r_p, r_s) plane. As the primary sheave closes, the operating point moves along that curve and samples the local ratio-change field point by point.

Figure C.3 overlays three representative belt-length-compatible paths on the local ratio-change surface. Each path corresponds to a different belt-length/reference-geometry choice. The figure shows that two designs with similar overall ratio aperture can still distribute their ratio change very differently over the available shift travel. One path may enter the steep small-r_p region early, concentrating a large amount of ratio change into a short portion of the stroke. Another may spend more of the stroke in the flatter region, distributing the same overall change more gradually.

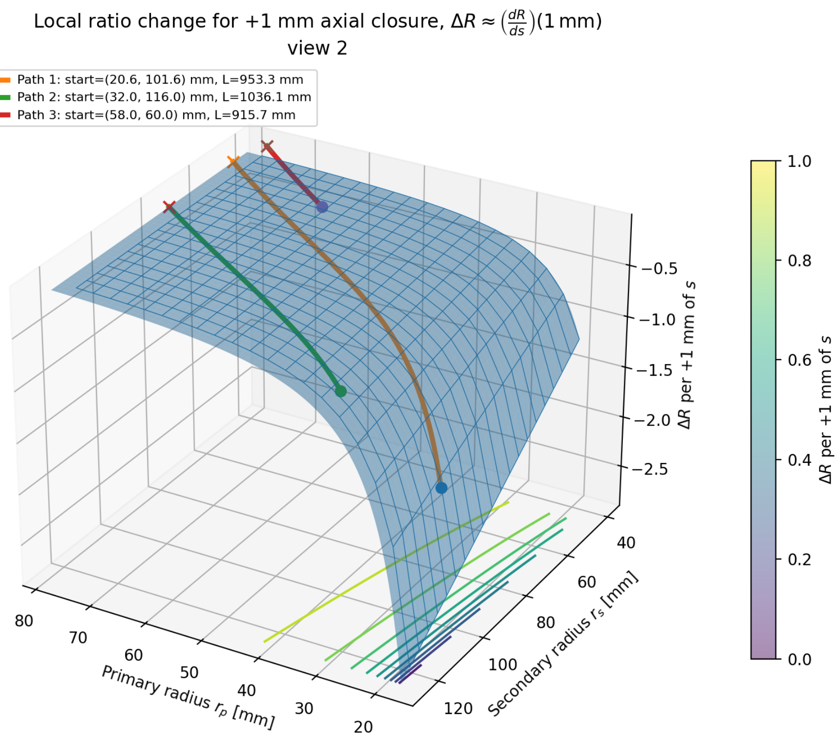
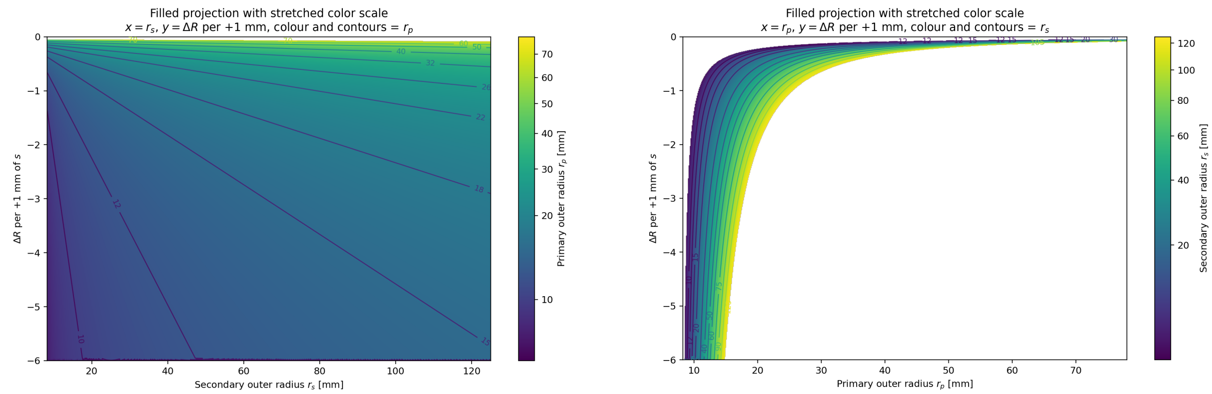


Figure C.3: Representative fixed-belt-length paths through the local ratio-change surface. Each coloured curve is a geometrically compatible pulley-radius path for a different belt-length/reference-geometry choice. The surface value gives the local change in geometric ratio associated with an additional 1 mm of primary closure at that point.

This is the key design value of the map: it does not merely say what overall ratio range is available. It shows *where along the shift stroke* that ratio change is spent.

Separating the roles of the two pulley radii The two projections in Figure C.4 help isolate the individual effects of r_p and r_s. In Figure C.4(a), the horizontal axis is the secondary radius and the contours identify the primary radius. For a fixed primary radius, increasing r_s increases the geometric ratio and makes the local ratio change more negative. The effect is visible, but the contour spacing shows that it remains secondary to the influence of the primary radius.

Figure C.4(b) makes the primary-radius effect more explicit. Every secondary-radius contour bends sharply downward as r_p approaches its lower limit, then flattens at larger radius. This is the visual form of the $1/r_p$ dependence in Equation (C.0.7). Thus, the small-primary-radius region is not only where the geometric reduction is large; it is also where the ratio becomes most sensitive to additional axial motion.



(a) Local ratio change against secondary radius. Colour and contours identify the primary radius.

(b) Local ratio change against primary radius. Colour and contours identify the secondary radius.

Figure C.4: Two projections of the local ratio-change surface for a 1 mm primary closure. The projections separate the contributions of the two radii and show the dominant sensitivity to small primary radius.

Design interpretation Taken together, Figure C.1 and Figure C.2–Figure C.4 give a more complete geometric design picture than endpoint ratios alone.

The radius-plane view answers the question of feasibility. It makes clear how a target ratio aperture depends on the selected belt length and the available center-to-center distance. A longer or shorter belt can move the admissible family enough to change both the reachable ratio range and the portion of the plane through which the CVT shifts. Likewise, a packaging change in center distance can move the same belt into a very different family of compatible radius pairs.

The ratio-change view answers the question of distribution. Once a design selects its belt and installed geometry, the compatible path through the plane determines where ratio change is concentrated. A path that reaches very small r_p can produce large ratio change in little axial travel, which may be attractive when launch or climbing performance demands strong reduction with limited shift stroke. A path that stays farther from that corner distributes ratio change more gently and may be easier to control or to package within a desired sheave travel.

This does *not* mean that the smallest possible r_p is always best. The present plots are geometric only. They show the leverage of the geometry, but not its cost. Experimental work by Chen et al. (1998) showed that, for their rubber-belt CVT, the highest transmission efficiency occurred near a 1:1 speed ratio, reminding us that the most attractive geometric reduction state is not automatically the most efficient operating state. More directly, Childs and Cowburn (1987) identified additional loss penalties at small pulley radii in V-belt drives, with belt-carcass warping and shearing proposed as important contributors. In other words, the same small-r_p region that offers strong geometric leverage is also the region where the belt is bent and seated most severely.

For that reason, these plots are best used as an early design lens rather than a final design verdict. They help screen combinations of belt length, center distance, and pulley radii by showing not only how much ratio range is available, but how that range is distributed over the stroke. Later traction, loss, and dynamic analyses can then be used to judge whether the geometrically attractive part of the design space is physically supportable (Gerbert, 1996; Bertini et al., 2014; Messick, 2018).

C.0.4 Acceleration of the Geometric CVT Ratio

The ratio rate can change for two reasons. The primary sheave may accelerate, and the pulley geometry continues to change as the belt moves. Differentiate Equation (C.0.2) term by term to capture both effects.

For the first term, $\dot{r}_s/r_p$, use the quotient rule:

$$\frac{d}{dt}\left(\frac{\dot{r}_s}{r_p}\right) = \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s\dot{r}_p}{r_p^2}. \tag{C.0.9}$$

For the second term, $r_s\dot{r}_p/r_p^2$, all three factors may change. Using the product rule,

$$\frac{d}{dt}\left(\frac{r_s\dot{r}_p}{r_p^2}\right) = \frac{\dot{r}_s\dot{r}_p}{r_p^2} + \frac{r_s\ddot{r}_p}{r_p^2} - \frac{2r_s\dot{r}_p^2}{r_p^3}. \tag{C.0.10}$$

Subtracting Equation (C.0.10) from Equation (C.0.9) gives

$$\begin{aligned}\ddot{R}_g &= \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s\dot{r}_p}{r_p^2} - \left[\frac{\dot{r}_s\dot{r}_p}{r_p^2} + \frac{r_s\ddot{r}_p}{r_p^2} - \frac{2r_s\dot{r}_p^2}{r_p^3}\right] \\ &= \frac{\ddot{r}_s}{r_p} - \frac{2\dot{r}_s\dot{r}_p}{r_p^2} - \frac{r_s\ddot{r}_p}{r_p^2} + \frac{2r_s\dot{r}_p^2}{r_p^3}. \end{aligned} \tag{C.0.11}$$

The main derivation gives the required coupled secondary motion in **????**:

$$\dot{r}_s = -\frac{\phi_p}{\phi_s}\dot{r}_p, \tag{C.0.12}$$

$$\ddot{r}_s = -\frac{\phi_p}{\phi_s}\ddot{r}_p - \frac{8\pi^2}{\phi_s^3\sqrt{C^2-(r_s-r_p)^2}}\dot{r}_p^2. \tag{C.0.13}$$

Substituting these into Equation (C.0.11) gives

$$\begin{aligned}\ddot{R}_g &= \frac{1}{r_p}\left[-\frac{\phi_p}{\phi_s}\ddot{r}_p - \frac{8\pi^2}{\phi_s^3\sqrt{C^2-(r_s-r_p)^2}}\dot{r}_p^2\right] - \frac{2}{r_p^2}\left[-\frac{\phi_p}{\phi_s}\dot{r}_p\right]\dot{r}_p \\ &\quad - \frac{r_s}{r_p^2}\ddot{r}_p + \frac{2r_s}{r_p^3}\dot{r}_p^2 \\ &= -\frac{1}{r_p}\left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p}\right)\ddot{r}_p + \left[\frac{2}{r_p^2}\left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p}\right) - \frac{8\pi^2}{\phi_s^3 r_p\sqrt{C^2-(r_s-r_p)^2}}\right]\dot{r}_p^2 \\ &= -\frac{1}{r_p}\left(\frac{\phi_p}{\phi_s} + R_g\right)\ddot{r}_p + \left[\frac{2}{r_p^2}\left(\frac{\phi_p}{\phi_s} + R_g\right) - \frac{8\pi^2}{\phi_s^3 r_p\sqrt{C^2-(r_s-r_p)^2}}\right]\dot{r}_p^2. \end{aligned} \tag{C.0.14}$$

Finally, use the primary radius rate and acceleration from Equations (3.1.36) and (3.1.66):

Acceleration of Geometric CVT Ratio **(C.0.15)**

$$\ddot{R}_g = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ -\dfrac{\ddot{s}}{2\tan\beta\, r_p}\left(\dfrac{\phi_p}{\phi_s} + R_g\right) + \dfrac{\dot{s}^2}{2\tan^2\beta\, r_p^2}\left(\dfrac{\phi_p}{\phi_s} + R_g\right) - \dfrac{2\pi^2\dot{s}^2}{\phi_s^3 r_p \tan^2\beta\sqrt{C^2 - (r_s - r_p)^2}}, & s_{\mathrm{dz}} < s \le s_{\mathrm{max}}. \end{cases}$$

The ratio acceleration has two sources. The term proportional to $\ddot{s}$ is the direct conversion of primary-sheave acceleration into ratio acceleration. Its magnitude is scaled by the local ratio sensitivity dR_g/ds, so the same axial acceleration produces a larger ratio response where the geometric ratio changes rapidly with shift.

The terms proportional to $\dot{s}^2$ arise even when the primary sheave moves at constant axial speed. They reflect the curvature of the geometric relation $R_g(s)$: as the belt moves through the sheaves, the ratio change produced by each additional increment of axial travel need not remain constant. In the usual upshift direction, the large ratio sensitivity associated with small primary radius progressively weakens as the primary radius grows. The ratio therefore often changes rapidly at the beginning of the shift and then eases off, even without a change in axial shift speed.

Appendix D

Construction of Primary Flyweight Geometry Maps

The primary actuator dynamics derived in the main text do not require a particular flyweight shape directly. They require the mechanism maps

$$q_f = q_f(x_p), \qquad \frac{dq_f}{dx_p}, \qquad \frac{d^2 q_f}{dx_p^2}, \qquad J_{f,p} = J_{f,p}(x_p), \qquad \frac{dJ_{f,p}}{dx_p}, \tag{D.0.1}$$

together with the constant pivot-axis inertia I_f. This appendix constructs those quantities for the fixed-pivot roller flyweight used by the present primary mechanism.

The construction follows the physical mechanism directly. A ramp attached to the movable sheave defines the contact surface. The finite roller radius offsets that surface to a roller-center locus. Intersecting that locus with the rigid flyweight-arm circle determines the assembled flyweight pose and therefore $q_f(x_p)$. Differentiating the same geometric constraint supplies the motion ratios required by the acceleration-level actuator model. Finally, the flyweight mass distribution is evaluated at the resulting pose to obtain I_f, $J_{f,p}(x_p)$, and $dJ_{f,p}/dx_p$.

No independent initial flyweight angle is introduced. The assembled pose is a consequence of the pivot, ramp, roller, and arm geometry.

D.1 Fixed-Pivot Flyweight Geometry

Consider the axial–radial plane of one primary flyweight. Let O denote the shaft centre, let $+\boldsymbol{e}_x$ point in the positive primary closing direction, and let $+\boldsymbol{e}_r$ point radially outward. The flyweight pivots about the fixed point

$$\boldsymbol{r}_P = \begin{bmatrix} x_P \\ r_P \end{bmatrix}, \tag{D.1.1}$$

which is fixed to the pulley body. The movable sheave carries the physical ramp. The roller centre is denoted by C, the physical roller–ramp contact point by K, and the reference point at the start of the active ramp by A.

The rigid distance from the pivot to the roller centre is

$$\boxed{L_f = |PC|}, \tag{D.1.2}$$

and the roller radius is R_{roll}. The flyweight angle q_f is measured from the positive axial reference to the line from P to C, with increasing q_f corresponding to outward flyweight rotation. Thus

$$\boldsymbol{r}_C(q_f) = \boldsymbol{r}_P + L_f \begin{bmatrix} \cos q_f \\ \sin q_f \end{bmatrix}. \tag{D.1.3}$$

The arm is therefore constrained to a circle of radius L_f about P.

D.1.1 Physical Ramp Representation

Introduce a scalar ramp coordinate ξ, with A corresponding to $\xi = 0$. Let $h(\xi)$ be the radial profile measured from A, and let $d_x \in \{-1, +1\}$ specify whether increasing ξ proceeds in the positive or negative physical axial direction. For movable-sheave closure x_p, a point on the physical ramp is

$$\boxed{\boldsymbol{r}_K(\xi, x_p) = \begin{bmatrix} x_A + x_p + d_x\xi \\ r_A + h(\xi) \end{bmatrix}.} \tag{D.1.4}$$

The present Baja parameterization uses $d_x = -1$: increasing profile coordinate proceeds axially toward the pivot while the ramp rises radially outward.

Ramp angle is measured from the axial reference line, so $0°$ denotes an axial ramp. For a graph of the form in Equation (D.1.4), the local unsigned ramp angle satisfies

$$\alpha(\xi) = \tan^{-1} |h'(\xi)|. \tag{D.1.5}$$

Accordingly, a $20°$ ramp is nearly axial rather than nearly radial.

The unnormalized tangent is

$$\boldsymbol{t}(\xi) = \begin{bmatrix} d_x \\ h'(\xi) \end{bmatrix}, \qquad \|\boldsymbol{t}\| = \sqrt{1 + [h'(\xi)]^2}. \tag{D.1.6}$$

With $\sigma_n \in \{-1, +1\}$ selecting the physical side of the ramp on which the roller lies, a compatible unit normal is

$$\boxed{\boldsymbol{n}(\xi) = \sigma_n \frac{\begin{bmatrix} -h'(\xi) \\ d_x \end{bmatrix}}{\sqrt{1 + [h'(\xi)]^2}}.} \tag{D.1.7}$$

The sign σ_n is fixed once from the assembled mechanism and is then used throughout the active ramp range.

The profile h may be assembled from straight, circular, spline, or other smooth segments. Those shapes define only the physical contact surface; they do not themselves prescribe the flyweight motion or force.

D.1.2 Finite-Radius Roller-Centre Locus

The roller centre does not lie on the physical ramp. At contact it is offset by one roller radius along the selected ramp normal:

$$\boxed{\boldsymbol{r}_C(\xi, x_p) = \boldsymbol{r}_K(\xi, x_p) + R_{\text{roll}}\boldsymbol{n}(\xi).} \tag{D.1.8}$$

This finite-radius offset is the roller-centre path used by the mechanism. The point K remains the physical contact point; C is the centre of the roller bearing.

The contact problem can now be viewed geometrically as the intersection of the roller-centre offset curve with the rigid circle of radius L_f about P. A point-contact or zero-radius follower approximation is not used for the present mechanism.

D.1.3 Rigid-Arm Contact Constraint and Assembled Branch

A roller-centre location is compatible with the rigid flyweight when

$$\boxed{g(\xi, x_p) = \|\boldsymbol{r}_C(\xi, x_p) - \boldsymbol{r}_P\|^2 - L_f^2 = 0.} \tag{D.1.9}$$

For any fixed x_p, every root of Equation (D.1.9) is a mathematical flyweight configuration that satisfies the rigid arm and roller contact geometry.

Define

$$a = x_C - x_P, \qquad b = r_C - r_P. \tag{D.1.10}$$

Then

$$a^2 + b^2 = L_f^2, \tag{D.1.11}$$

and the corresponding flyweight angle is

$$\boxed{q_f(x_p) = \operatorname{atan2}(b, a).} \tag{D.1.12}$$

Thus the starting angle is recovered from the assembled geometry rather than specified as an independent parameter.

Multiple mathematical roots can exist at the same x_p. At the beginning of the declared operating interval, $x_p = x_{p,\min}$, enumerate the admissible contact configurations and initialize the assembled branch with

$$\boxed{q_f(x_{p,\min}) = \min_i q_{f,i}(x_{p,\min}).} \tag{D.1.13}$$

This smallest-q_f rule is an *initial assembly rule only*. For increasing or decreasing closure thereafter, the same solution branch is followed continuously. The flyweight is not permitted to jump to a disconnected arm orientation simply because another root has a smaller angle at a later position.

Alternate arm orientations that separately satisfy Equation (D.1.9) are not themselves a contact failure. They are mathematically distinct assembled configurations. A different condition occurs when the *same selected roller pose* contacts two distinct locations of the ramp simultaneously; that true two-point contact lies outside the present single-contact mechanism.

For the selected branch the geometry chain is therefore

$$\boxed{h(\xi) \longrightarrow \boldsymbol{r}_C(\xi, x_p) \longrightarrow g(\xi, x_p) = 0 \longrightarrow q_f(x_p).} \tag{D.1.14}$$

D.2 Contact-Branch Kinematics

The main actuator dynamics require $q_f'(x_p)$ and $q_f''(x_p)$. These quantities follow from the same physical contact constraint rather than from an independent fitted motion law.

Along the selected branch,

$$g(\xi(x_p), x_p) = 0. \tag{D.2.1}$$

Differentiating with respect to x_p gives

$$g_\xi \xi' + g_x = 0, \tag{D.2.2}$$

where $g_x \equiv \partial g/\partial x_p$, and therefore

$$\boxed{\xi' = -\frac{g_x}{g_\xi}.} \tag{D.2.3}$$

Differentiating once more gives

$$g_{xx} + 2g_{x\xi}\xi' + g_{\xi\xi}(\xi')^2 + g_\xi \xi'' = 0, \tag{D.2.4}$$

and hence

$$\boxed{\xi'' = -\frac{g_{xx} + 2g_{x\xi}\xi' + g_{\xi\xi}(\xi')^2}{g_\xi}.} \tag{D.2.5}$$

These relations require

$$\boxed{g_\xi \neq 0} \tag{D.2.6}$$

along the selected branch. A zero value corresponds to a fold or dead-centre at which the local contact map cannot be continued smoothly in x_p.

For compactness write

$$\boldsymbol{c}(\xi, x_p) \equiv \boldsymbol{r}_C(\xi, x_p). \tag{D.2.7}$$

Its total first and second derivatives along the selected branch are

$$\boxed{\boldsymbol{c}' = \boldsymbol{c}_x + \boldsymbol{c}_\xi \xi',} \tag{D.2.8}$$

and

$$\boxed{\boldsymbol{c}'' = \boldsymbol{c}_{xx} + 2\boldsymbol{c}_{x\xi}\xi' + \boldsymbol{c}_{\xi\xi}(\xi')^2 + \boldsymbol{c}_\xi \xi''.} \tag{D.2.9}$$

The explicit x_p-dependence is particularly simple because the ramp carrier undergoes rigid axial translation, but the chain-rule form above keeps the construction general and makes the geometric origin of every derivative clear.

Using a and b from Equation (D.1.10), the first angle derivative is

$$\boxed{q_f' = \frac{ab' - ba'}{L_f^2}.} \tag{D.2.10}$$

Because $a^2 + b^2 = L_f^2$ is constant, the second derivative reduces to

$$\boxed{q_f'' = \frac{ab'' - ba''}{L_f^2}.} \tag{D.2.11}$$

The angle, motion ratio, and change of motion ratio are therefore all direct consequences of the ramp, roller, pivot, and arm geometry.

D.2.1 Smoothness of the Dynamic Ramp Representation

The finite roller-centre curve contains the normal $\boldsymbol{n}(h')$. Its first profile derivative therefore contains h'', while its second profile derivative contains h'''. Since $q_f''(x_p)$ depends on $\boldsymbol{c}_{\xi\xi}$, the acceleration-level mechanism map requires the active ramp representation to satisfy

$$\boxed{h \in C^3} \tag{D.2.12}$$

through piecewise junctions used by the production dynamic map. A discontinuity in slope, curvature, or the third derivative can contaminate the acceleration terms even when the physical profile itself appears visually continuous.

When idealized or measured ramp pieces do not meet this requirement directly, a short artificial smoothing segment may be introduced explicitly as a model approximation. The provisional Baja ramp used during development consists of

$$20^\circ \text{ linear over } 5\,\text{mm} \longrightarrow 3\,\text{mm } C^3 \text{ blend} \longrightarrow 35^\circ \to 10^\circ \text{ circular over approximately } 30\,\text{mm}. \tag{D.2.13}$$

The blend is represented by a sixth-order polynomial transition selected so that derivatives through third order match the neighbouring segments. The 3 mm blend length is a modelling choice, not a measured hardware dimension. A sharp-corner finite-roller construction may still be useful for geometry diagnostics, but it is not admitted directly into the production map while q_f'' is retained.

D.3 Flyweight Mass Properties

Once $q_f(x_p)$ is known, the remaining mechanism quantities follow from the flyweight mass distribution. Introduce body-fixed coordinates relative to the pivot P: u along the nominal arm from the pivot toward the roller centre, v perpendicular to u in the axial–radial mechanism plane, and z circumferentially along the pivot axis.

For one flyweight define

$$m_f = \int dm, \tag{D.3.1}$$

$$M_u = \int u\,dm, \qquad M_v = \int v\,dm, \tag{D.3.2}$$

$$S_u = \int u^2\,dm, \qquad S_v = \int v^2\,dm, \tag{D.3.3}$$

$$S_{uv} = \int uv\,dm, \qquad S_z = \int z^2\,dm. \tag{D.3.4}$$

These first and second moments may be obtained from a simplified analytic mass model, direct measurement, or CAD.

For n_f identical flyweights, the inertia about their individual pivot axes is

$$\boxed{I_f = n_f(S_u + S_v).} \tag{D.3.5}$$

This is the inertia multiplying the relative pivot acceleration in the primary actuator balance.

At flyweight angle q_f, a mass element has radial shaft coordinate

$$r = r_P + u \sin q_f + v \cos q_f. \tag{D.3.6}$$

Including its circumferential offset z, integration gives the total flyweight inertia about the primary shaft:

$$\boxed{\begin{aligned} J_{f,p}(q_f) = n_f \Big[& m_f r_P^2 + 2r_P(M_u \sin q_f + M_v \cos q_f) \\ & + S_u \sin^2 q_f + S_v \cos^2 q_f \\ & + 2S_{uv} \sin q_f \cos q_f + S_z \Big]. \end{aligned}} \tag{D.3.7}$$

Composition with the geometry map gives

$$\boxed{J_{f,p}(x_p) = J_{f,p}(q_f(x_p)),} \tag{D.3.8}$$

and

$$\boxed{\frac{dJ_{f,p}}{dx_p} = \frac{dJ_{f,p}}{dq_f} q_f'.} \tag{D.3.9}$$

The same mass description therefore supplies both the pivot inertia and the configuration-dependent shaft-axis inertia required by the dynamic actuator.

D.3.1 Present Baja Mass Approximation

The present parameterization does not attempt to reproduce the exact outline and centroidal inertia of every flyweight component. For each flyweight, the arm/body mass m_a is represented as a uniform slender member extending from $u = 0$ to $u = L_f$, while the hardware carried at the outer end is collected into a concentrated tip mass m_t at $u = L_f$, $v = z = 0$. The tip package is

$$m_t = m_{\text{roller/bearing}} + m_{\text{bolt}} + m_{\text{nut/washer}} + m_{\text{fixed hardware}} + m_{\text{tuning}}. \tag{D.3.10}$$

For the uniform arm,

$$M_{u,a} = \frac{m_a L_f}{2}, \qquad S_{u,a} = \frac{m_a L_f^2}{3}, \tag{D.3.11}$$

and for the concentrated tip package,

$$M_{u,t} = m_t L_f, \qquad S_{u,t} = m_t L_f^2. \tag{D.3.12}$$

Hence

$$\boxed{M_u = \frac{m_a L_f}{2} + m_t L_f,} \tag{D.3.13}$$

and

$$\boxed{S_u = \frac{m_a L_f^2}{3} + m_t L_f^2.} \tag{D.3.14}$$

with

$$M_v = S_v = S_{uv} = S_z = 0, \qquad m_f = m_a + m_t. \tag{D.3.15}$$

The resulting pivot inertia is

$$\boxed{I_f = n_f \left(\frac{1}{3} m_a L_f^2 + m_t L_f^2 \right).} \tag{D.3.16}$$

The corresponding shaft-axis inertia map is

$$\boxed{J_{f,p}(q_f) = n_f \left[(m_a + m_t)r_P^2 + 2r_P M_u \sin q_f + S_u \sin^2 q_f\right].} \tag{D.3.17}$$

Its configuration derivative is

$$\boxed{\frac{dJ_{f,p}}{dq_f} = 2n_f \cos q_f \left(r_P M_u + S_u \sin q_f\right),} \tag{D.3.18}$$

so the required axial derivative becomes

$$\boxed{\frac{dJ_{f,p}}{dx_p} = 2n_f \cos q_f \left(r_P M_u + S_u \sin q_f\right) q_f'.} \tag{D.3.19}$$

This reduction is an explicit approximation to the hardware mass distribution. It neglects the centroidal inertia associated with the finite roller/bearing radius, bolt diameter and length, nut and washer dimensions, tuning-weight dimensions, arm thickness and nonuniform outline, holes and reliefs, and any off-centre placement of tip hardware not represented by the roller-centre station. These features are not assumed physically absent; they are omitted when reducing the real flyweight to the mass moments above.

For the present assembly, the tuning mass is of order > 100 g, while the measured arm/body mass is approximately 13.646 g per flyweight. The parallel-axis contribution of the tip package is therefore expected to dominate many small centroidal corrections, although this observation does not make the omitted terms identically zero. If greater fidelity is needed, the same governing equations accept CAD- or measurement-derived $M_u, M_v, S_u, S_v, S_{uv}, S_z$ without changing the actuator dynamics.

D.3.2 Present Geometry Values

The provisional geometry used during development is summarized below. These values identify the mechanism currently being parameterized; the ramp blend and unverified tip masses should not be interpreted as final metrology.

Quantity	Present value
Number of flyweights, n_f	3
Pivot radius, r_P	1.675 in = 42.545 mm
Pivot-to-roller-centre length, L_f	1.241 in = 31.5214 mm
Roller radius, R_{roll}	6.5 mm
Ramp-reference offset from pivot	$\Delta x_A = 1.5$ in, $\Delta r_A = 0.2776$ in
Measured arm/body mass, m_a	13.646 g per flyweight
Profile direction	$d_x = -1$
Provisional active ramp	20° linear over 5 mm, 3 mm C^3 blend, then circular 35° → 10° over approximately 30 mm

Table D.1: Present fixed-pivot flyweight geometry and provisional ramp data. The 3 mm smoothing length is a modelling choice rather than a measured hardware dimension.

The roller/bearing, bolt, nut/washer, fixed tip-hardware, and tuning masses are entered individually when measured and then collected into m_t through Equation (D.3.10).

D.4 Resulting Mechanism Interface

The construction above supplies exactly the quantities required by the primary actuator dynamics:

$$\boxed{q_f(x_p), \quad q_f'(x_p), \quad q_f''(x_p), \quad I_f, \quad J_{f,p}(x_p), \quad J_{f,p}'(x_p).} \tag{D.4.1}$$

For the fixed-pivot mechanism, the complete chain is

$$\boxed{\text{ramp geometry} \to \text{roller-centre locus} \to \text{rigid-arm branch} \to q_f, q_f', q_f'' \to \text{mass moments} \to I_f, J_{f,p}, J_{f,p}'.} \tag{D.4.2}$$

Substitution into the main actuator result in Equation (3.2.11) gives

$$\boxed{F_{p,\mathrm{fly}} = \frac{1}{2}\omega_p^2 J_{f,p}' - I_f (q_f')^2 \ddot{x}_p - I_f q_f' q_f'' \dot{x}_p^2.} \tag{D.4.3}$$

The three terms are respectively the centrifugal generalized closing drive, the pivot inertia reflected into the axial coordinate, and the inertial term created by variation of the mechanism motion ratio.

The same configuration-dependent mass distribution contributes to the owning shaft through

$$\boxed{J_{f,p}\dot{\omega}_p + J_{f,p}'\dot{x}_p\,\omega_p,} \tag{D.4.4}$$

which is the local x_p-coordinate form of the primary angular-momentum term used in Equation (3.3.4). Under the assumed circumferential symmetry, the corresponding flyweight kinetic energy may be written

$$\boxed{T_f = \frac{1}{2}J_{f,p}\omega_p^2 + \frac{1}{2}I_f (q_f'\dot{x}_p)^2.} \tag{D.4.5}$$

Replacing the simplified mass moments with higher-fidelity CAD or measured moments changes only the supplied maps, not these governing relations.

D.5 Mechanism Admissibility and Assumptions

The maps in Equation (D.4.1) are valid only while the geometric, mass, and unilateral-contact assumptions of the mechanism remain satisfied. The conditions below define the admissible mechanism used by the present model.

D.5.1 Geometry and Branch Admissibility

All physical dimensions and masses must be finite, with

$$L_f > 0, \qquad R_{\mathrm{roll}} > 0, \qquad r_P > 0, \qquad n_f \geq 1, \tag{D.5.1}$$

and nonnegative component masses. The declared ramp domain and primary travel interval must both be nonempty.

At $x_p = x_{p,\min}$, at least one contact solution of Equation (D.1.9) must exist. The physical branch is selected there using Equation (D.1.13) and must then continue continuously through the entire declared travel. Loss of that branch or a jump to a disconnected arm orientation is inadmissible.

The selected branch must remain regular:

$$\boxed{g_\xi \neq 0, \qquad 0 < q_f'(x_p) < \infty, \qquad |q_f''(x_p)| < \infty.} \tag{D.5.2}$$

The sign requirement corresponds to the present fixed-pivot mechanism, for which primary closure rotates the flyweight outward. A zero or negative motion ratio indicates a dead-centre, reversal, or incompatible branch for this mechanism definition.

The finite roller offset must also remain regular. Let κ_n denote the signed ramp curvature consistent with the selected normal. The normal-offset regularity factor must satisfy

$$\boxed{1 - R_{\text{roll}}\kappa_n \neq 0} \tag{D.5.3}$$

through the selected contact range. Operation close to this singular condition should be treated as geometrically sensitive even before equality is reached.

The selected contact coordinate must remain within the physical ramp extent for the full primary travel. In practice, the active ramp should extend beyond the selected contact path with an appropriate geometric and manufacturing margin.

The rigid arm condition itself is exact up to numerical tolerance:

$$\boxed{\|\boldsymbol{r}_C - \boldsymbol{r}_P\| = L_f.} \tag{D.5.4}$$

It is not an approximate fitted arm length.

D.5.2 No Penetration or Simultaneous Double Contact

For the selected roller pose, no second portion of the physical ramp may penetrate the roller. Conceptually, for every distinct ramp coordinate $\tilde{\xi}$ away from the selected contact,

$$\boxed{\left\|\boldsymbol{r}_C - \boldsymbol{r}_K(\tilde{\xi}, x_p)\right\| \geq R_{\text{roll}}.} \tag{D.5.5}$$

A strict violation indicates interference. Equality at a second distinct ramp location indicates true simultaneous two-point contact and lies outside the present single-contact model. This condition is different from the existence of multiple arm orientations that separately solve the contact equation.

D.5.3 Mass and Inertia Admissibility

The supplied mass description must satisfy

$$\boxed{I_f \geq 0, \qquad J_{f,p}(x_p) > 0} \tag{D.5.6}$$

throughout the selected branch. These conditions follow naturally for the nonnegative arm-plus-tip approximation. If arbitrary CAD moments are supplied, they must correspond to a physically realizable nonnegative mass distribution.

Flyweight mass and inertia represented here must not be included again in the movable-sheave translating mass, the constant rigid pulley rotational inertia, or another actuator inertia term. This separation prevents double counting of the same kinetic energy.

D.5.4 Unilateral Contact-Force Admissibility

Geometric compatibility alone does not guarantee that the ramp can maintain the constrained roller contact during every dynamic state. Under the sign convention used in the main text, compressive ramp contact requires

$$\boxed{F_{p,\mathrm{fly}} \geq 0.} \tag{D.5.7}$$

If $F_{p,\mathrm{fly}} < 0$, the constrained solution would require the ramp to pull on the roller. That state is physically inadmissible. The result is not replaced by $\max(0, F_{p,\mathrm{fly}})$, since clipping the force would hide a change in mechanism topology.

A complete treatment of such a state would introduce flyweight lift-off, free-flight, re-contact, and any associated impact dynamics. Those modes are not retained here. Until such an extension is introduced, negative contact force is treated as an explicit admissibility failure. The condition $F_{p,\mathrm{fly}} = 0$ marks the unilateral-contact boundary.

D.5.5 Mechanism and Contact Assumptions

The fixed-pivot construction uses the following assumptions:

- the flyweight pivot is fixed relative to the pulley body;
- the ramp translates rigidly with the movable sheave;
- the pivot-to-roller-centre length is rigid and constant;
- the roller has finite radius and the flyweight motion is planar in the axial–radial plane;
- repeated flyweights are identical and arranged with sufficient circumferential symmetry that omitted cross and gyroscopic terms cancel;
- structural deformation of the arm, pivot, roller, ramp, and carrier is neglected;
- backlash and manufacturing clearance in the kinematic constraint are neglected;
- the dynamic ramp representation is C^3 over the active contact range;
- ramp contact is an ideal geometric, frictionless constraint in the flyweight force law;
- roller-bearing loss, roller spin inertia, local Hertzian compliance, and wear are neglected;
- only one active roller–ramp contact is retained; and
- the selected contact must remain compressive.

For the present Baja mass approximation, the arm/body is additionally treated as a uniform slender member and all outer hardware is concentrated at the roller-centre station. The finite dimensions and centroidal inertias listed in Appendix D.3.1 are consequently omitted. A higher fidelity mass description may replace this reduction without changing the geometry or dynamics developed above.

D.6 Relation to Existing Actuator Models

The geometry and mass reductions used here are not intended as novelty claims by themselves. Detailed CVT actuator models already use measured geometry, centre-of-mass locations, contact construction, and mechanical advantage to recover flyweight thrust. For example, Messick (2018) derives primary thrust from measured/CAD flyweight geometry under quasi-equilibrium, Kim et al. (2002) retain detailed roller/contact geometry and contact friction in an equilibrium actuator treatment, and Skinner (2020) resolves the flyweight, supporting link, and ramp reactions with explicit free bodies while representing the flyweight and link centrifugal loading through their mass and working radii. These approaches establish the use of reduced mass and geometry information for detailed actuator force prediction.

If the pivot inertia is neglected and the mass distribution is reduced to a point or centre-of-mass description at radius $r_f(x_p)$, then

$$J_{f,p} = m_f r_f^2 \tag{D.6.1}$$

and the centrifugal term reduces to

$$\boxed{\frac{1}{2}\omega_p^2 \frac{dJ_{f,p}}{dx_p} = m_f r_f \omega_p^2 \frac{dr_f}{dx_p}.} \tag{D.6.2}$$

This is the familiar centrifugal force-times-motion-ratio form used by quasi-static actuator descriptions. The extension developed here is that the same geometry and mass maps are consumed by the transient balance while retaining the flyweight's pivoting inertia, changing mechanism motion ratio, and configuration-dependent shaft angular momentum. The distinction is therefore dynamic coupling of the actuator to the shift and rotational equations, not the existence of curved ramps, finite rollers, or detailed flyweight geometry.